

Canonical W configurations

species: electron

source configurations: 1156

canonical configurations collected: 74

canonical display criterion: exactly 3 retained final-state components

retained-component cutoff: fraction $\geq 2.0\%$

normalized amplitude display: bar length = $|A_{\text{tilde}}|$, bar color/text = phase

canonical configurations by polarization cluster:

P2: 18

P3: 22

P5: 15

P6: 19

all retained-component count distribution:

3 components: 74 configurations

4 components: 109 configurations

5 components: 29 configurations

6 components: 832 configurations

7 components: 109 configurations

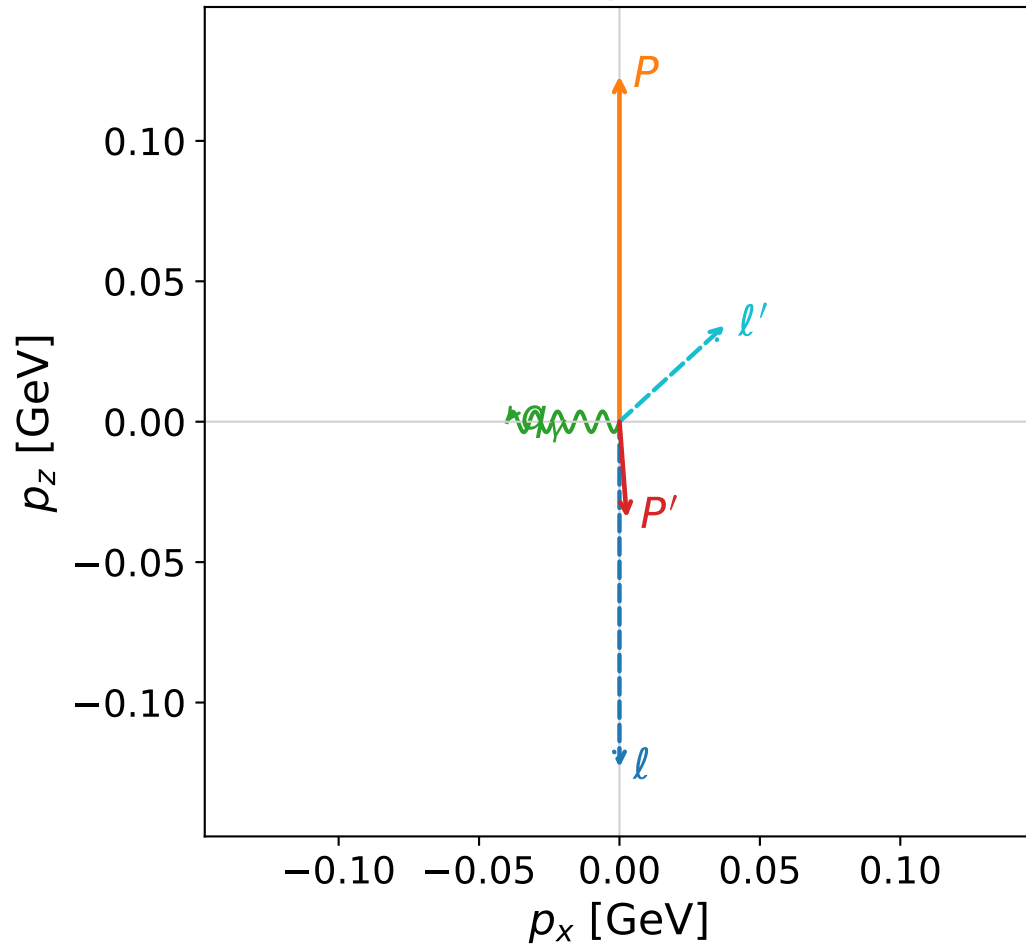
8 components: 3 configurations

CSV index: Output\GradientPhaseSpaceScan\electron\Data\W\dw\canonical\canonical_dw_configurations_electron.csv

Each following page is one complete momentum, kinematic-summary, and normalized-amplitude configuration.

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

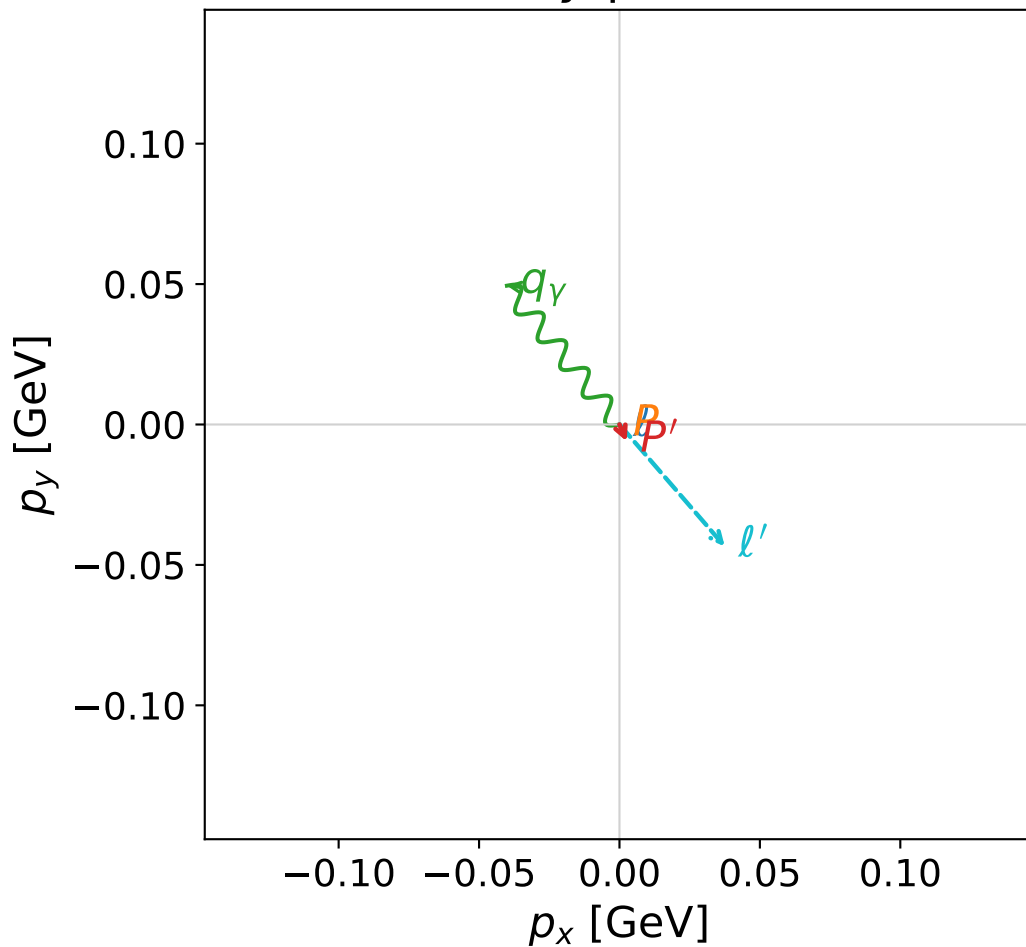


dw_mixing_angles_polarization_02_local_minimum_379
 $D_W=0.00121395$
 region: polarization_cluster_02_local_minimum_379
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0379
 lepton mixing angle: $\alpha_e=0.79117094$ rad
 incoming lepton: $0.703013|+\rangle + 0.711177|-\rangle$
 proton mixing angle: $\alpha_p=1.7484036$ rad
 incoming proton: $-0.176675|+\rangle + 0.984269|-\rangle$

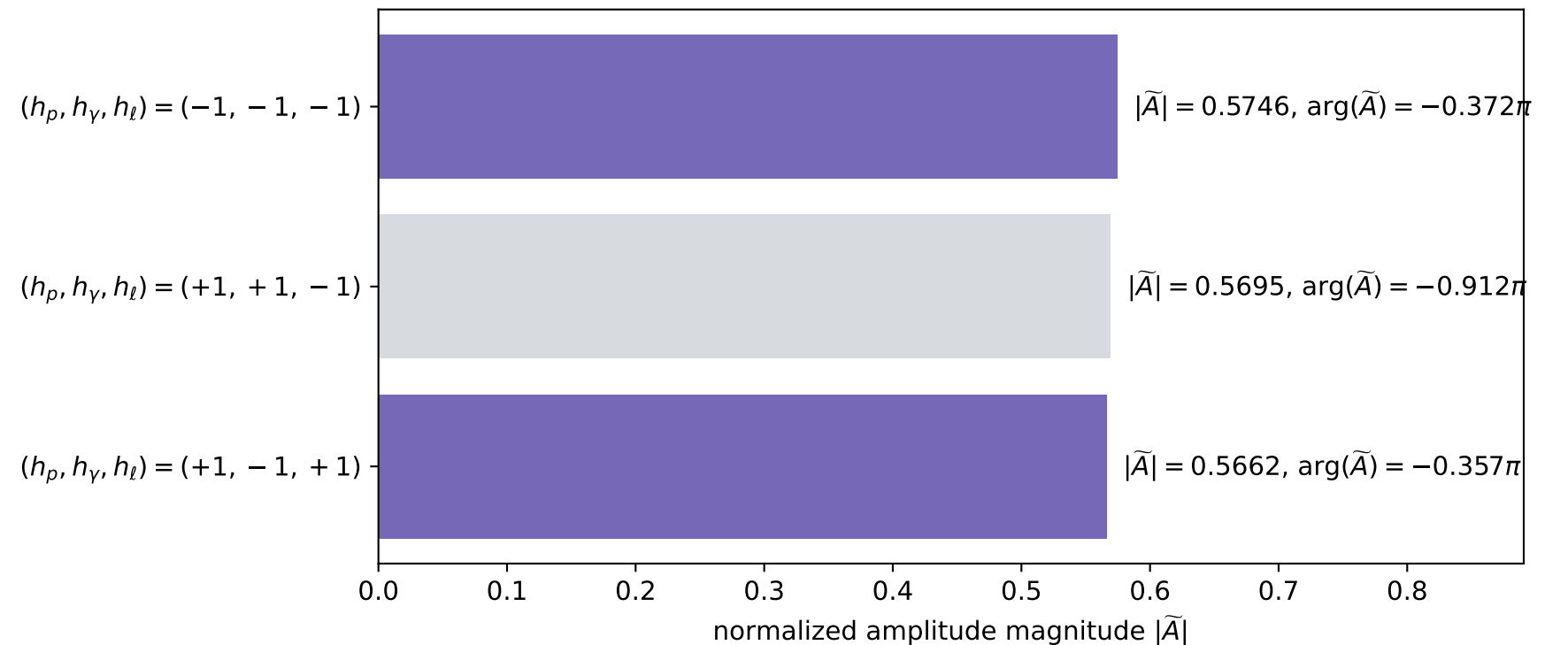
$s=1.14298$, $\sqrt{s}=1.0691$, $|\vec{P}|=0.123061$, $|\vec{P}'|=0.0348211$
 $\theta_{p'}=2.95231$, $\theta_\gamma=1.5721$, $\phi_{p'}=5.11662$, $\phi_\gamma=2.25205$
 $E_\gamma=0.063631$, $\theta_{l'}=1.03207$, $\phi_{l'}=5.42489$
 $Q^2=0.0248839$, $x_B=0.171454$, $t=-0.0247191$, $W^2=1.00009$, $y=0.551566$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E=0.1231$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.1231$
 P $E=0.9460$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1231$
 l' $E=0.0668$ $p_x=0.0375$ $p_y=-0.0434$ $p_z=0.0343$
 P' $E=0.9386$ $p_x=0.0026$ $p_y=-0.0060$ $p_z=-0.0342$
 q_γ $E=0.0636$ $p_x=-0.0401$ $p_y=0.0494$ $p_z=-0.0001$

x--y plane

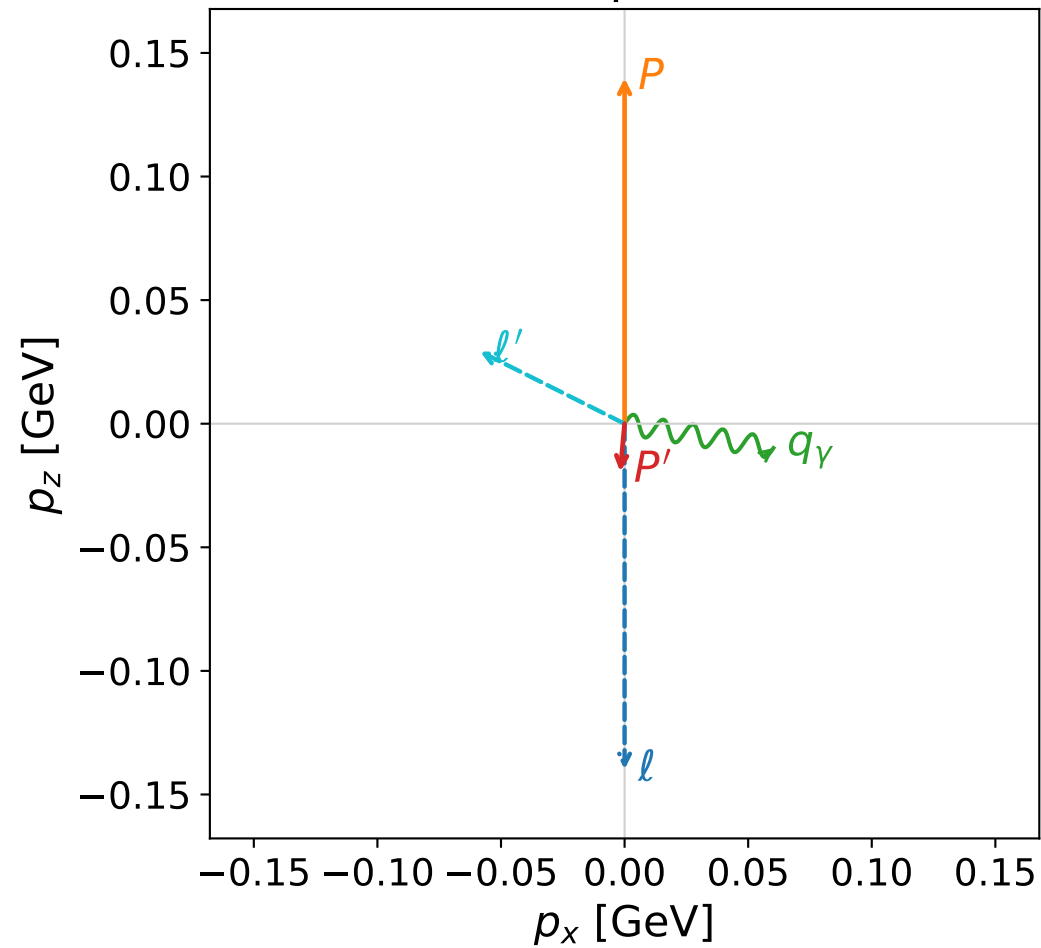


Leading normalized final-state amplitudes
 (N=3, retained=97.5%)

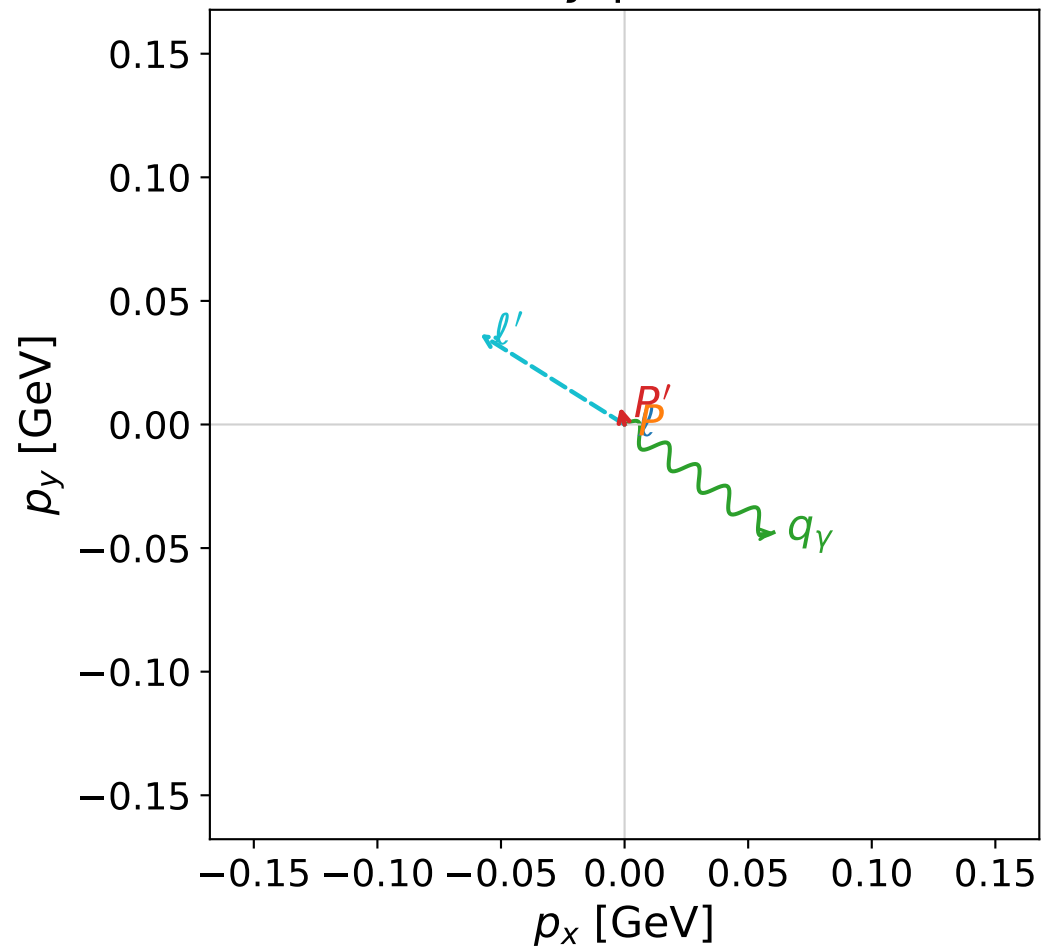


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



x--y plane

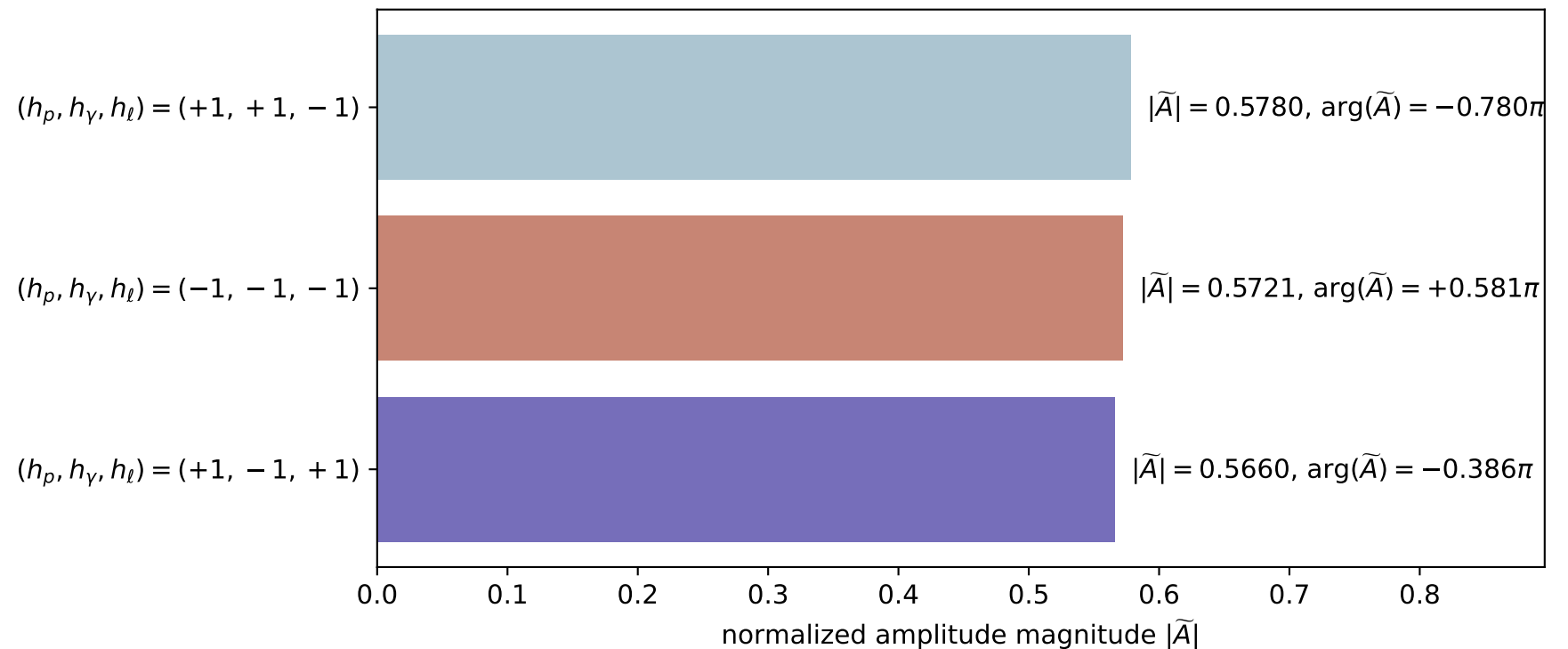


dw_mixing_angles_polarization_02_local_minimum_442
 $D_W=0.00176181$
 region: polarization_cluster_02_local_minimum_442
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0442
 lepton mixing angle: $\alpha_e=0.80580383$ rad
 incoming lepton: $0.692532|+\rangle + 0.721388|-\rangle$
 proton mixing angle: $\alpha_p=1.3763217$ rad
 incoming proton: $0.193251|+\rangle + 0.981149|-\rangle$

$s=1.18413$, $\sqrt{s}=1.08818$, $|\vec{P}|=0.139813$, $|\vec{P}'|=0.0206987$
 $\theta_{p'}=2.78097$, $\theta_\gamma=1.70231$, $\phi_{p'}=1.81532$, $\phi_\gamma=5.65562$
 $E_\gamma=0.0750484$, $\theta_{\ell'}=1.17019$, $\phi_{\ell'}=2.5822$
 $Q^2=0.0291113$, $x_B=0.170854$, $t=-0.0252891$, $W^2=1.02112$, $y=0.55996$

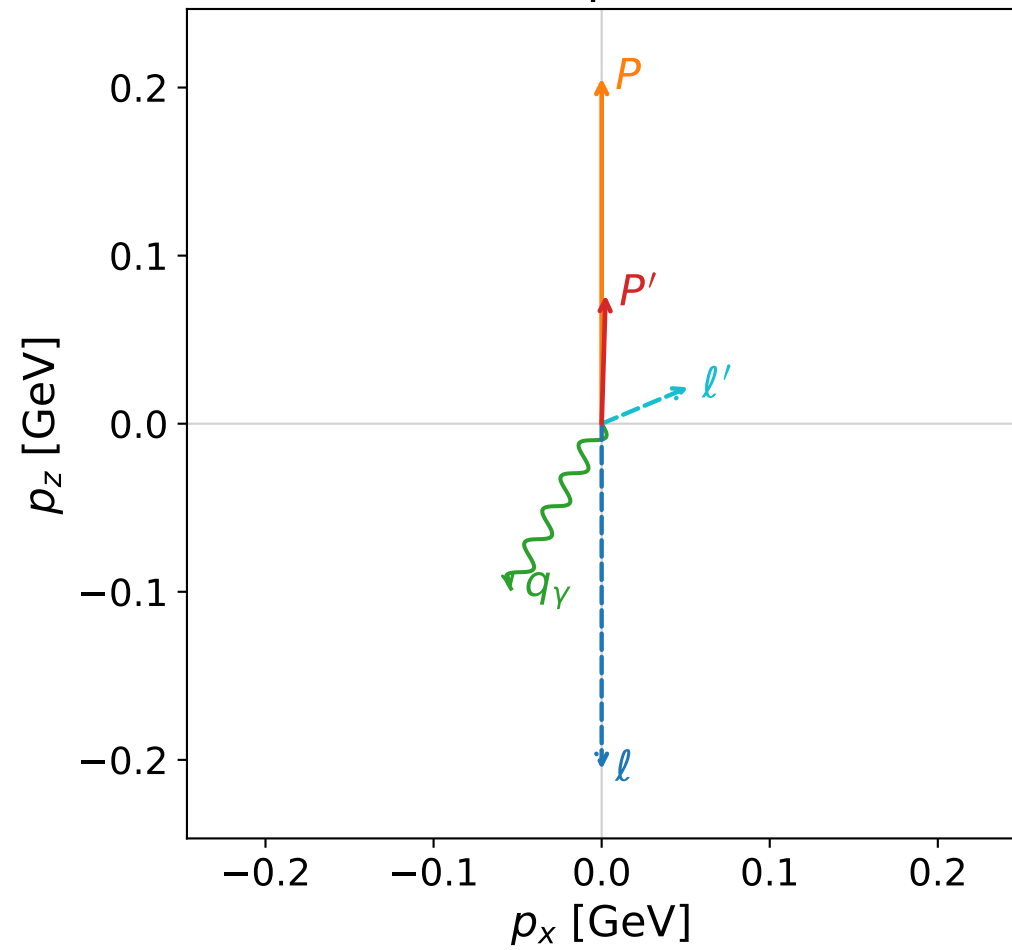
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1398$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1398$
 P $E= 0.9484$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1398$
 ℓ' $E= 0.0749$ $p_x= -0.0585$ $p_y= 0.0366$ $p_z= 0.0292$
 P' $E= 0.9382$ $p_x= -0.0018$ $p_y= 0.0071$ $p_z= -0.0194$
 q_γ $E= 0.0750$ $p_x= 0.0602$ $p_y= -0.0437$ $p_z= -0.0098$

Leading normalized final-state amplitudes
 (N=3, retained=98.2%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

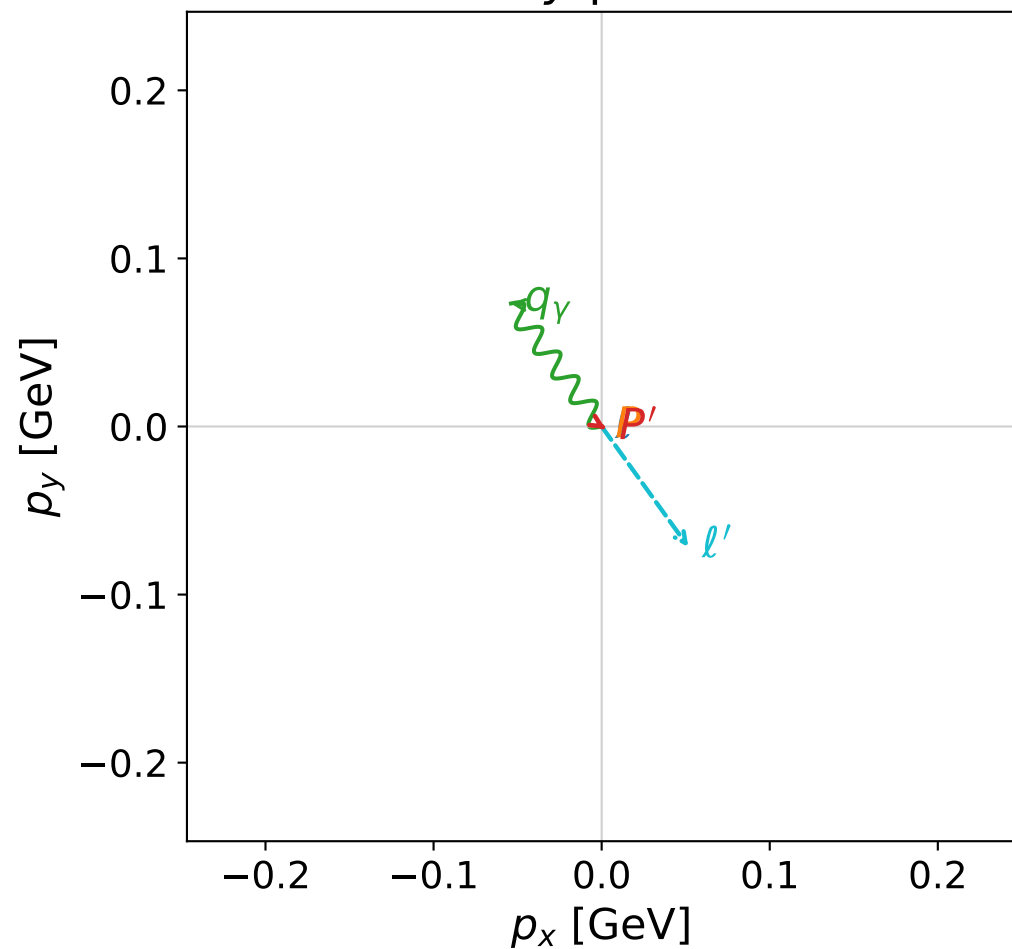


dw_mixing_angles_polarization_02_local_minimum_590
 $D_W=0.00392354$
 region: polarization_cluster_02_local_minimum_590
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0590
 lepton mixing angle: $\alpha_e=0.78087704$ rad
 incoming lepton: $0.710296|+\rangle + 0.703903|-\rangle$
 proton mixing angle: $\alpha_p=1.6435266$ rad
 incoming proton: $-0.0726662|+\rangle + 0.997356|-\rangle$

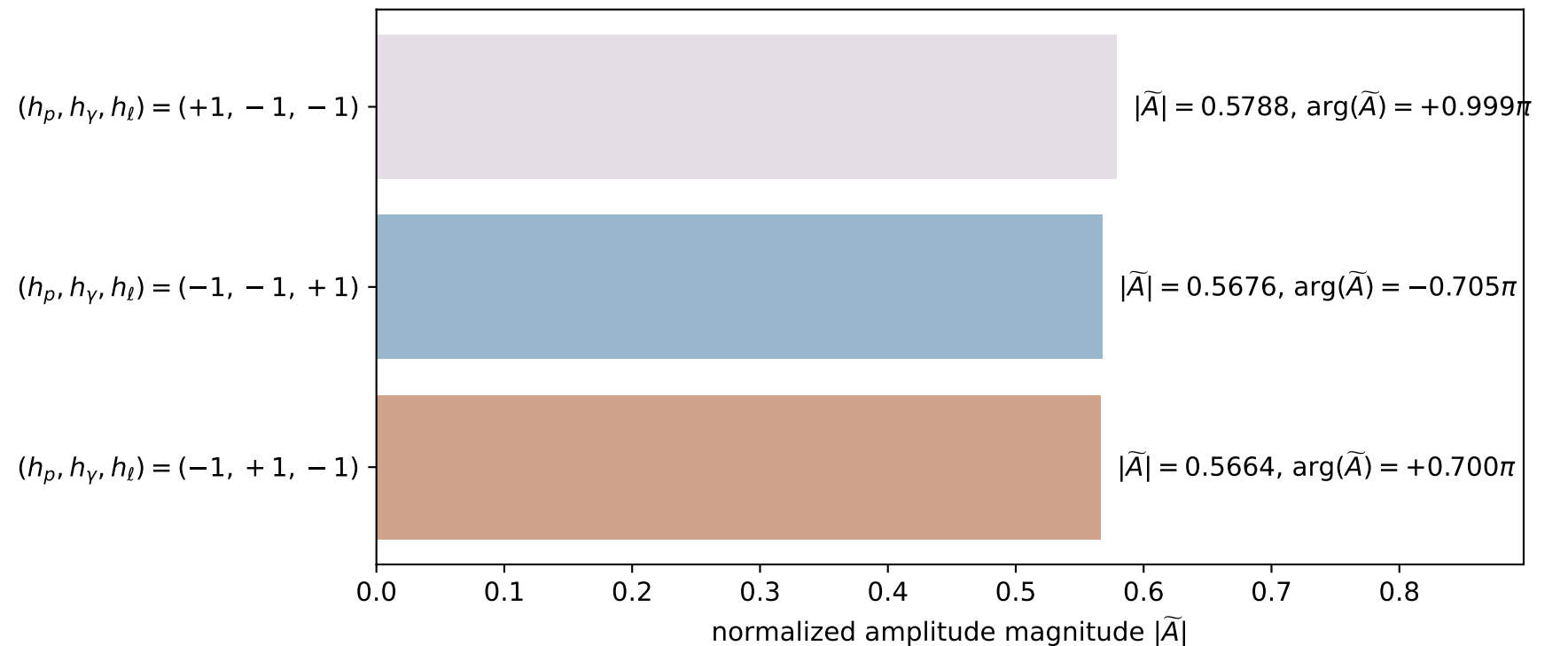
$s=1.35928$, $\sqrt{s}=1.16588$, $|\vec{P}|=0.205609$, $|\vec{P}'|=0.0765314$
 $\theta_{p'}=0.036252$, $\theta_\gamma=2.39442$, $\phi_{p'}=5.72703$, $\phi_\gamma=2.20606$
 $E_\gamma=0.133793$, $\theta_{l'}=1.33026$, $\phi_{l'}=5.33602$
 $Q^2=0.0463199$, $x_B=0.147689$, $t=-0.0163149$, $W^2=1.14715$, $y=0.654171$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.2056$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2056$
 P $E= 0.9603$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.2056$
 l' $E= 0.0910$ $p_x= 0.0516$ $p_y= -0.0717$ $p_z= 0.0217$
 P' $E= 0.9411$ $p_x= 0.0024$ $p_y= -0.0015$ $p_z= 0.0765$
 q_γ $E= 0.1338$ $p_x= -0.0540$ $p_y= 0.0732$ $p_z= -0.0982$

x--y plane

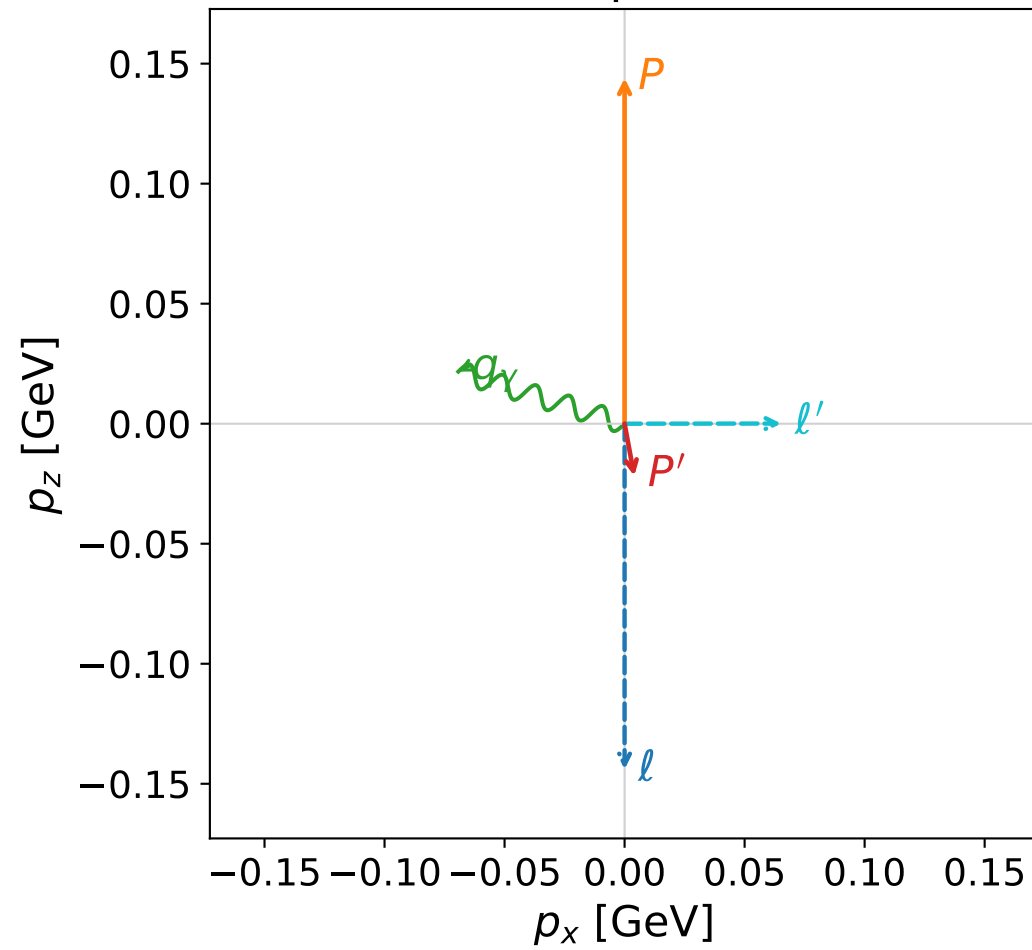


Leading normalized final-state amplitudes
 (N=3, retained=97.8%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

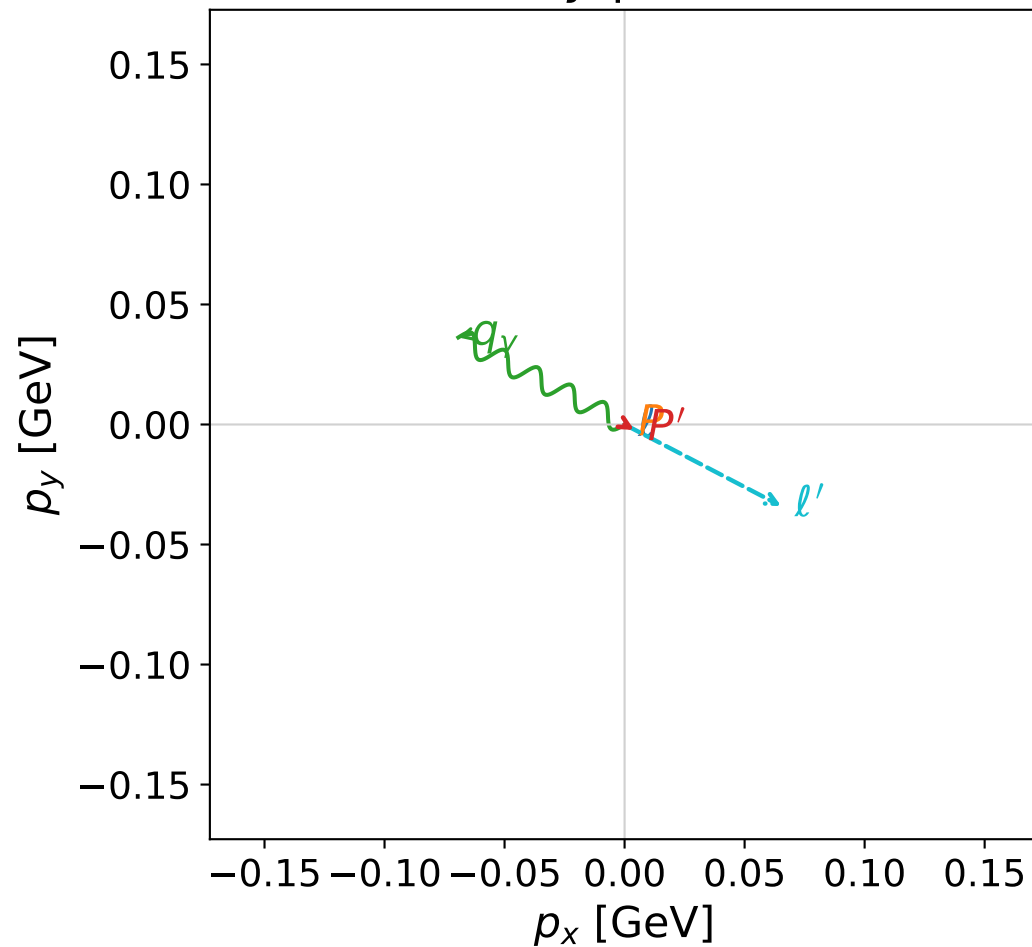


dw_mixing_angles_polarization_02_local_minimum_592
 $D_W=0.00392984$
 region: polarization_cluster_02_local_minimum_592
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0592
 lepton mixing angle: $\alpha_e=0.77959771$ rad
 incoming lepton: $0.711196|+\rangle +0.702993|-\rangle$
 proton mixing angle: $\alpha_p=1.6772067$ rad
 incoming proton: $-0.10621|+\rangle +0.994344|-\rangle$

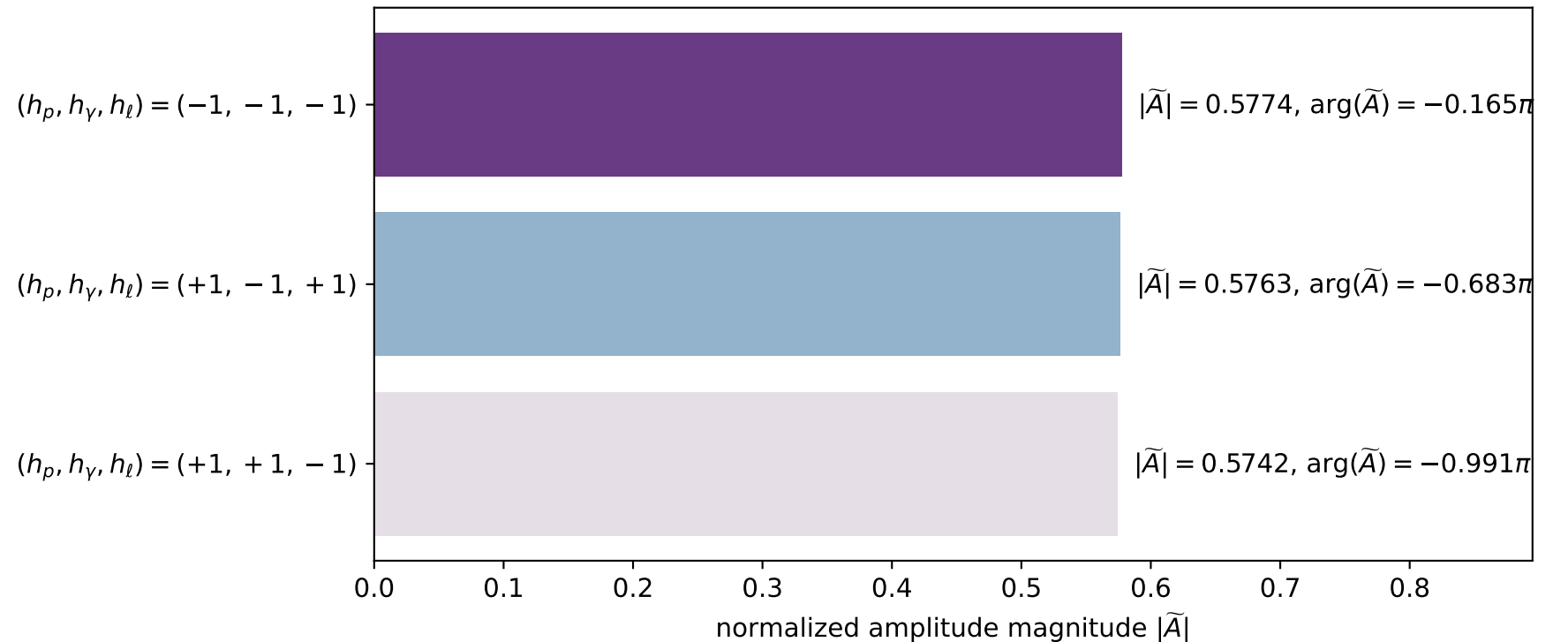
$s=1.19455$, $\sqrt{s}=1.09296$, $|\vec{P}|=0.14397$, $|\vec{P}'|=0.022253$
 $\theta_{p'}=2.93211$, $\theta_\gamma=1.30008$, $\phi_{p'}=5.76646$, $\phi_\gamma=2.65937$
 $E_\gamma=0.0811348$, $\theta_{l'}=1.56985$, $\phi_{l'}=5.80313$
 $Q^2=0.0211997$, $x_B=0.121059$, $t=-0.0273752$, $W^2=1.03376$, $y=0.55645$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E=0.1440$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.1440$
 P $E=0.9490$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1440$
 l' $E=0.0736$ $p_x=0.0652$ $p_y=-0.0340$ $p_z=0.0001$
 P' $E=0.9383$ $p_x=0.0040$ $p_y=-0.0023$ $p_z=-0.0218$
 q_γ $E=0.0811$ $p_x=-0.0693$ $p_y=0.0363$ $p_z=0.0217$

x--y plane

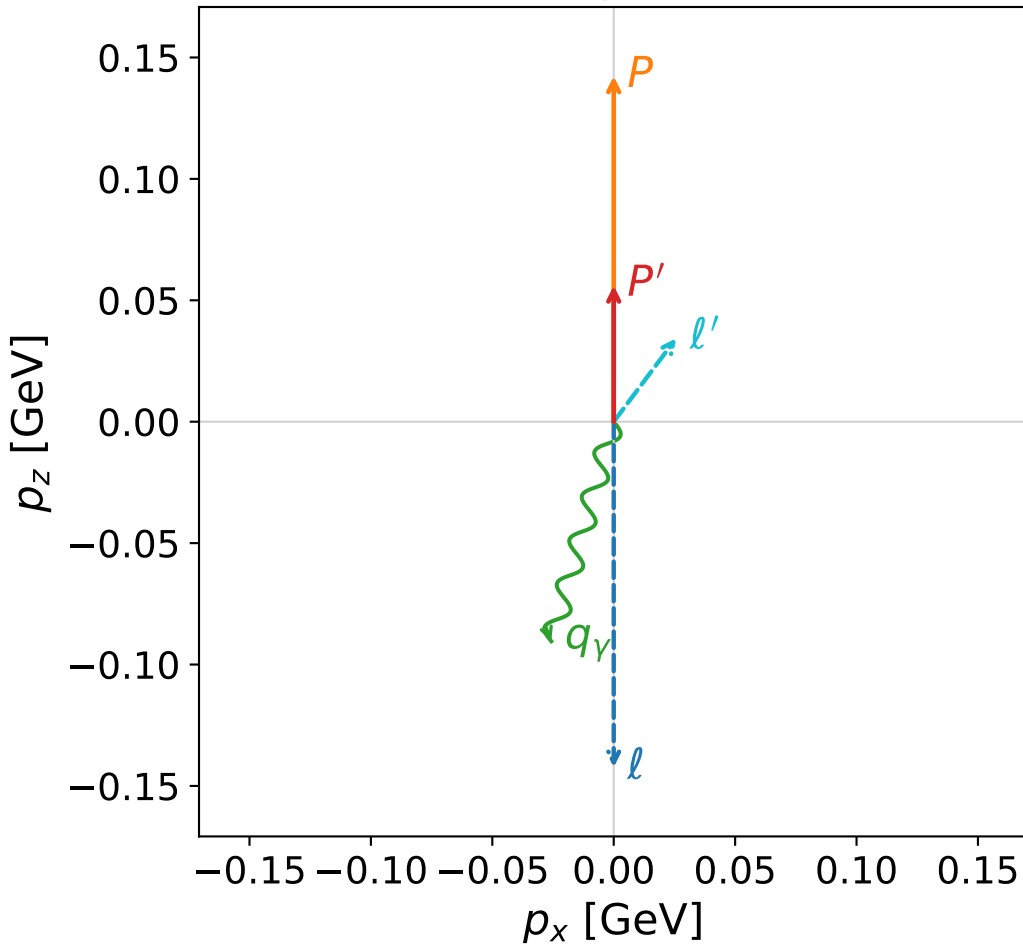


Leading normalized final-state amplitudes
(N=3, retained=99.5%)

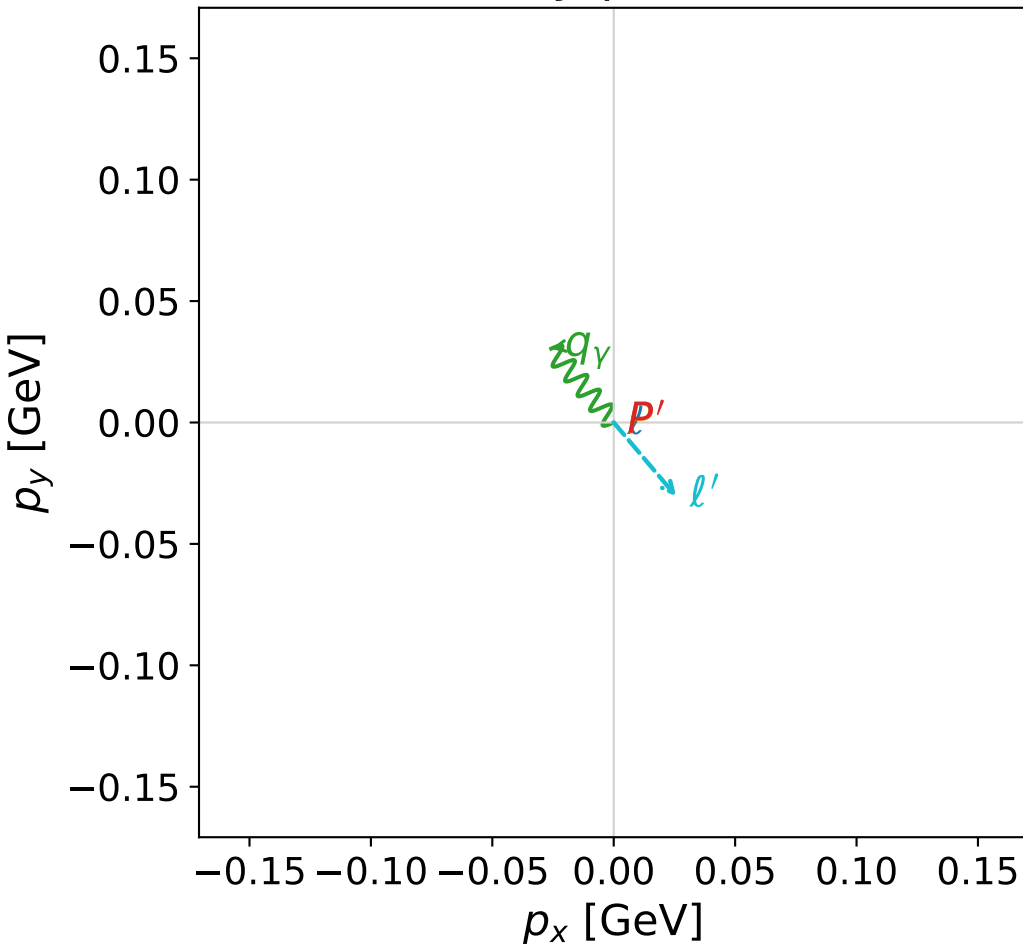


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



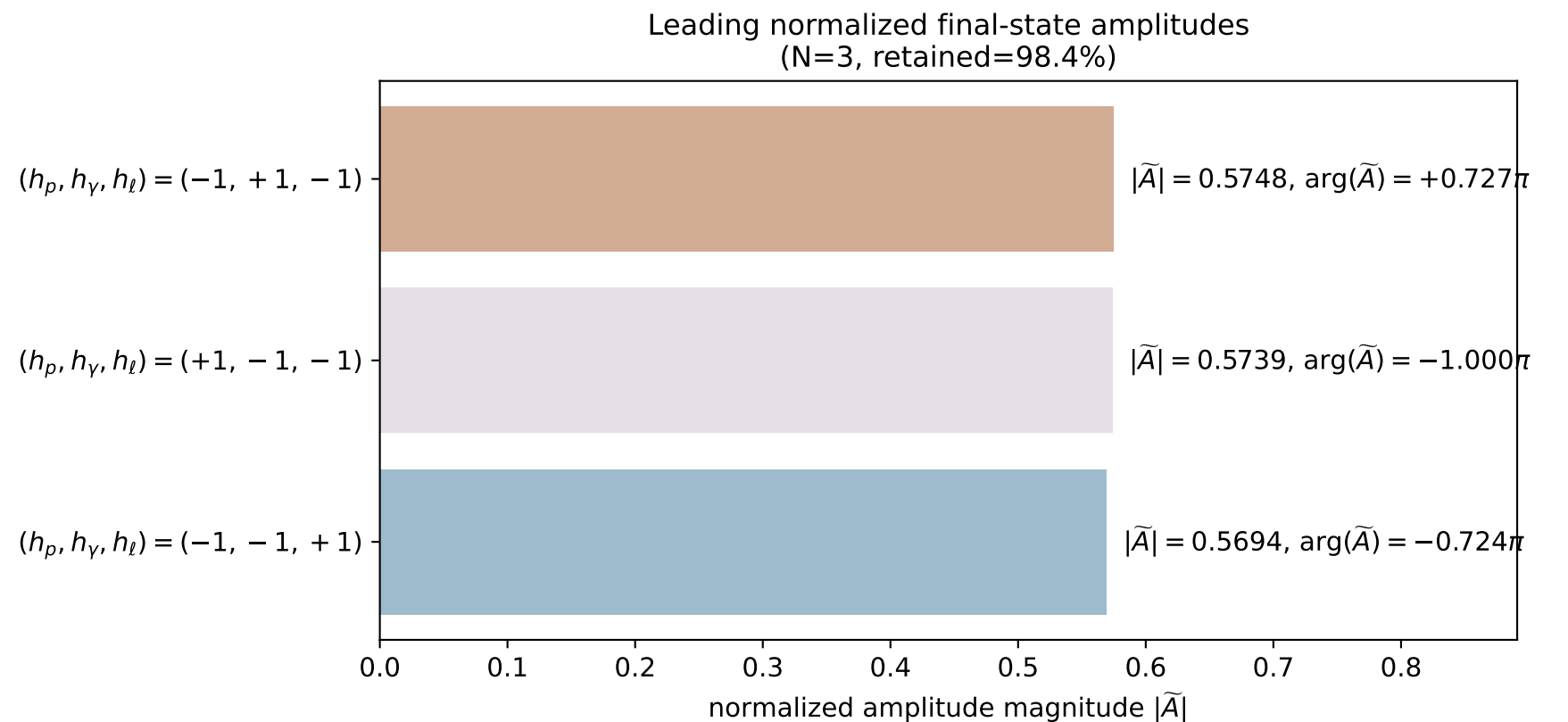
x--y plane



dw_mixing_angles_polarization_02_local_minimum_693
 $D_W=0.00687147$
 region: polarization_cluster_02_local_minimum_693
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0693
 lepton mixing angle: $\alpha_e=0.78199713$ rad
 incoming lepton: $0.709508|+\rangle +0.704698|-\rangle$
 proton mixing angle: $\alpha_p=1.6520513$ rad
 incoming proton: $-0.0811656|+\rangle +0.996701|-\rangle$

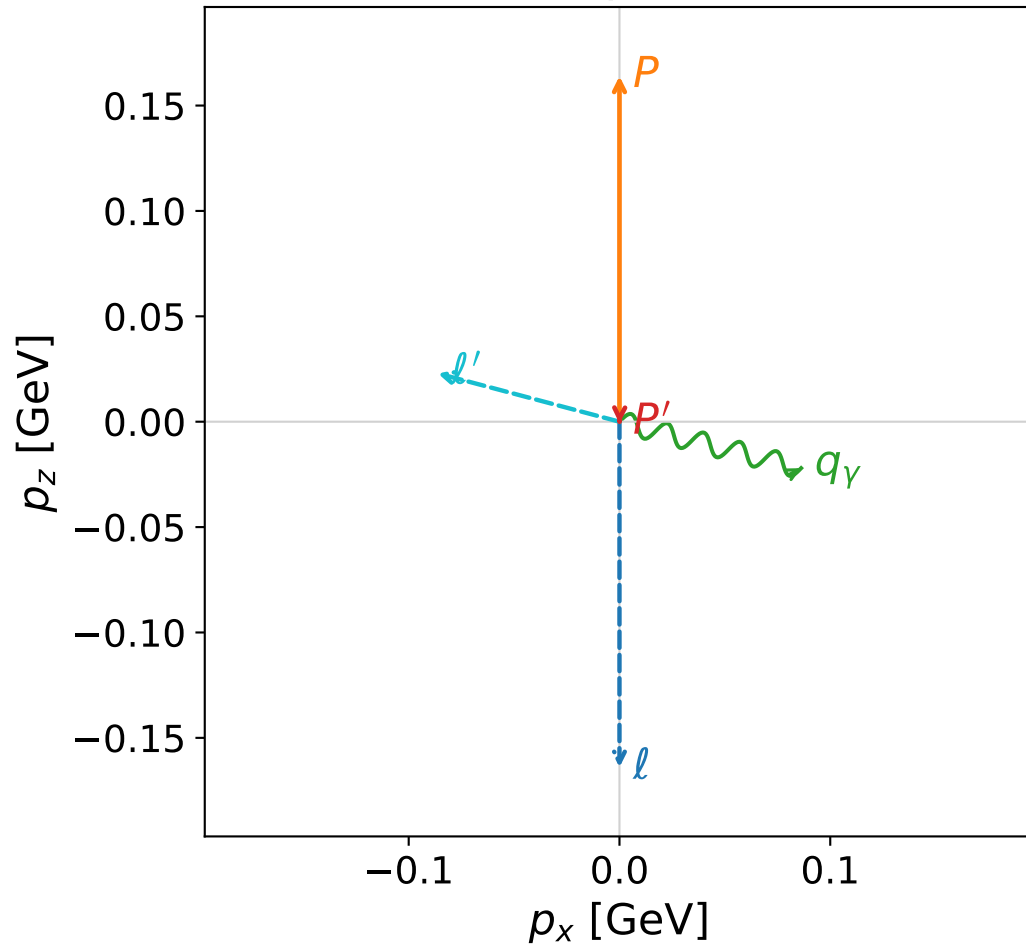
$s=1.19042$, $\sqrt{s}=1.09106$, $|\vec{P}|=0.142324$, $|\vec{P}'|=0.0560568$
 $\theta_{P'}=0$, $\theta_\gamma=2.72655$, $\phi_{P'}=0.420211$, $\phi_\gamma=2.2728$
 $E_\gamma=0.0987891$, $\theta_{l'}=0.859265$, $\phi_{l'}=5.41439$
 $Q^2=0.0247481$, $x_B=0.112214$, $t=-0.00735997$, $W^2=1.07564$, $y=0.710119$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.1423$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1423$
 P $E= 0.9487$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1423$
 l' $E= 0.0526$ $p_x= 0.0257$ $p_y= -0.0304$ $p_z= 0.0343$
 P' $E= 0.9397$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0561$
 q_γ $E= 0.0988$ $p_x= -0.0257$ $p_y= 0.0304$ $p_z= -0.0904$



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

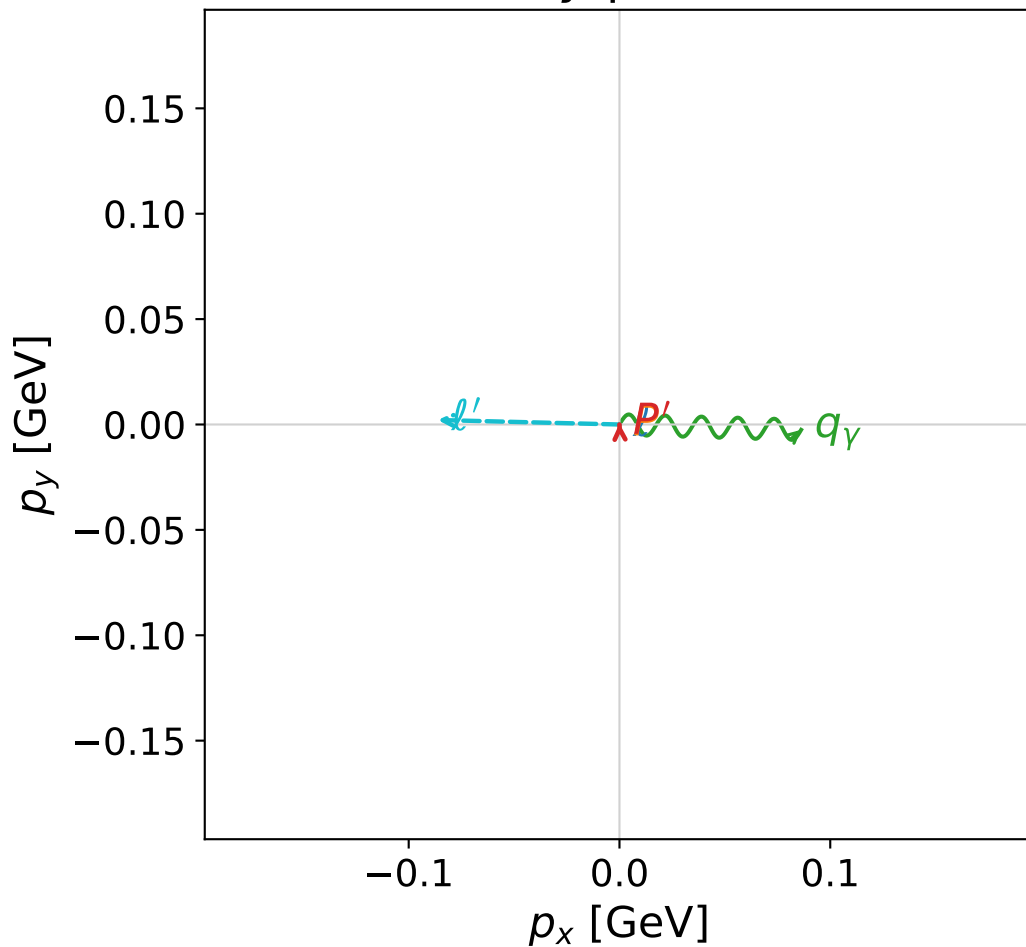


dw_mixing_angles_polarization_02_local_minimum_713
 $D_W=0.00788043$
 region: polarization_cluster_02_local_minimum_713
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0713
 lepton mixing angle: $\alpha_e=0.80739396$ rad
 incoming lepton: $0.691384|+\rangle + 0.722488|-\rangle$
 proton mixing angle: $\alpha_p=1.3762109$ rad
 incoming proton: $0.19336|+\rangle + 0.981128|-\rangle$

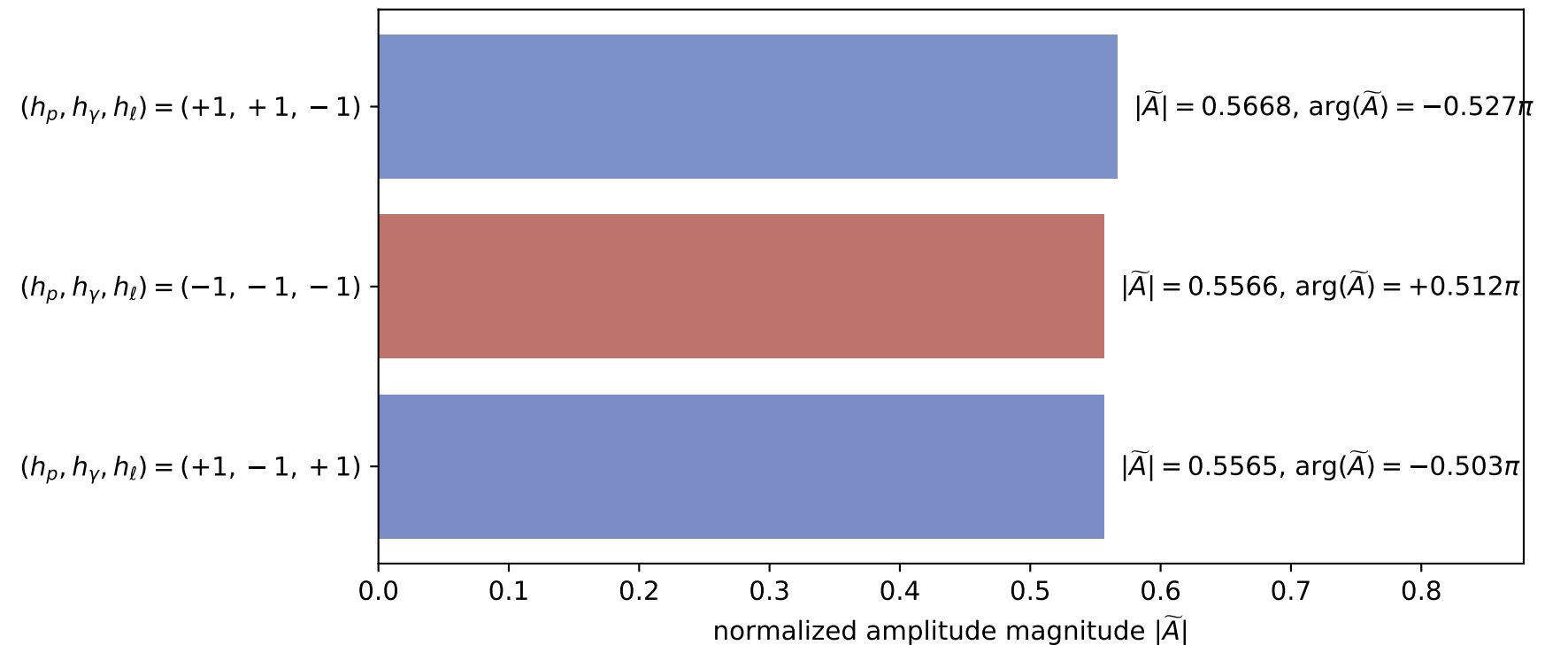
$s=1.24588$, $\sqrt{s}=1.11619$, $|\vec{P}|=0.163967$, $|\vec{P}'|=0.000947889$
 $\theta_{p'}=2.69809$, $\theta_\gamma=1.82098$, $\phi_{p'}=1.60646$, $\phi_\gamma=6.25378$
 $E_\gamma=0.0889989$, $\theta_{\ell'}=1.31124$, $\phi_{\ell'}=3.11689$
 $Q^2=0.0367552$, $x_B=0.180452$, $t=-0.0269645$, $W^2=1.04677$, $y=0.556456$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1640$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1640$
 P $E= 0.9522$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1640$
 ℓ' $E= 0.0892$ $p_x= -0.0862$ $p_y= 0.0021$ $p_z= 0.0229$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= 0.0004$ $p_z= -0.0009$
 q_γ $E= 0.0890$ $p_x= 0.0862$ $p_y= -0.0025$ $p_z= -0.0220$

x--y plane

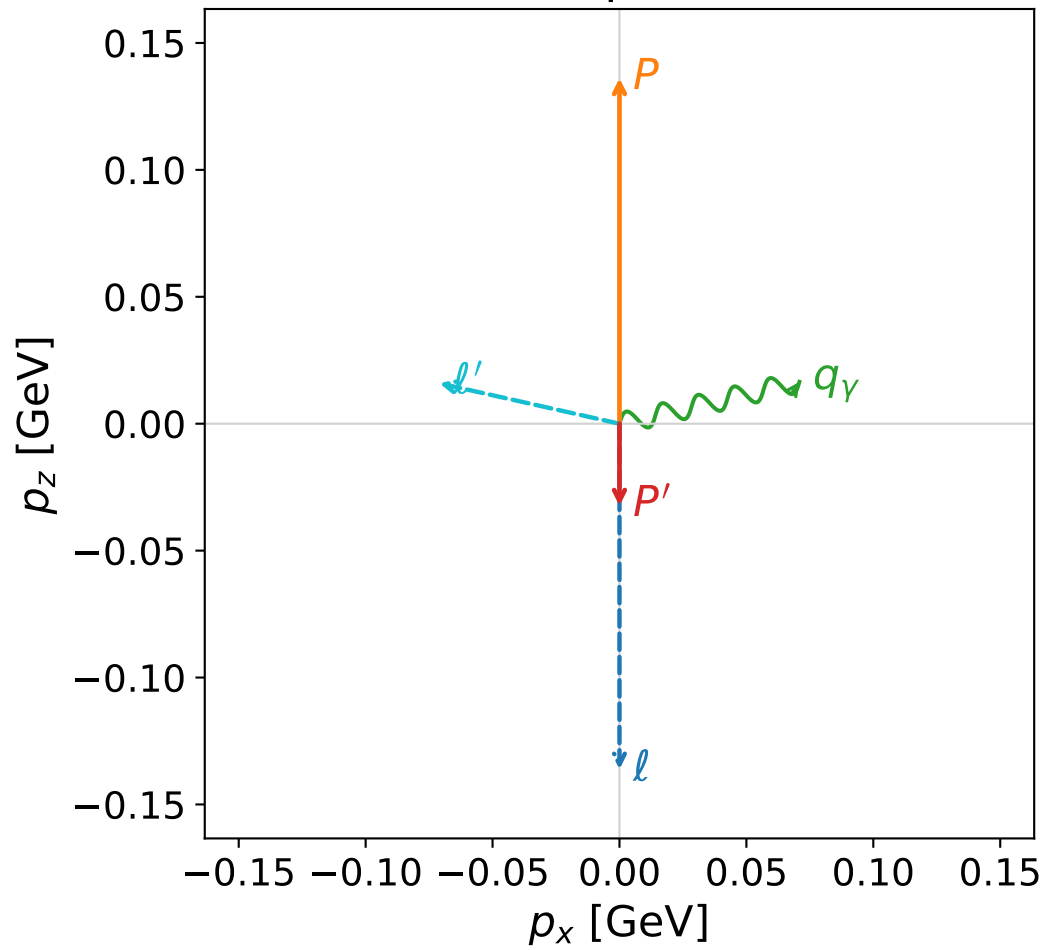


Leading normalized final-state amplitudes
 (N=3, retained=94.1%)

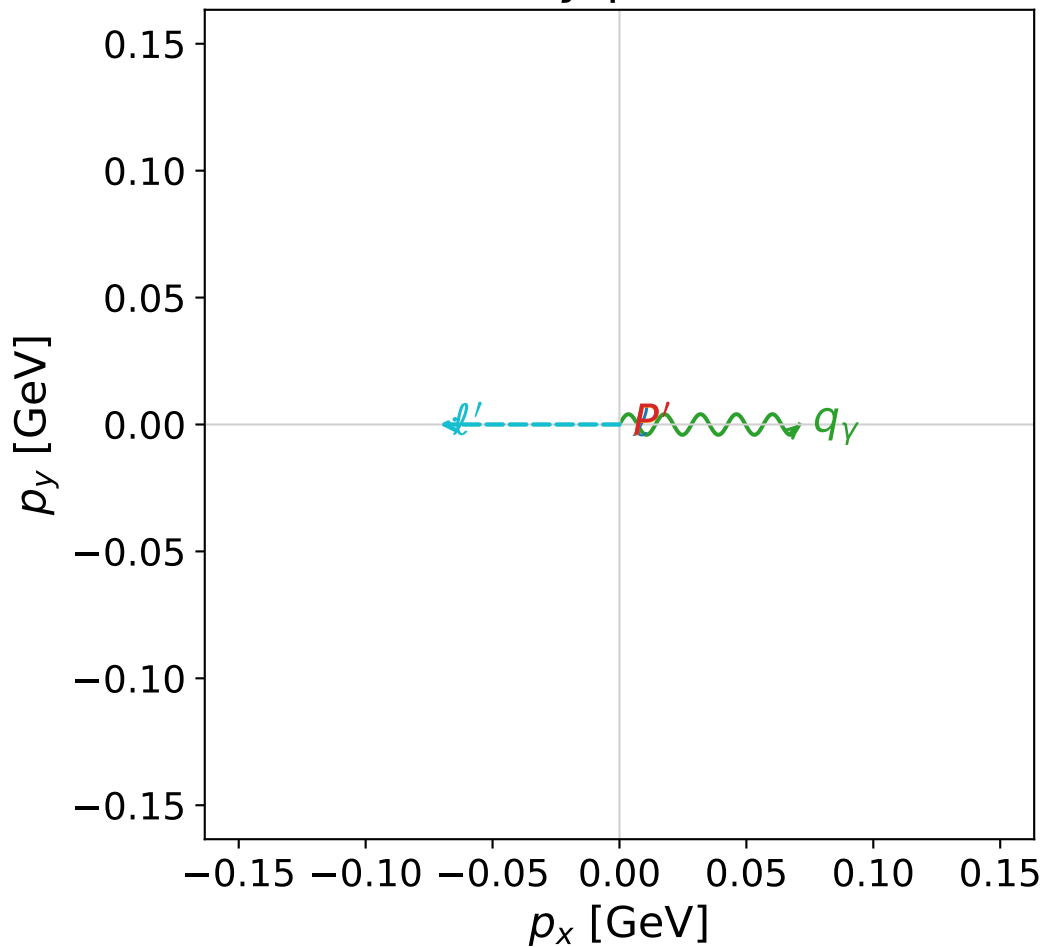


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



x--y plane



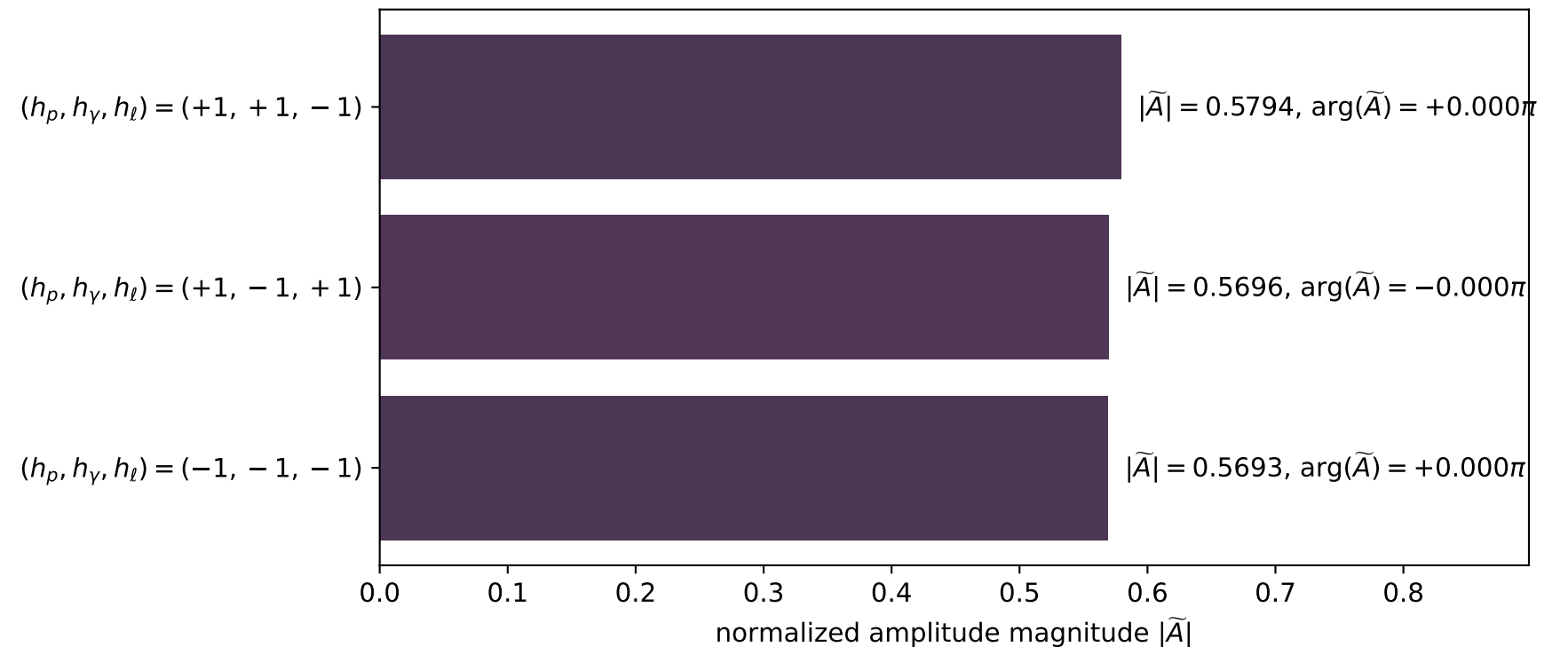
dw_mixing_angles_polarization_02_local_minimum_726
 $D_W=0.00815666$
 region: polarization_cluster_02_local_minimum_726
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0726
 lepton mixing angle: $\alpha_e=0.80690904$ rad
 incoming lepton: $0.691734|+\rangle + 0.722153|-\rangle$
 proton mixing angle: $\alpha_p=1.3699192$ rad
 incoming proton: $0.199529|+\rangle + 0.979892|-\rangle$

$s=1.17502$, $\sqrt{s}=1.08398$, $|\vec{P}|=0.136152$, $|\vec{P}'|=0.0323792$
 $\theta_{p'}=3.14159$, $\theta_\gamma=1.34233$, $\phi_{p'}=4.50934$, $\phi_\gamma=7.32617e-06$
 $E_\gamma=0.0727764$, $\theta_{\ell'}=1.35019$, $\phi_{\ell'}=3.1416$
 $Q^2=0.0241106$, $x_B=0.149026$, $t=-0.0283168$, $W^2=1.01752$, $y=0.548109$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

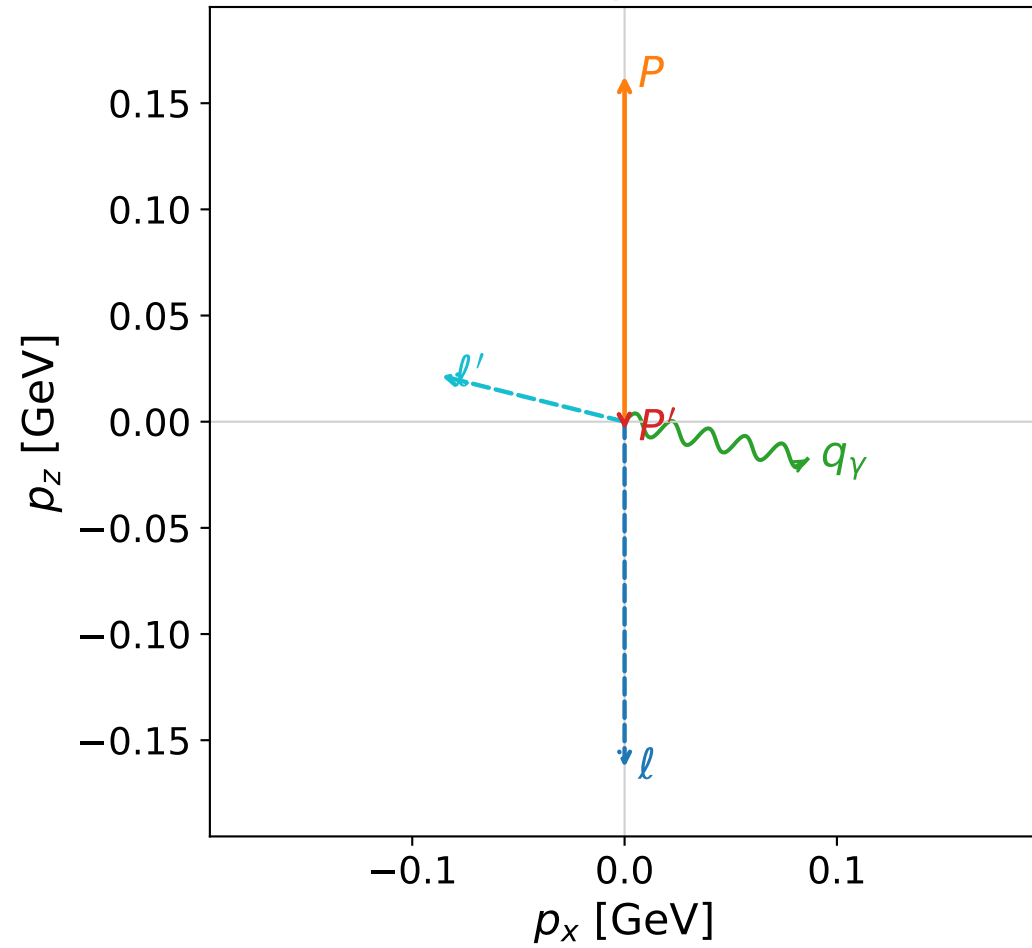
ℓ	$E=$	0.1362	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	-0.1362
P	$E=$	0.9478	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	0.1362
ℓ'	$E=$	0.0726	$p_x=$	-0.0709	$p_y=$	-0.0000	$p_z=$	0.0159
P'	$E=$	0.9386	$p_x=$	-0.0000	$p_y=$	-0.0000	$p_z=$	-0.0324
q_γ	$E=$	0.0728	$p_x=$	0.0709	$p_y=$	0.0000	$p_z=$	0.0165

Leading normalized final-state amplitudes
 (N=3, retained=98.4%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

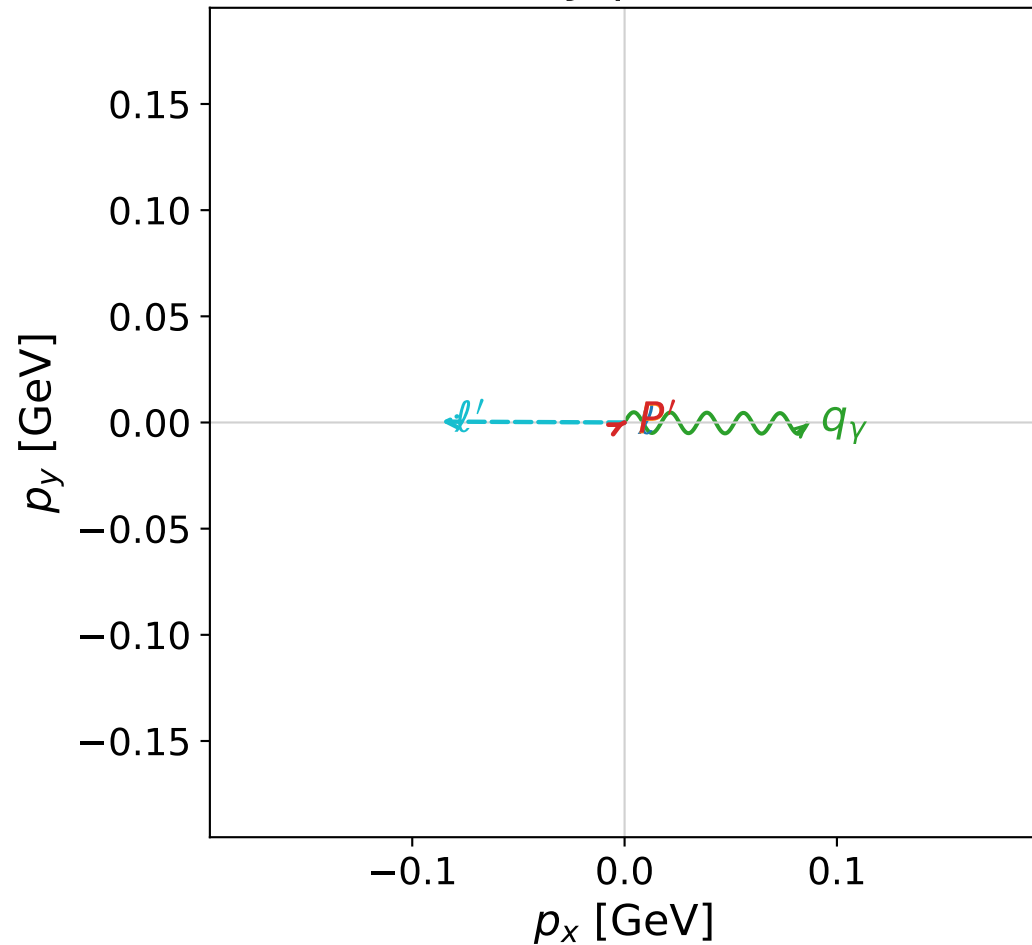


dw_mixing_angles_polarization_02_local_minimum_729
 $D_W=0.00833546$
 region: polarization_cluster_02_local_minimum_729
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0729
 lepton mixing angle: $\alpha_e=0.80706385$ rad
 incoming lepton: $0.691622|+\rangle + 0.72226|-\rangle$
 proton mixing angle: $\alpha_p=1.3715292$ rad
 incoming proton: $0.197951|+\rangle + 0.980212|-\rangle$

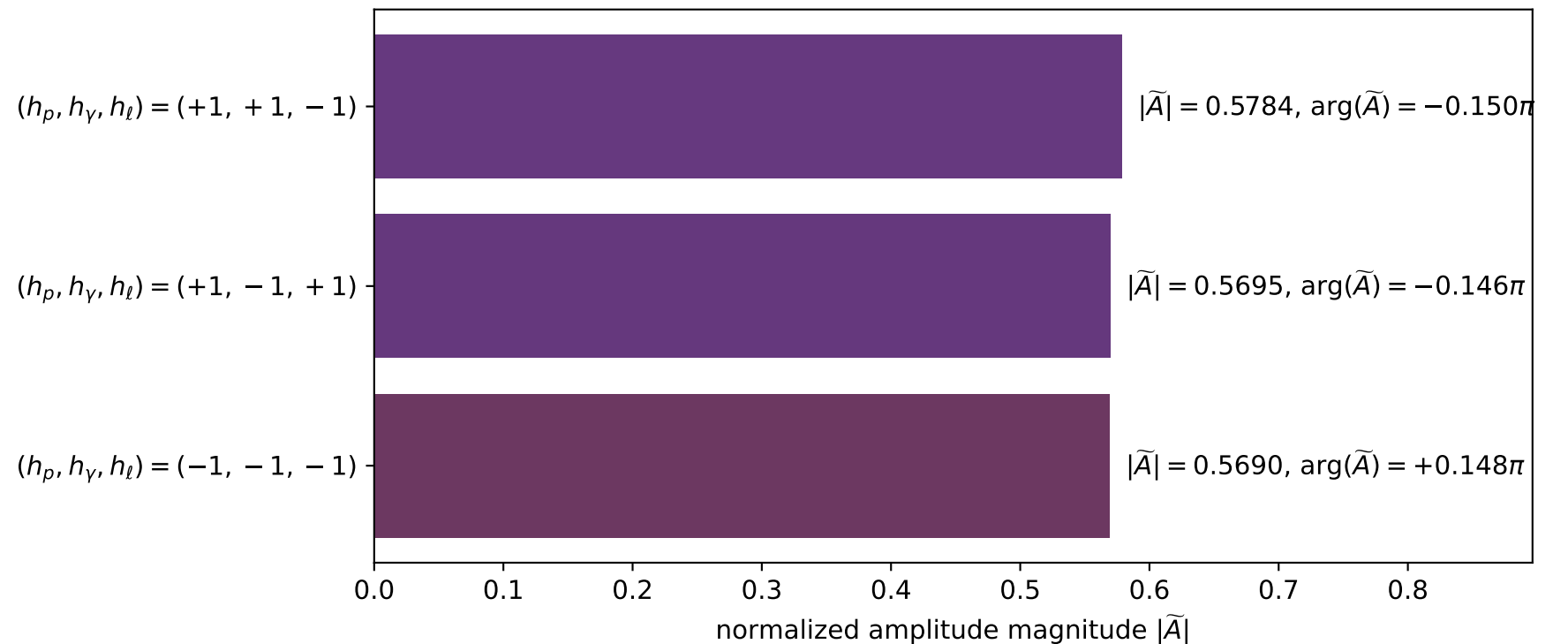
$s=1.24272$, $\sqrt{s}=1.11477$, $|\vec{P}|=0.162758$, $|\vec{P}'|=0.00392398$
 $\theta_{p'}=3.09273$, $\theta_\gamma=1.77317$, $\phi_{p'}=0.464333$, $\phi_\gamma=6.27731$
 $E_\gamma=0.087864$, $\theta_{\ell'}=1.32561$, $\phi_{\ell'}=3.13672$
 $Q^2=0.0359632$, $x_B=0.179252$, $t=-0.027585$, $W^2=1.04451$, $y=0.552885$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E=0.1628$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.1628$
 P $E=0.9520$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1628$
 ℓ' $E=0.0889$ $p_x=-0.0862$ $p_y=0.0004$ $p_z=0.0216$
 P' $E=0.9380$ $p_x=0.0002$ $p_y=0.0001$ $p_z=-0.0039$
 q_γ $E=0.0879$ $p_x=0.0861$ $p_y=-0.0005$ $p_z=-0.0177$

x--y plane

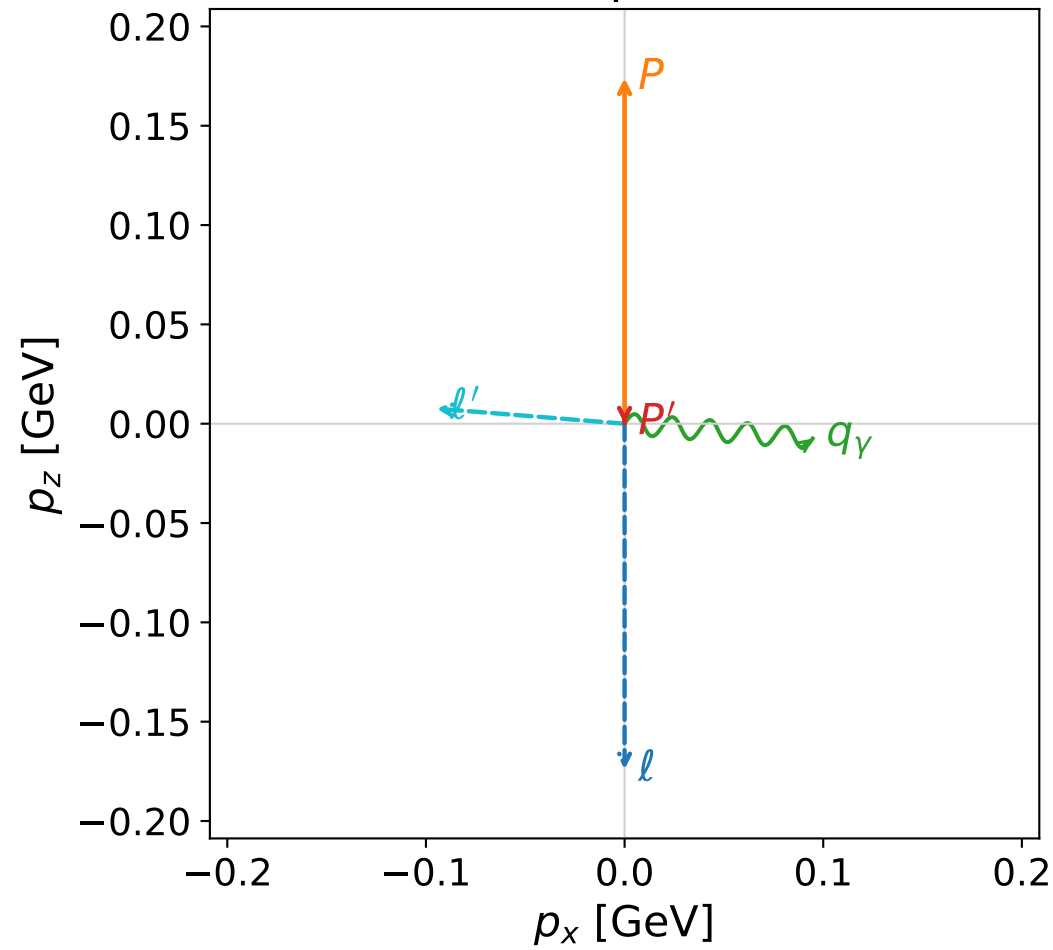


Leading normalized final-state amplitudes
 (N=3, retained=98.3%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

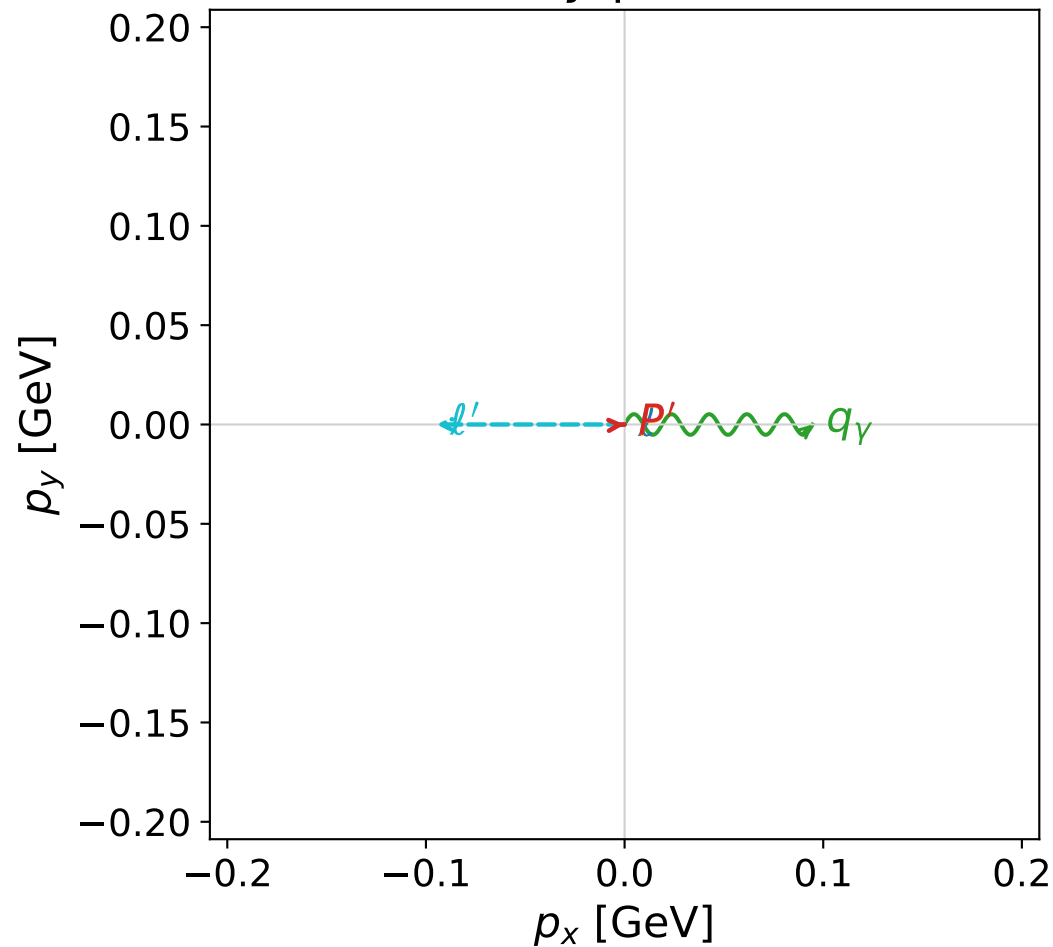


dw_mixing_angles_polarization_02_local_minimum_739
 $D_W=0.0085628$
 region: polarization_cluster_02_local_minimum_739
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0739
 lepton mixing angle: $\alpha_e=0.7888655$ rad
 incoming lepton: $0.704651|+\rangle +0.709554|-\rangle$
 proton mixing angle: $\alpha_p=1.3637025$ rad
 incoming proton: $0.205617|+\rangle +0.978633|-\rangle$

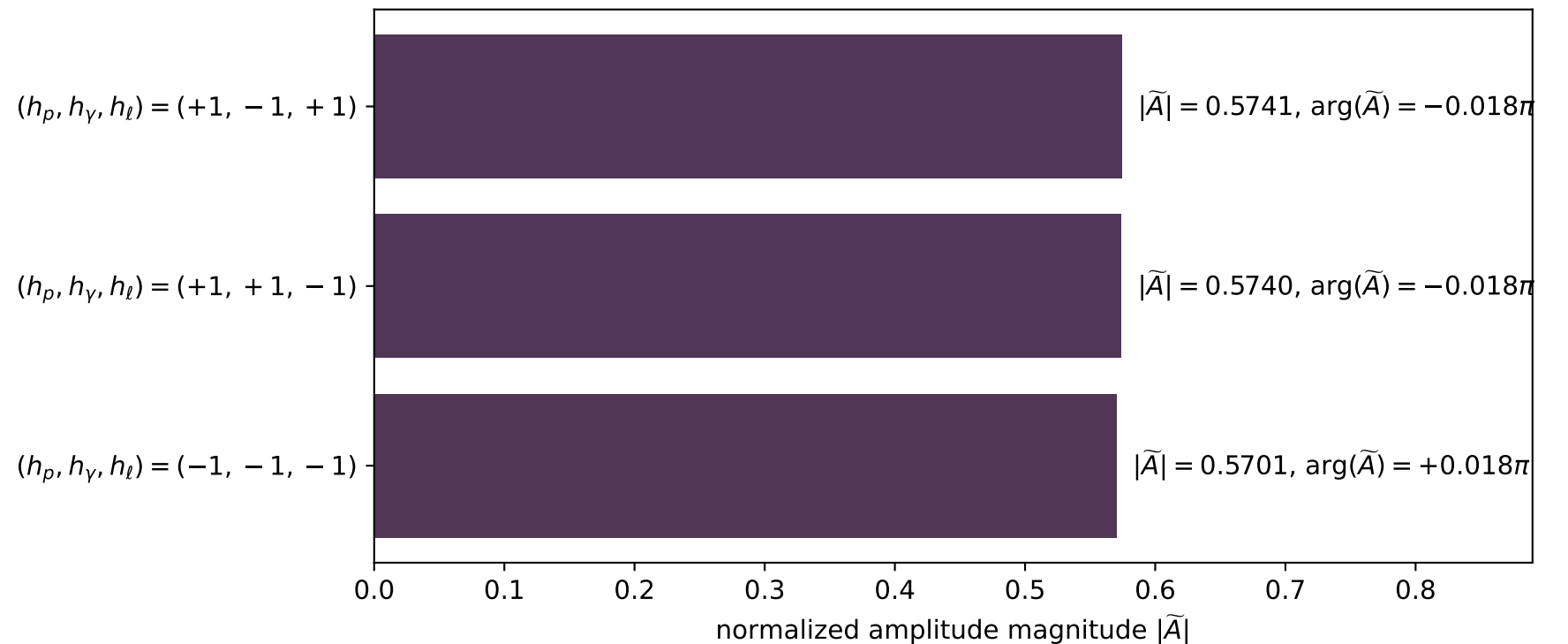
$s=1.27231$, $\sqrt{s}=1.12797$, $|\vec{P}|=0.173971$, $|\vec{P}'|=0.000134712$
 $\theta_{p'}=3.14126$, $\theta_\gamma=1.64947$, $\phi_{p'}=0.0552426$, $\phi_\gamma=1.07418e-06$
 $E_\gamma=0.0949782$, $\theta_{l'}=1.49071$, $\phi_{l'}=3.14159$
 $Q^2=0.0356947$, $x_B=0.166898$, $t=-0.0300569$, $W^2=1.05802$, $y=0.54494$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.1740$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1740$
 P $E= 0.9540$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1740$
 l' $E= 0.0950$ $p_x= -0.0947$ $p_y= -0.0000$ $p_z= 0.0076$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0001$
 q_γ $E= 0.0950$ $p_x= 0.0947$ $p_y= 0.0000$ $p_z= -0.0075$

x--y plane

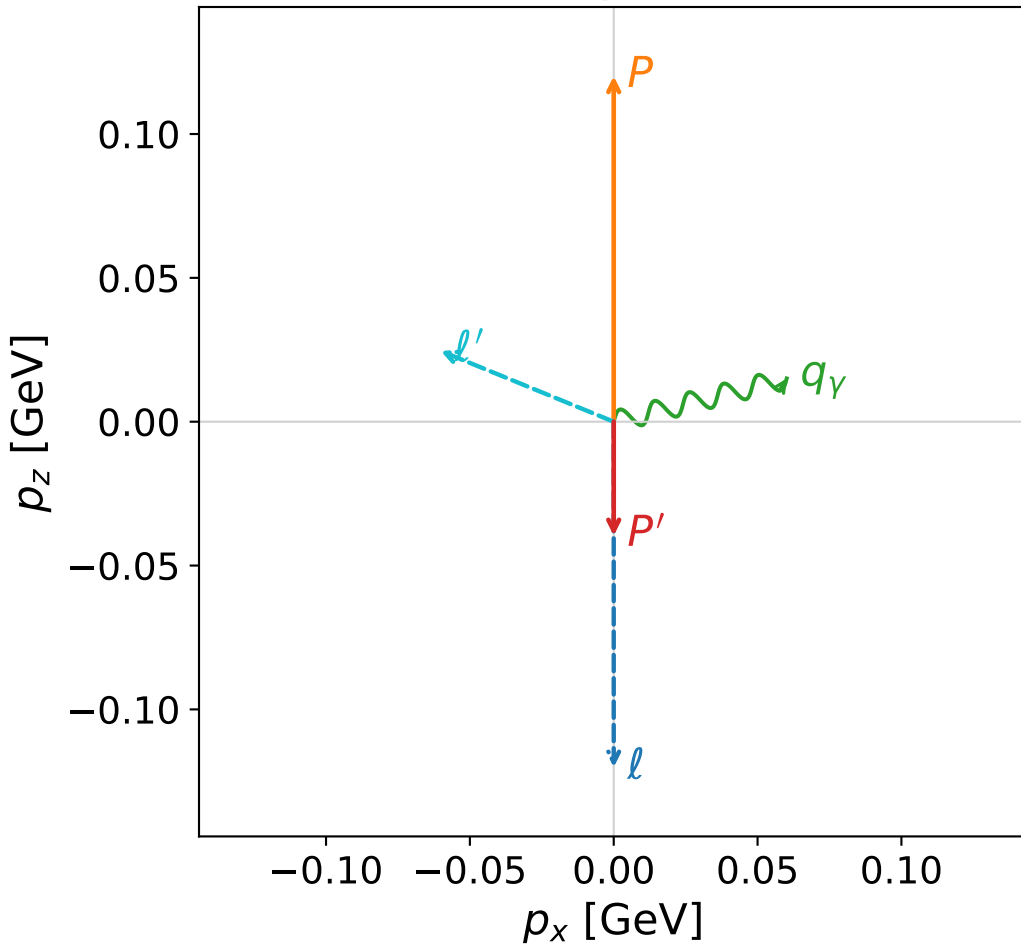


Leading normalized final-state amplitudes
 (N=3, retained=98.4%)

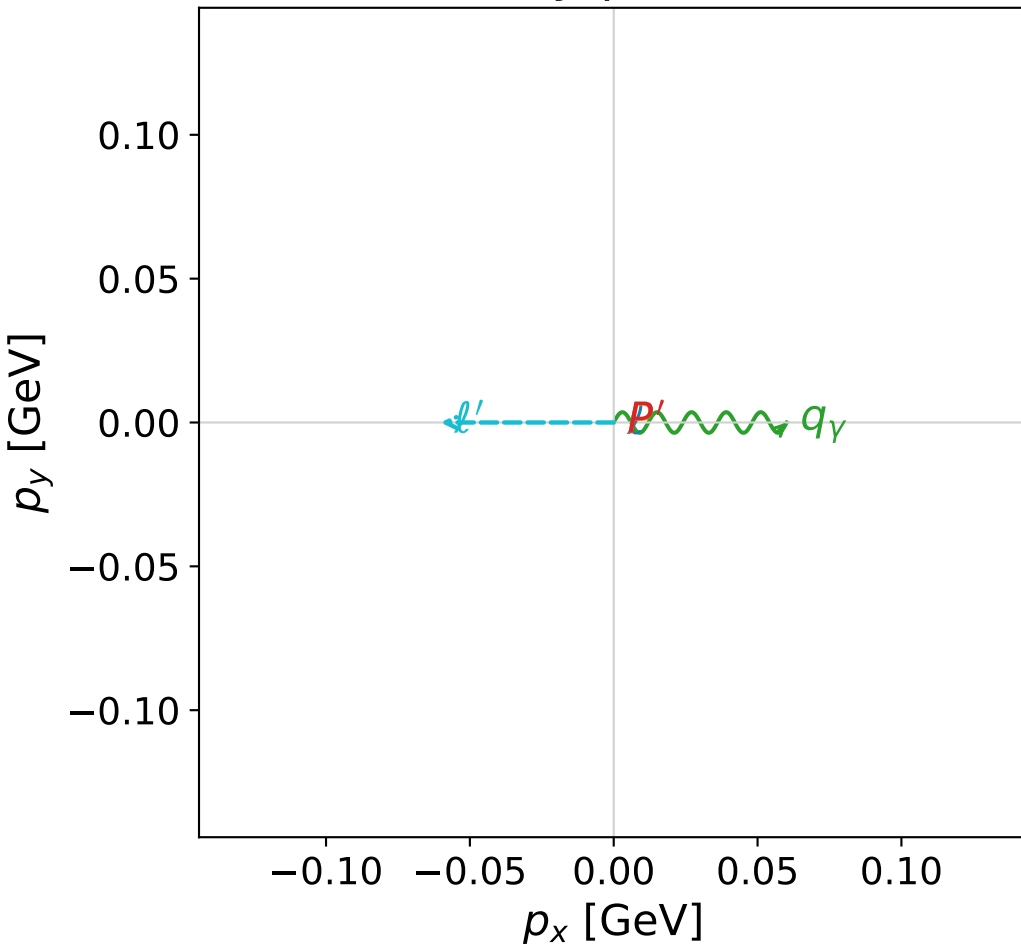


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



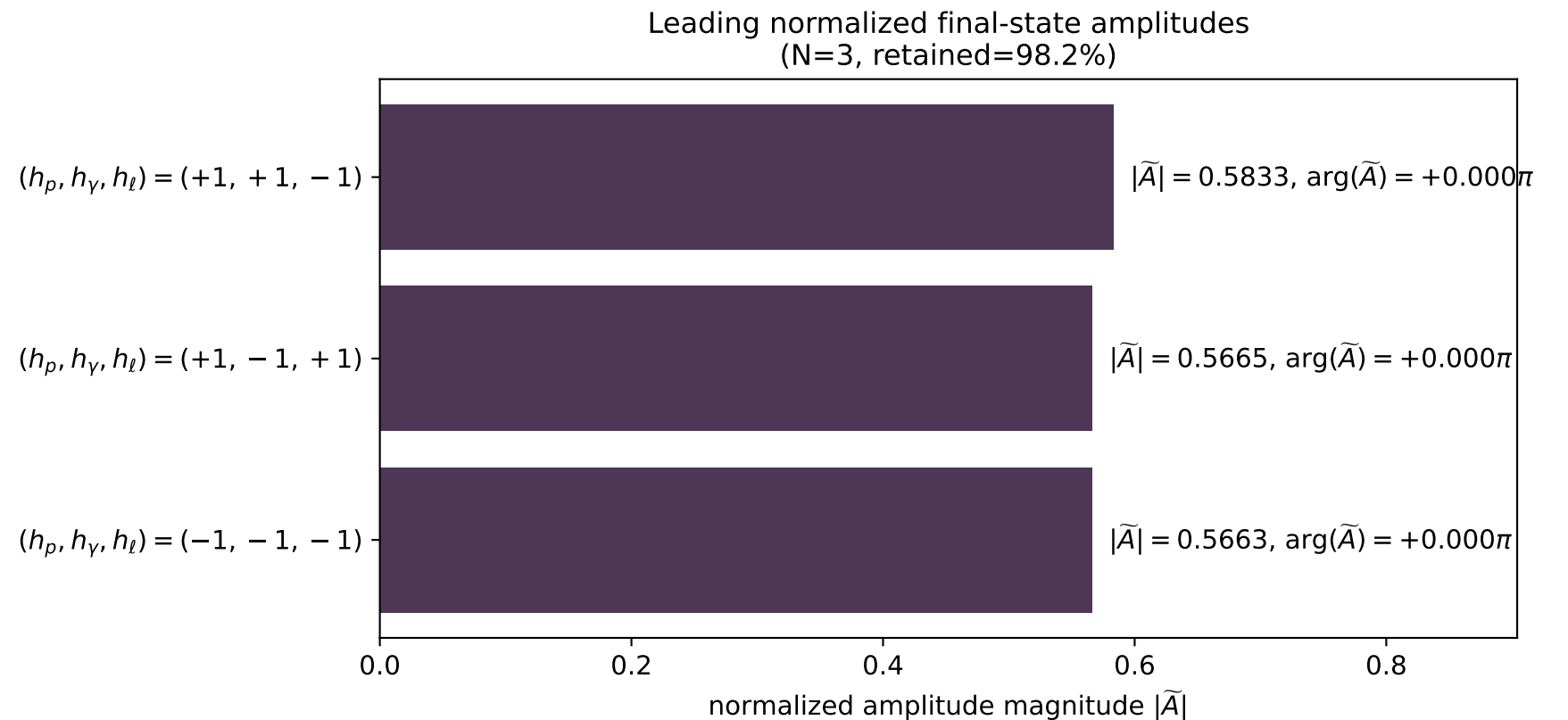
x--y plane



dw_mixing_angles_polarization_02_local_minimum_767
 $D_W=0.00946605$
 region: polarization_cluster_02_local_minimum_767
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0767
 lepton mixing angle: $\alpha_e=0.82066942$ rad
 incoming lepton: $0.681732|+\rangle +0.731602|-\rangle$
 proton mixing angle: $\alpha_p=1.3944256$ rad
 incoming proton: $0.175458|+\rangle +0.984487|-\rangle$

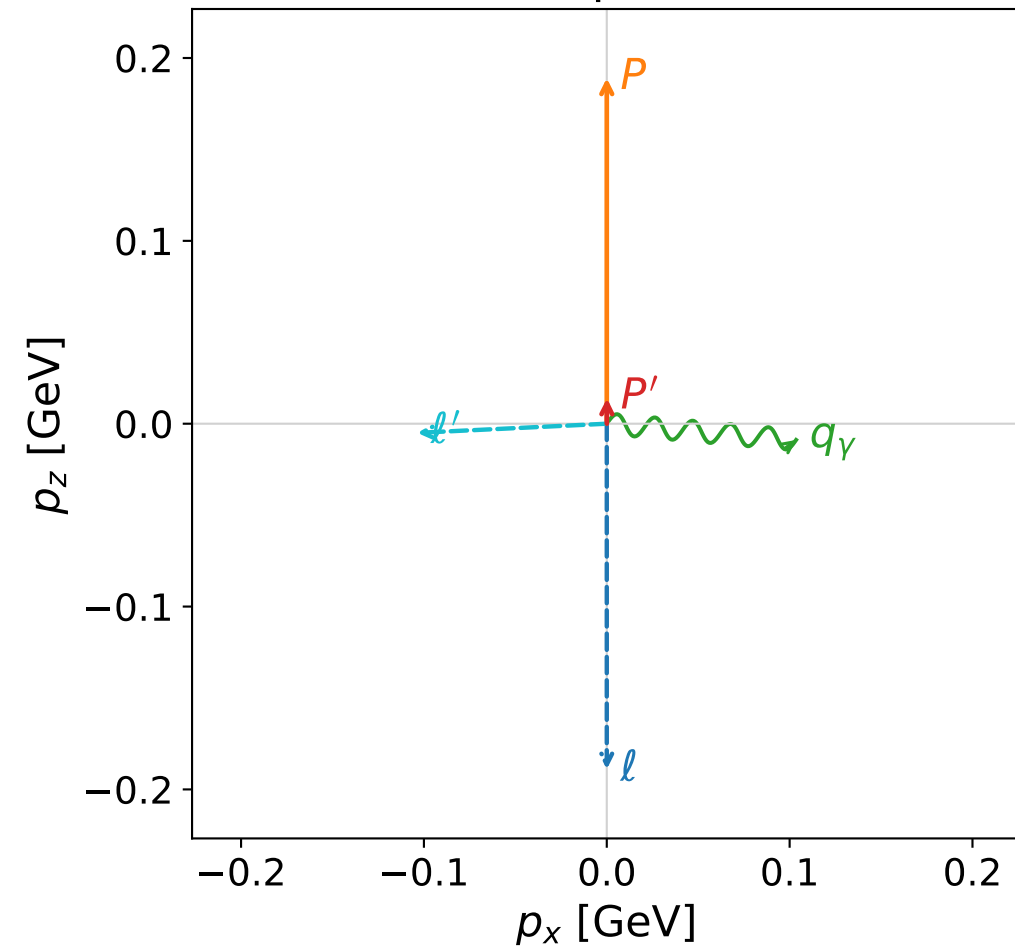
$s=1.1359$, $\sqrt{s}=1.06579$, $|\vec{P}|=0.120126$, $|\vec{P}'|=0.0396205$
 $\theta_{p'}=3.14159$, $\theta_\gamma=1.32582$, $\phi_{p'}=0$, $\phi_\gamma=0$
 $E_\gamma=0.061984$, $\theta_{l'}=1.18266$, $\phi_{l'}=3.14159$
 $Q^2=0.0215155$, $x_B=0.154685$, $t=-0.0254725$, $W^2=0.997421$, $y=0.543202$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E=0.1201$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.1201$
 P $E=0.9457$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1201$
 l' $E=0.0650$ $p_x=-0.0601$ $p_y=-0.0000$ $p_z=0.0246$
 P' $E=0.9388$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.0396$
 q_γ $E=0.0620$ $p_x=0.0601$ $p_y=0.0000$ $p_z=0.0150$

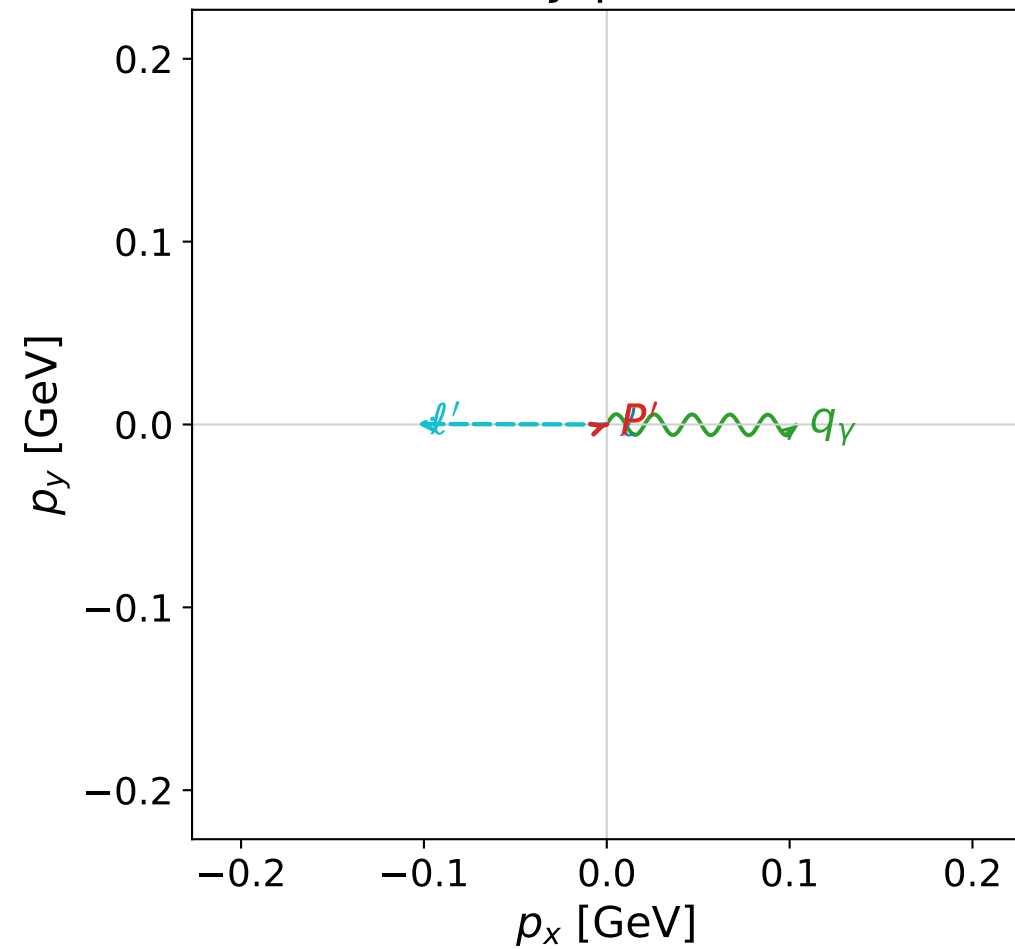


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



x--y plane

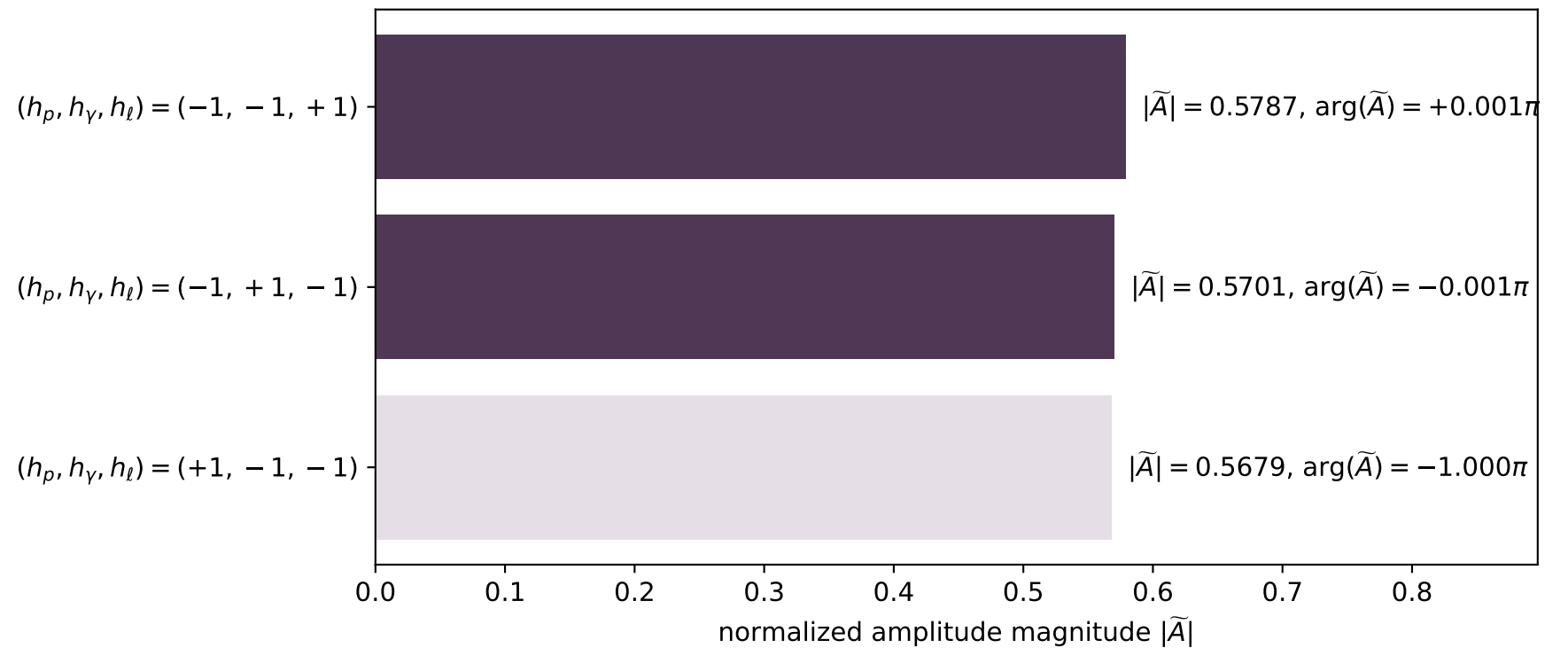


dw_mixing_angles_polarization_02_local_minimum_788
 $D_W=0.0100564$
 region: polarization_cluster_02_local_minimum_788
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0788
 lepton mixing angle: $\alpha_e=0.77394525$ rad
 incoming lepton: $0.715159|+\rangle +0.698962|-\rangle$
 proton mixing angle: $\alpha_p=1.3656881$ rad
 incoming proton: $0.203673|+\rangle +0.979039|-\rangle$

$s=1.31297$, $\sqrt{s}=1.14585$, $|\vec{P}|=0.188999$, $|\vec{P}'|=0.0138893$
 $\theta_{P'}=0.0119919$, $\theta_\gamma=1.65635$, $\phi_{P'}=0.302166$, $\phi_\gamma=6.28013$
 $E_\gamma=0.103924$, $\theta_{l'}=1.61905$, $\phi_{l'}=3.13902$
 $Q^2=0.037352$, $x_B=0.160621$, $t=-0.0303122$, $W^2=1.07504$, $y=0.536901$

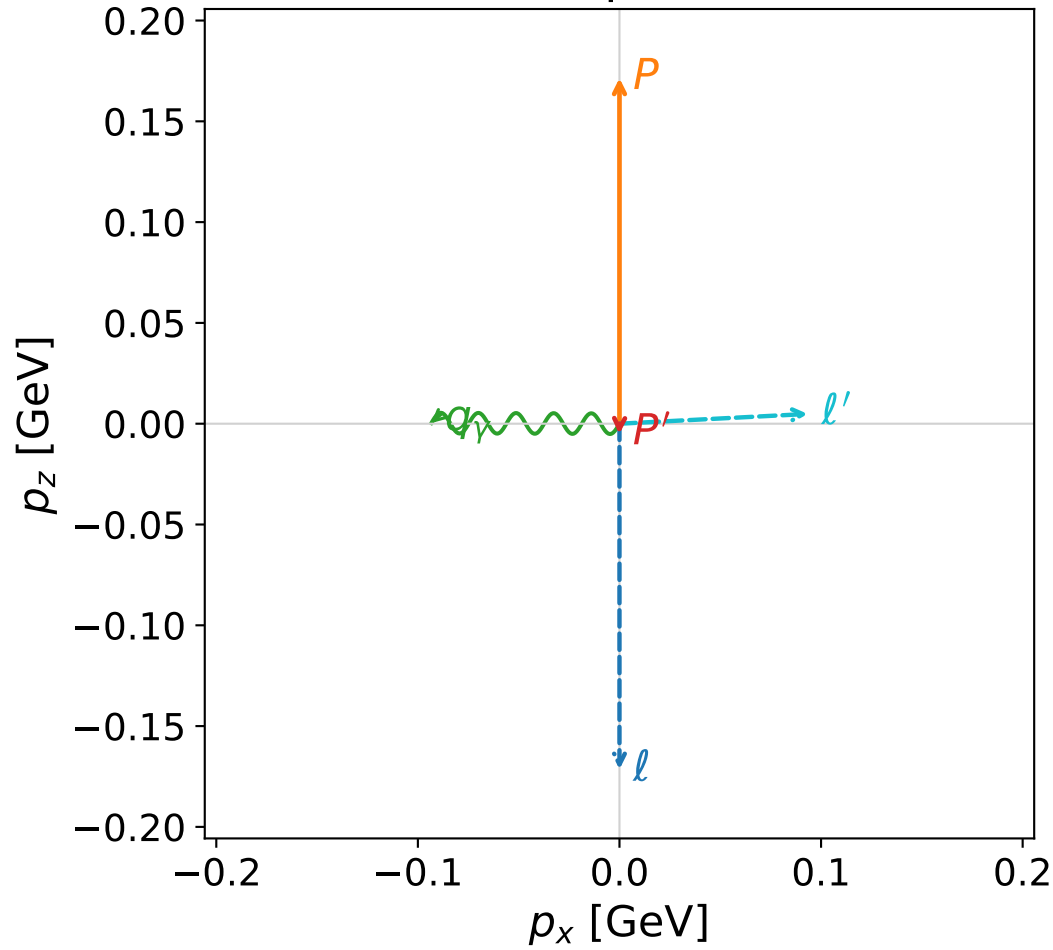
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E=0.1890$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.1890$
 P $E=0.9569$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1890$
 l' $E=0.1038$ $p_x=-0.1037$ $p_y=0.0003$ $p_z=-0.0050$
 P' $E=0.9381$ $p_x=0.0002$ $p_y=0.0000$ $p_z=0.0139$
 q_γ $E=0.1039$ $p_x=0.1035$ $p_y=-0.0003$ $p_z=-0.0089$

Leading normalized final-state amplitudes
 (N=3, retained=98.2%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



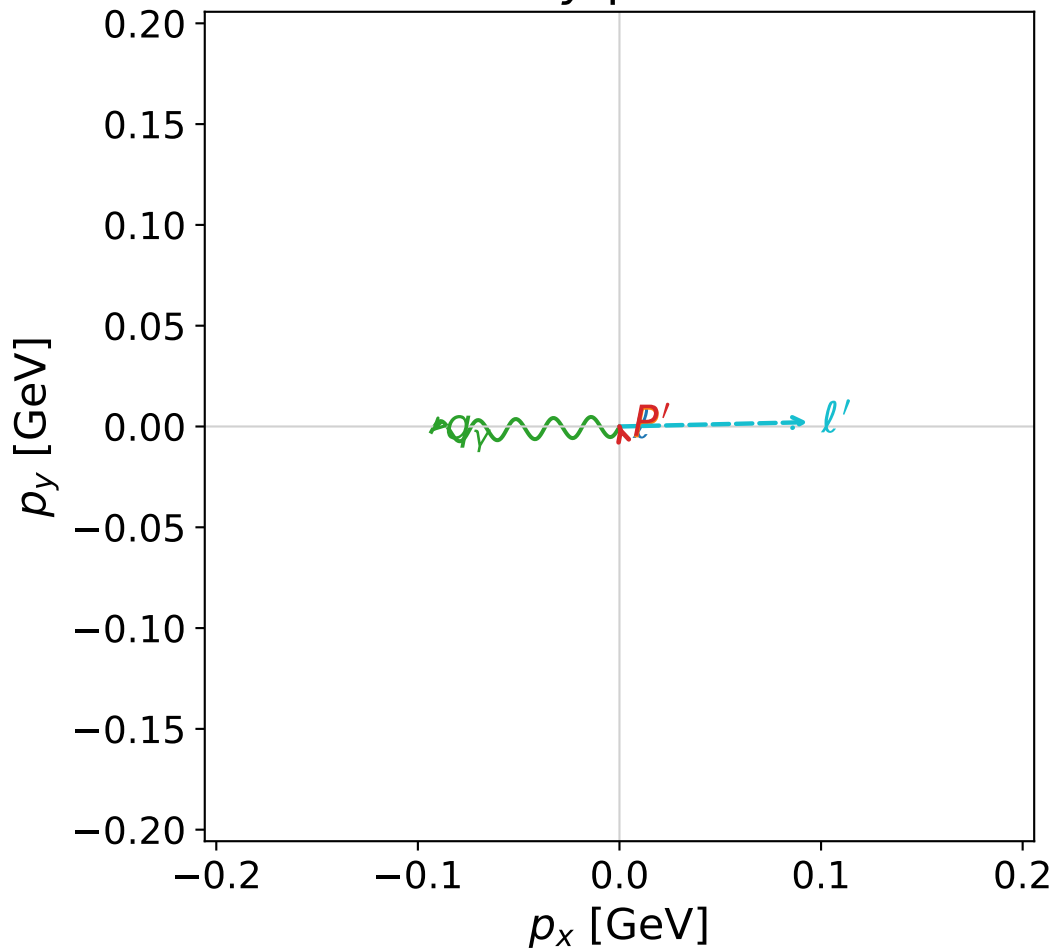
dw_mixing_angles_polarization_02_local_minimum_790
 $D_W=0.0101139$
 region: polarization_cluster_02_local_minimum_790
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0790
 lepton mixing angle: $\alpha_e=0.77782162$ rad
 incoming lepton: $0.712444|+\rangle + 0.701729|-\rangle$
 proton mixing angle: $\alpha_p=1.7730066$ rad
 incoming proton: $-0.200835|+\rangle + 0.979625|-\rangle$

$s=1.26556$, $\sqrt{s}=1.12497$, $|\vec{P}|=0.171433$, $|\vec{P}'|=0.00518528$
 $\theta_{p'}=3.05207$, $\theta_\gamma=1.56744$, $\phi_{p'}=1.85608$, $\phi_\gamma=3.16934$
 $E_\gamma=0.0933554$, $\theta_{\ell'}=1.51895$, $\phi_{\ell'}=0.0229511$
 $Q^2=0.0337555$, $x_B=0.161607$, $t=-0.0309459$, $W^2=1.05496$, $y=0.541524$

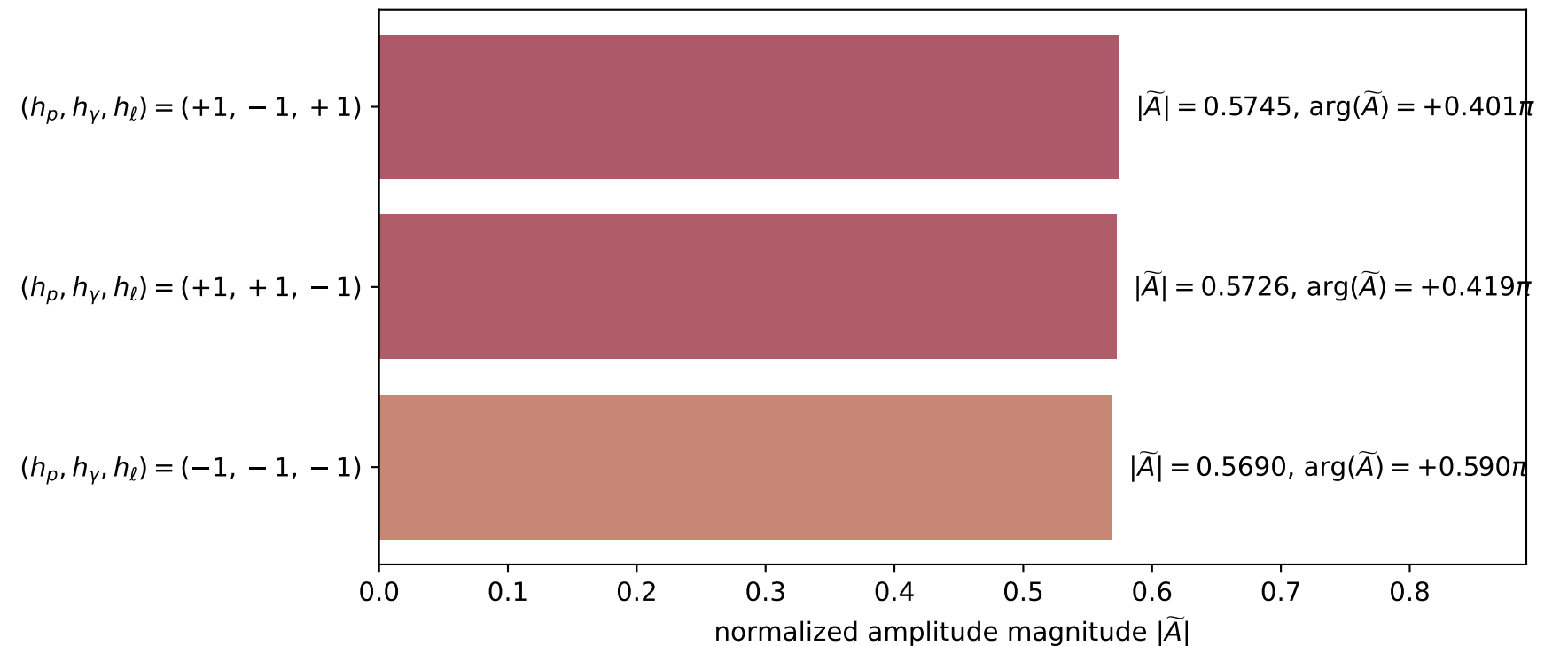
four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ	$E=$	0.1714	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	-0.1714
P	$E=$	0.9535	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	0.1714
ℓ'	$E=$	0.0936	$p_x=$	0.0934	$p_y=$	0.0021	$p_z=$	0.0049
P'	$E=$	0.9380	$p_x=$	-0.0001	$p_y=$	0.0004	$p_z=$	-0.0052
q_γ	$E=$	0.0934	$p_x=$	-0.0933	$p_y=$	-0.0026	$p_z=$	0.0003

x--y plane

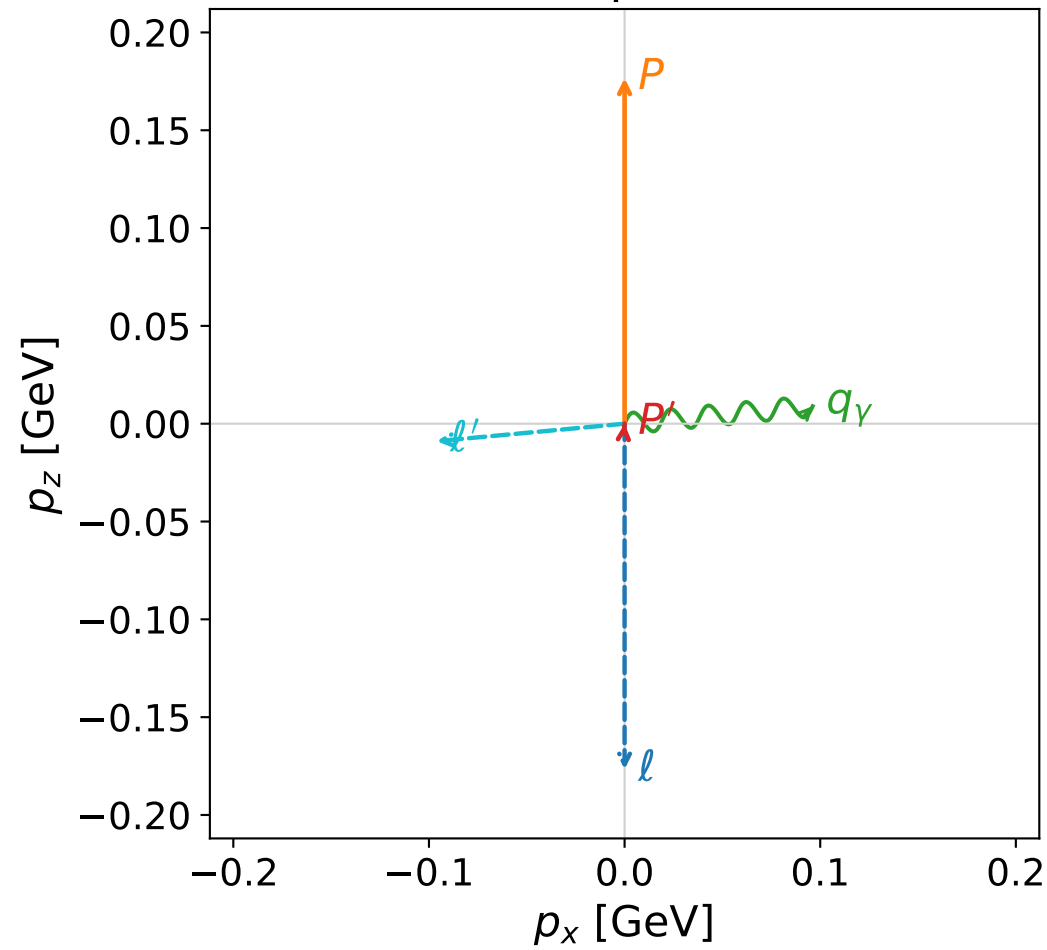


Leading normalized final-state amplitudes
 (N=3, retained=98.2%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

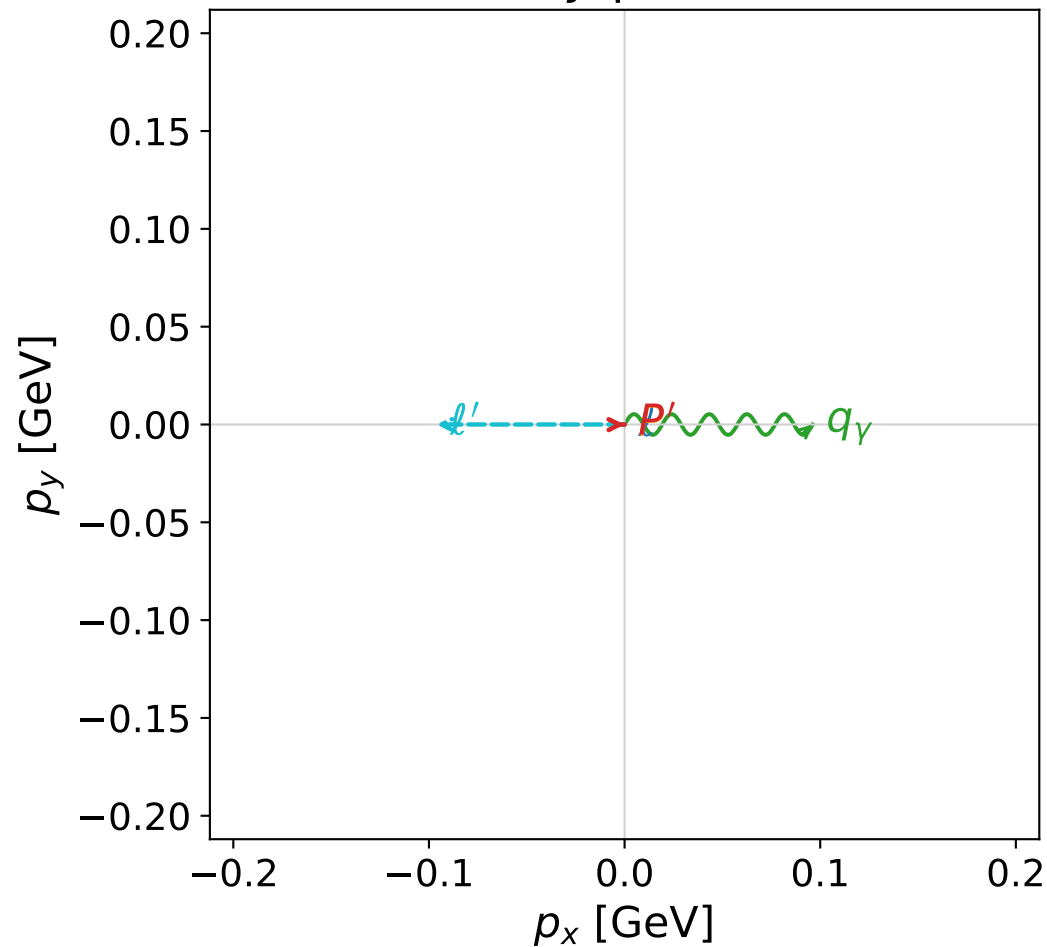


dw_mixing_angles_polarization_02_local_minimum_797
 $D_W=0.0103184$
 region: polarization_cluster_02_local_minimum_797
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0797
 lepton mixing angle: $\alpha_e=0.76992886$ rad
 incoming lepton: $0.71796|+\rangle + 0.696084|-\rangle$
 proton mixing angle: $\alpha_p=1.3669543$ rad
 incoming proton: $0.202433|+\rangle + 0.979296|-\rangle$

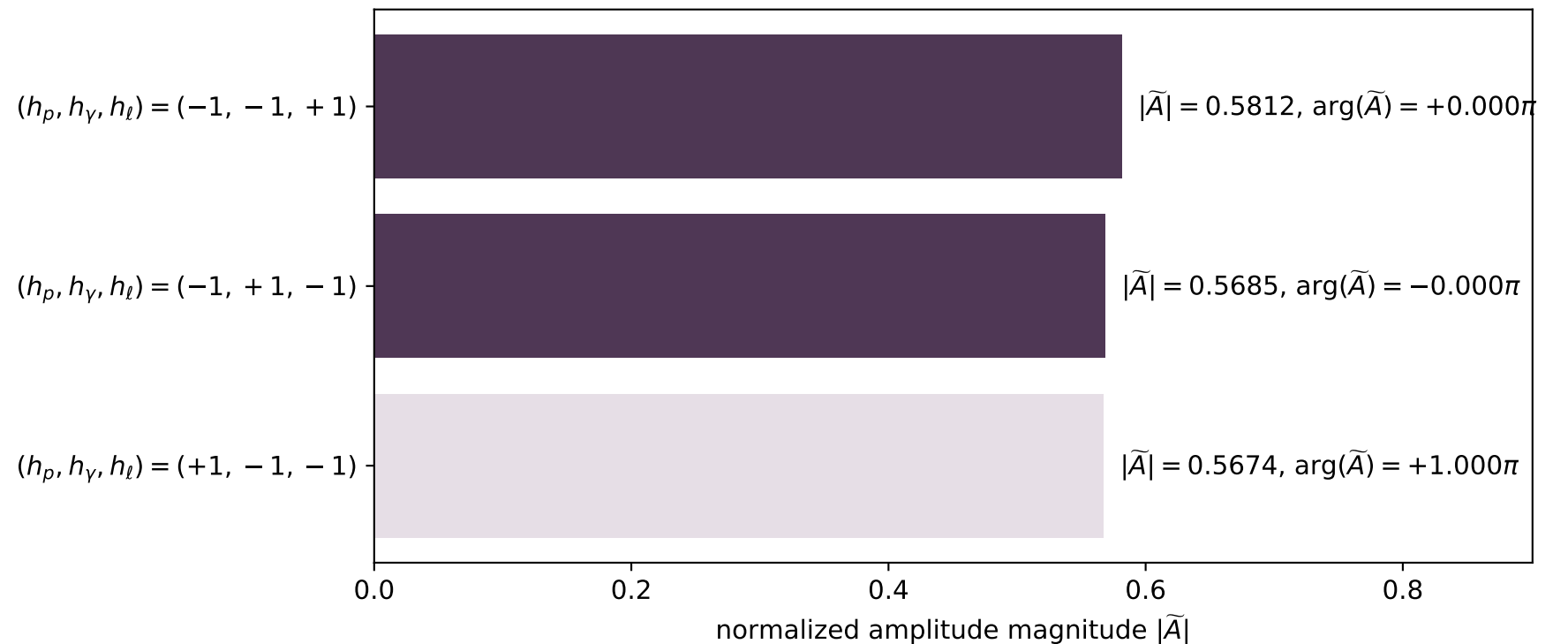
$s=1.27948$, $\sqrt{s}=1.13114$, $|\vec{P}|=0.176653$, $|\vec{P}'|=9.29299e-07$
 $\theta_{p'}=0.0778937$, $\theta_Y=1.47805$, $\phi_{p'}=0.00922162$, $\phi_Y=6.28316$
 $E_Y=0.0965708$, $\theta_{\ell'}=1.66355$, $\phi_{\ell'}=3.14156$
 $Q^2=0.0309589$, $x_B=0.145946$, $t=-0.030934$, $W^2=1.06101$, $y=0.530791$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1767$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1767$
 P $E= 0.9545$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1767$
 ℓ' $E= 0.0966$ $p_x= -0.0962$ $p_y= 0.0000$ $p_z= -0.0089$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0000$
 q_Y $E= 0.0966$ $p_x= 0.0962$ $p_y= -0.0000$ $p_z= 0.0089$

x--y plane

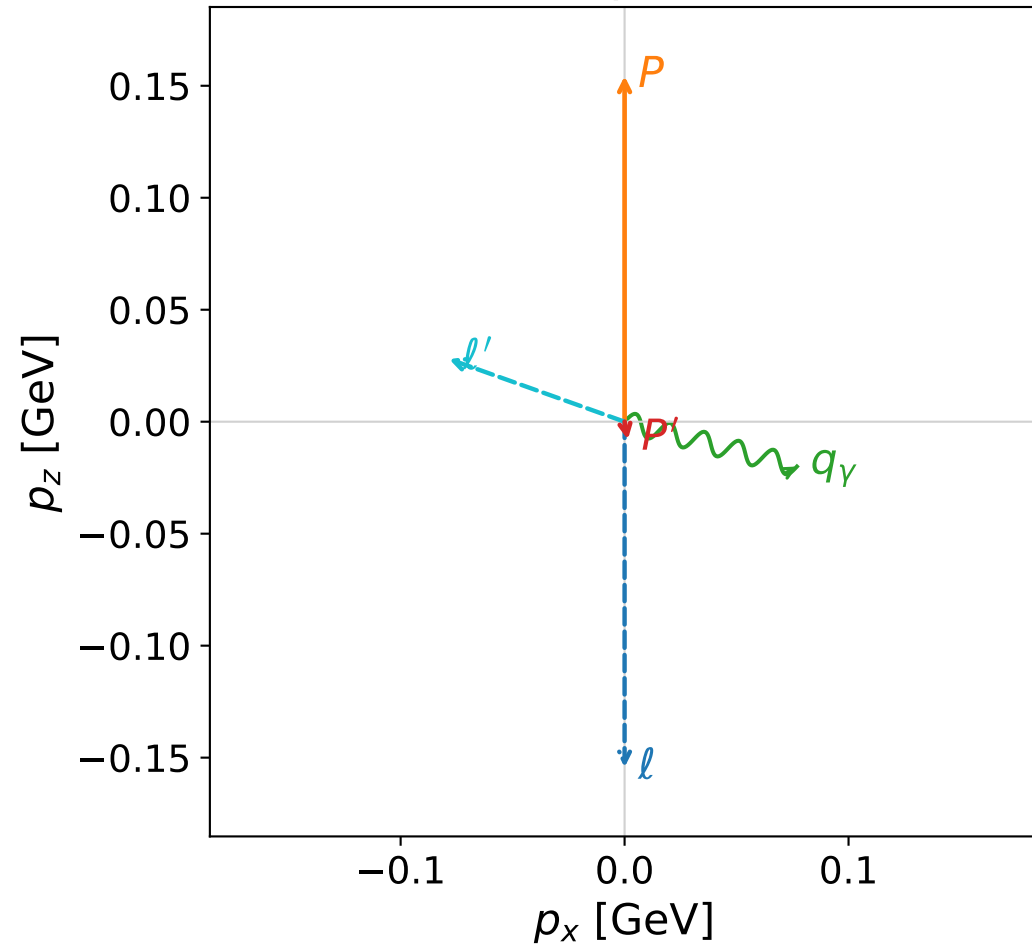


Leading normalized final-state amplitudes
 (N=3, retained=98.3%)

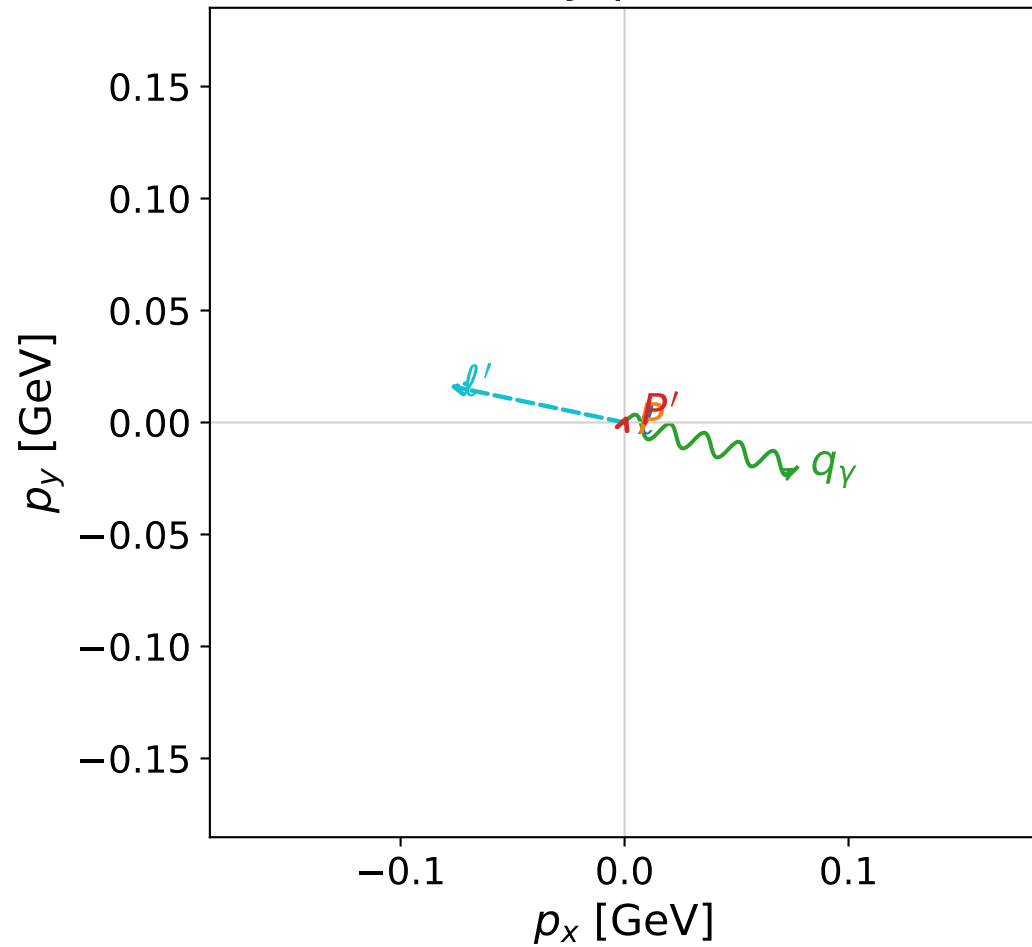


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



x--y plane

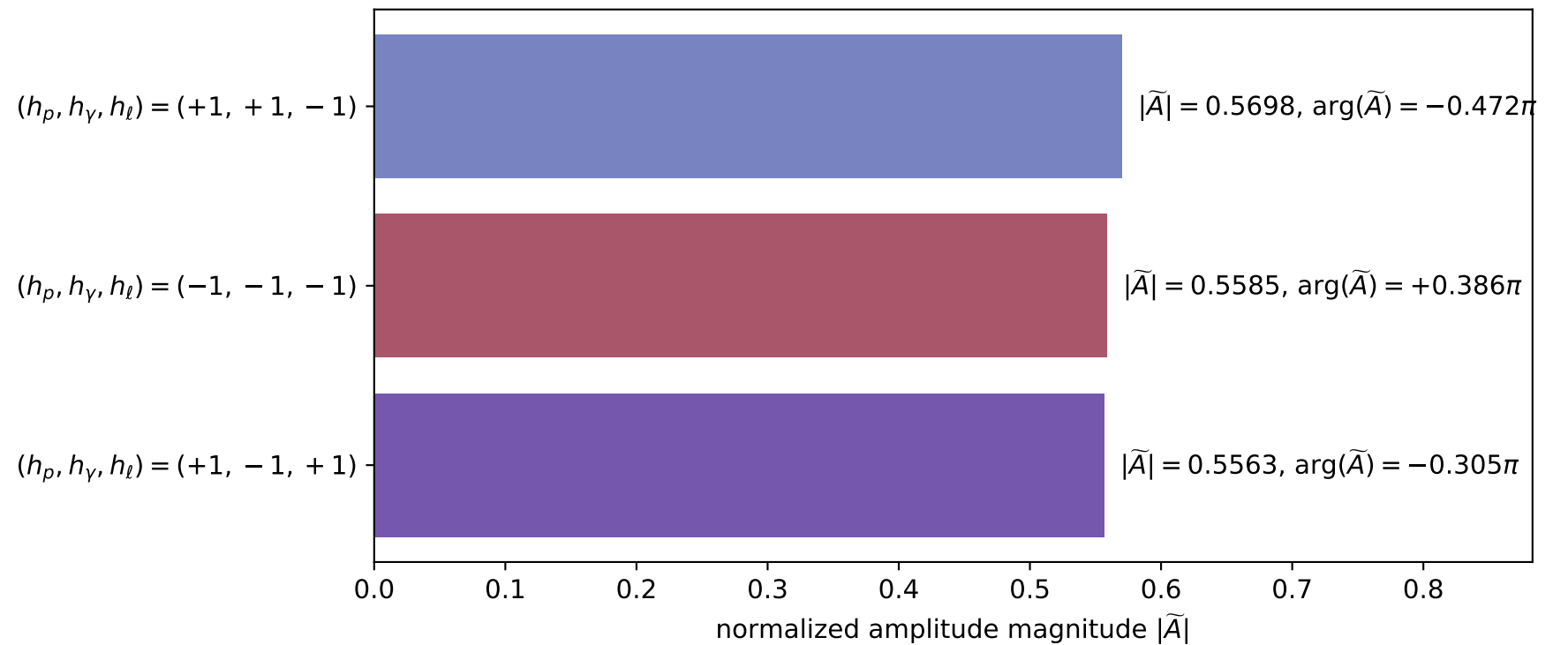


dw_mixing_angles_polarization_02_local_minimum_798
 $D_W=0.0103804$
 region: polarization_cluster_02_local_minimum_798
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0798
 lepton mixing angle: $\alpha_e=0.81788713$ rad
 incoming lepton: $0.683764|+\rangle + 0.729703|-\rangle$
 proton mixing angle: $\alpha_p=1.3602032$ rad
 incoming proton: $0.20904|+\rangle + 0.977907|-\rangle$

$s=1.22085$, $\sqrt{s}=1.10492$, $|\vec{P}|=0.154311$, $|\vec{P}'|=0.00891841$
 $\theta_{p'}=2.67511$, $\theta_\gamma=1.81721$, $\phi_{p'}=1.20325$, $\phi_\gamma=6.02686$
 $E_\gamma=0.0820325$, $\theta_{\ell'}=1.23478$, $\phi_{\ell'}=2.93505$
 $Q^2=0.0348188$, $x_B=0.184883$, $t=-0.026192$, $W^2=1.03335$, $y=0.552277$

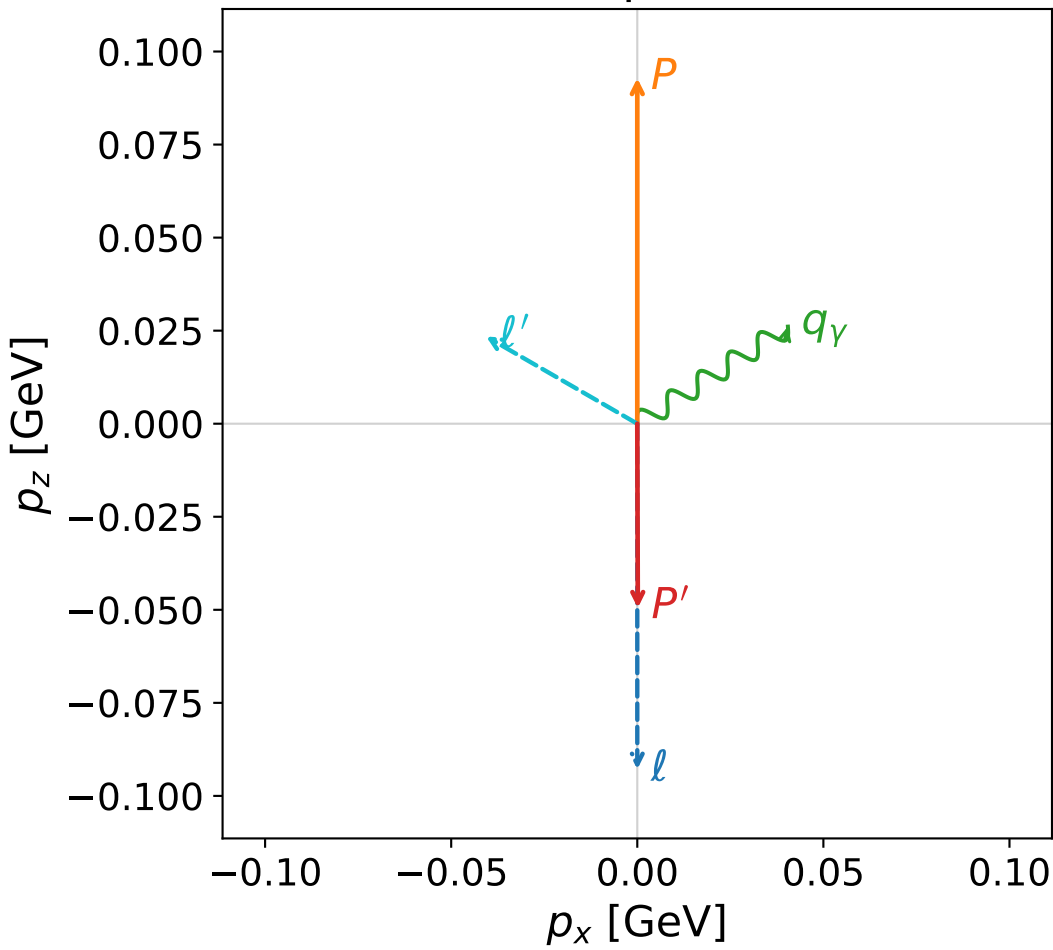
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1543$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1543$
 P $E= 0.9506$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1543$
 ℓ' $E= 0.0848$ $p_x= -0.0784$ $p_y= 0.0164$ $p_z= 0.0280$
 P' $E= 0.9380$ $p_x= 0.0014$ $p_y= 0.0037$ $p_z= -0.0080$
 q_γ $E= 0.0820$ $p_x= 0.0770$ $p_y= -0.0202$ $p_z= -0.0200$

Leading normalized final-state amplitudes
 (N=3, retained=94.6%)

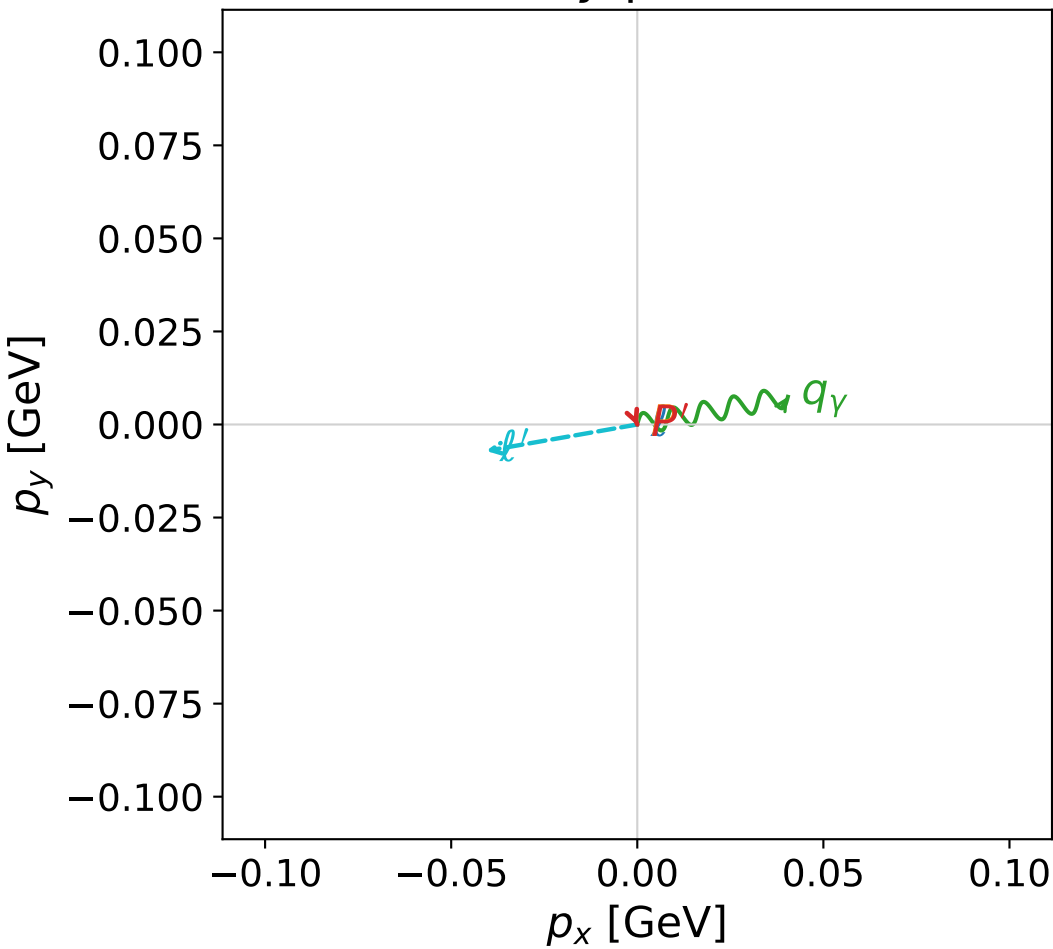


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



x--y plane



dw_mixing_angles_polarization_02_local_minimum_834

D_W=0.0114814

region: polarization_cluster_02_local_minimum_834

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0834

lepton mixing angle: $\alpha_e=0.82148265$ rad

incoming lepton: $0.681136|+\rangle + 0.732157|-\rangle$

proton mixing angle: $\alpha_p=1.4392388$ rad

incoming proton: $0.131178|+\rangle + 0.991359|-\rangle$

$s=1.07213$, $\sqrt{s}=1.03544$, $|\vec{P}|=0.0928531$, $|\vec{P}'|=0.0495096$

$\theta_{p'}=3.13403$, $\theta_Y=1.00582$, $\phi_{p'}=5.09593$, $\phi_Y=0.182049$

$E_Y=0.0487168$, $\theta_{\ell'}=1.05404$, $\phi_{\ell'}=3.31475$

$Q^2=0.0131554$, $x_B=0.122659$, $t=-0.0202561$, $W^2=0.97394$, $y=0.557761$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ $E= 0.0929$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0929$

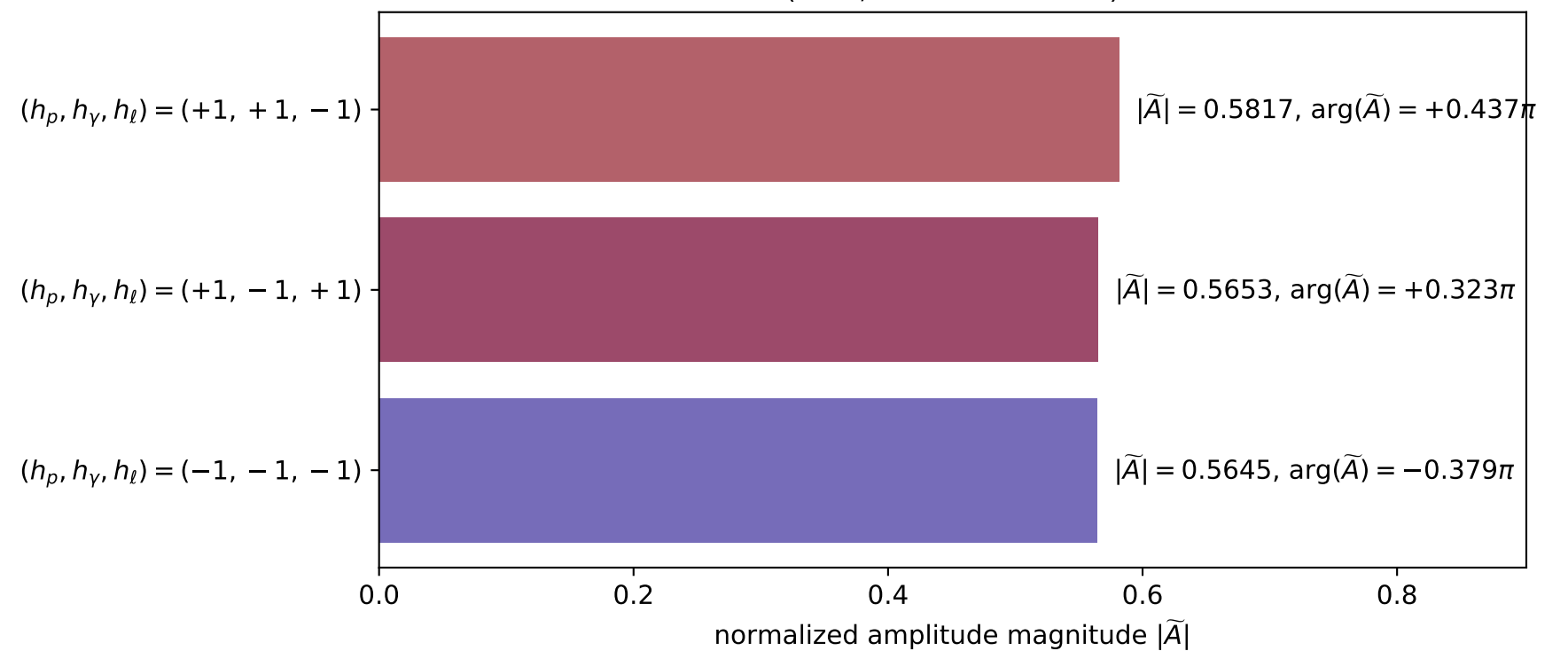
P $E= 0.9426$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0929$

ℓ' $E= 0.0474$ $p_x= -0.0406$ $p_y= -0.0071$ $p_z= 0.0234$

P' $E= 0.9393$ $p_x= 0.0001$ $p_y= -0.0003$ $p_z= -0.0495$

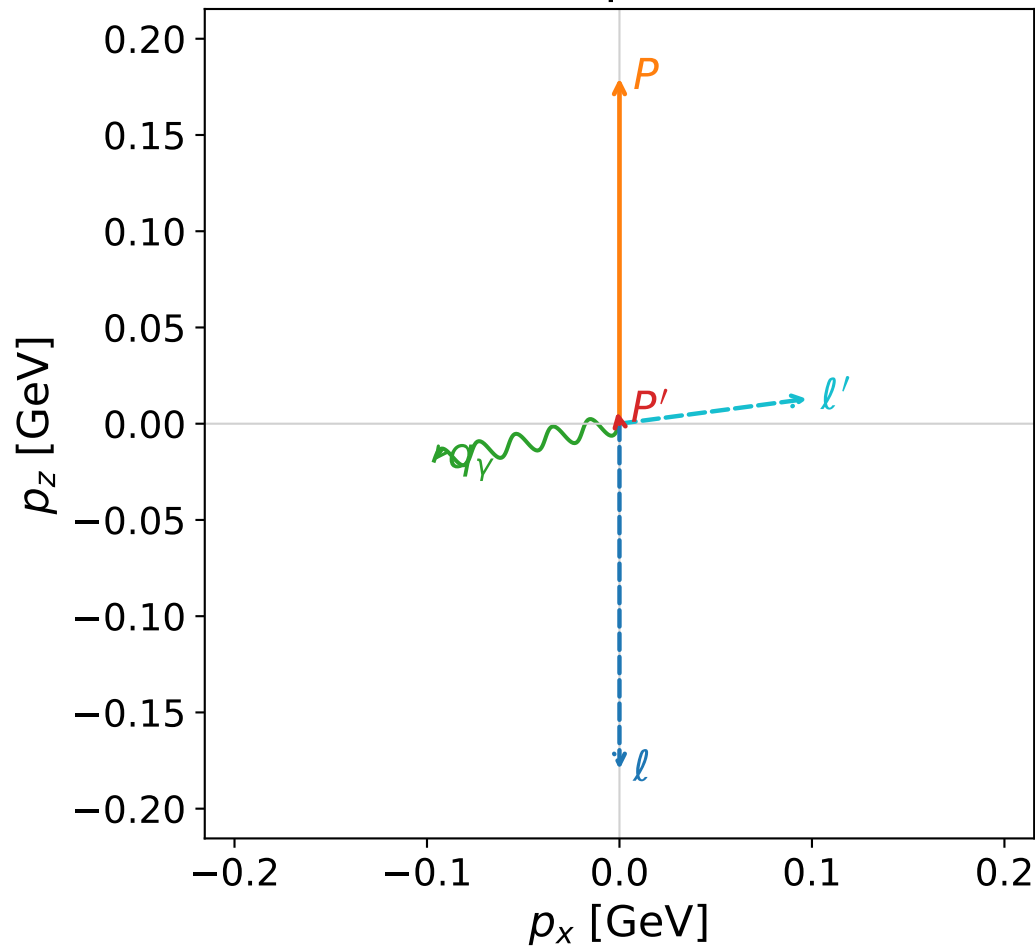
q_Y $E= 0.0487$ $p_x= 0.0405$ $p_y= 0.0074$ $p_z= 0.0261$

Leading normalized final-state amplitudes
(N=3, retained=97.7%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



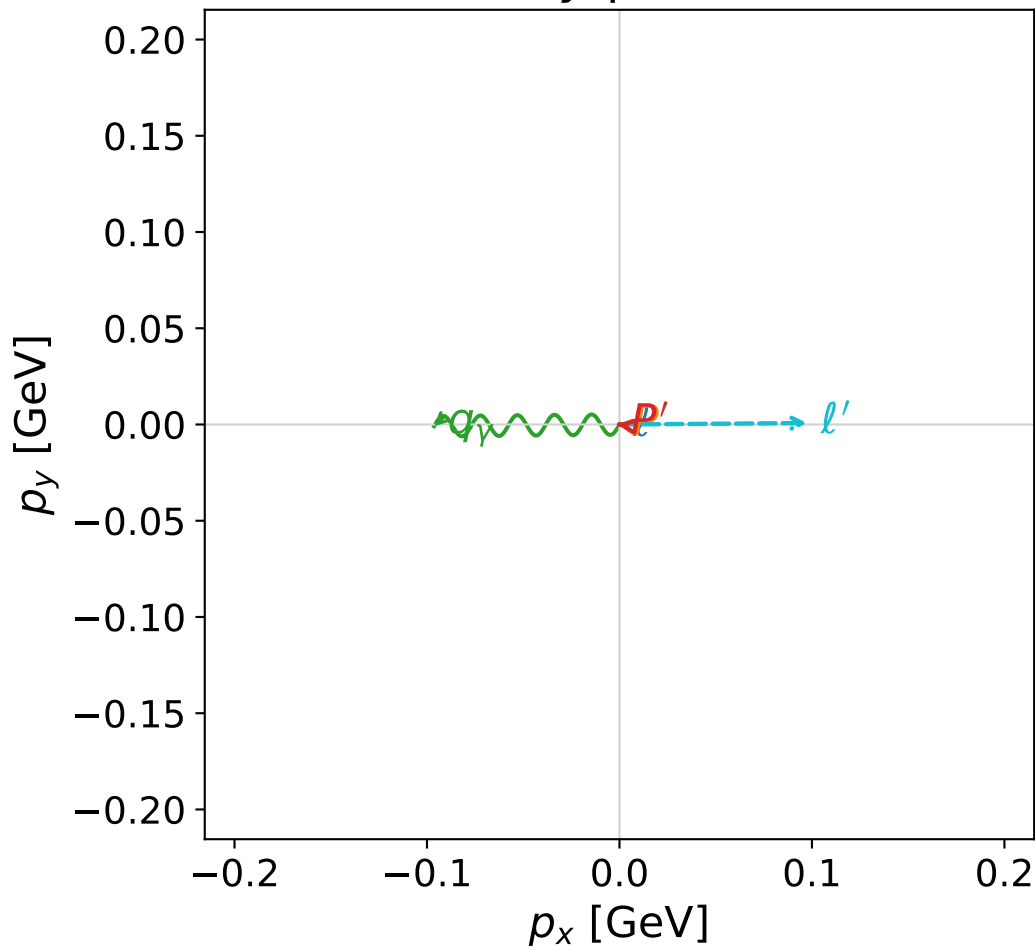
dw_mixing_angles_polarization_02_local_minimum_837
 $D_W=0.0115901$
 region: polarization_cluster_02_local_minimum_837
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0837
 lepton mixing angle: $\alpha_e=0.78523567$ rad
 incoming lepton: $0.707222|+\rangle + 0.706992|-\rangle$
 proton mixing angle: $\alpha_p=1.7803765$ rad
 incoming proton: $-0.208049|+\rangle + 0.978118|-\rangle$

$s=1.28726$, $\sqrt{s}=1.13458$, $|\vec{P}|=0.179546$, $|\vec{P}'|=0.00639123$
 $\theta_{p'}=0.190187$, $\theta_\gamma=1.76725$, $\phi_{p'}=3.00613$, $\phi_\gamma=3.15113$
 $E_\gamma=0.0981989$, $\theta_{l'}=1.43934$, $\phi_{l'}=0.00774615$
 $Q^2=0.0399477$, $x_B=0.178192$, $t=-0.0297348$, $W^2=1.06408$, $y=0.550253$

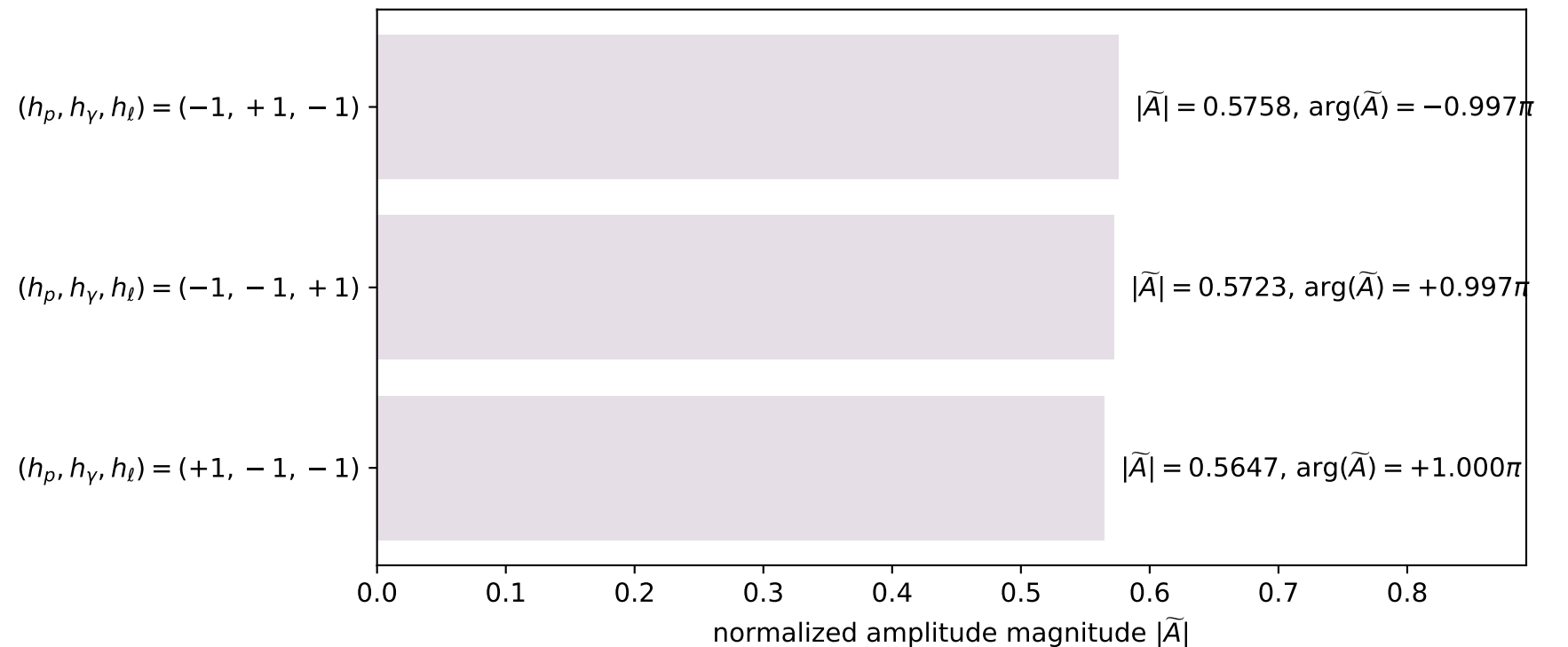
four-momenta $[E, p_x, p_y, p_z]$ GeV:

l	$E=$	0.1795	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	-0.1795
P	$E=$	0.9550	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	0.1795
l'	$E=$	0.0984	$p_x=$	0.0975	$p_y=$	0.0008	$p_z=$	0.0129
P'	$E=$	0.9380	$p_x=$	-0.0012	$p_y=$	0.0002	$p_z=$	0.0063
q_γ	$E=$	0.0982	$p_x=$	-0.0963	$p_y=$	-0.0009	$p_z=$	-0.0192

x--y plane

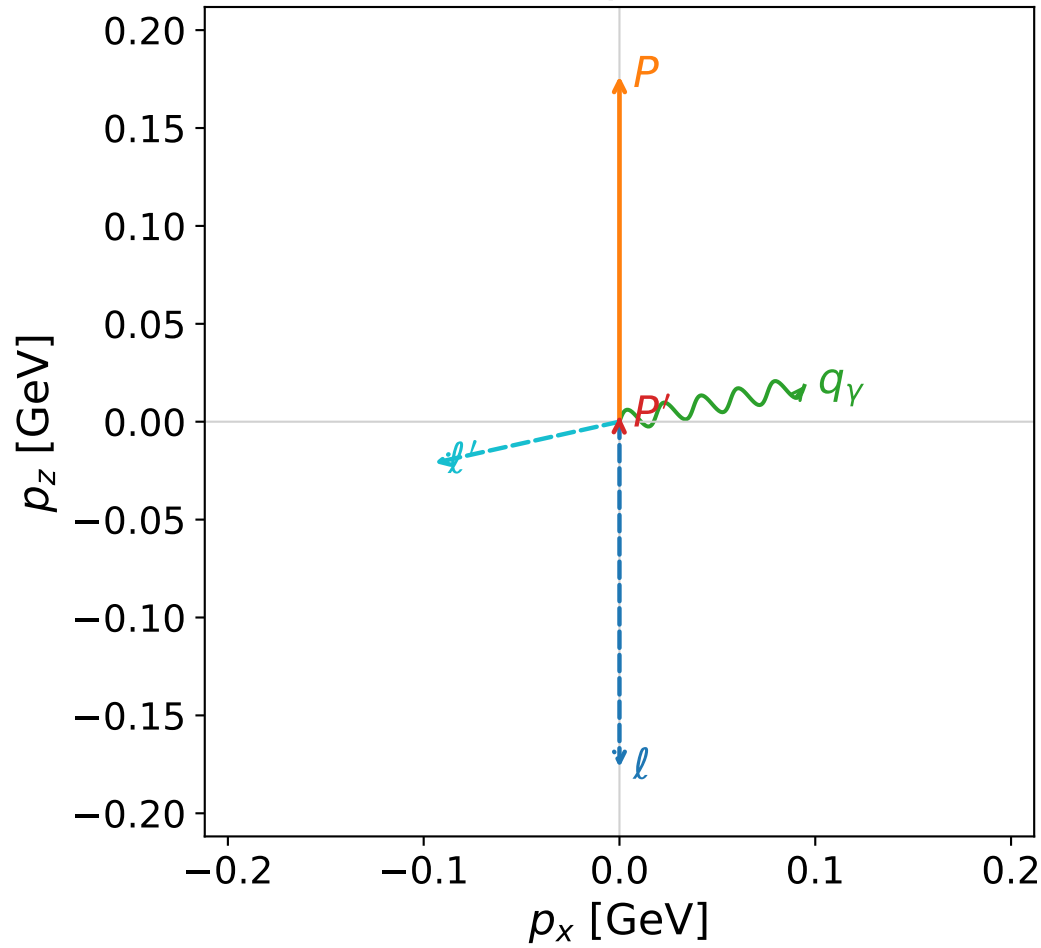


Leading normalized final-state amplitudes
(N=3, retained=97.8%)

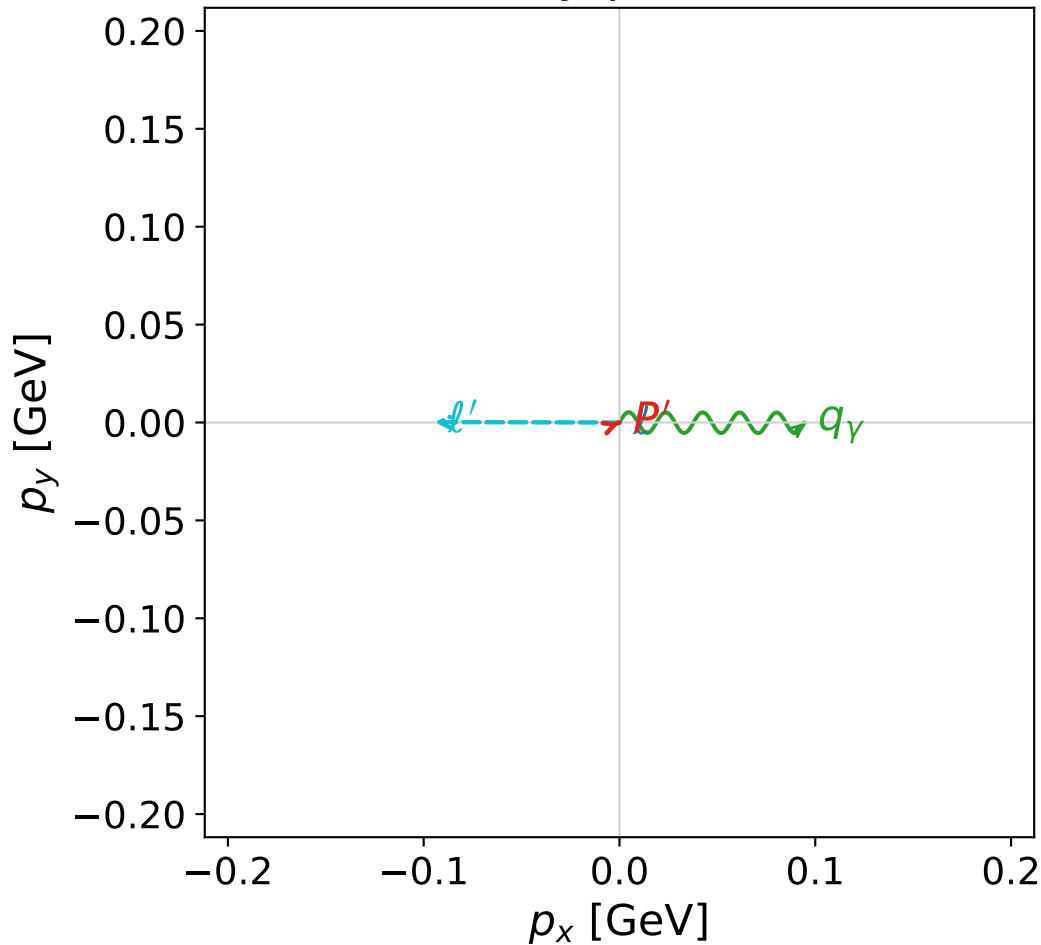


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



x--y plane



dw_mixing_angles_polarization_02_local_minimum_872

D_W=0.0124982

region: polarization_cluster_02_local_minimum_872

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0872

lepton mixing angle: $\alpha_e=0.75872549$ rad

incoming lepton: $0.725713|+\rangle + 0.687997|-\rangle$

proton mixing angle: $\alpha_p=1.3797863$ rad

incoming proton: $0.189851|+\rangle + 0.981813|-\rangle$

$s=1.2791$, $\sqrt{s}=1.13098$, $|\vec{P}|=0.176511$, $|\vec{P}'|=0.00273129$

$\theta_{p'}=0.0285799$, $\theta_Y=1.37942$, $\phi_{p'}=0.367067$, $\phi_Y=6.28093$

$E_Y=0.0961711$, $\theta_{\ell'}=1.78973$, $\phi_{\ell'}=3.13964$

$Q^2=0.0267504$, $x_B=0.129195$, $t=-0.0299289$, $W^2=1.06015$, $y=0.518595$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ $E= 0.1765$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1765$

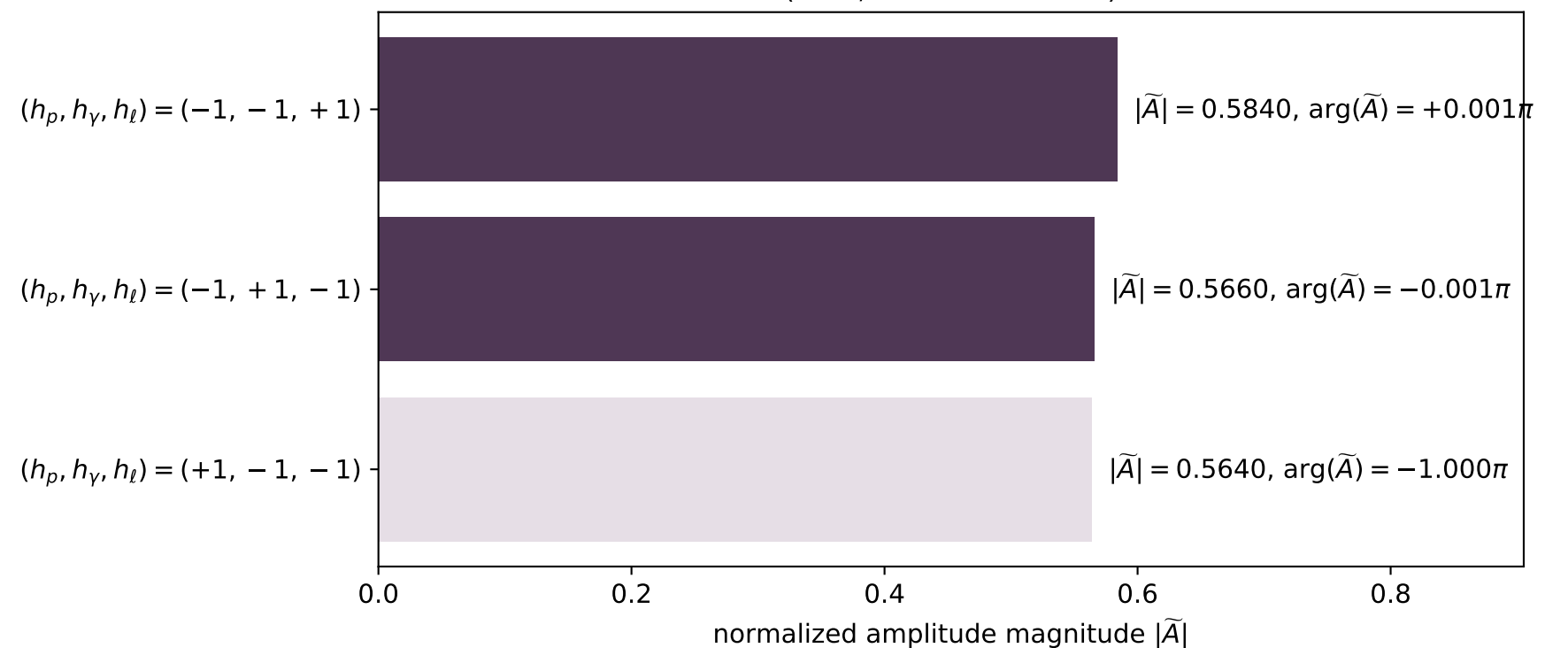
P $E= 0.9545$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1765$

ℓ' $E= 0.0968$ $p_x= -0.0945$ $p_y= 0.0002$ $p_z= -0.0210$

P' $E= 0.9380$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= 0.0027$

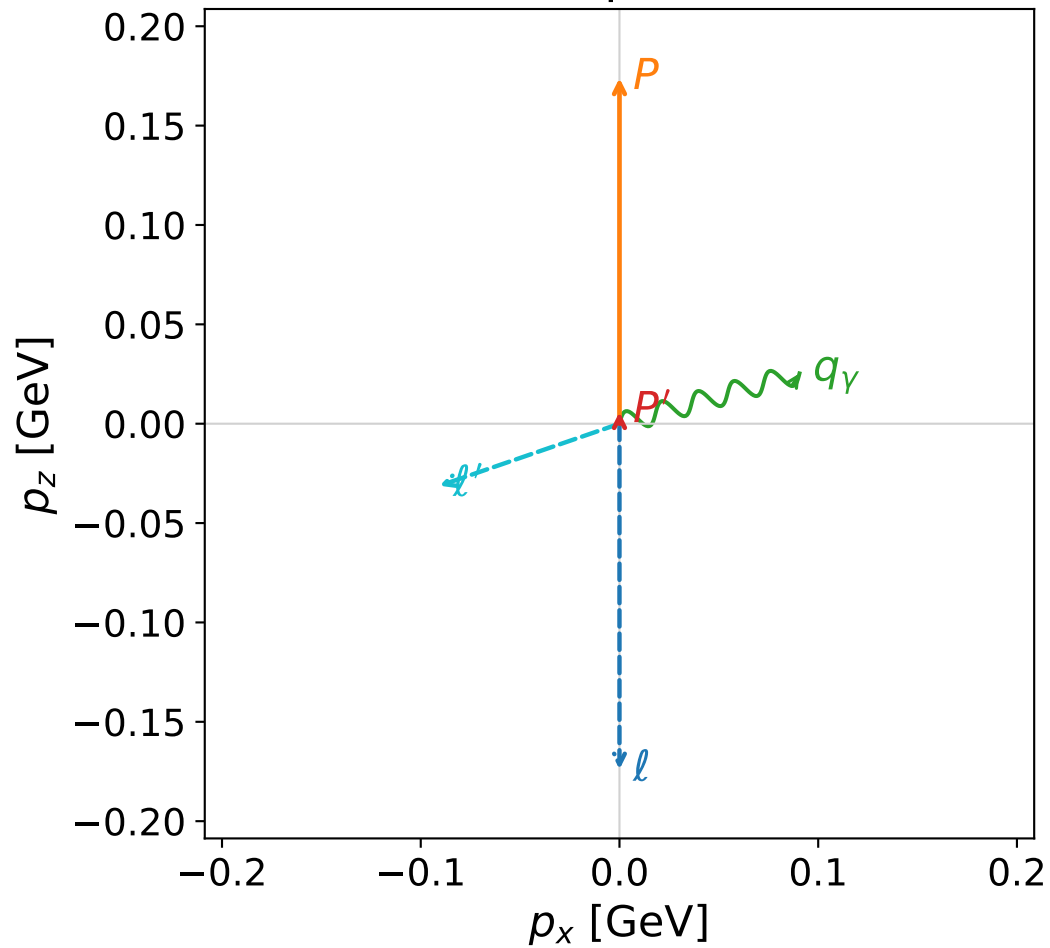
q_Y $E= 0.0962$ $p_x= 0.0944$ $p_y= -0.0002$ $p_z= 0.0183$

Leading normalized final-state amplitudes
(N=3, retained=97.9%)

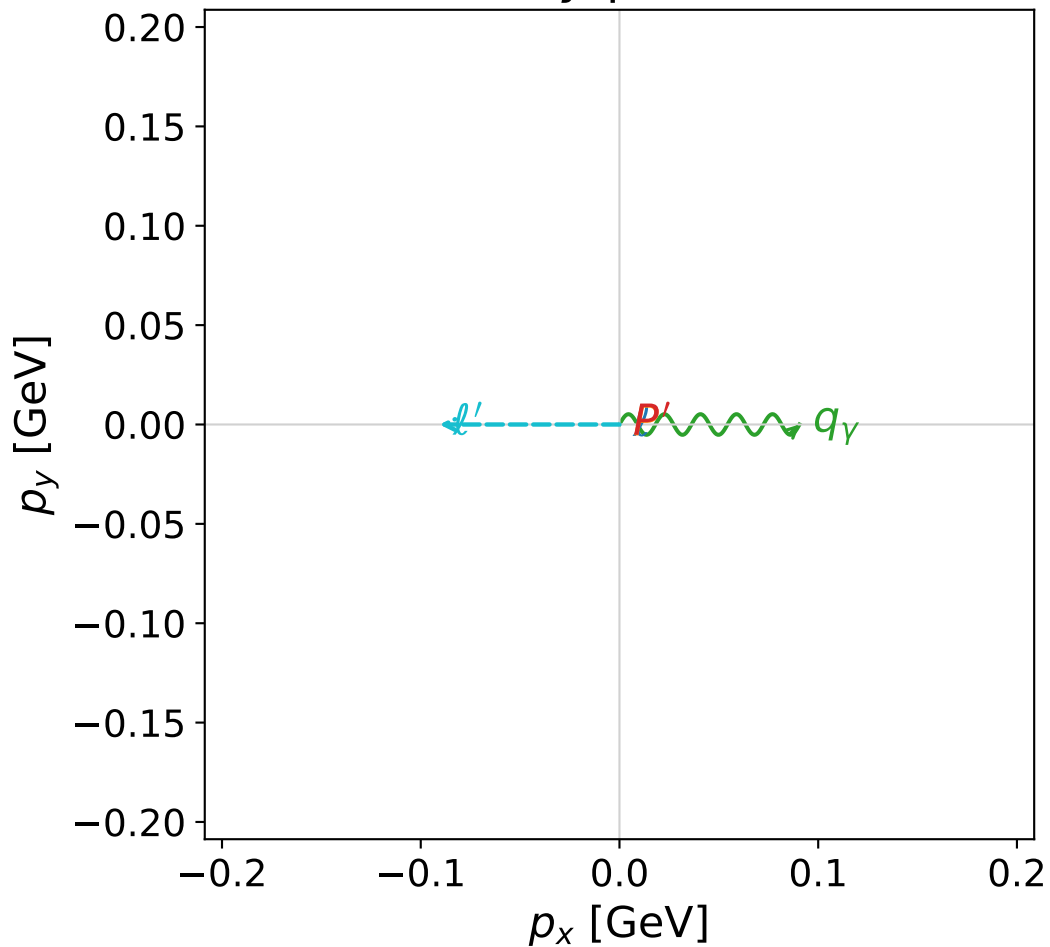


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



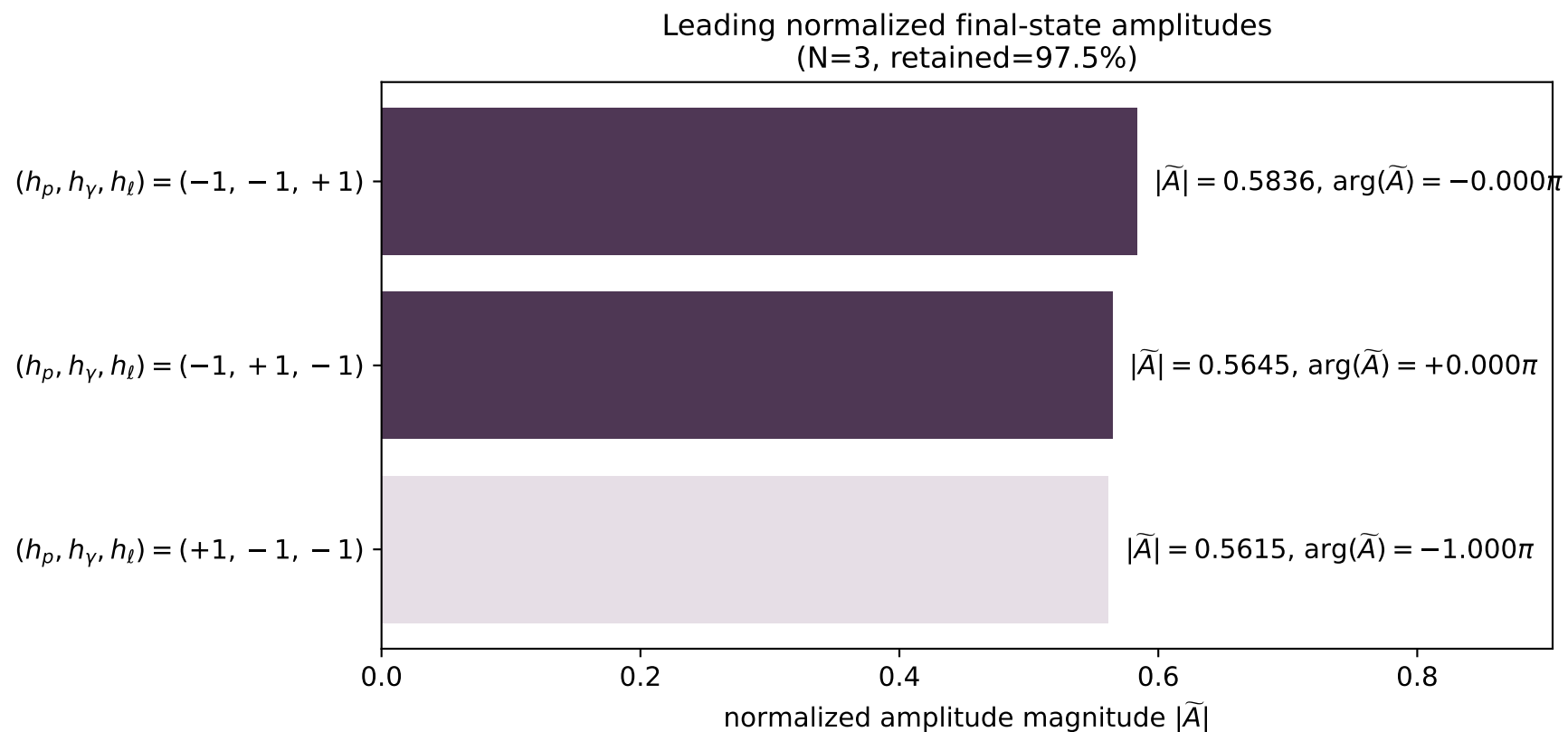
x--y plane



dw_mixing_angles_polarization_02_local_minimum_928
 $D_W=0.0148606$
 region: polarization_cluster_02_local_minimum_928
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0928
 lepton mixing angle: $\alpha_e=0.75409916$ rad
 incoming lepton: $0.728889|+\rangle + 0.684632|-\rangle$
 proton mixing angle: $\alpha_p=1.4076007$ rad
 incoming proton: $0.162472|+\rangle + 0.986713|-\rangle$

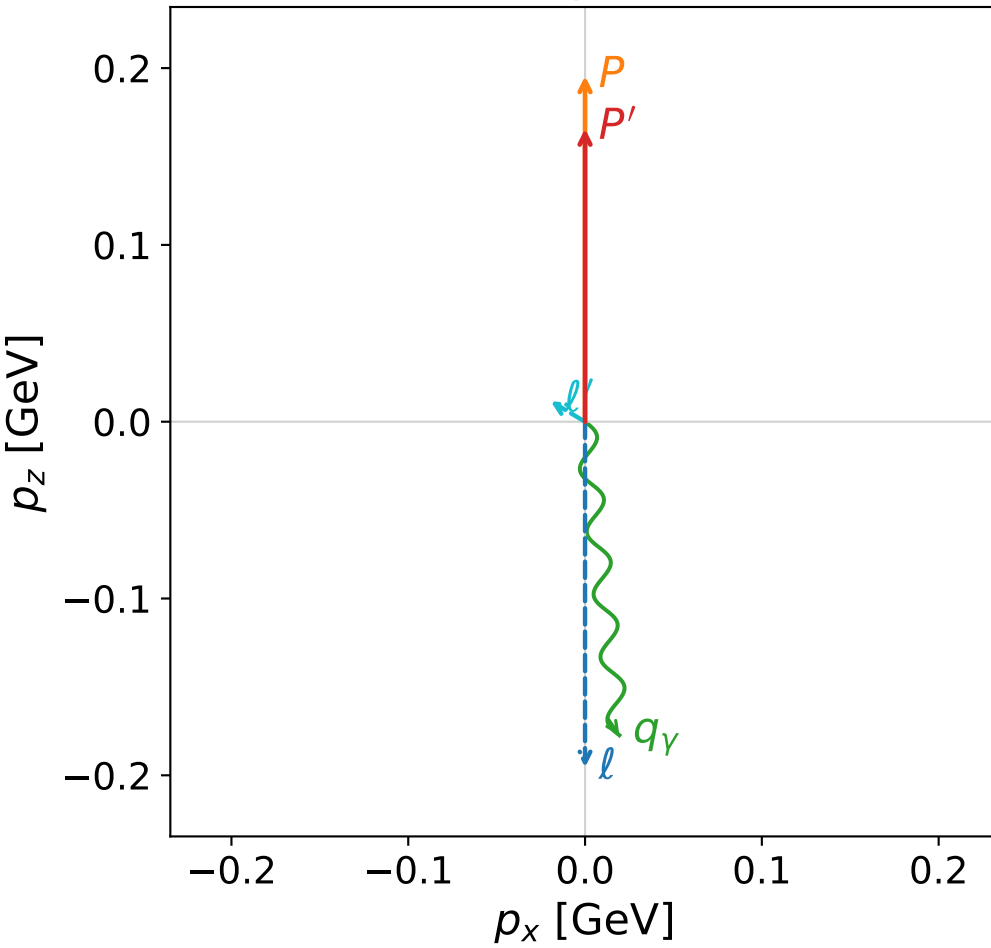
$s=1.27212$, $\sqrt{s}=1.12788$, $|\vec{P}|=0.173898$, $|\vec{P}'|=0.00591441$
 $\theta_{p'}=0$, $\theta_\gamma=1.2983$, $\phi_{p'}=0.768168$, $\phi_\gamma=2.86224e-05$
 $E_\gamma=0.0940503$, $\theta_{\ell'}=1.90278$, $\phi_{\ell'}=3.14162$
 $Q^2=0.0224623$, $x_B=0.113101$, $t=-0.0279635$, $W^2=1.05599$, $y=0.506292$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1739$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1739$
 P $E= 0.9540$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1739$
 ℓ' $E= 0.0958$ $p_x= -0.0906$ $p_y= -0.0000$ $p_z= -0.0312$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0059$
 q_γ $E= 0.0941$ $p_x= 0.0906$ $p_y= 0.0000$ $p_z= 0.0253$



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

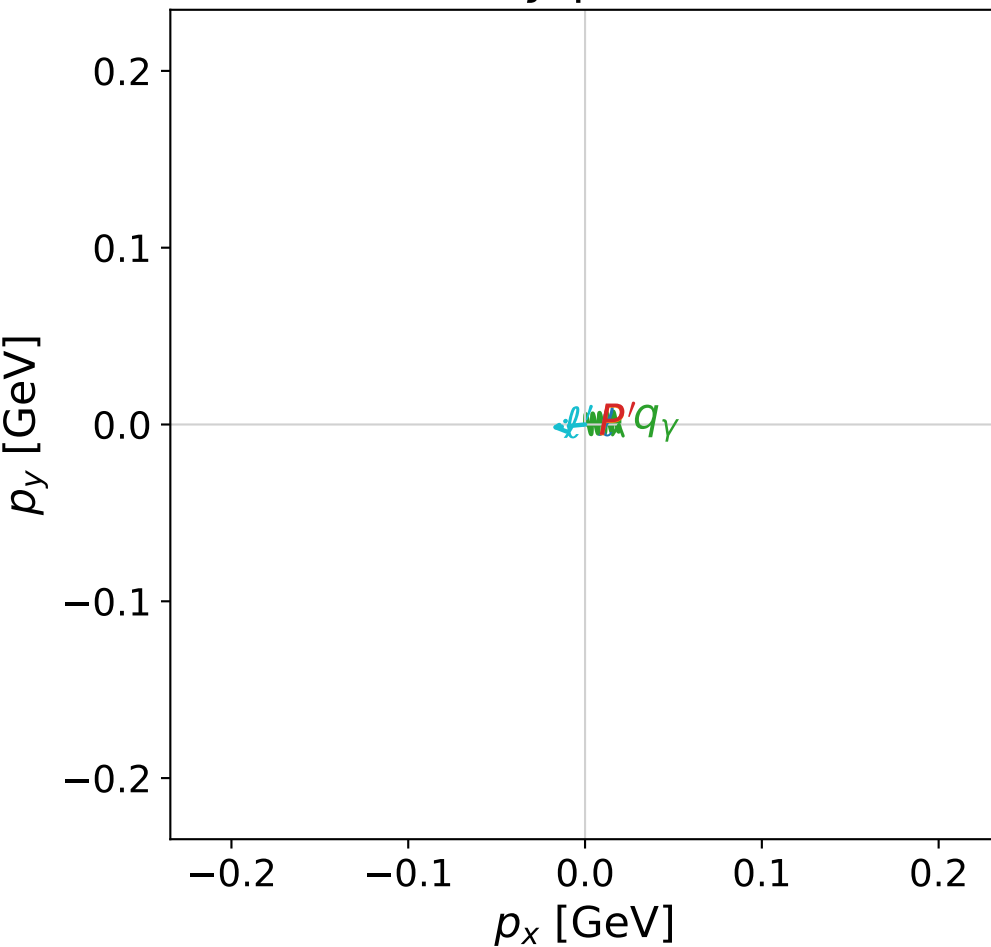


dw_mixing_angles_polarization_03_local_minimum_378
 $D_W=0.0012061$
 region: polarization_cluster_03_local_minimum_378
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0378
 lepton mixing angle: $\alpha_e=0.8336692$ rad
 incoming lepton: $0.672164|+\rangle +0.740403|-\rangle$
 proton mixing angle: $\alpha_p=3.1197285$ rad
 incoming proton: $-0.999761|+\rangle +0.0218624|-\rangle$

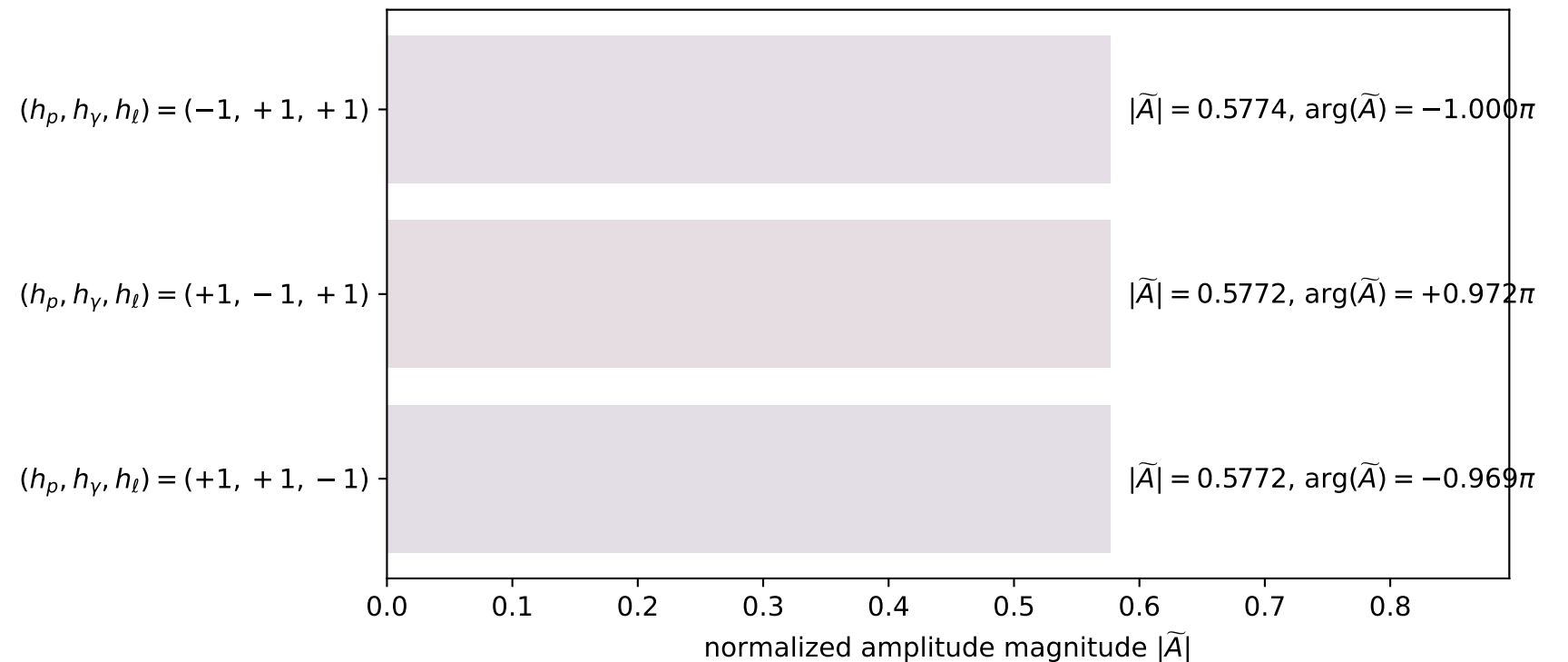
$s=1.33084$, $\sqrt{s}=1.15362$, $|\vec{P}|=0.195471$, $|\vec{P}'|=0.16574$
 $\theta_{p'}=0$, $\theta_\gamma=3.03202$, $\phi_{p'}=0.000172031$, $\phi_\gamma=0.0936696$
 $E_\gamma=0.178397$, $\theta_{\ell'}=1.03482$, $\phi_{\ell'}=3.23526$
 $Q^2=0.0134016$, $x_B=0.0325252$, $t=-0.000852303$, $W^2=1.27848$, $y=0.913613$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E=0.1955$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.1955$
 P $E=0.9582$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1955$
 ℓ' $E=0.0227$ $p_x=-0.0194$ $p_y=-0.0018$ $p_z=0.0116$
 P' $E=0.9525$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1657$
 q_γ $E=0.1784$ $p_x=0.0194$ $p_y=0.0018$ $p_z=-0.1773$

x--y plane

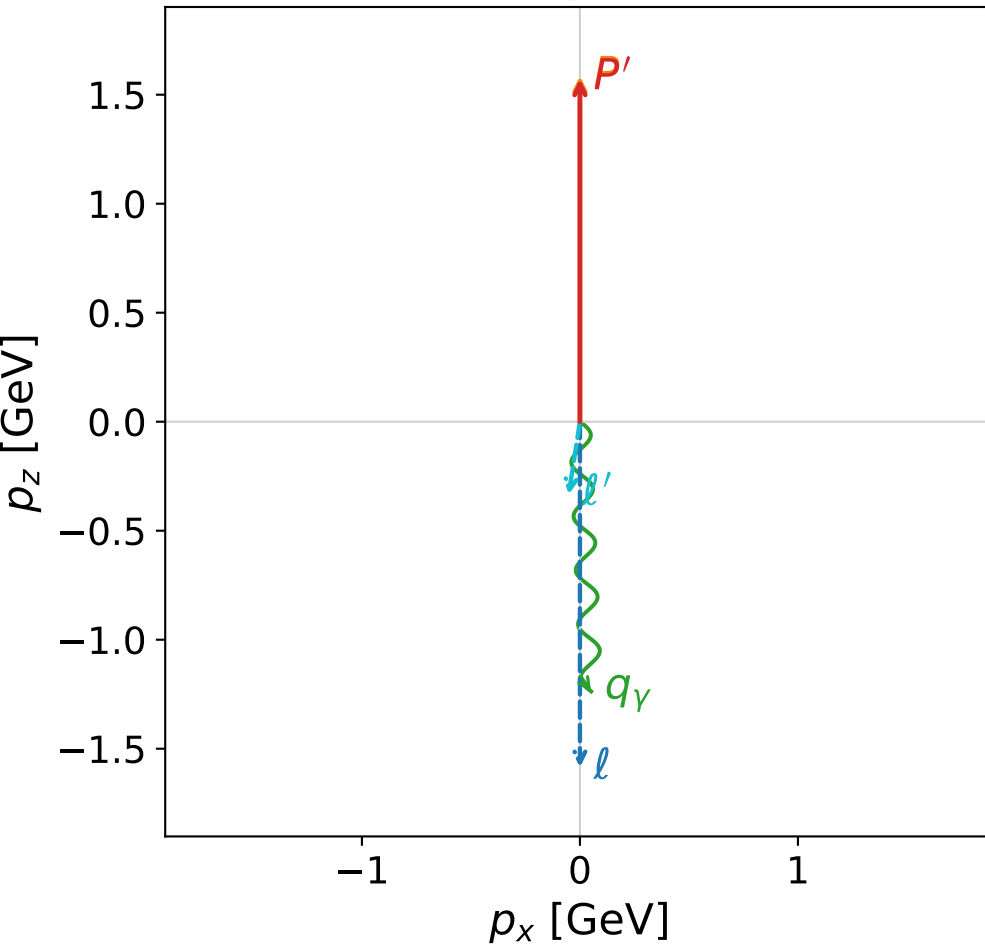


Leading normalized final-state amplitudes
 (N=3, retained=100.0%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



dw_mixing_angles_polarization_03_local_minimum_583

$D_W=0.00377731$

region: polarization_cluster_03_local_minimum_583

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0583

lepton mixing angle: $\alpha_e=0.79847104$ rad

incoming lepton: $0.697803|+\rangle + 0.71629|-\rangle$

proton mixing angle: $\alpha_p=3.0412699$ rad

incoming proton: $-0.994972|+\rangle + 0.100155|-\rangle$

$s=11.746$, $\sqrt{s}=3.42724$, $|\vec{P}|=1.58526$, $|\vec{P}'|=1.57167$

$\theta_{p'}=0$, $\theta_\gamma=3.04754$, $\phi_{p'}=2.35408$, $\phi_\gamma=5.17152$

$E_\gamma=1.24274$, $\theta_{l'}=2.80583$, $\phi_{l'}=2.02993$

$Q^2=0.0627118$, $x_B=0.00737703$, $t=-4.81958e-05$, $W^2=9.31808$, $y=0.782334$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

l $E= 1.5853$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.5853$

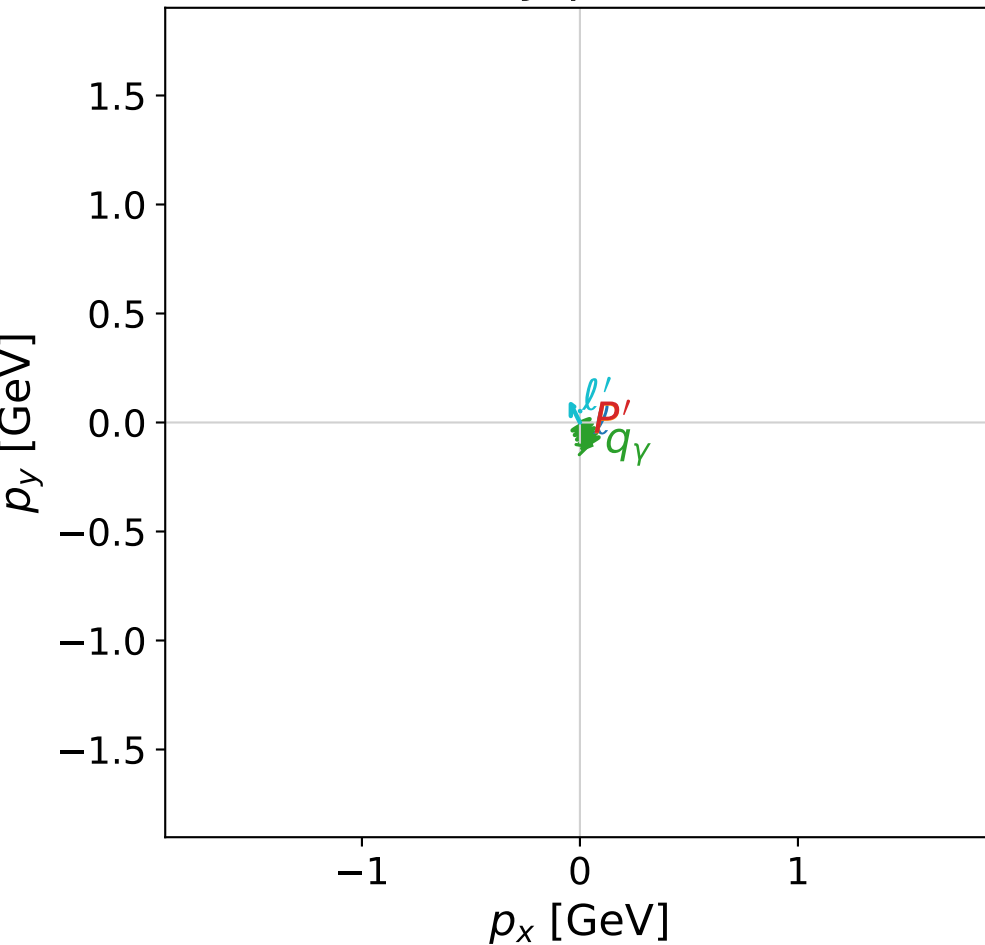
P $E= 1.8420$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.5853$

l' $E= 0.3542$ $p_x= -0.0517$ $p_y= 0.1046$ $p_z= -0.3344$

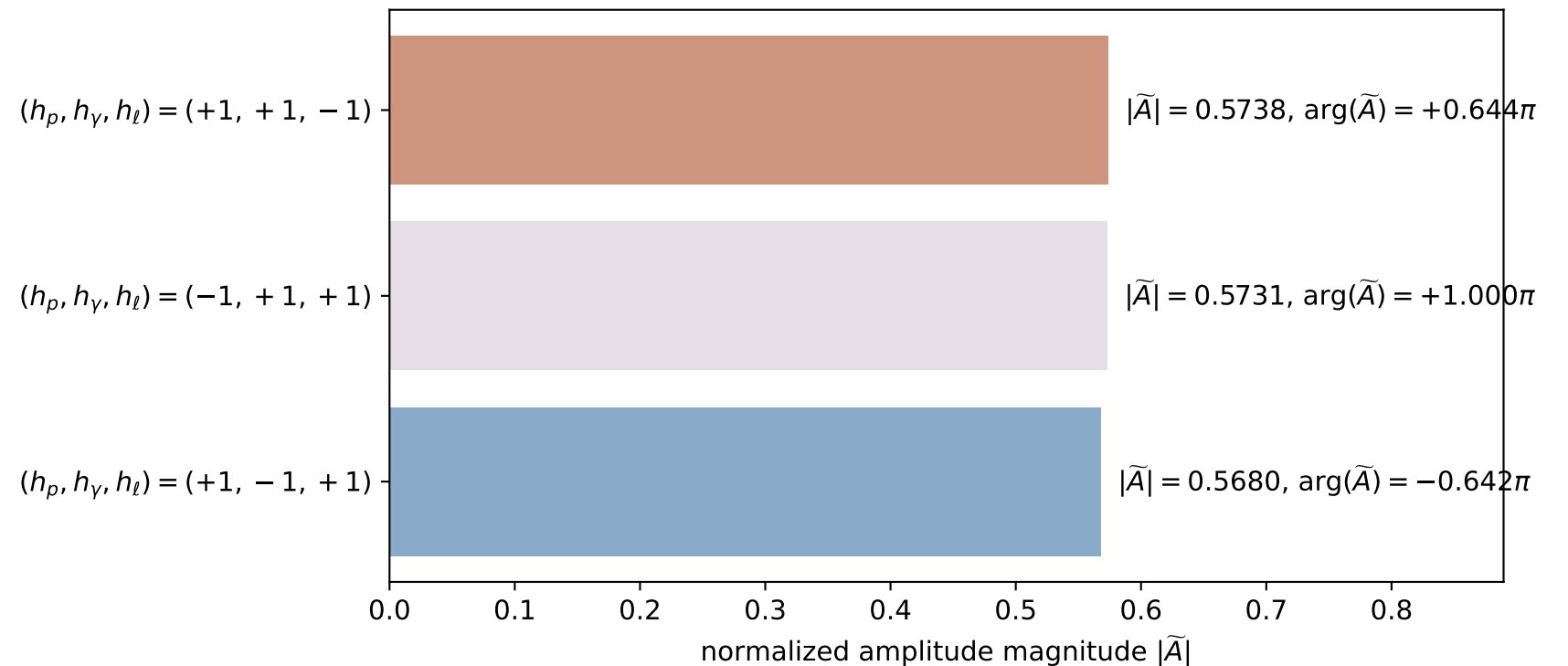
P' $E= 1.8303$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.5717$

q_γ $E= 1.2427$ $p_x= 0.0517$ $p_y= -0.1046$ $p_z= -1.2372$

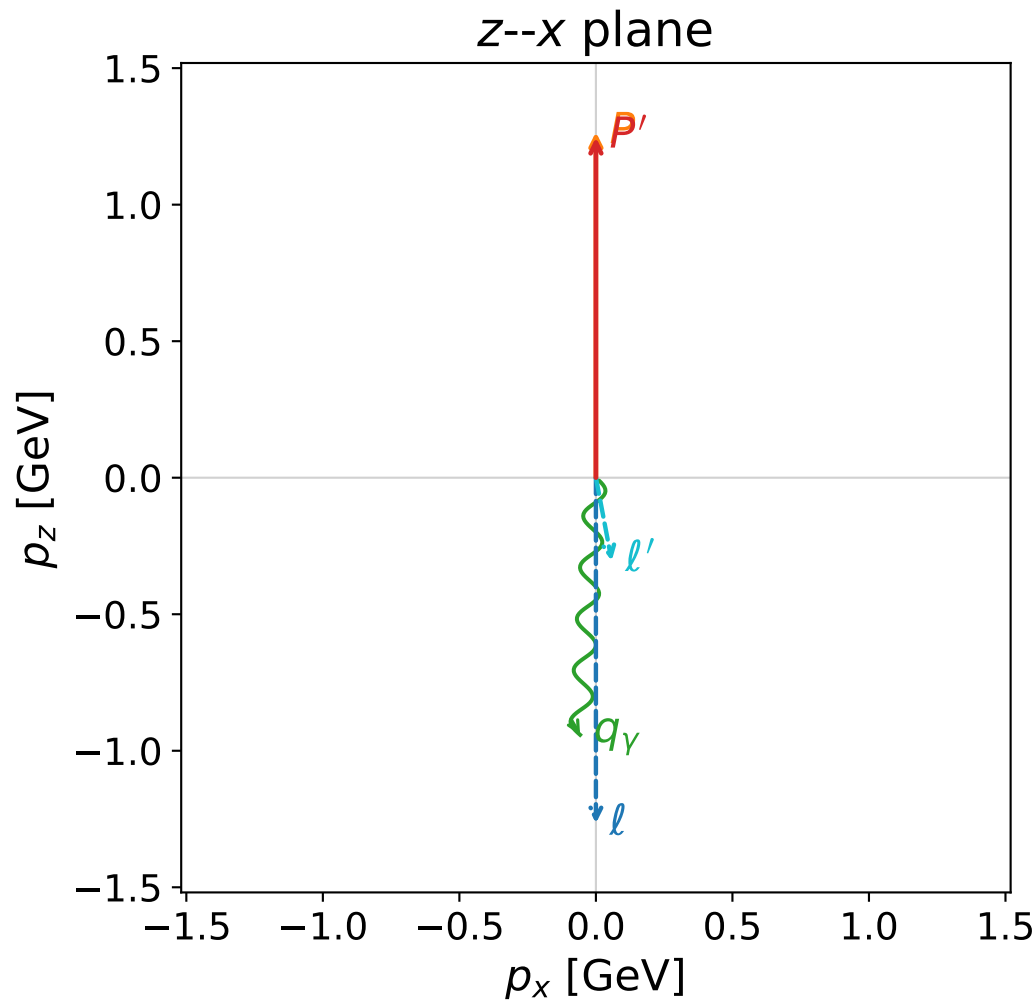
x--y plane



Leading normalized final-state amplitudes
(N=3, retained=98.0%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

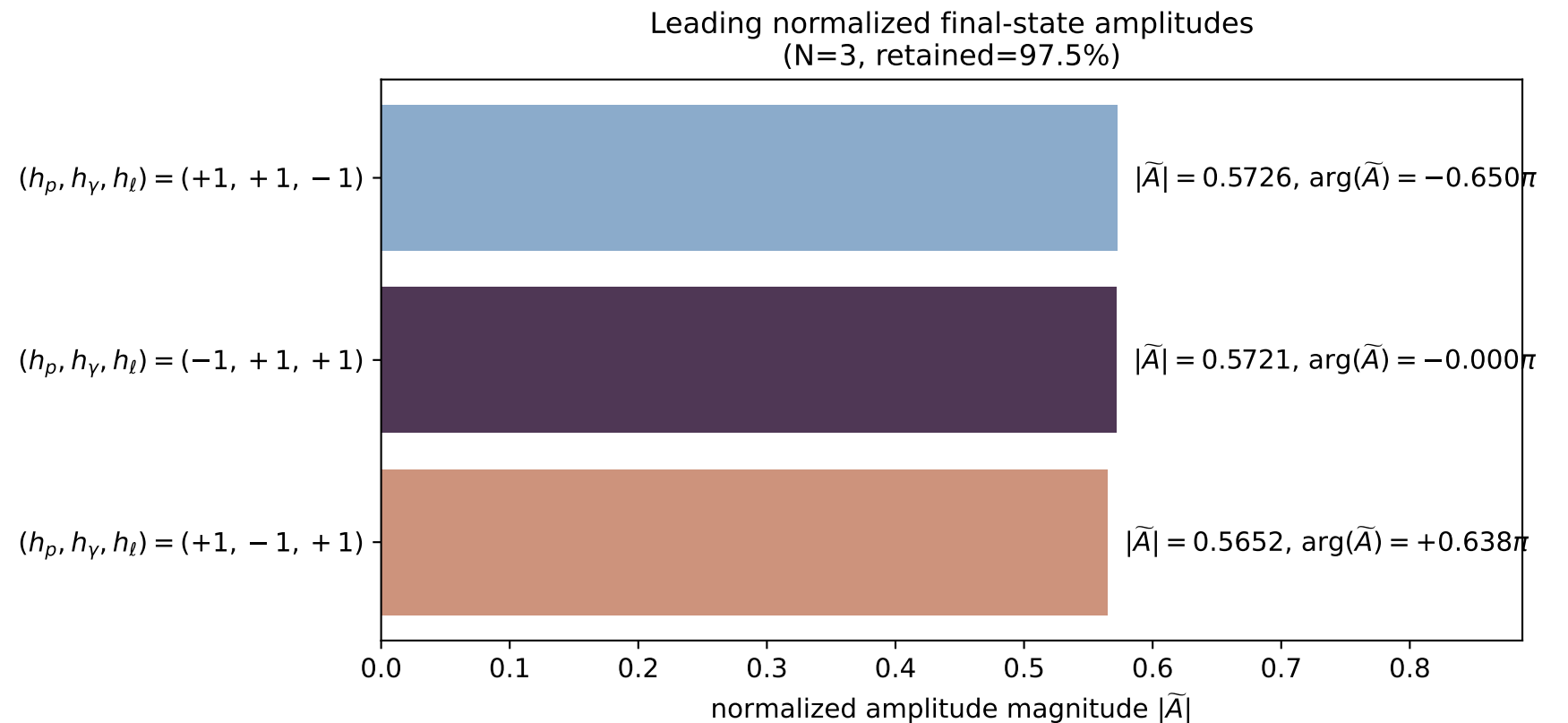
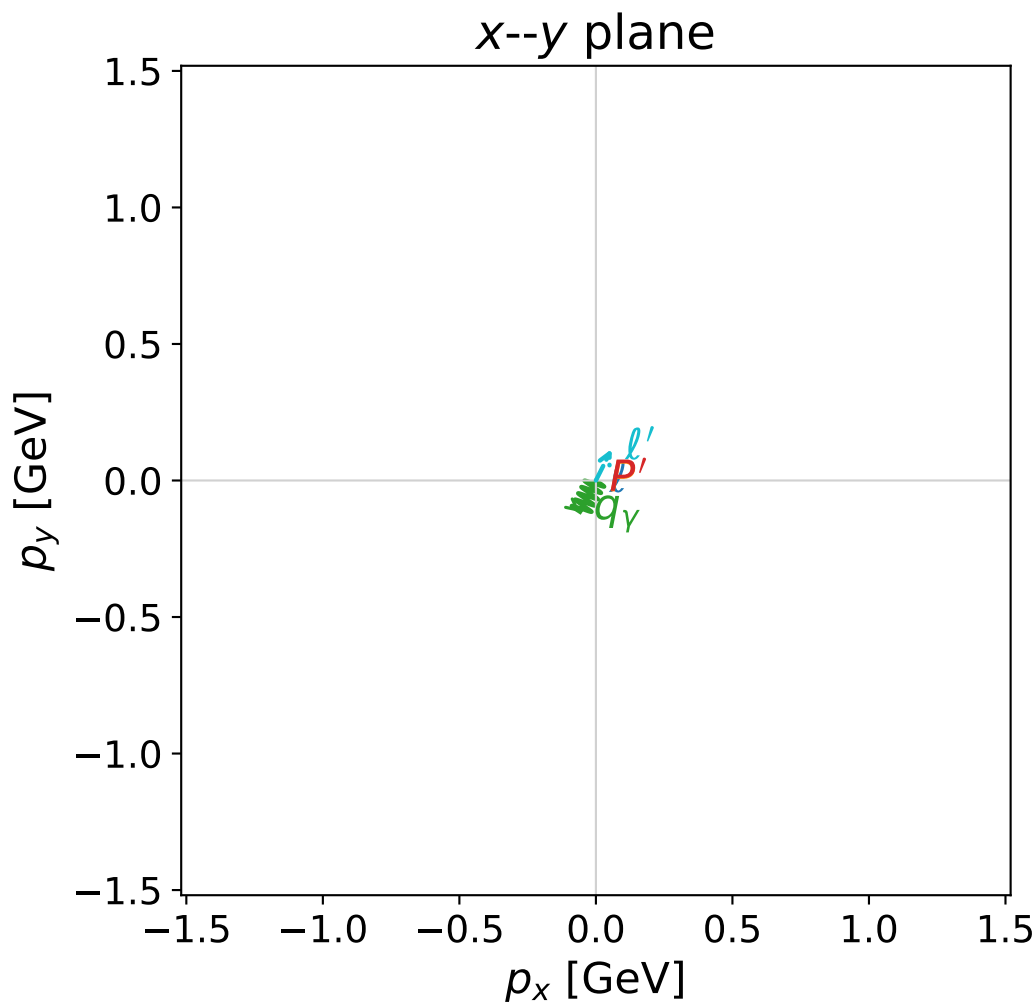


dw_mixing_angles_polarization_03_local_minimum_644
 $D_W=0.00536814$
 region: polarization_cluster_03_local_minimum_644
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0644
 lepton mixing angle: $\alpha_e=0.79589385$ rad
 incoming lepton: $0.699646|+\rangle +0.714489|-\rangle$
 proton mixing angle: $\alpha_p=0.11066879$ rad
 incoming proton: $0.993882|+\rangle +0.110443|-\rangle$

$s=8.07157$, $\sqrt{s}=2.84105$, $|\vec{P}|=1.26568$, $|\vec{P}'|=1.24622$
 $\theta_{p'}=0$, $\theta_\gamma=3.00532$, $\phi_{p'}=4.67144$, $\phi_\gamma=4.24742$
 $E_\gamma=0.95051$, $\theta_{\ell'}=2.74054$, $\phi_{\ell'}=1.10583$
 $Q^2=0.066438$, $x_B=0.012352$, $t=-0.000135646$, $W^2=6.19213$, $y=0.747905$

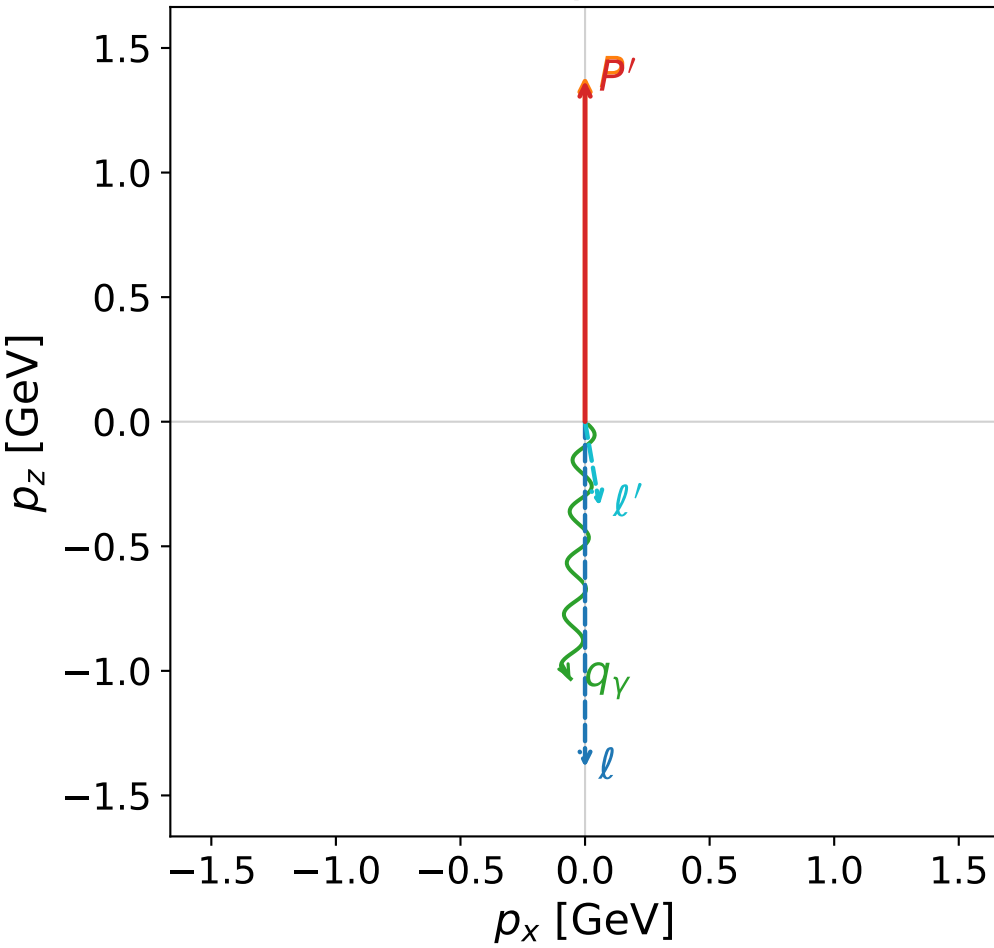
four-momenta $[E, p_x, p_y, p_z]$ GeV:

Particle	E	p_x	p_y	p_z
ℓ	1.2657	0.0000	0.0000	-1.2657
P	1.5754	0.0000	0.0000	1.2657
ℓ'	0.3308	0.0579	0.1154	-0.3045
P'	1.5598	-0.0000	-0.0000	1.2462
q_γ	0.9505	-0.0579	-0.1154	-0.9417

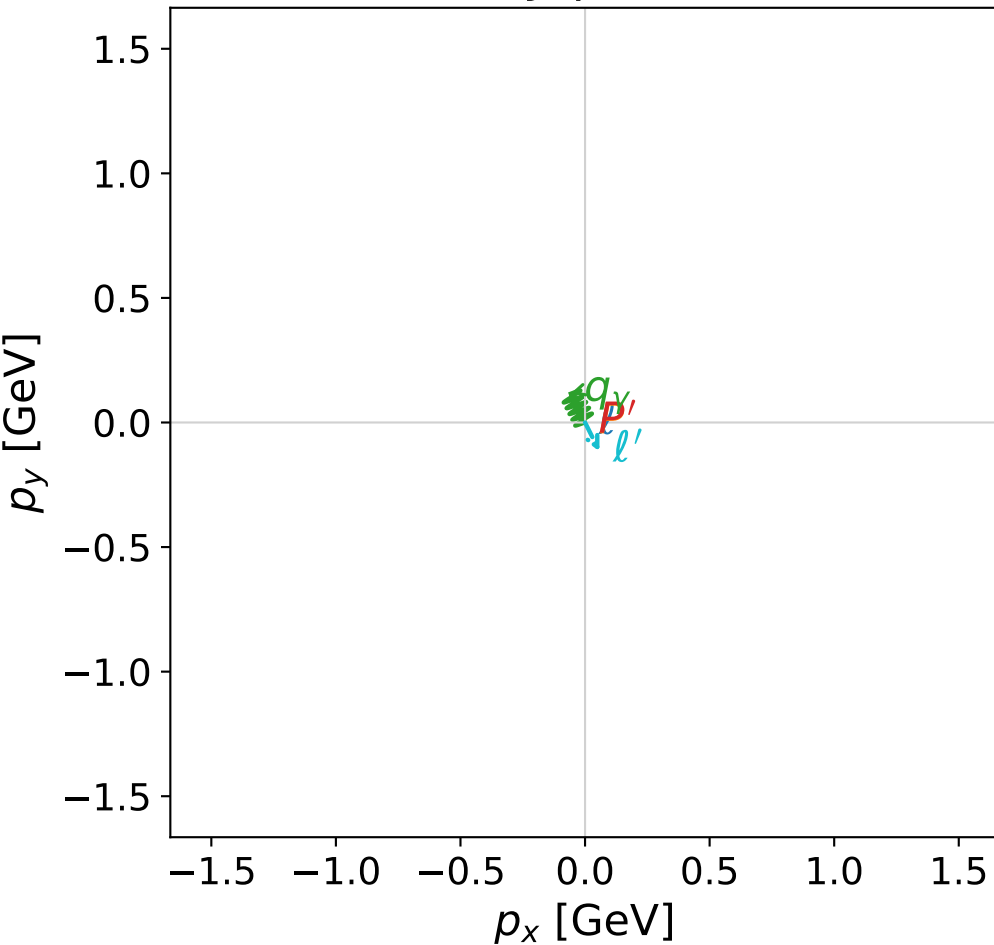


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



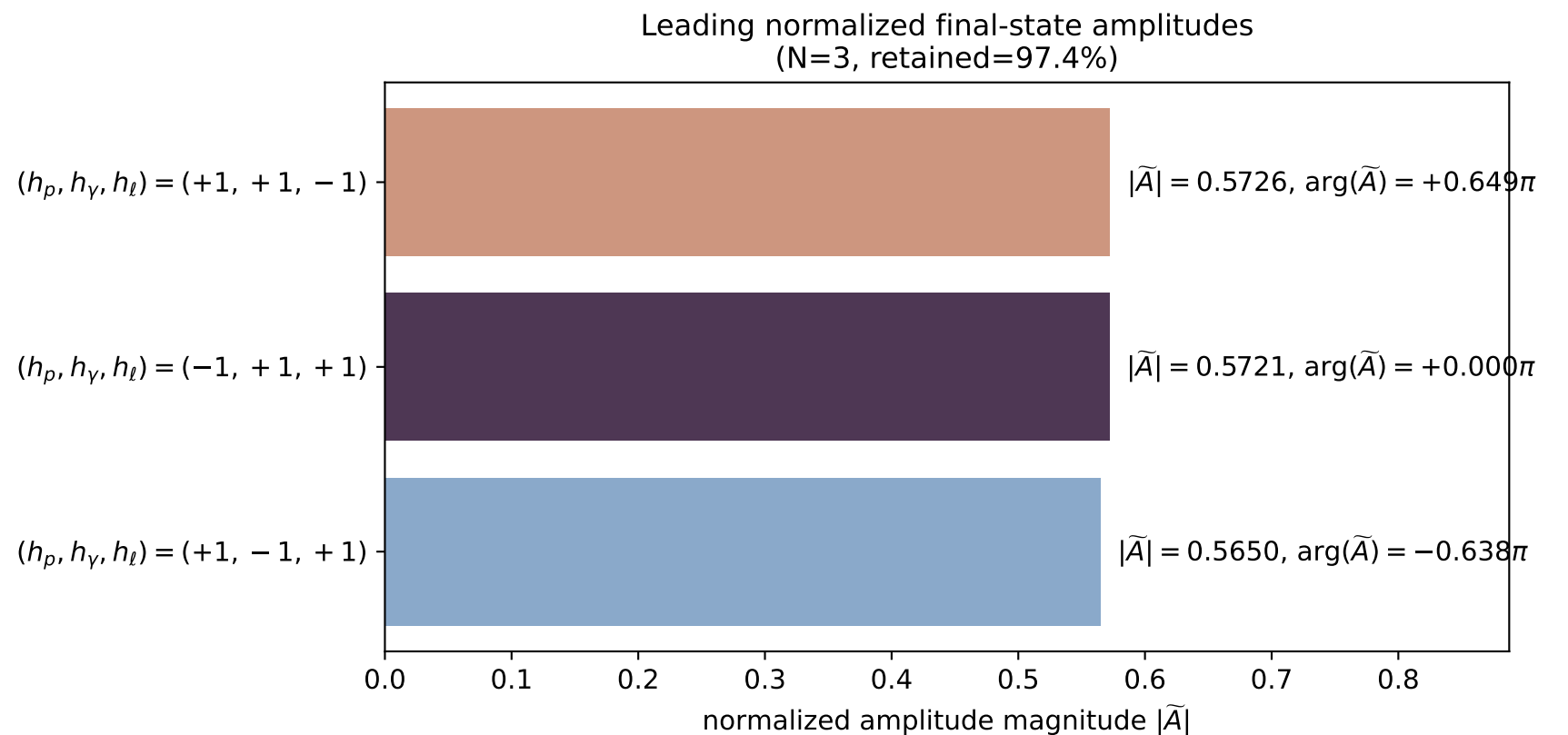
x--y plane



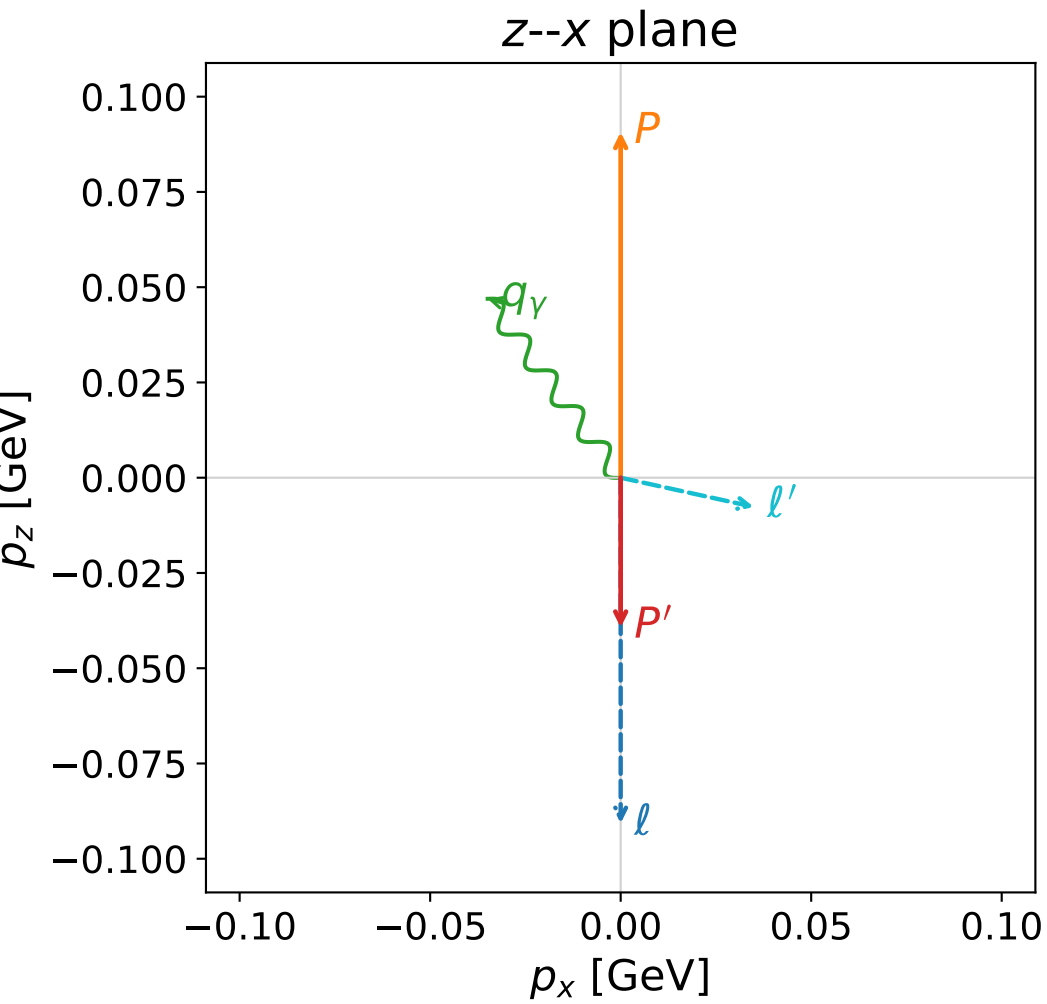
dw_mixing_angles_polarization_03_local_minimum_648
 $D_W=0.0054095$
 region: polarization_cluster_03_local_minimum_648
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0648
 lepton mixing angle: $\alpha_e=0.79612311$ rad
 incoming lepton: $0.699483|+\rangle +0.71465|-\rangle$
 proton mixing angle: $\alpha_p=0.11130992$ rad
 incoming proton: $0.993811|+\rangle +0.11108|-\rangle$

$s=9.37335$, $\sqrt{s}=3.06159$, $|\vec{P}|=1.38711$, $|\vec{P}'|=1.36953$
 $\theta_{P'}=0$, $\theta_\gamma=3.01671$, $\phi_{P'}=0.61743$, $\phi_\gamma=2.03429$
 $E_\gamma=1.04008$, $\theta_{l'}=2.77514$, $\phi_{l'}=5.17588$
 $Q^2=0.0665991$, $x_B=0.0104943$, $t=-9.77838e-05$, $W^2=7.15947$, $y=0.747186$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.3871$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.3871$
 P $E= 1.6745$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.3871$
 l' $E= 0.3616$ $p_x= 0.0579$ $p_y= -0.1159$ $p_z= -0.3375$
 P' $E= 1.6600$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.3695$
 q_γ $E= 1.0401$ $p_x= -0.0579$ $p_y= 0.1159$ $p_z= -1.0320$



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

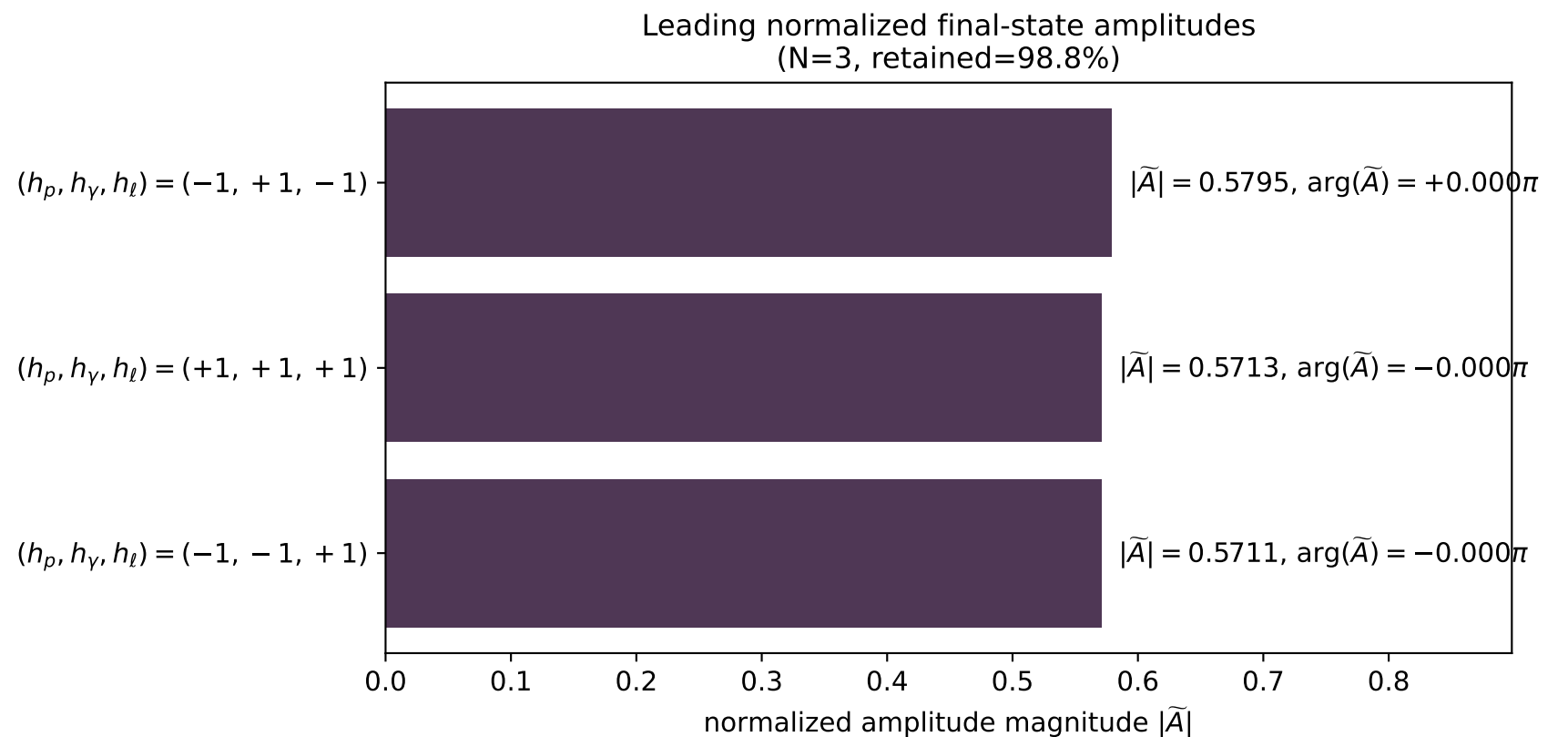
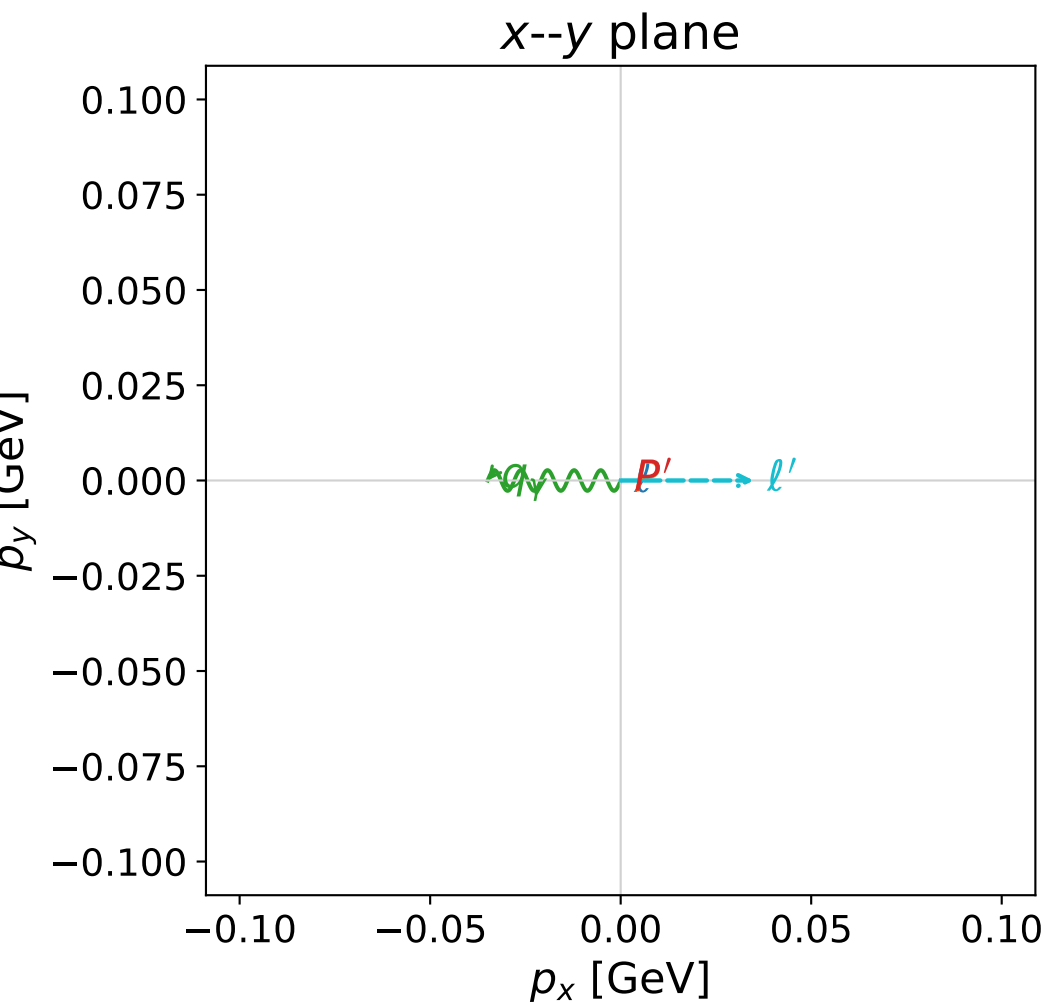


dw_mixing_angles_polarization_03_local_minimum_651
 $D_W=0.00554297$
 region: polarization_cluster_03_local_minimum_651
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0651
 lepton mixing angle: $\alpha_e=0.79607924$ rad
 incoming lepton: $0.699514|+\rangle + 0.714619|-\rangle$
 proton mixing angle: $\alpha_p=0.15671114$ rad
 incoming proton: $0.987746|+\rangle + 0.156071|-\rangle$

$s=1.06724$, $\sqrt{s}=1.03307$, $|\vec{P}|=0.0906976$, $|\vec{P}'|=0.0393062$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0.639302$, $\phi_{p'}=4.0112$, $\phi_\gamma=3.1416$
 $E_\gamma=0.0585098$, $\theta_{\ell'}=1.78649$, $\phi_{\ell'}=8.94111e-06$
 $Q^2=0.00509536$, $x_B=0.0429455$, $t=-0.0168884$, $W^2=0.993396$, $y=0.633133$

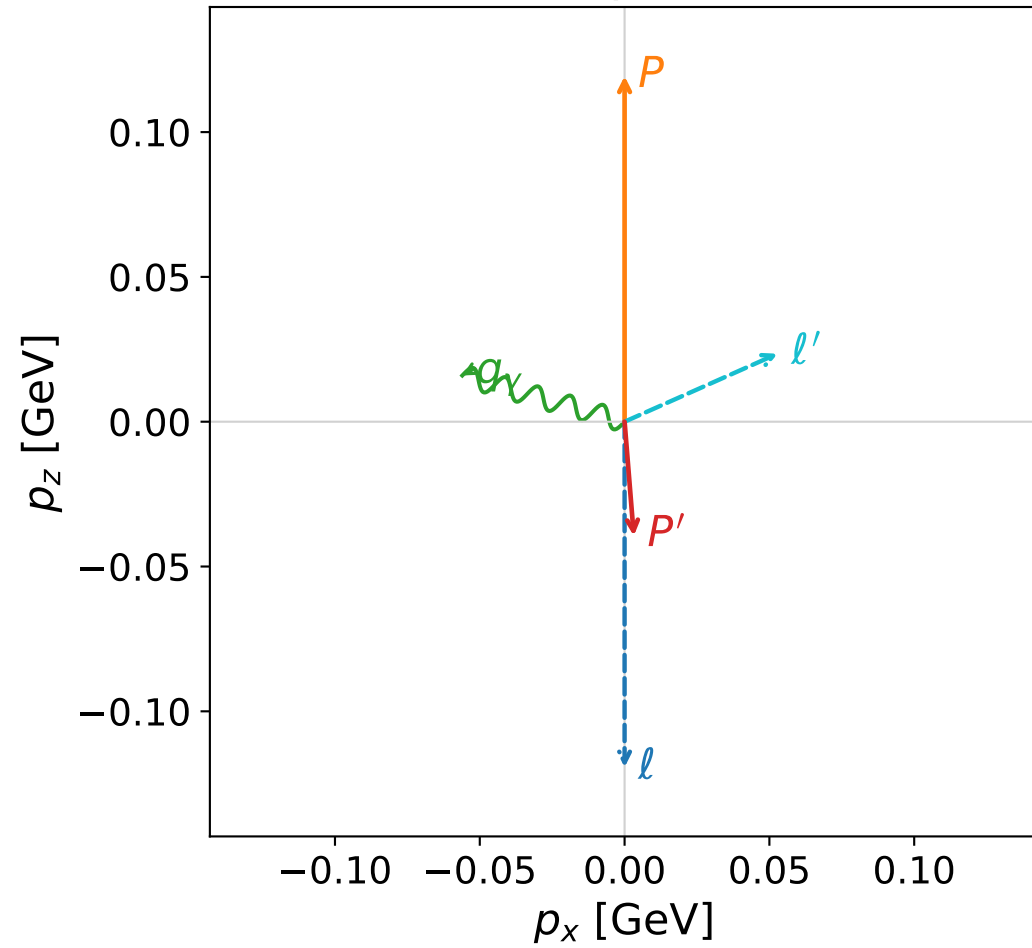
four-momenta $[E, p_x, p_y, p_z]$ GeV:

Particle	E	p_x	p_y	p_z
ℓ	0.0907	0.0000	0.0000	-0.0907
P	0.9424	0.0000	0.0000	0.0907
ℓ'	0.0357	0.0349	0.0000	-0.0076
P'	0.9388	-0.0000	-0.0000	-0.0393
q_γ	0.0585	-0.0349	-0.0000	0.0470



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

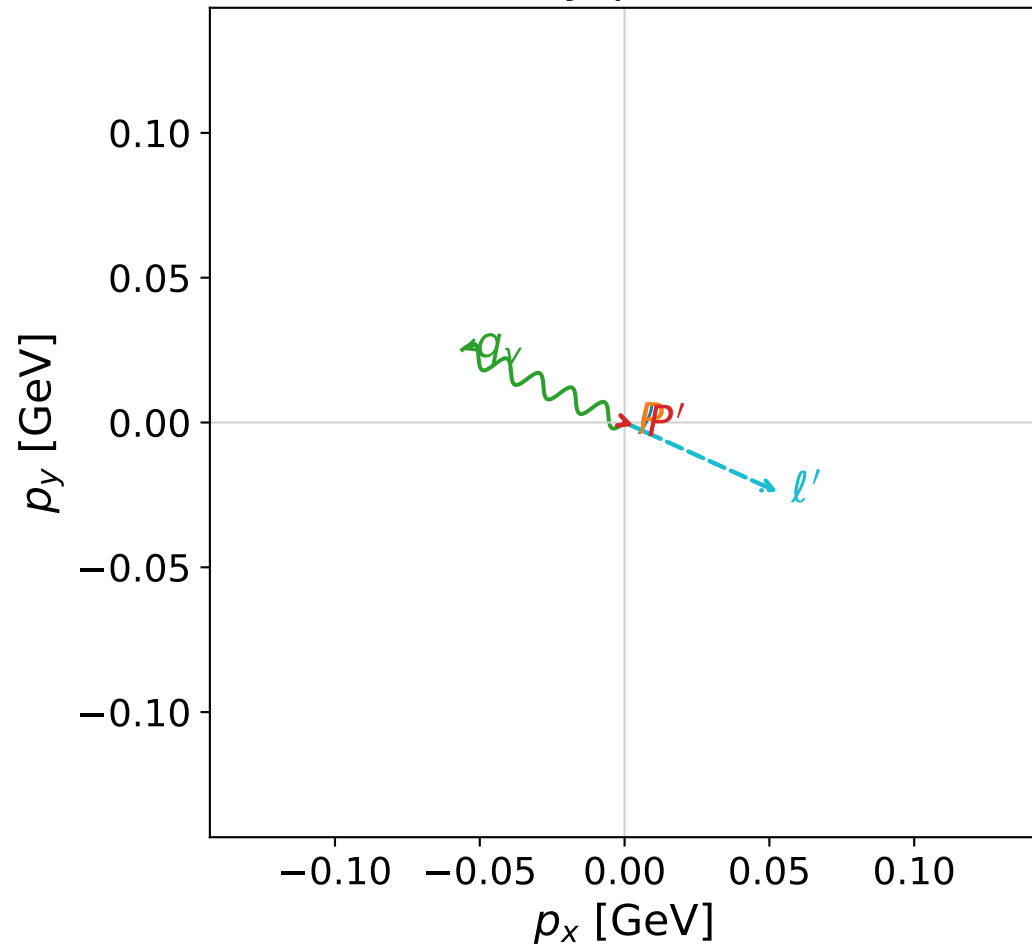


dw_mixing_angles_polarization_03_local_minimum_657
 $D_W=0.0057073$
 region: polarization_cluster_03_local_minimum_657
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0657
 lepton mixing angle: $\alpha_e=0.77071917$ rad
 incoming lepton: $0.71741|+\rangle + 0.696651|-\rangle$
 proton mixing angle: $\alpha_p=0.10578762$ rad
 incoming proton: $0.99441|+\rangle + 0.10559|-\rangle$

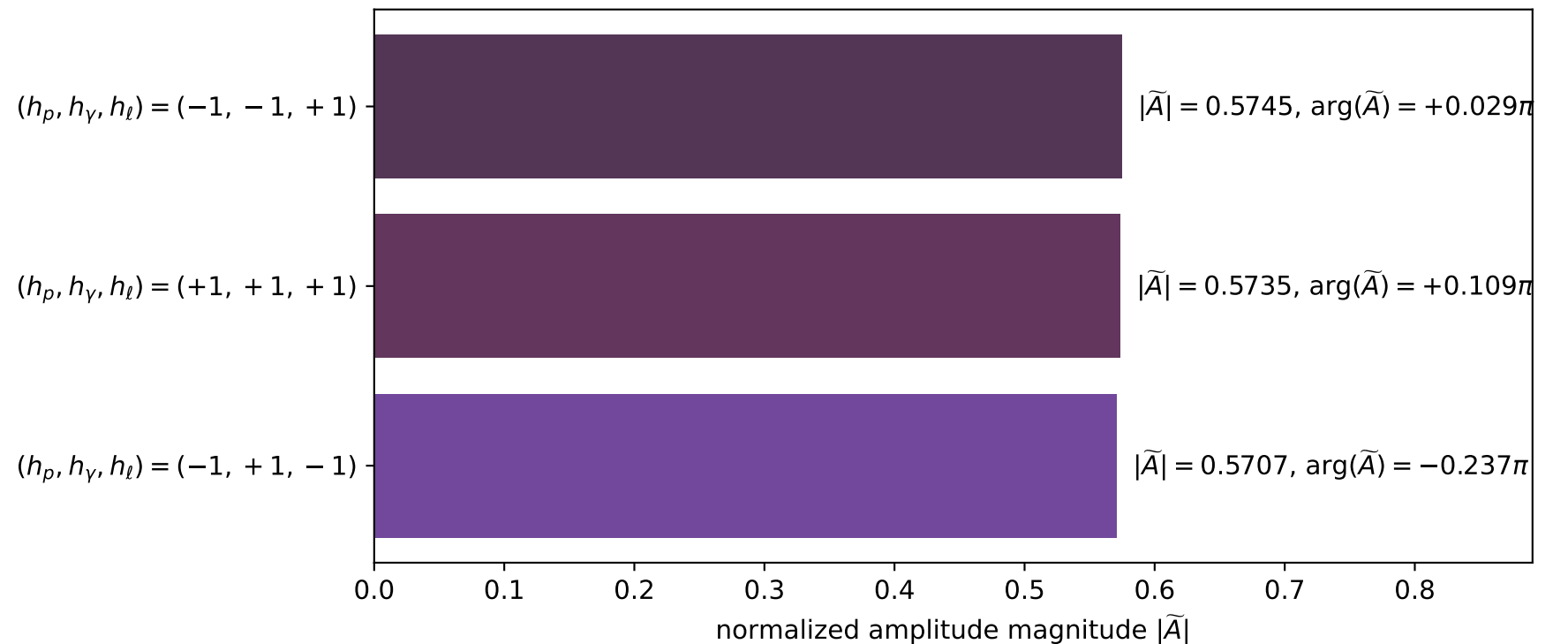
$s=1.13399$, $\sqrt{s}=1.06489$, $|\vec{P}|=0.11933$, $|\vec{P}'|=0.0395815$
 $\theta_{p'}=3.05612$, $\theta_\gamma=1.31674$, $\phi_{p'}=5.94507$, $\phi_\gamma=2.72054$
 $E_\gamma=0.0634418$, $\theta_{\ell'}=1.18619$, $\phi_{\ell'}=5.8573$
 $Q^2=0.0205499$, $x_B=0.14539$, $t=-0.0251732$, $W^2=1.00064$, $y=0.556145$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1193$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1193$
 P $E= 0.9456$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1193$
 ℓ' $E= 0.0626$ $p_x= 0.0529$ $p_y= -0.0240$ $p_z= 0.0235$
 P' $E= 0.9388$ $p_x= 0.0032$ $p_y= -0.0011$ $p_z= -0.0394$
 q_γ $E= 0.0634$ $p_x= -0.0560$ $p_y= 0.0251$ $p_z= 0.0159$

x--y plane

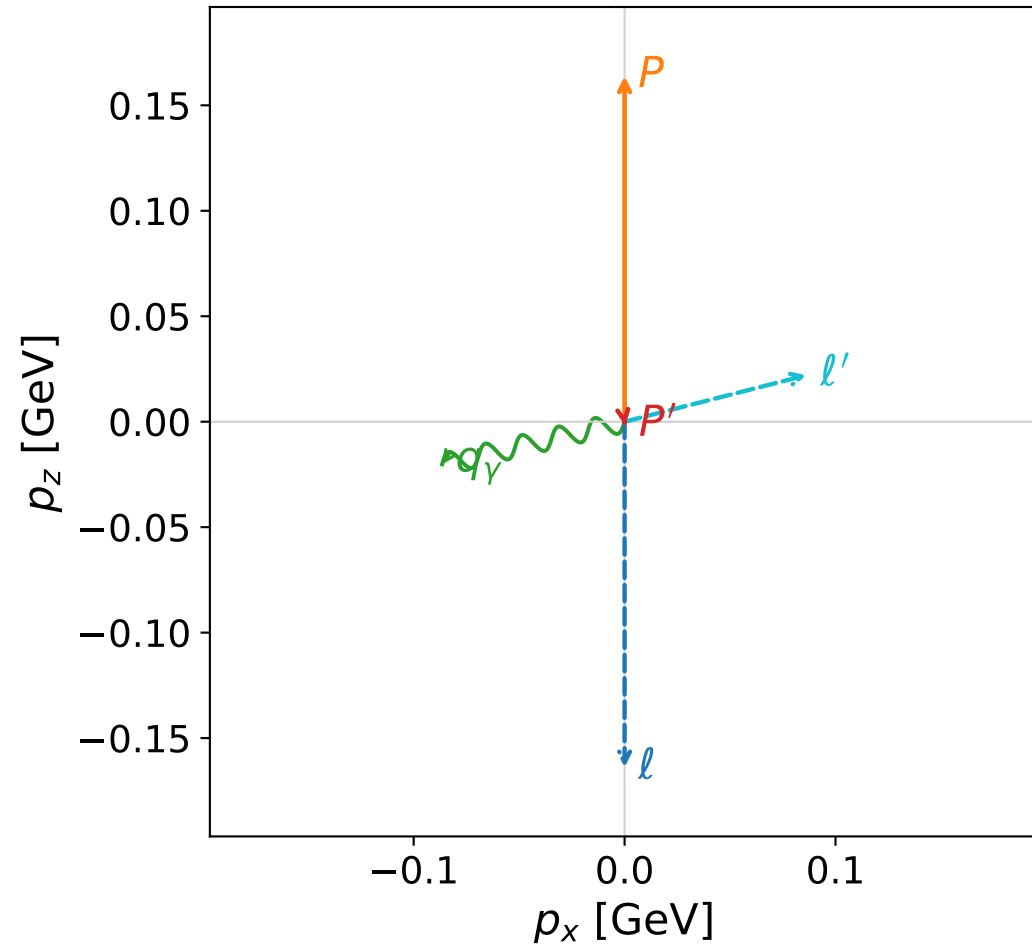


Leading normalized final-state amplitudes
(N=3, retained=98.5%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

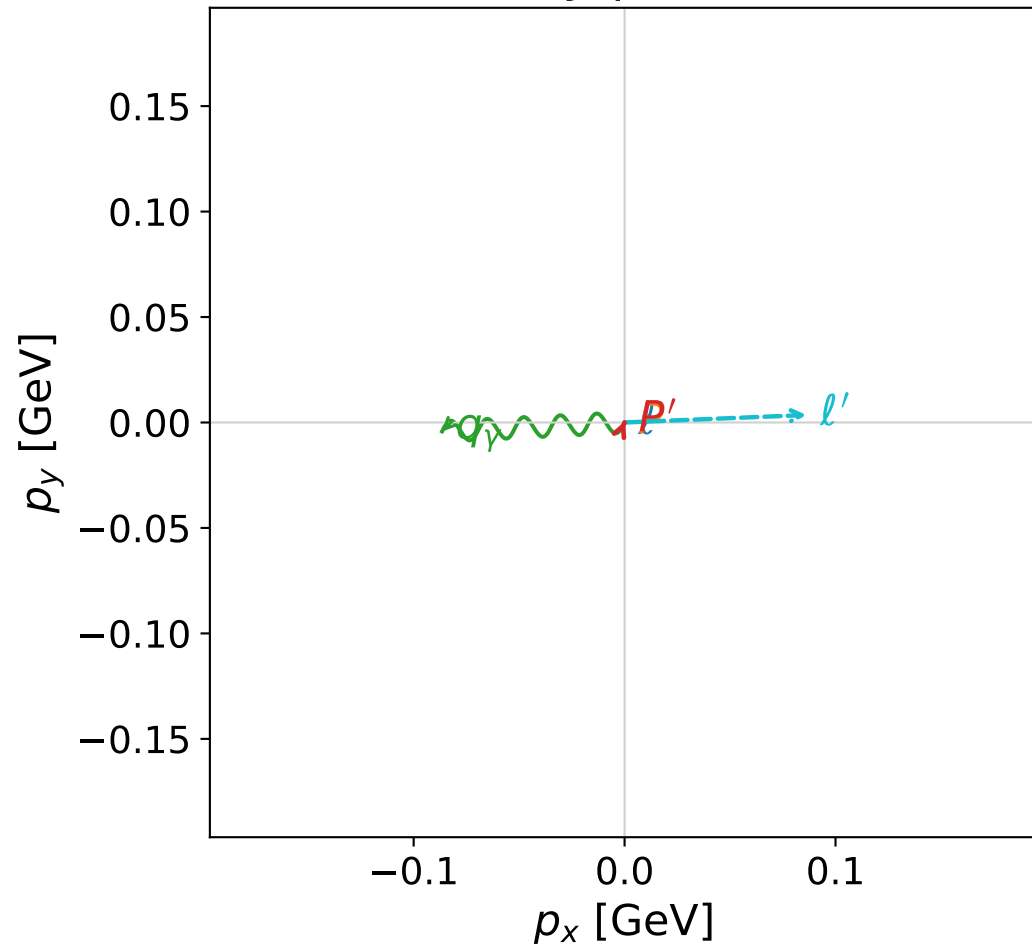


dw_mixing_angles_polarization_03_local_minimum_704
 $D_W=0.00743594$
 region: polarization_cluster_03_local_minimum_704
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0704
 lepton mixing angle: $\alpha_e=0.7651863$ rad
 incoming lepton: $0.721253|+\rangle + 0.692671|-\rangle$
 proton mixing angle: $\alpha_p=0.19293922$ rad
 incoming proton: $0.981445|+\rangle + 0.191744|-\rangle$

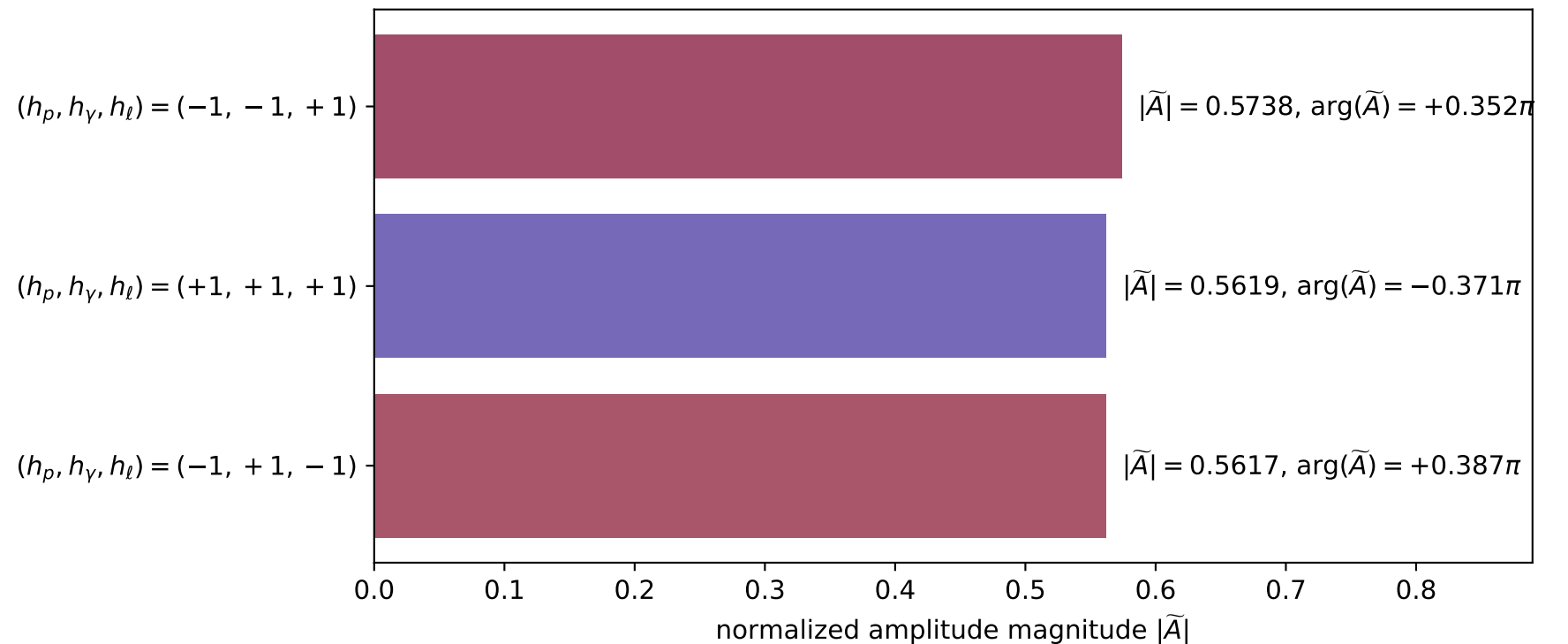
$s=1.24557$, $\sqrt{s}=1.11605$, $|\vec{P}|=0.163846$, $|\vec{P}'|=0.00200459$
 $\theta_{p'}=2.76927$, $\theta_\gamma=1.79959$, $\phi_{p'}=1.16918$, $\phi_\gamma=3.19046$
 $E_\gamma=0.0889545$, $\theta_{\ell'}=1.32079$, $\phi_{\ell'}=0.0412666$
 $Q^2=0.0364176$, $x_B=0.179153$, $t=-0.0272598$, $W^2=1.0467$, $y=0.555822$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E=0.1638$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.1638$
 P $E=0.9522$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1638$
 l' $E=0.0891$ $p_x=0.0862$ $p_y=0.0036$ $p_z=0.0220$
 P' $E=0.9380$ $p_x=0.0003$ $p_y=0.0007$ $p_z=-0.0019$
 q_γ $E=0.0890$ $p_x=-0.0865$ $p_y=-0.0042$ $p_z=-0.0202$

x--y plane

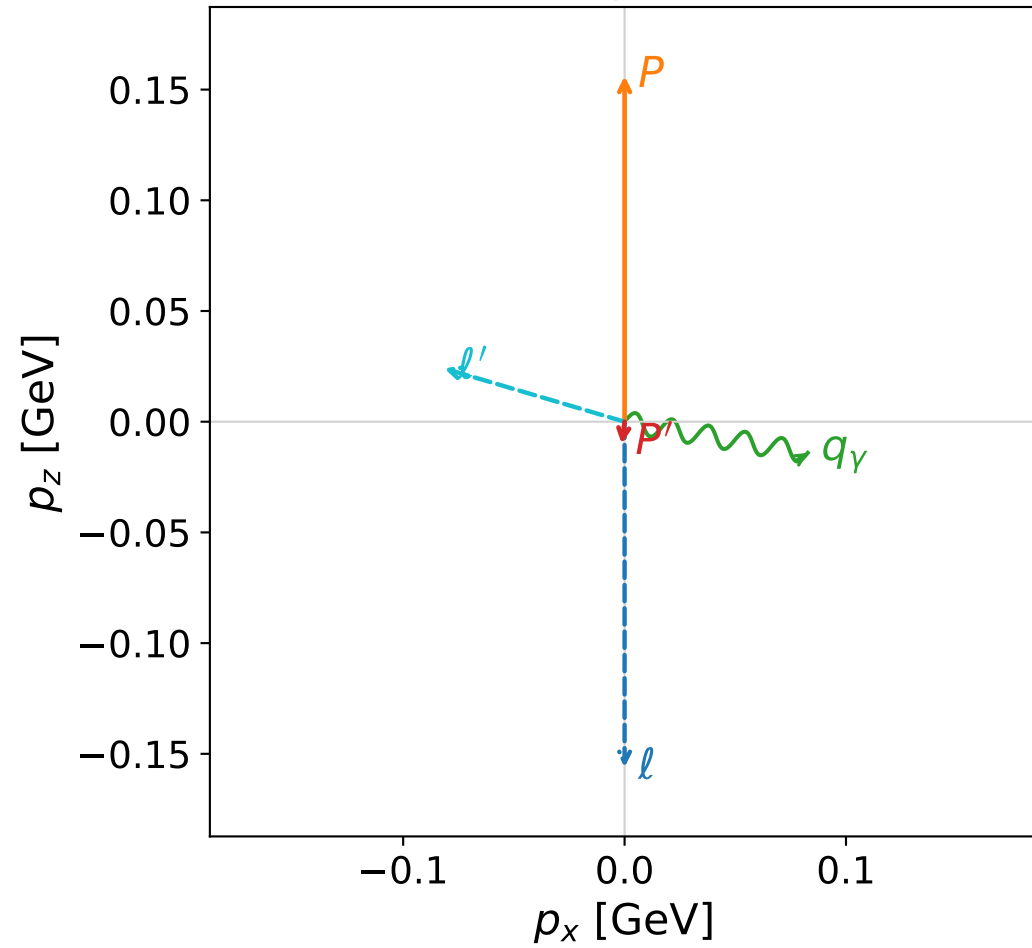


Leading normalized final-state amplitudes
 (N=3, retained=96.1%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

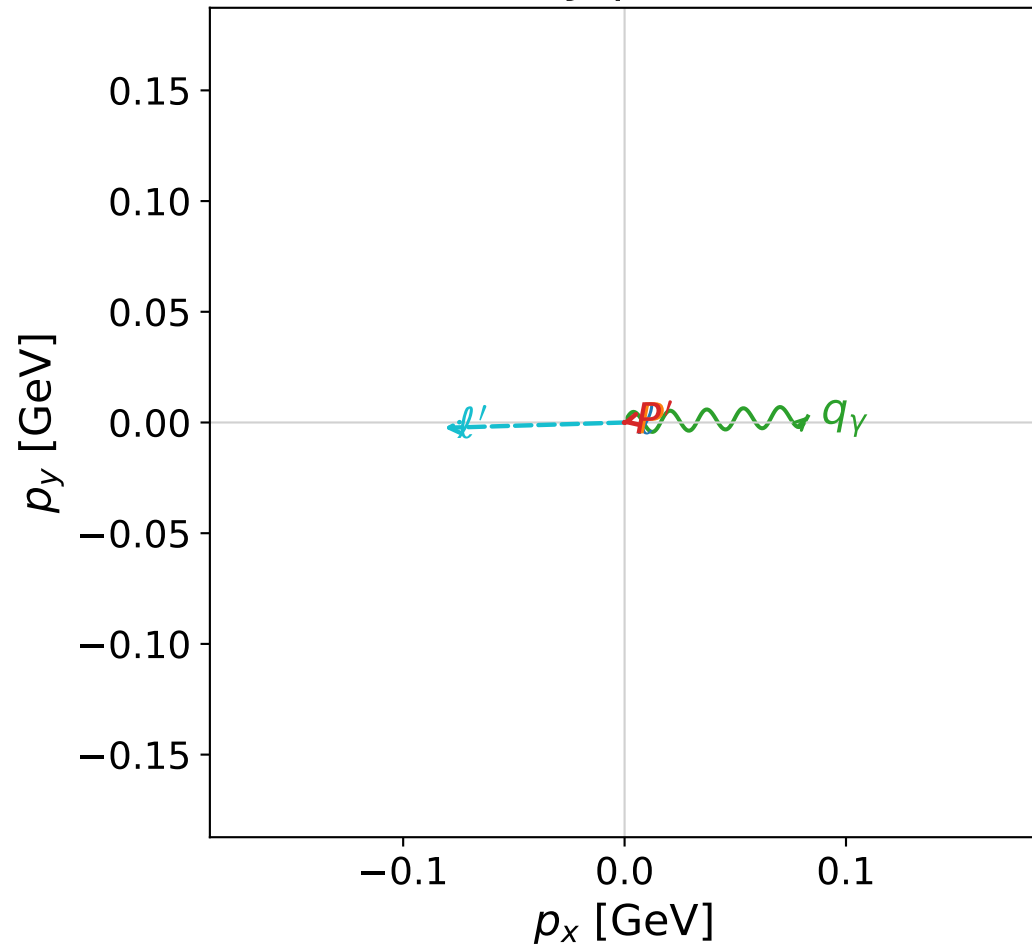


dw_mixing_angles_polarization_03_local_minimum_707
 $D_W=0.00749259$
 region: polarization_cluster_03_local_minimum_707
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0707
 lepton mixing angle: $\alpha_e=0.77624153$ rad
 incoming lepton: $0.713552|+\rangle + 0.700603|-\rangle$
 proton mixing angle: $\alpha_p=2.9784057$ rad
 incoming proton: $-0.986715|+\rangle + 0.162464|-\rangle$

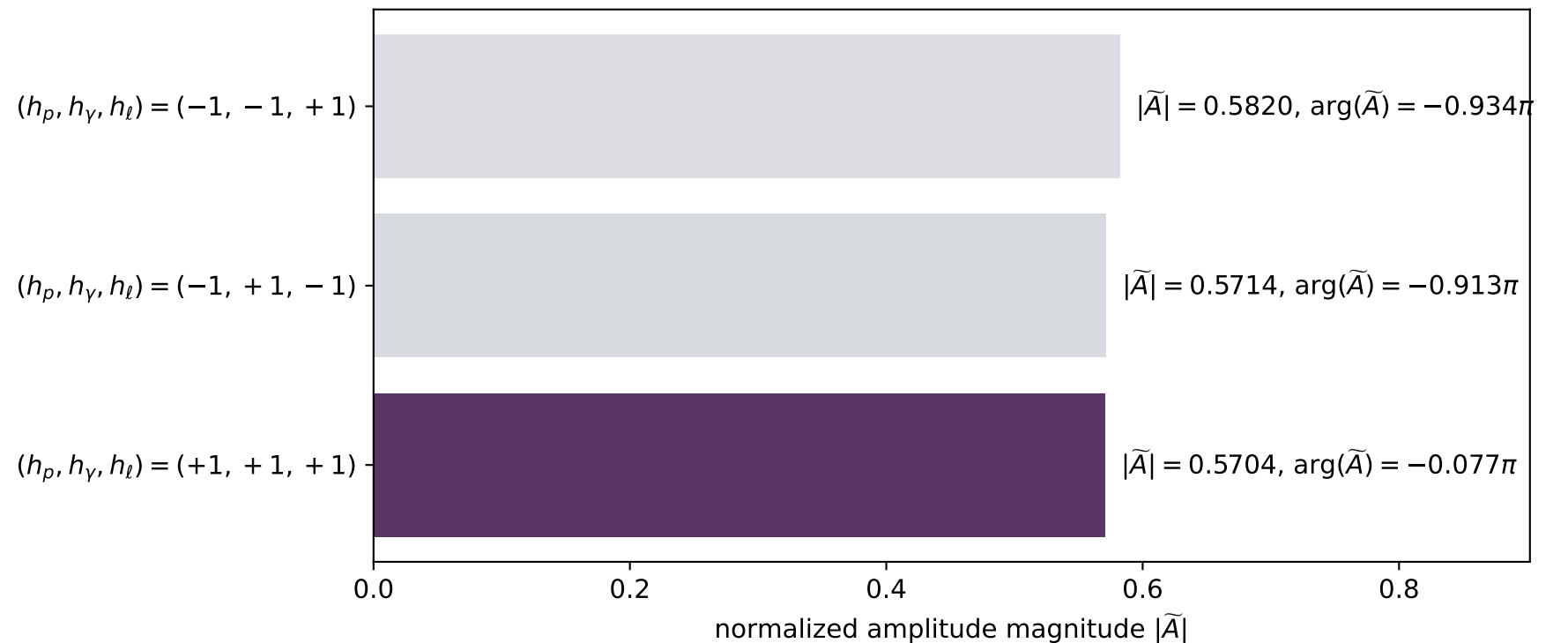
$s=1.22541$, $\sqrt{s}=1.10698$, $|\vec{P}|=0.156084$, $|\vec{P}'|=0.0100748$
 $\theta_{p'}=3.01025$, $\theta_\gamma=1.73909$, $\phi_{p'}=3.38298$, $\phi_\gamma=0.0329562$
 $E_\gamma=0.0839642$, $\theta_{\ell'}=1.28379$, $\phi_{\ell'}=3.1712$
 $Q^2=0.0340311$, $x_B=0.177718$, $t=-0.0274168$, $W^2=1.0373$, $y=0.554133$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1561$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1561$
 P $E= 0.9509$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1561$
 ℓ' $E= 0.0850$ $p_x= -0.0815$ $p_y= -0.0024$ $p_z= 0.0241$
 P' $E= 0.9381$ $p_x= -0.0013$ $p_y= -0.0003$ $p_z= -0.0100$
 q_γ $E= 0.0840$ $p_x= 0.0827$ $p_y= 0.0027$ $p_z= -0.0141$

x--y plane

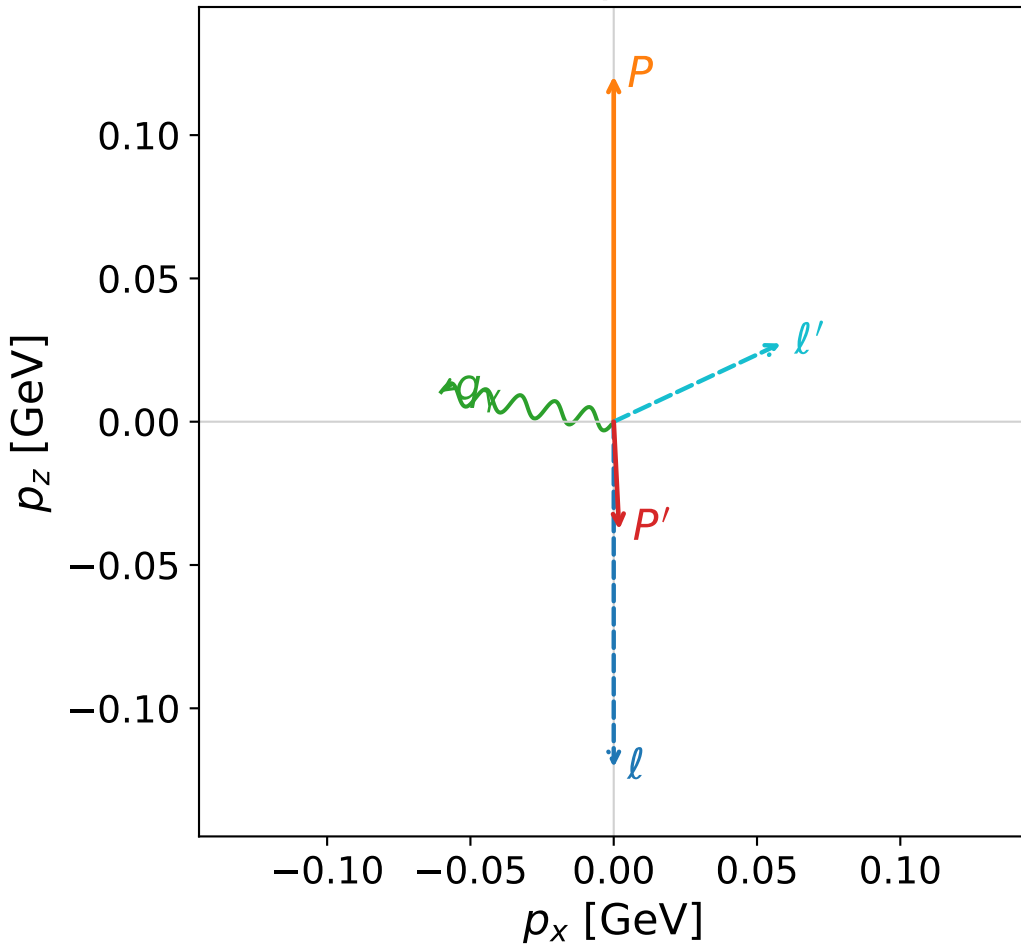


Leading normalized final-state amplitudes
 (N=3, retained=99.1%)

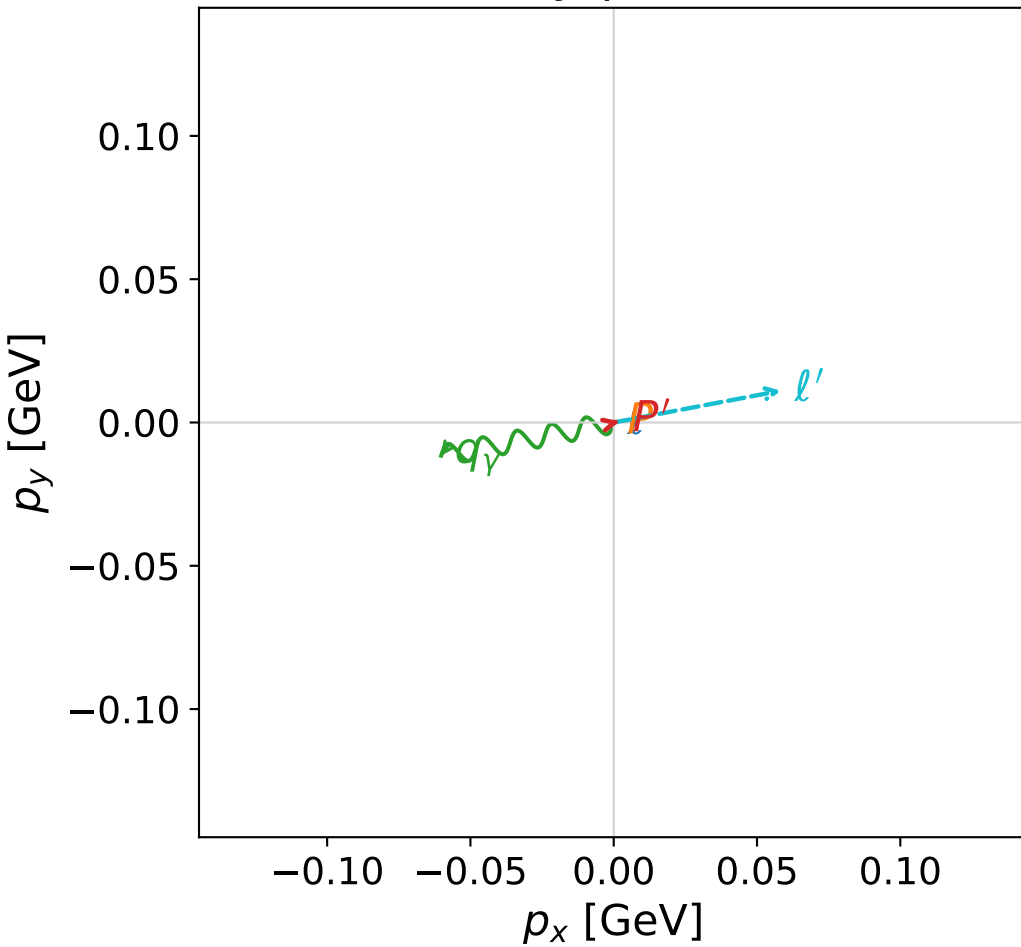


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



x--y plane

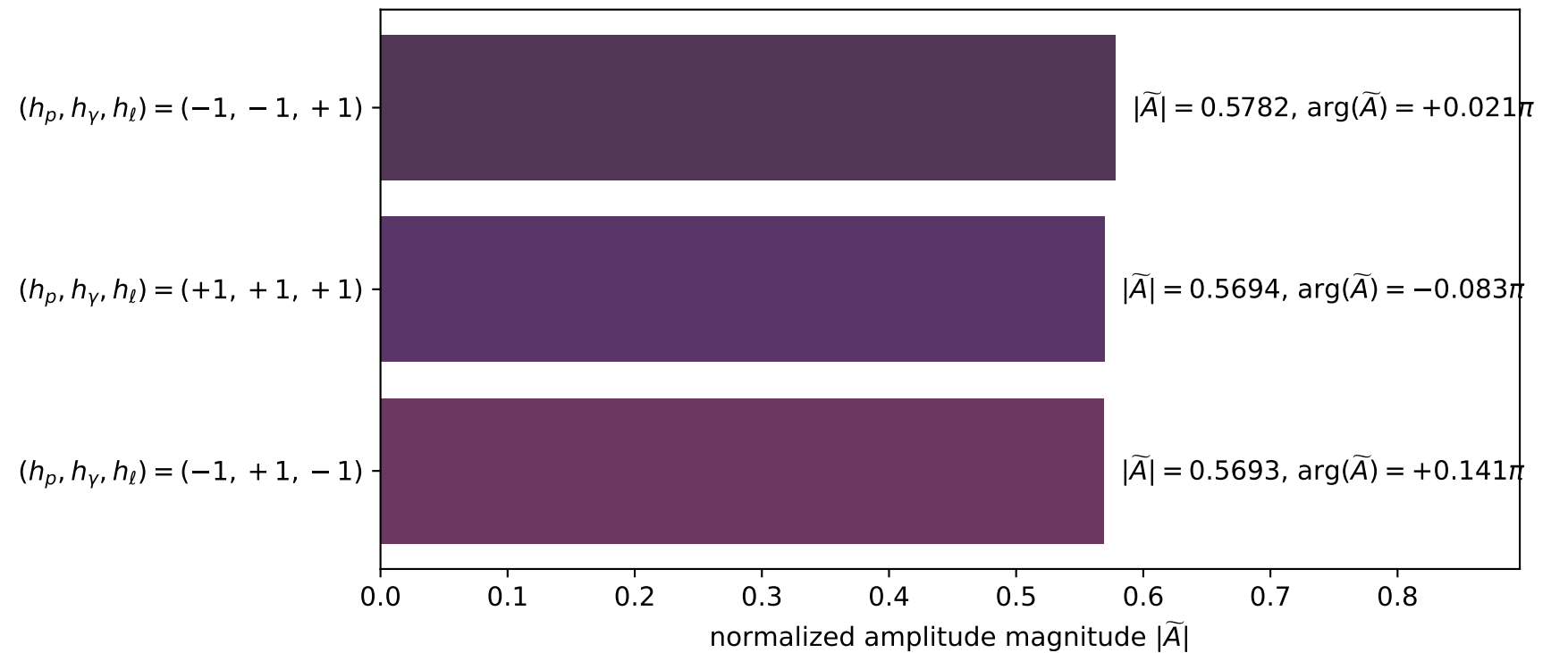


dw_mixing_angles_polarization_03_local_minimum_722
 $D_W=0.00805759$
 region: polarization_cluster_03_local_minimum_722
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0722
 lepton mixing angle: $\alpha_e=0.76206041$ rad
 incoming lepton: $0.723415|+\rangle +0.690413|-\rangle$
 proton mixing angle: $\alpha_p=0.11757213$ rad
 incoming proton: $0.993096|+\rangle +0.117301|-\rangle$

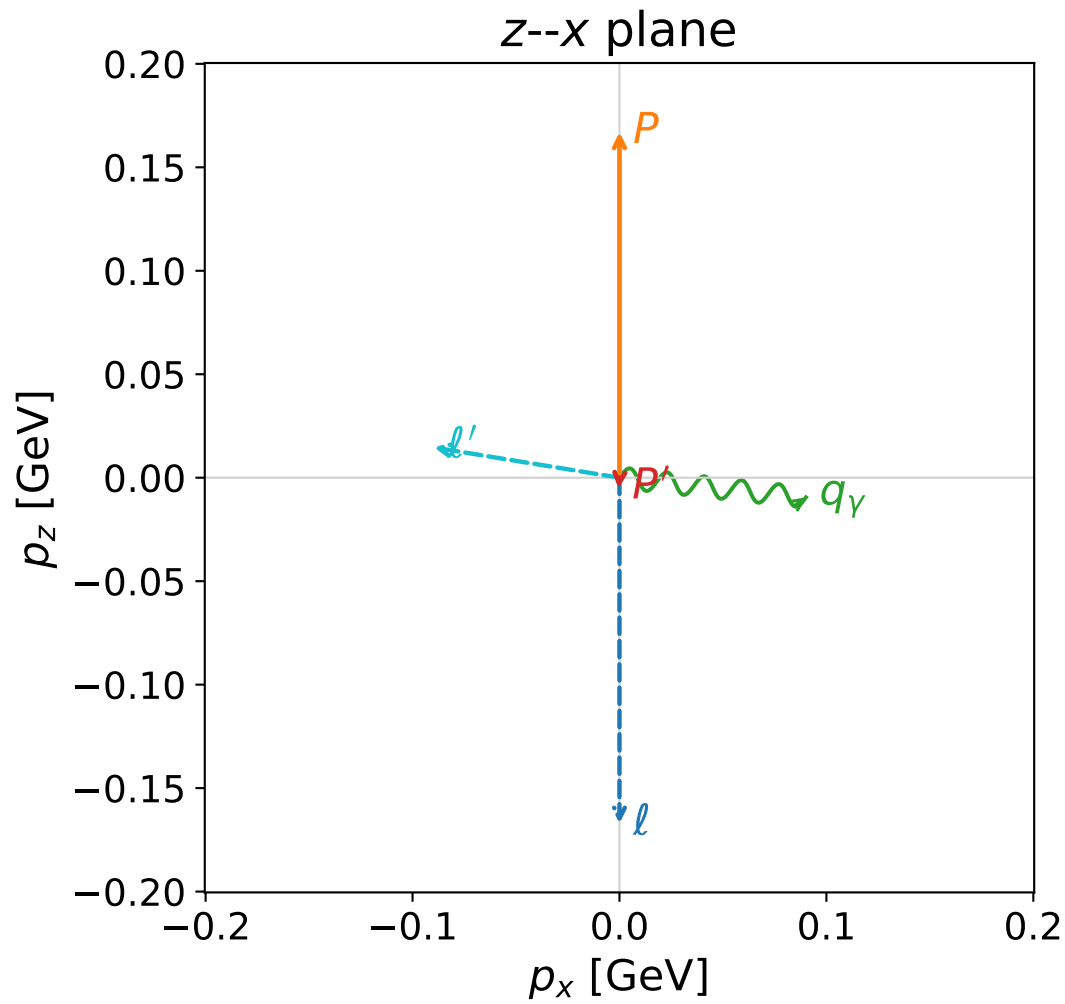
$s=1.13702$, $\sqrt{s}=1.06631$, $|\vec{P}|=0.120589$, $|\vec{P}'|=0.0377302$
 $\theta_{p'}=3.08974$, $\theta_\gamma=1.40324$, $\phi_{p'}=0.260976$, $\phi_\gamma=3.33166$
 $E_\gamma=0.0621959$, $\theta_{l'}=1.13973$, $\phi_{l'}=0.187734$
 $Q^2=0.0223475$, $x_B=0.159463$, $t=-0.0250042$, $W^2=0.997639$, $y=0.544938$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E=0.1206$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.1206$
 P $E=0.9457$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1206$
 l' $E=0.0654$ $p_x=0.0583$ $p_y=0.0111$ $p_z=0.0273$
 P' $E=0.9388$ $p_x=0.0019$ $p_y=0.0005$ $p_z=-0.0377$
 q_γ $E=0.0622$ $p_x=-0.0602$ $p_y=-0.0116$ $p_z=0.0104$

Leading normalized final-state amplitudes
 (N=3, retained=98.3%)



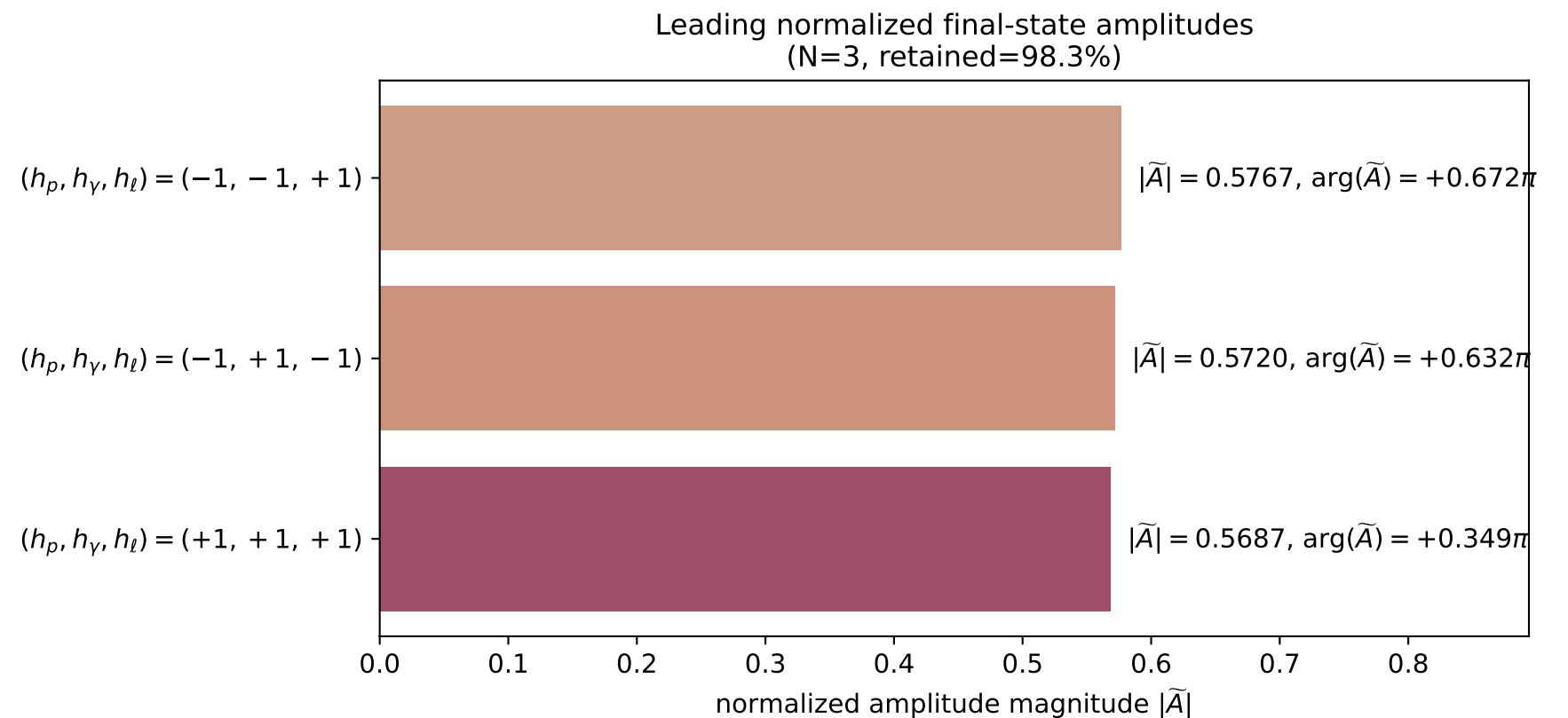
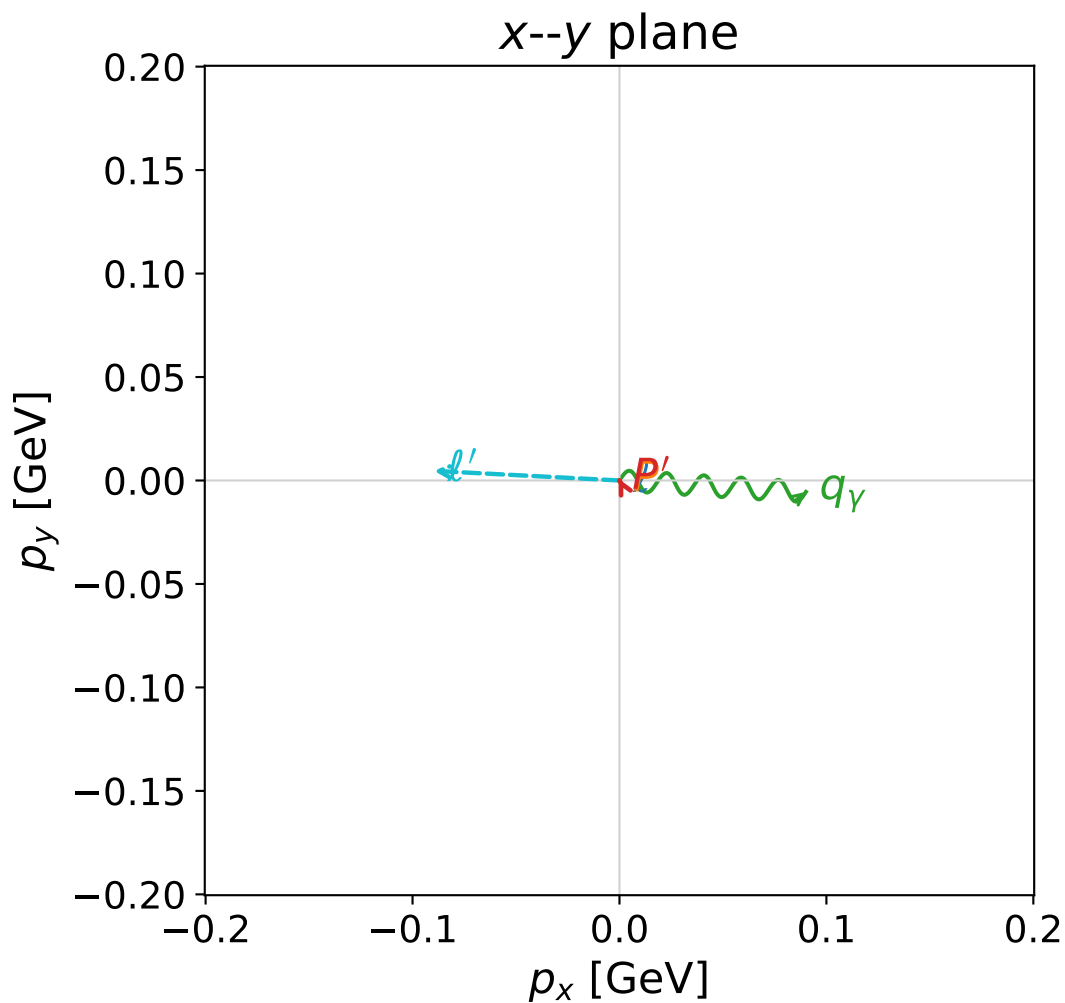
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)



```
dw_mixing_angles_polarization_03_local_minimum_735
D_W=0.00848145
region: polarization_cluster_03_local_minimum_735
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0735
lepton mixing angle: alpha_e=0.78322937 rad
incoming lepton: 0.708639|+> +0.705572|->
proton mixing angle: alpha_p=2.9506423 rad
incoming proton: -0.981824|+> +0.189792|->
```

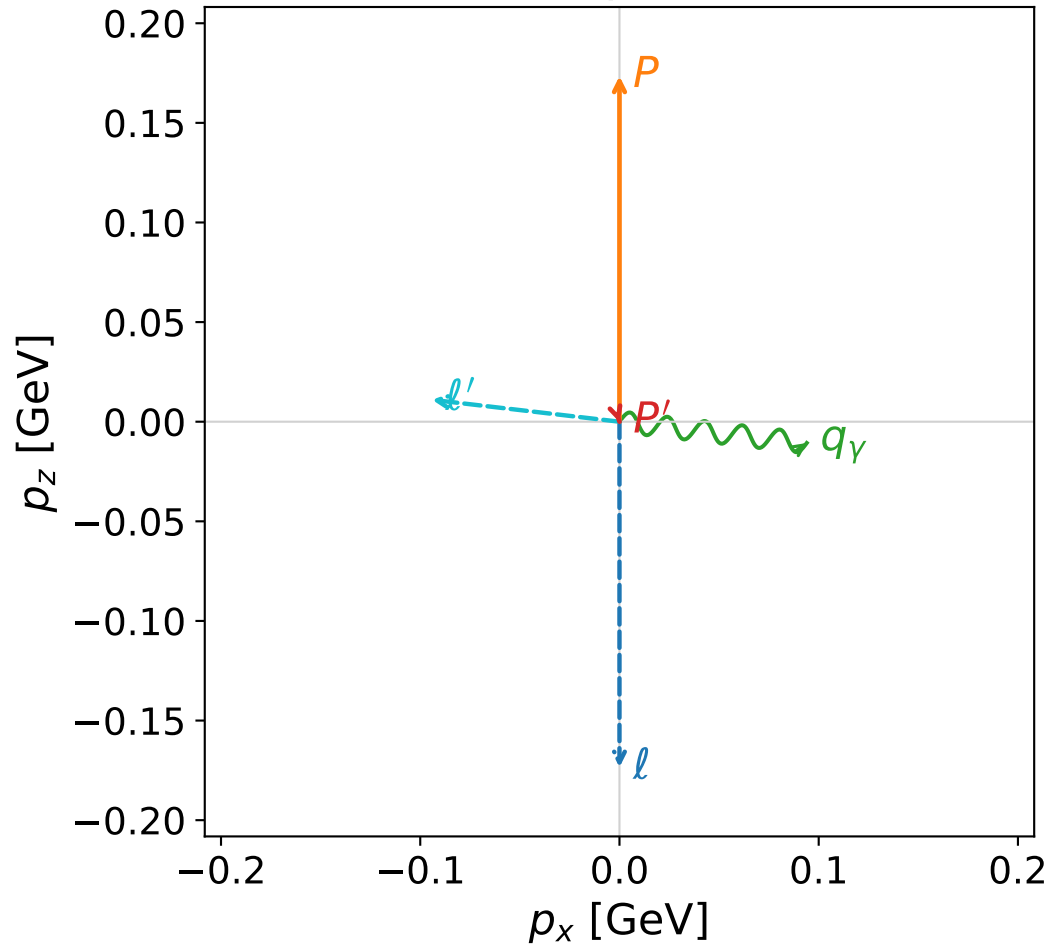
```
s=1.25386, sqrt(s)=1.11976, |P|=0.167008, |P'|=0.0052075
theta_p=2.95455, theta_gamma=1.67547, phi_p'=2.04328, phi_gamma=6.22212
E_gamma=0.0907752, theta_l'=1.4096, phi_l'=3.08981
Q^2=0.0352621, x_B=0.171549, t=-0.0294106, W^2=1.05013, y=0.549574
```

```
four-momenta [E, p_x, p_y, p_z] GeV:
l      E=  0.1670 p_x=  0.0000 p_y=  0.0000 p_z= -0.1670
P      E=  0.9528 p_x=  0.0000 p_y=  0.0000 p_z=  0.1670
l'     E=  0.0910 p_x= -0.0897 p_y=  0.0046 p_z=  0.0146
P'     E=  0.9380 p_x= -0.0004 p_y=  0.0009 p_z= -0.0051
q_gamma E=  0.0908 p_x=  0.0901 p_y= -0.0055 p_z= -0.0095
```



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

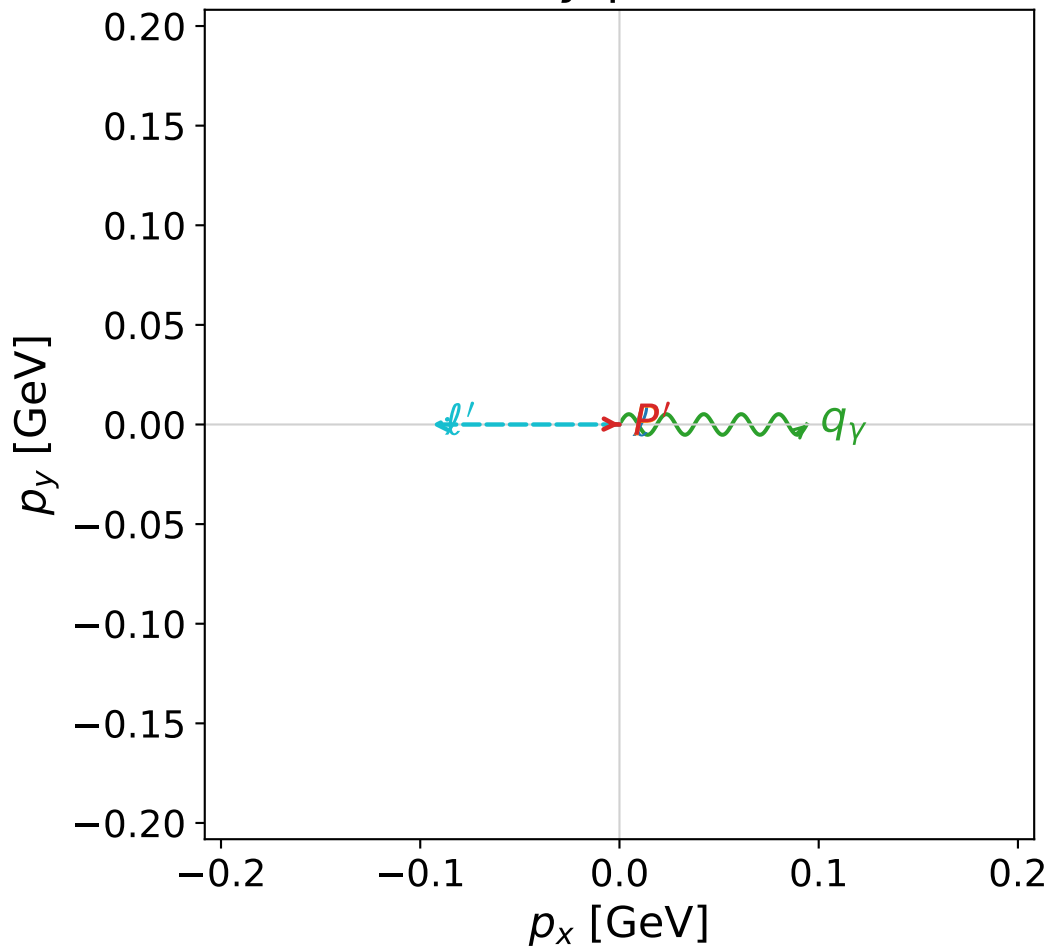


dw_mixing_angles_polarization_03_local_minimum_772
 $D_W=0.0095849$
 region: polarization_cluster_03_local_minimum_772
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0772
 lepton mixing angle: $\alpha_e=0.78701964$ rad
 incoming lepton: $0.705959|+\rangle + 0.708252|-\rangle$
 proton mixing angle: $\alpha_p=2.9424013$ rad
 incoming proton: $-0.980227|+\rangle + 0.197877|-\rangle$

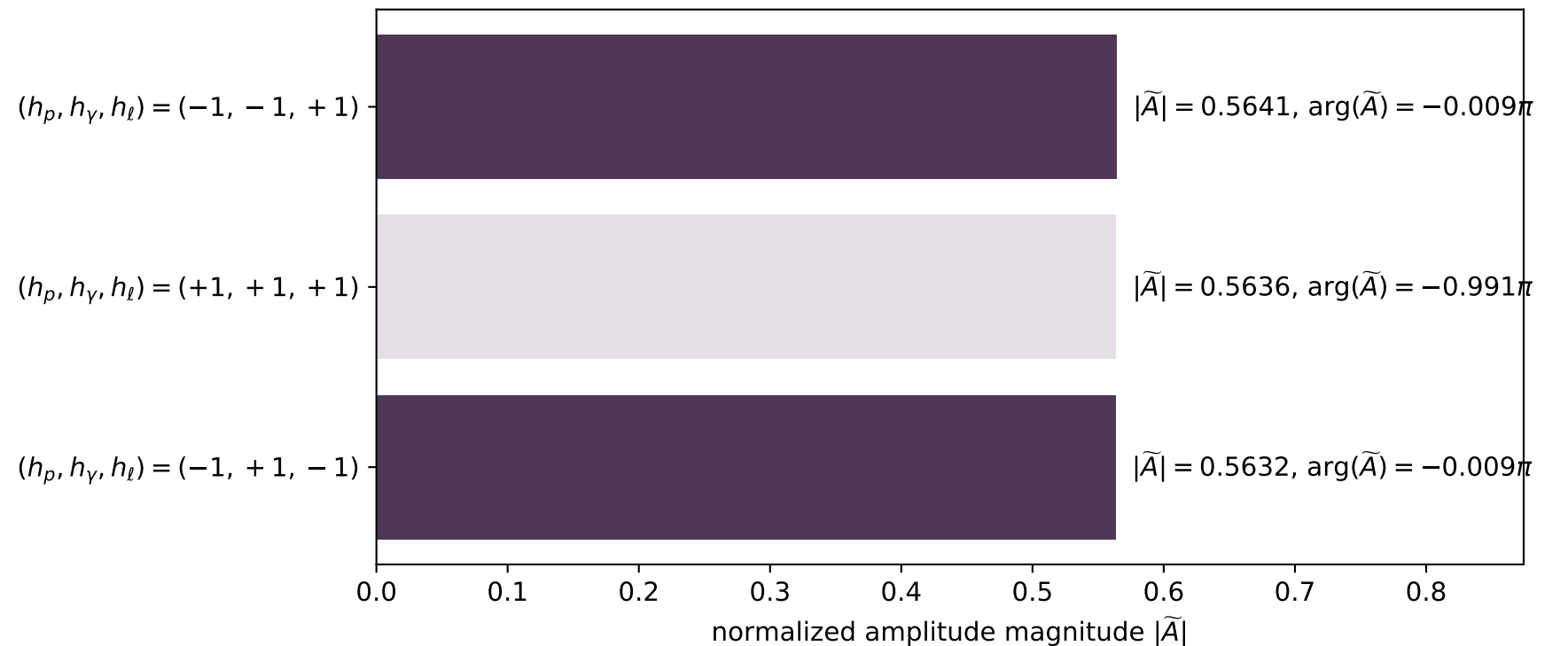
$s=1.27096$, $\sqrt{s}=1.12737$, $|\vec{P}|=0.173463$, $|\vec{P}'|=0.000335174$
 $\theta_{p'}=2.84197$, $\theta_\gamma=1.68397$, $\phi_{p'}=6.25528$, $\phi_\gamma=0.000170876$
 $E_\gamma=0.0946156$, $\theta_{\ell'}=1.45439$, $\phi_{\ell'}=3.14173$
 $Q^2=0.0366894$, $x_B=0.171315$, $t=-0.0299475$, $W^2=1.05732$, $y=0.547572$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1735$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1735$
 P $E= 0.9539$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1735$
 ℓ' $E= 0.0948$ $p_x= -0.0941$ $p_y= -0.0000$ $p_z= 0.0110$
 P' $E= 0.9380$ $p_x= 0.0001$ $p_y= -0.0000$ $p_z= -0.0003$
 q_γ $E= 0.0946$ $p_x= 0.0940$ $p_y= 0.0000$ $p_z= -0.0107$

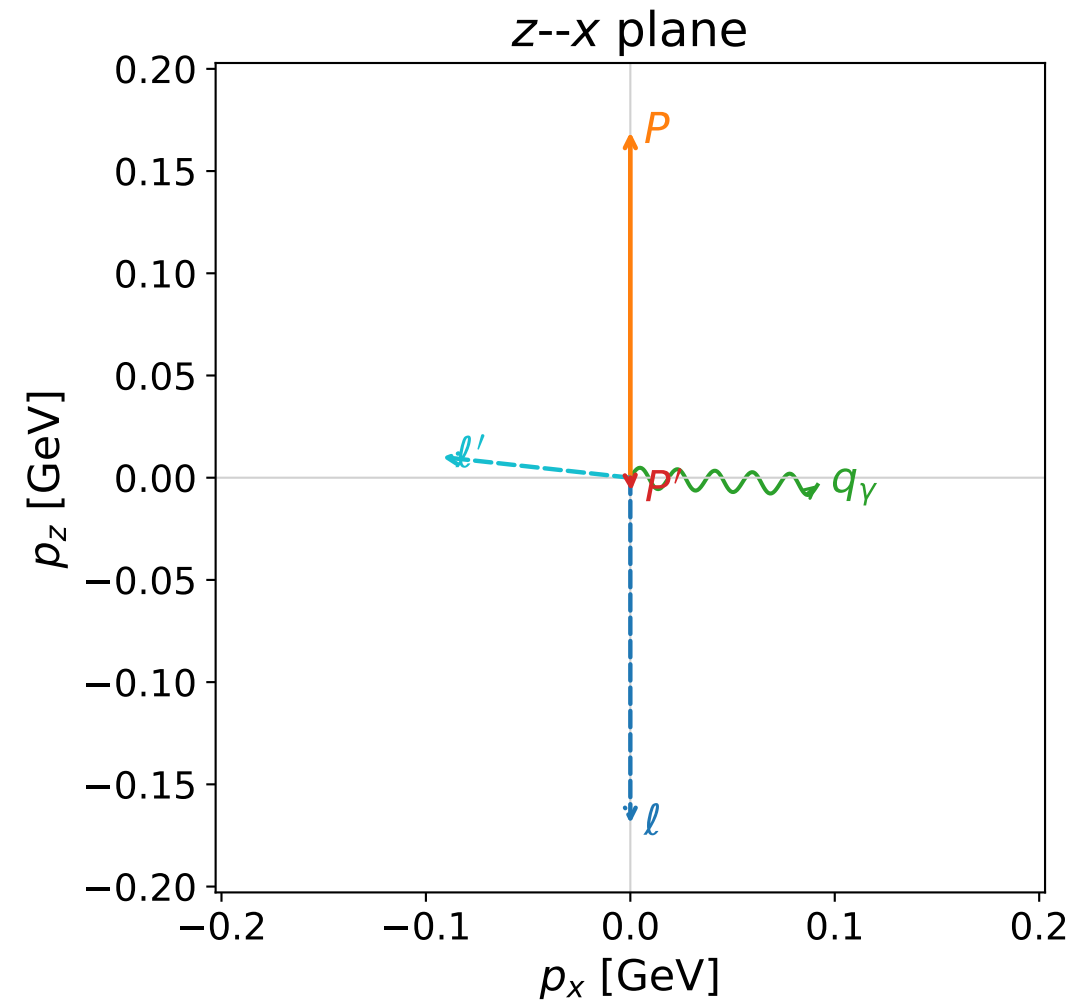
x--y plane



Leading normalized final-state amplitudes
 (N=3, retained=95.3%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

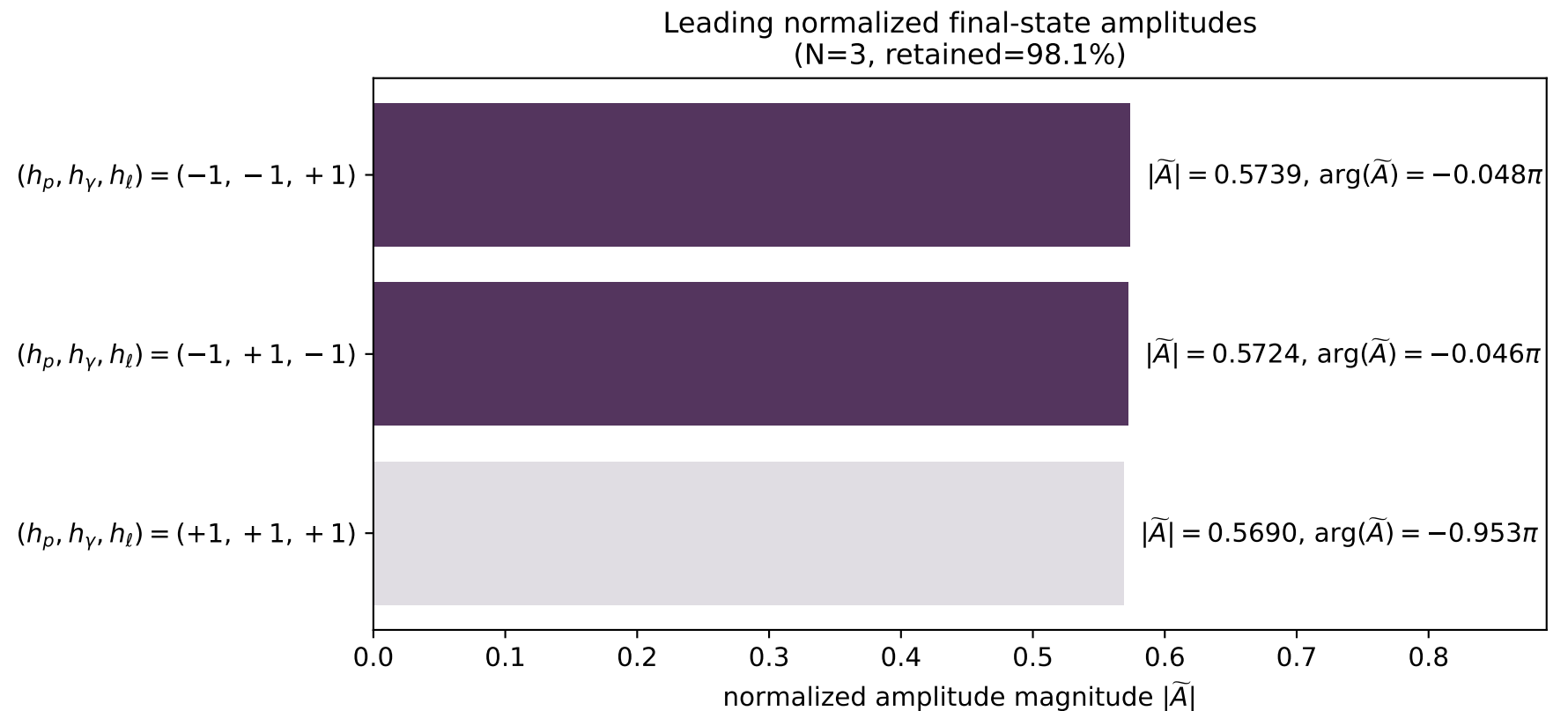
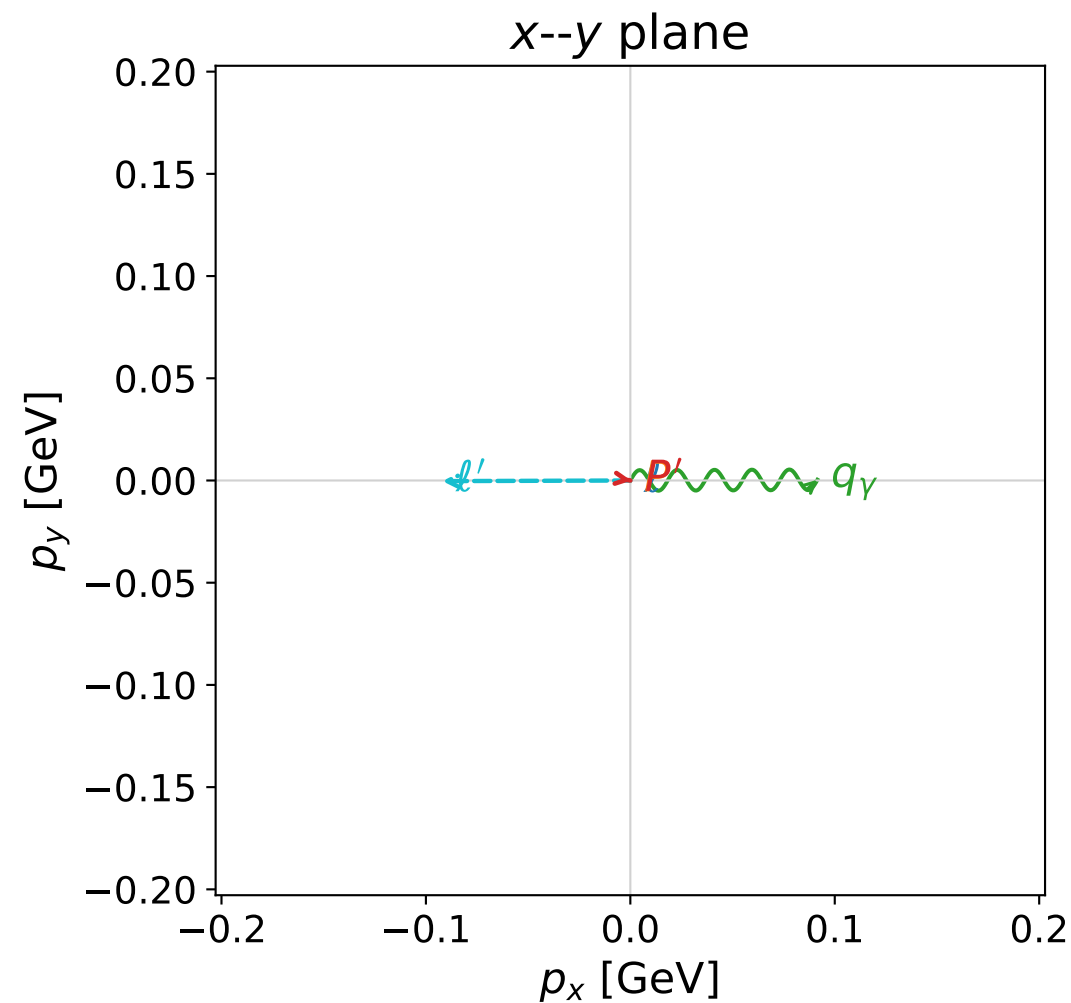


dw_mixing_angles_polarization_03_local_minimum_800
 $D_W=0.0103956$
 region: polarization_cluster_03_local_minimum_800
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0800
 lepton mixing angle: $\alpha_e=0.78689655$ rad
 incoming lepton: $0.706046|+\rangle + 0.708166|-\rangle$
 proton mixing angle: $\alpha_p=2.9368835$ rad
 incoming proton: $-0.97912|+\rangle + 0.203282|-\rangle$

$s=1.25926$, $\sqrt{s}=1.12217$, $|\vec{P}|=0.169055$, $|\vec{P}'|=0.00653172$
 $\theta_{p'}=3.07246$, $\theta_\gamma=1.61004$, $\phi_{p'}=6.13637$, $\phi_\gamma=0.00404817$
 $E_\gamma=0.0916075$, $\theta_{\ell'}=1.46132$, $\phi_{\ell'}=3.1449$
 $Q^2=0.0347062$, $x_B=0.16812$, $t=-0.0305978$, $W^2=1.05158$, $y=0.54409$

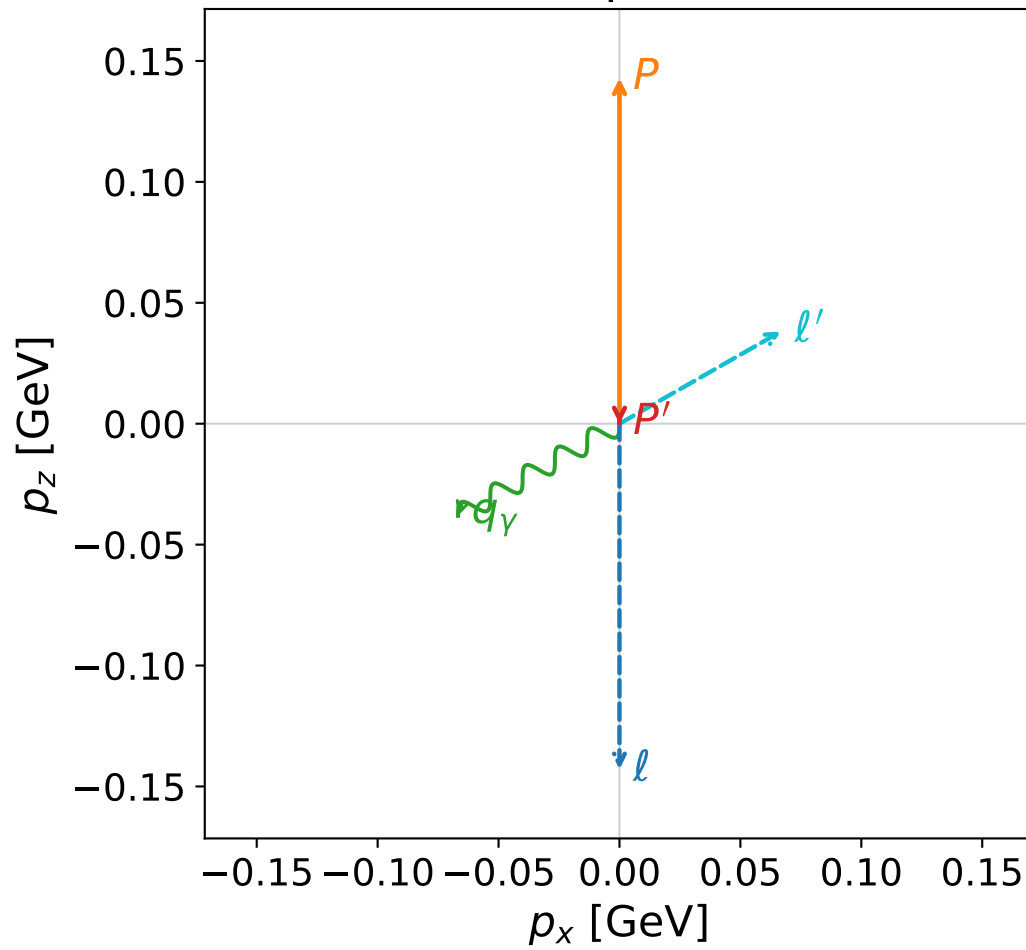
four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ	$E=$	0.1691	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	-0.1691
P	$E=$	0.9531	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	0.1691
ℓ'	$E=$	0.0925	$p_x=$	-0.0920	$p_y=$	-0.0003	$p_z=$	0.0101
P'	$E=$	0.9380	$p_x=$	0.0004	$p_y=$	-0.0001	$p_z=$	-0.0065
q_γ	$E=$	0.0916	$p_x=$	0.0915	$p_y=$	0.0004	$p_z=$	-0.0036



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

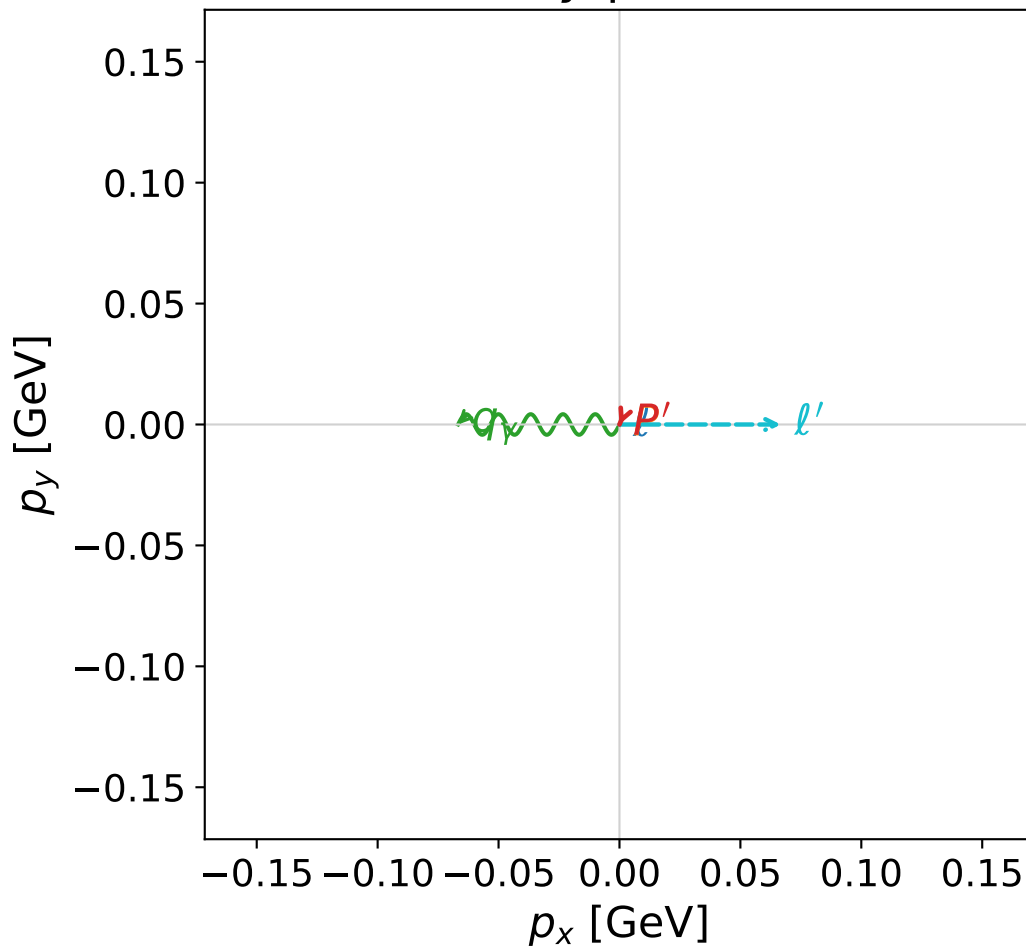


dw_mixing_angles_polarization_03_local_minimum_813
 $D_W=0.0107951$
 region: polarization_cluster_03_local_minimum_813
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0813
 lepton mixing angle: $\alpha_e=0.75167736$ rad
 incoming lepton: $0.730544|+\rangle + 0.682865|-\rangle$
 proton mixing angle: $\alpha_p=0.12679047$ rad
 incoming proton: $0.991973|+\rangle + 0.126451|-\rangle$

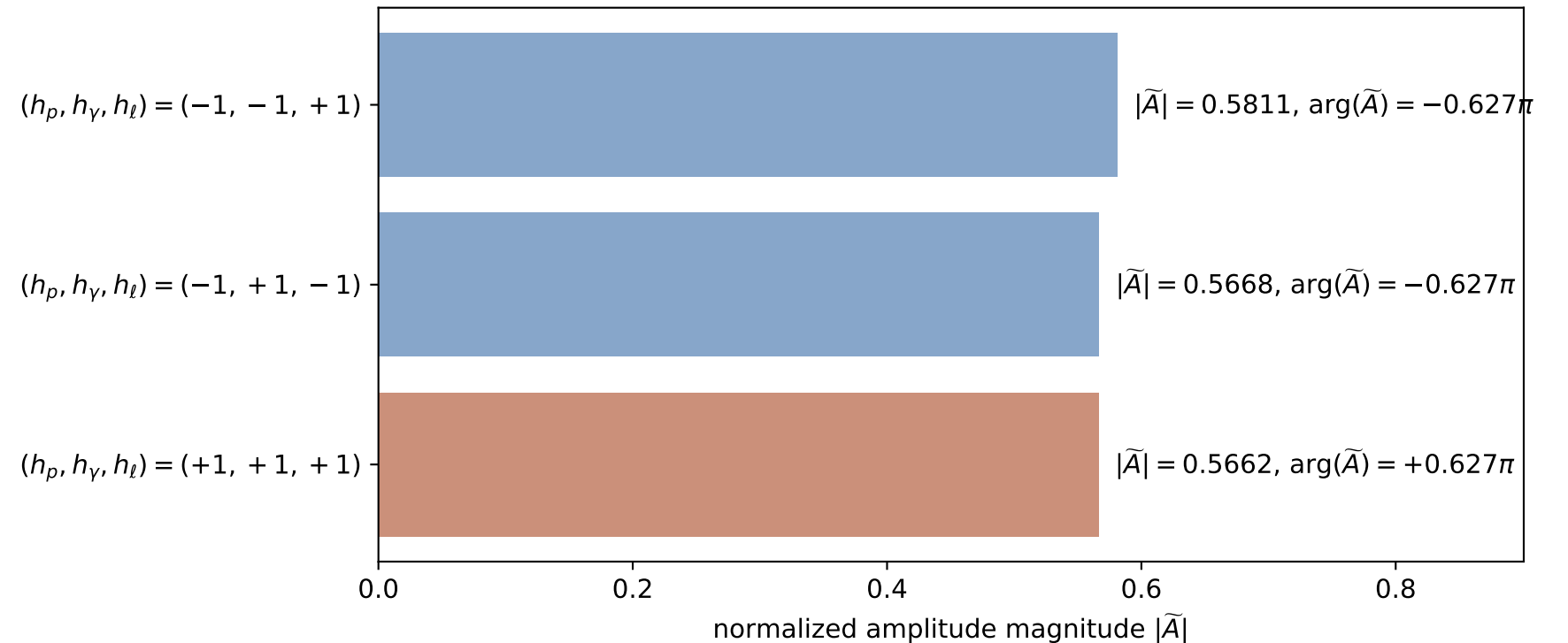
$s=1.19188$, $\sqrt{s}=1.09173$, $|\vec{P}|=0.142907$, $|\vec{P}'|=3.11807e-05$
 $\theta_{p'}=3.13792$, $\theta_\gamma=2.08925$, $\phi_{p'}=4.3132$, $\phi_\gamma=3.14151$
 $E_\gamma=0.076857$, $\theta_{l'}=1.05199$, $\phi_{l'}=6.28311$
 $Q^2=0.0328655$, $x_B=0.185632$, $t=-0.0203141$, $W^2=1.02403$, $y=0.567398$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.1429$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1429$
 P $E= 0.9488$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1429$
 l' $E= 0.0769$ $p_x= 0.0668$ $p_y= -0.0000$ $p_z= 0.0381$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.0000$
 q_γ $E= 0.0769$ $p_x= -0.0668$ $p_y= 0.0000$ $p_z= -0.0381$

x--y plane

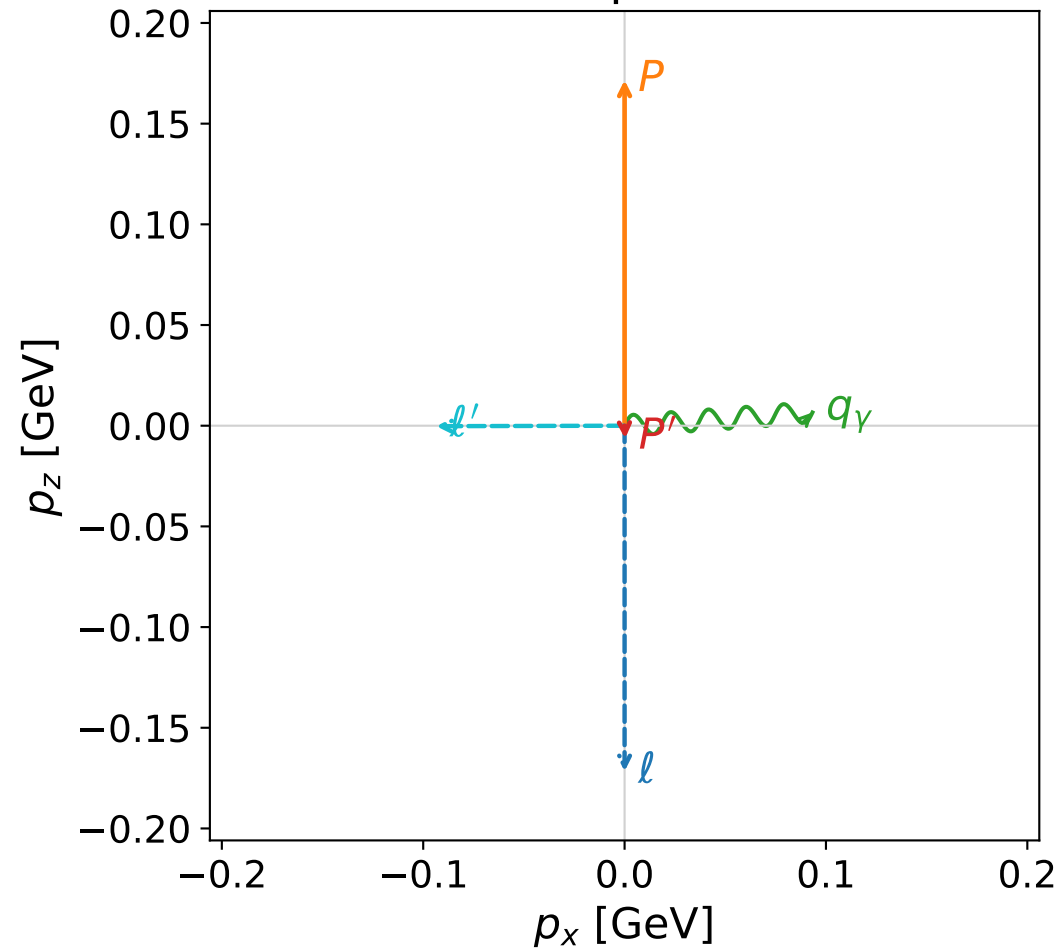


Leading normalized final-state amplitudes
 (N=3, retained=98.0%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



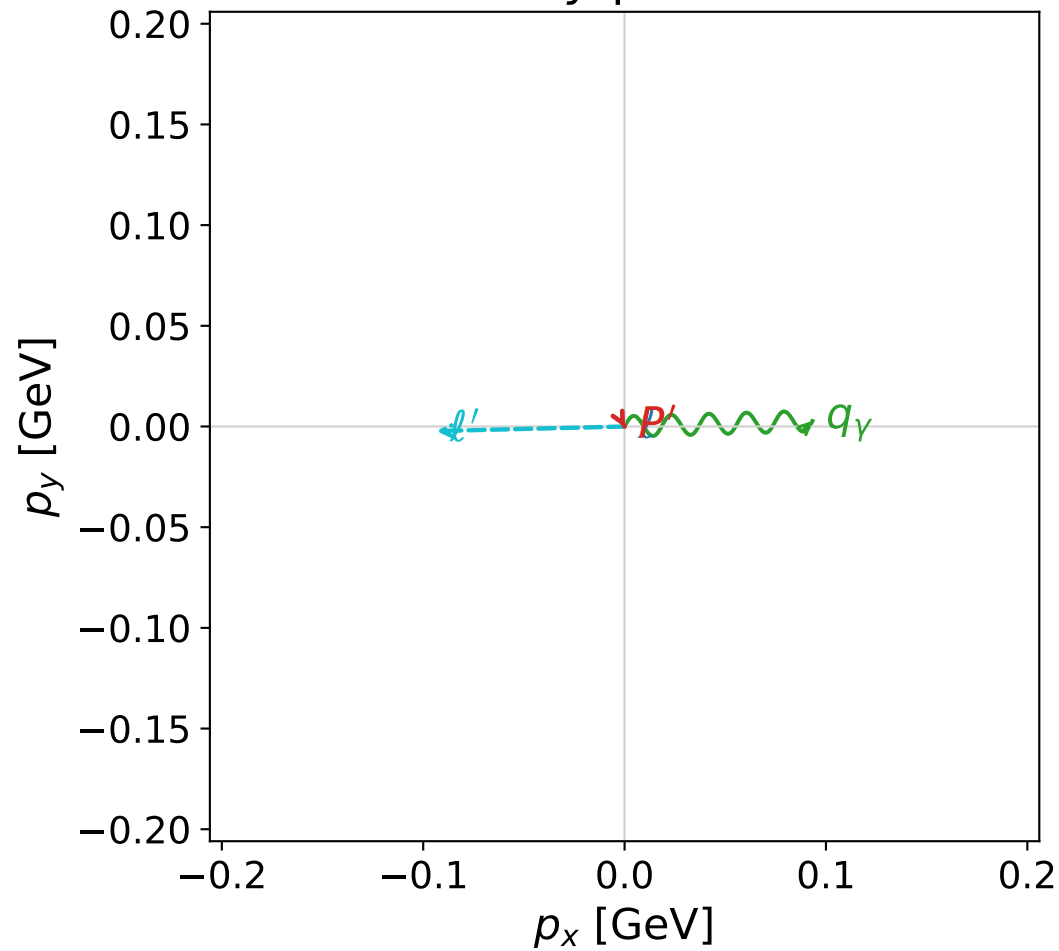
dw_mixing_angles_polarization_03_local_minimum_815
 $D_W=0.010864$
 region: polarization_cluster_03_local_minimum_815
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0815
 lepton mixing angle: $\alpha_e=0.79883948$ rad
 incoming lepton: $0.697539|+\rangle +0.716547|-\rangle$
 proton mixing angle: $\alpha_p=2.9379666$ rad
 incoming proton: $-0.97934|+\rangle +0.202222|-\rangle$

$s=1.26605$, $\sqrt{s}=1.12519$, $|\vec{P}|=0.171619$, $|\vec{P}'|=0.00636176$
 $\theta_{p'}=3.05711$, $\theta_\gamma=1.50047$, $\phi_{p'}=5.20611$, $\phi_\gamma=0.0292564$
 $E_\gamma=0.0935787$, $\theta_{\ell'}=1.57333$, $\phi_{\ell'}=3.16572$
 $Q^2=0.0320423$, $x_B=0.154318$, $t=-0.0314278$, $W^2=1.05544$, $y=0.53763$

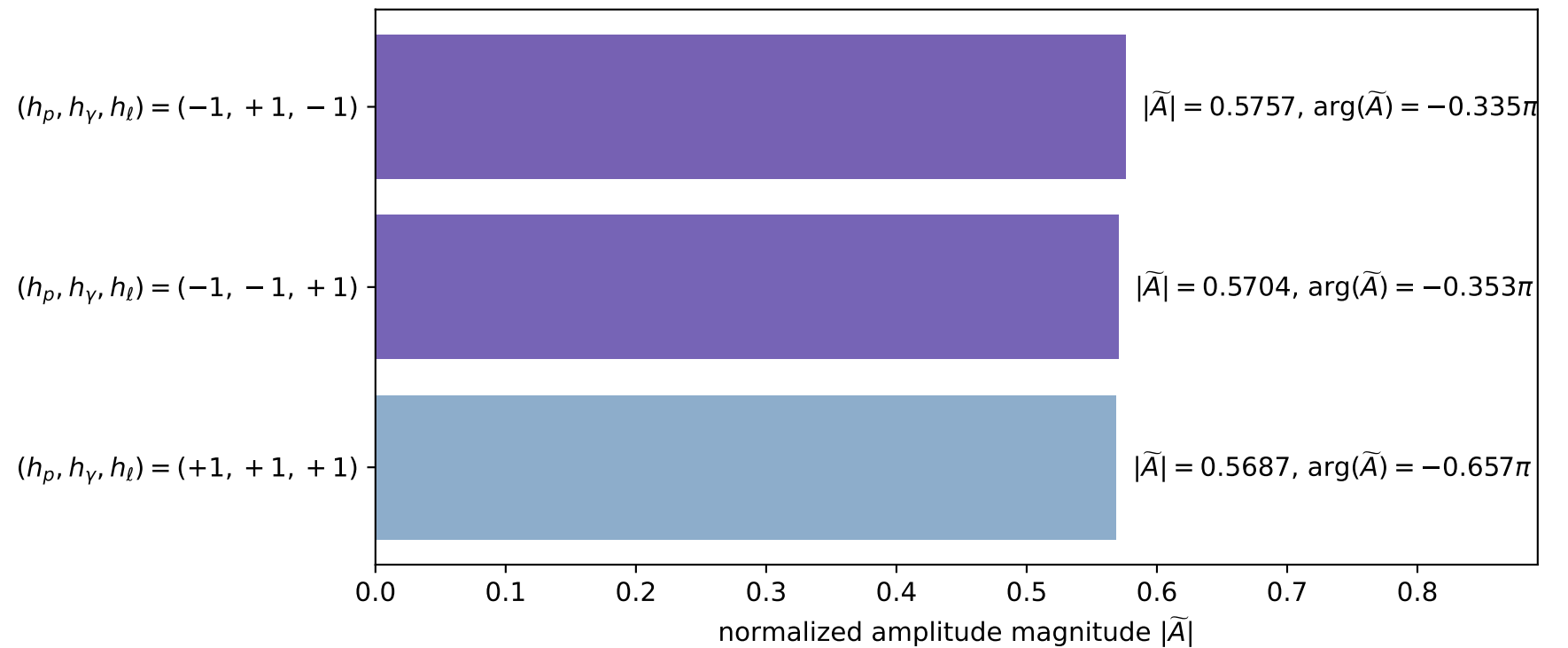
four-momenta $[E, p_x, p_y, p_z]$ GeV:

Particle	E	p_x	p_y	p_z
ℓ	0.1716	0.0000	0.0000	-0.1716
P	0.9536	0.0000	0.0000	0.1716
ℓ'	0.0936	-0.0936	-0.0023	-0.0002
P'	0.9380	0.0003	-0.0005	-0.0063
q_γ	0.0936	0.0933	0.0027	0.0066

x--y plane

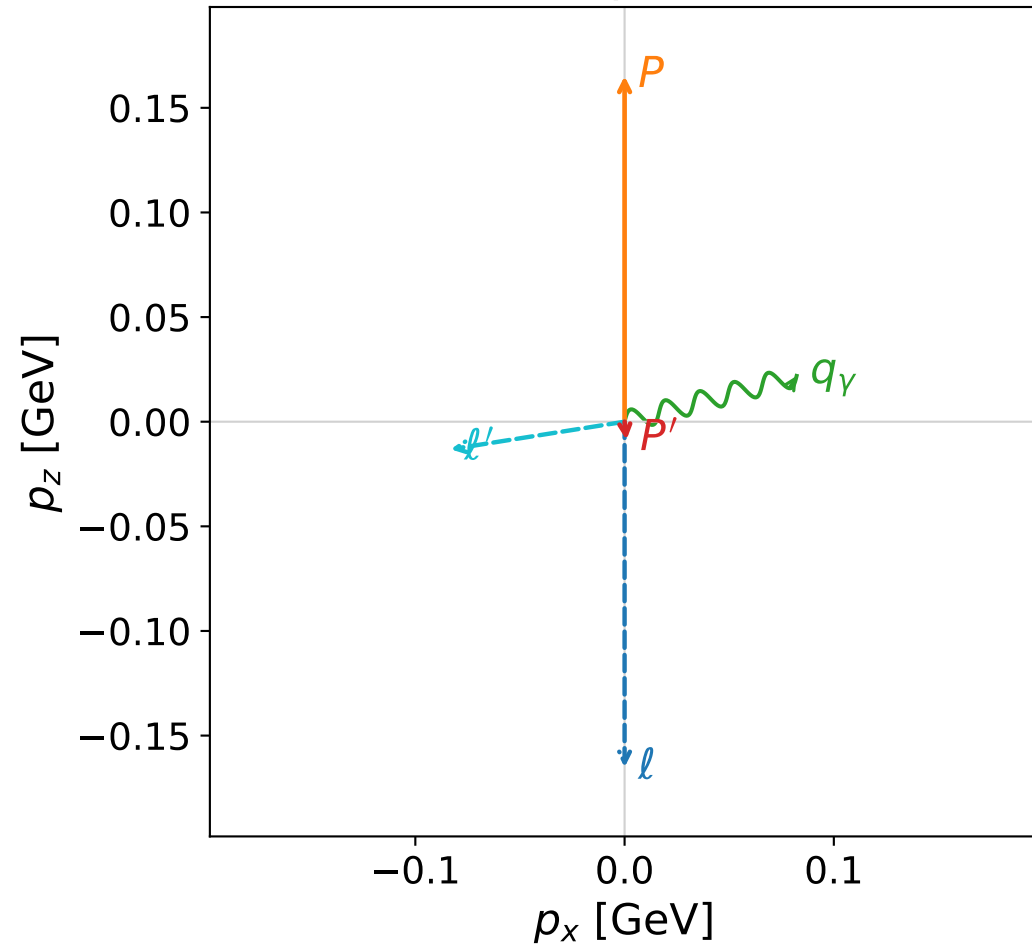


Leading normalized final-state amplitudes
(N=3, retained=98.0%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

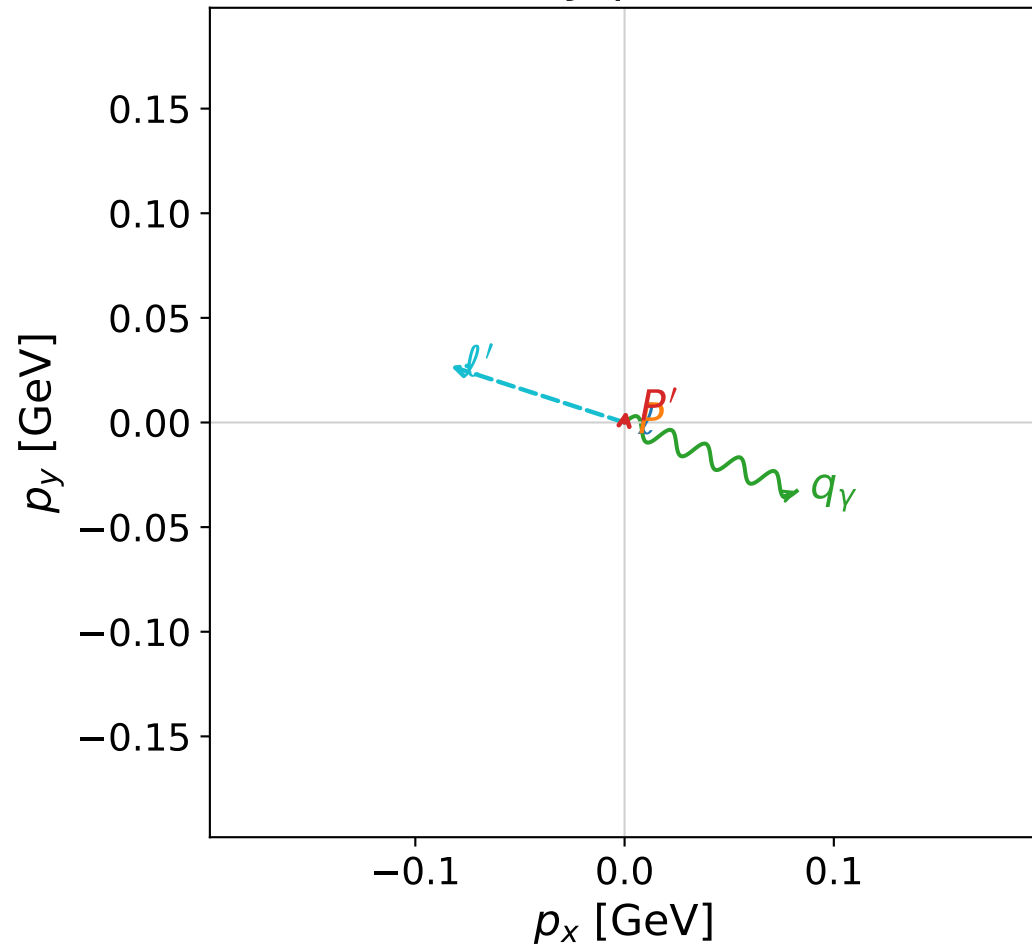


dw_mixing_angles_polarization_03_local_minimum_839
 $D_W=0.0116253$
 region: polarization_cluster_03_local_minimum_839
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0839
 lepton mixing angle: $\alpha_e=0.81111552$ rad
 incoming lepton: $0.68869|+\rangle + 0.725056|-\rangle$
 proton mixing angle: $\alpha_p=2.9294071$ rad
 incoming proton: $-0.977573|+\rangle + 0.210597|-\rangle$

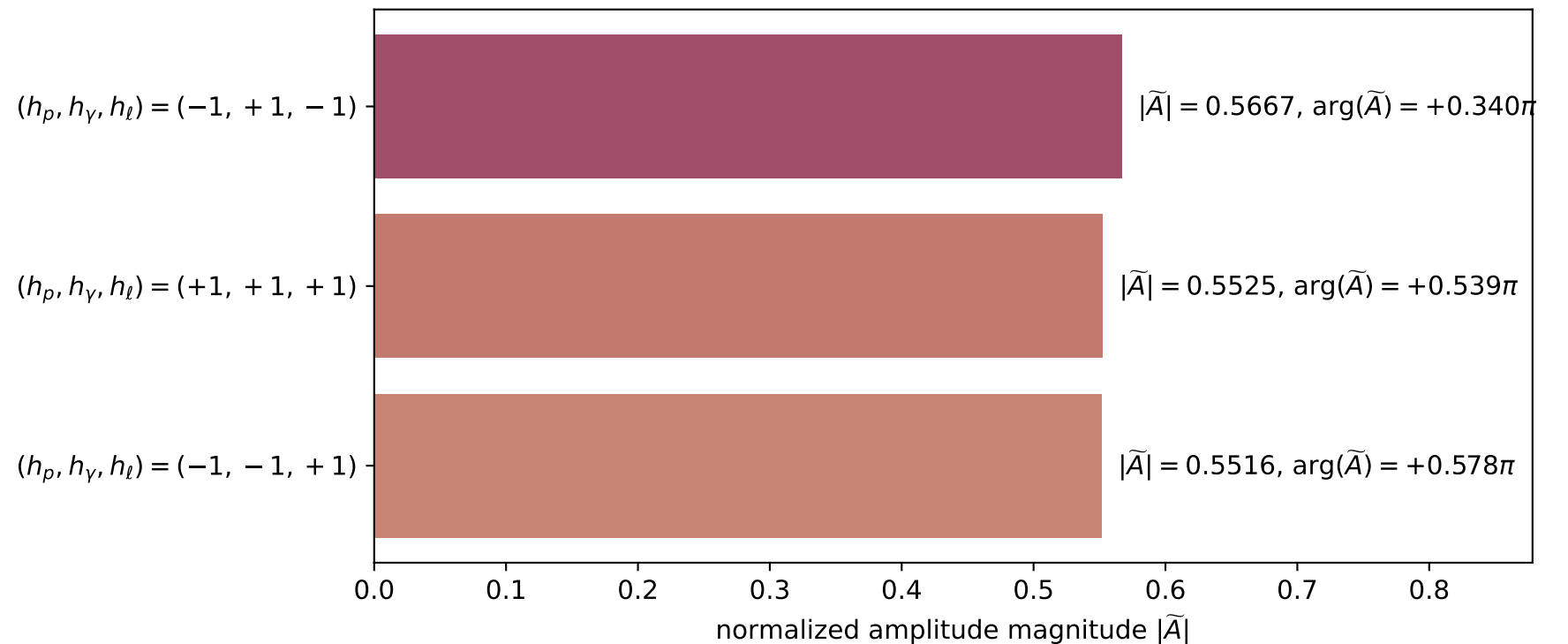
$s=1.24899$, $\sqrt{s}=1.11758$, $|\vec{P}|=0.165155$, $|\vec{P}'|=0.0105191$
 $\theta_{p'}=2.54466$, $\theta_\gamma=1.32836$, $\phi_{p'}=1.42117$, $\phi_\gamma=5.9038$
 $E_\gamma=0.0911783$, $\theta_{l'}=1.72064$, $\phi_{l'}=2.82816$
 $Q^2=0.024825$, $x_B=0.126333$, $t=-0.0300539$, $W^2=1.05152$, $y=0.532319$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.1652$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1652$
 P $E= 0.9524$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1652$
 l' $E= 0.0883$ $p_x= -0.0831$ $p_y= 0.0269$ $p_z= -0.0132$
 P' $E= 0.9381$ $p_x= 0.0009$ $p_y= 0.0058$ $p_z= -0.0087$
 q_γ $E= 0.0912$ $p_x= 0.0822$ $p_y= -0.0328$ $p_z= 0.0219$

x--y plane

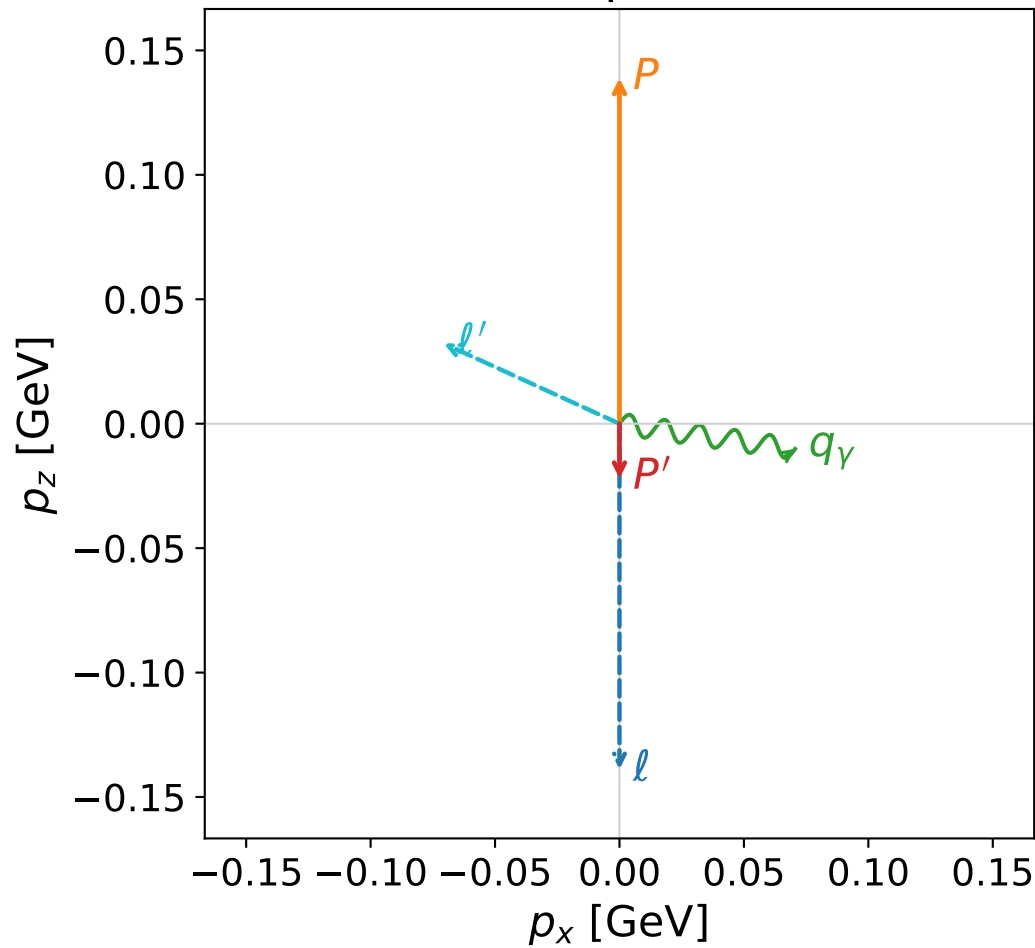


Leading normalized final-state amplitudes
 (N=3, retained=93.1%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



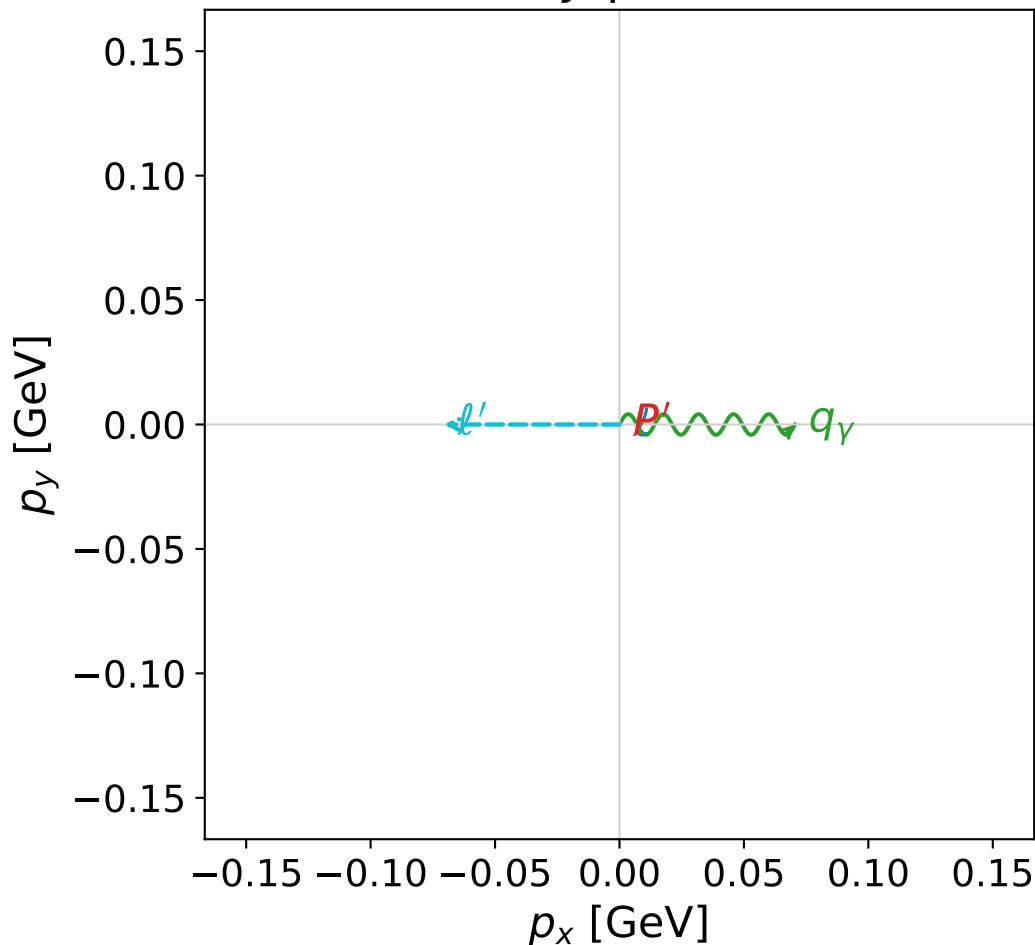
dw_mixing_angles_polarization_03_local_minimum_870
 $D_W=0.0123808$
 region: polarization_cluster_03_local_minimum_870
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0870
 lepton mixing angle: $\alpha_e=0.76538427$ rad
 incoming lepton: $0.721116|+\rangle + 0.692814|-\rangle$
 proton mixing angle: $\alpha_p=2.99225$ rad
 incoming proton: $-0.988869|+\rangle + 0.148788|-\rangle$

$s=1.18176$, $\sqrt{s}=1.08709$, $|\vec{P}|=0.138864$, $|\vec{P}'|=0.0220325$
 $\theta_{p'}=3.14159$, $\theta_\gamma=1.71361$, $\phi_{p'}=5.23833$, $\phi_\gamma=1.72592e-05$
 $E_\gamma=0.0712811$, $\theta_{\ell'}=1.14291$, $\phi_{\ell'}=3.14161$
 $Q^2=0.0304738$, $x_B=0.186058$, $t=-0.0257884$, $W^2=1.01316$, $y=0.542491$

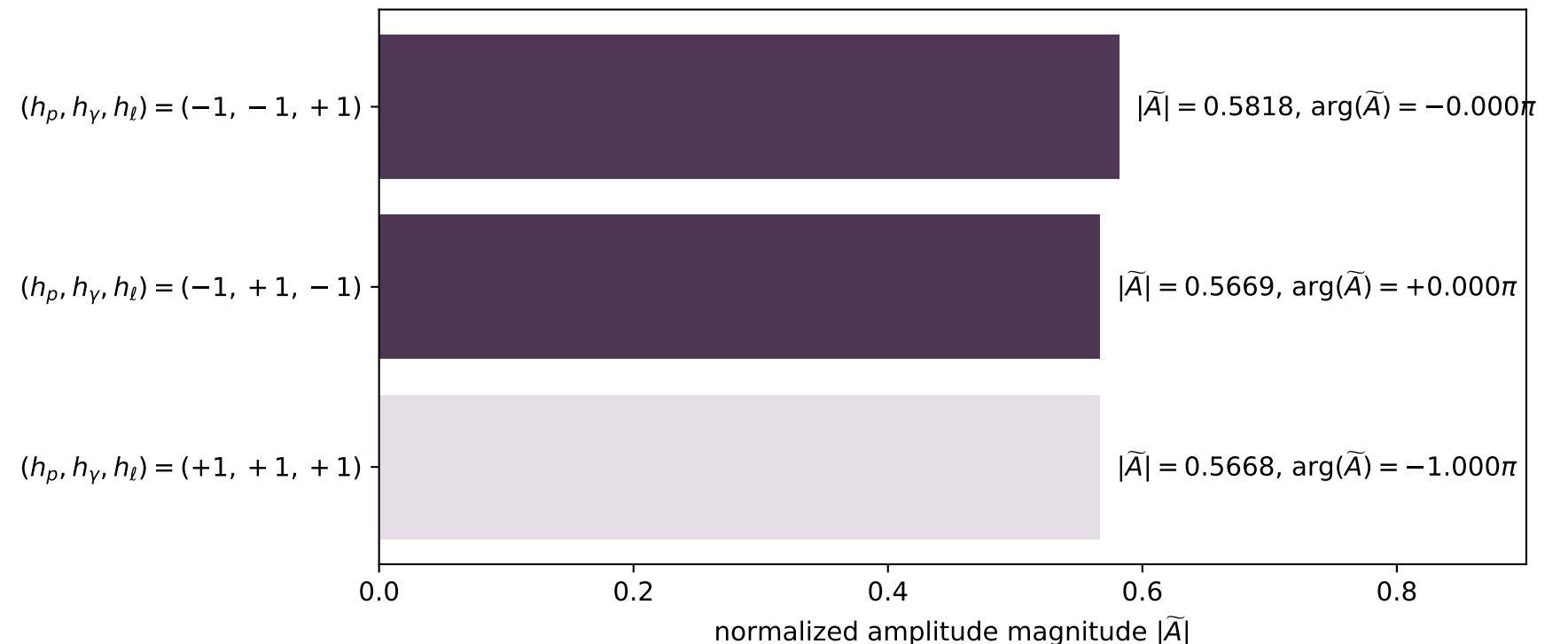
four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ	$E=$	0.1389	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	-0.1389
P	$E=$	0.9482	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	0.1389
ℓ'	$E=$	0.0775	$p_x=$	-0.0706	$p_y=$	-0.0000	$p_z=$	0.0322
P'	$E=$	0.9383	$p_x=$	0.0000	$p_y=$	-0.0000	$p_z=$	-0.0220
q_γ	$E=$	0.0713	$p_x=$	0.0706	$p_y=$	0.0000	$p_z=$	-0.0101

x--y plane

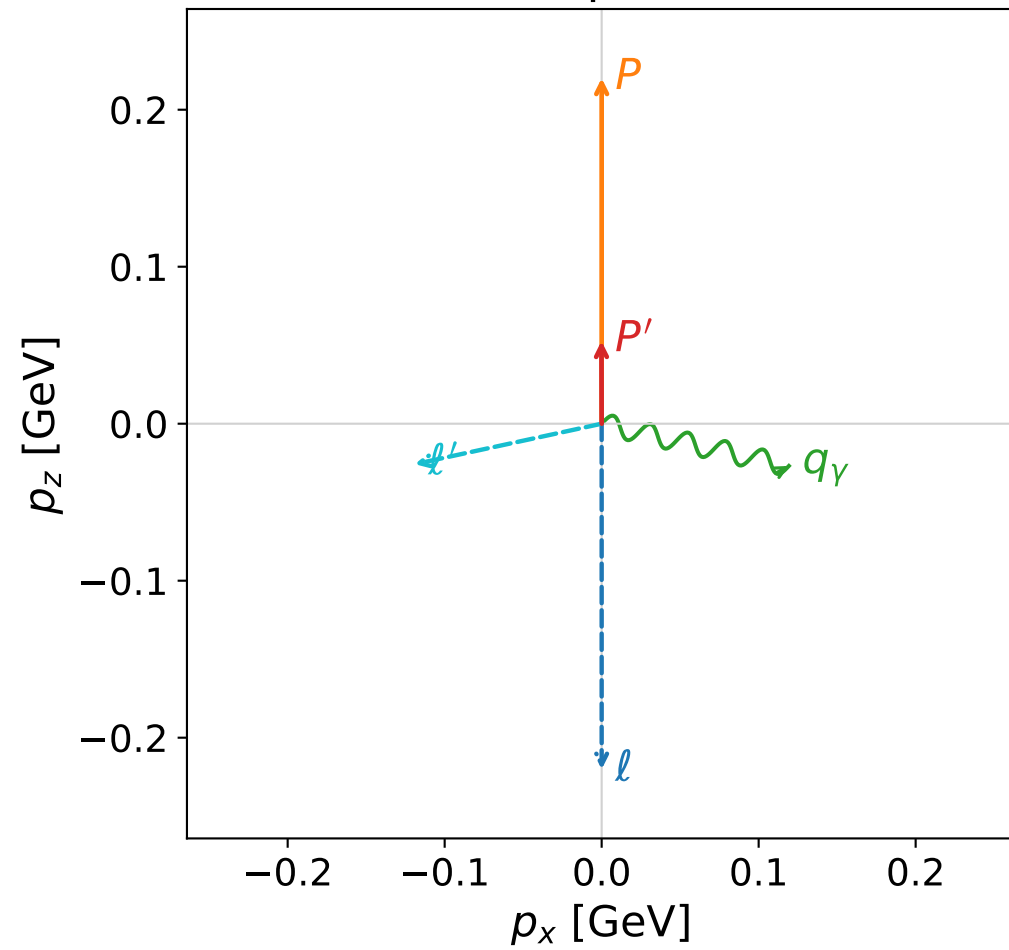


Leading normalized final-state amplitudes
(N=3, retained=98.1%)

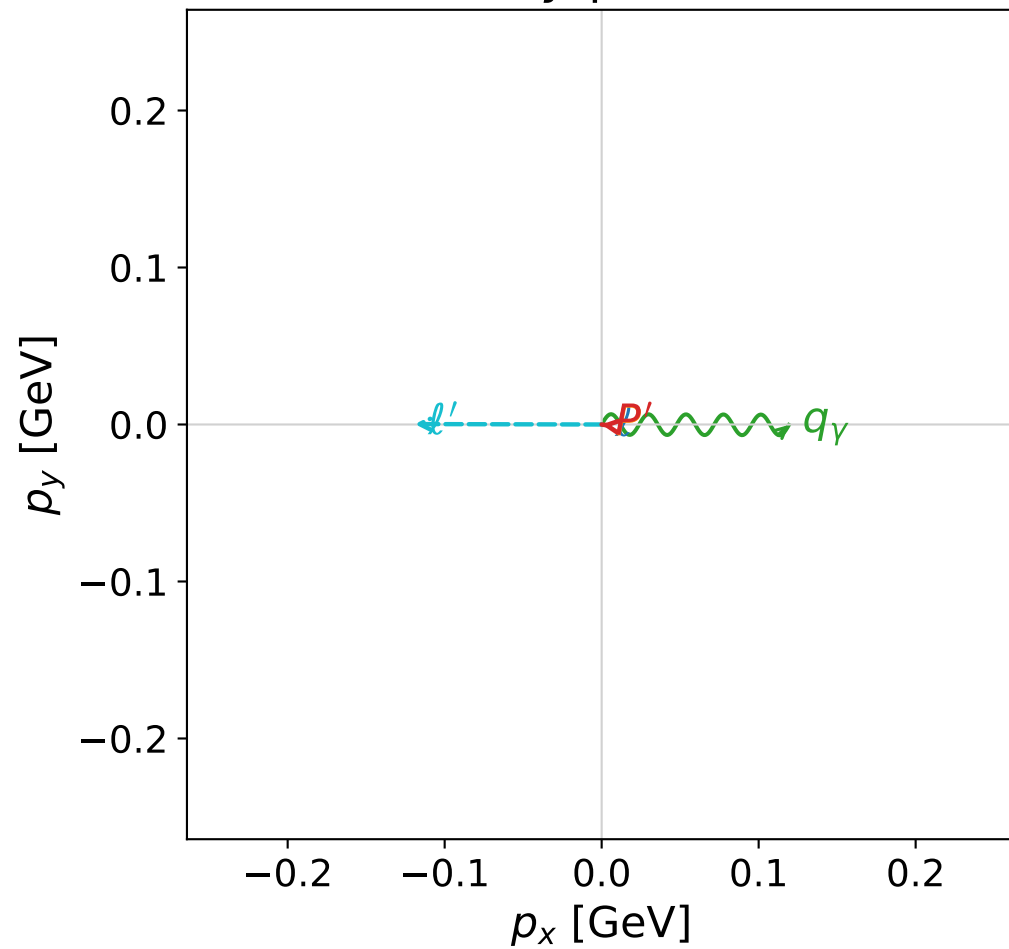


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



x--y plane



dw_mixing_angles_polarization_03_local_minimum_885

$D_W=0.0130708$

region: polarization_cluster_03_local_minimum_885

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0885

lepton mixing angle: $\alpha_e=0.81698053$ rad

incoming lepton: $0.684426|+\rangle + 0.729083|-\rangle$

proton mixing angle: $\alpha_p=2.9786241$ rad

incoming proton: $-0.98675|+\rangle + 0.162248|-\rangle$

$s=1.40095$, $\sqrt{s}=1.18362$, $|\vec{P}|=0.220131$, $|\vec{P}'|=0.0526706$

$\theta_{p'}=0.00278182$, $\theta_\gamma=1.79256$, $\phi_{p'}=3.00775$, $\phi_\gamma=6.28075$

$E_\gamma=0.122259$, $\theta_{l'}=1.78392$, $\phi_{l'}=3.13932$

$Q^2=0.0423086$, $x_B=0.153908$, $t=-0.0274667$, $W^2=1.11243$, $y=0.527529$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

l $E= 0.2201$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2201$

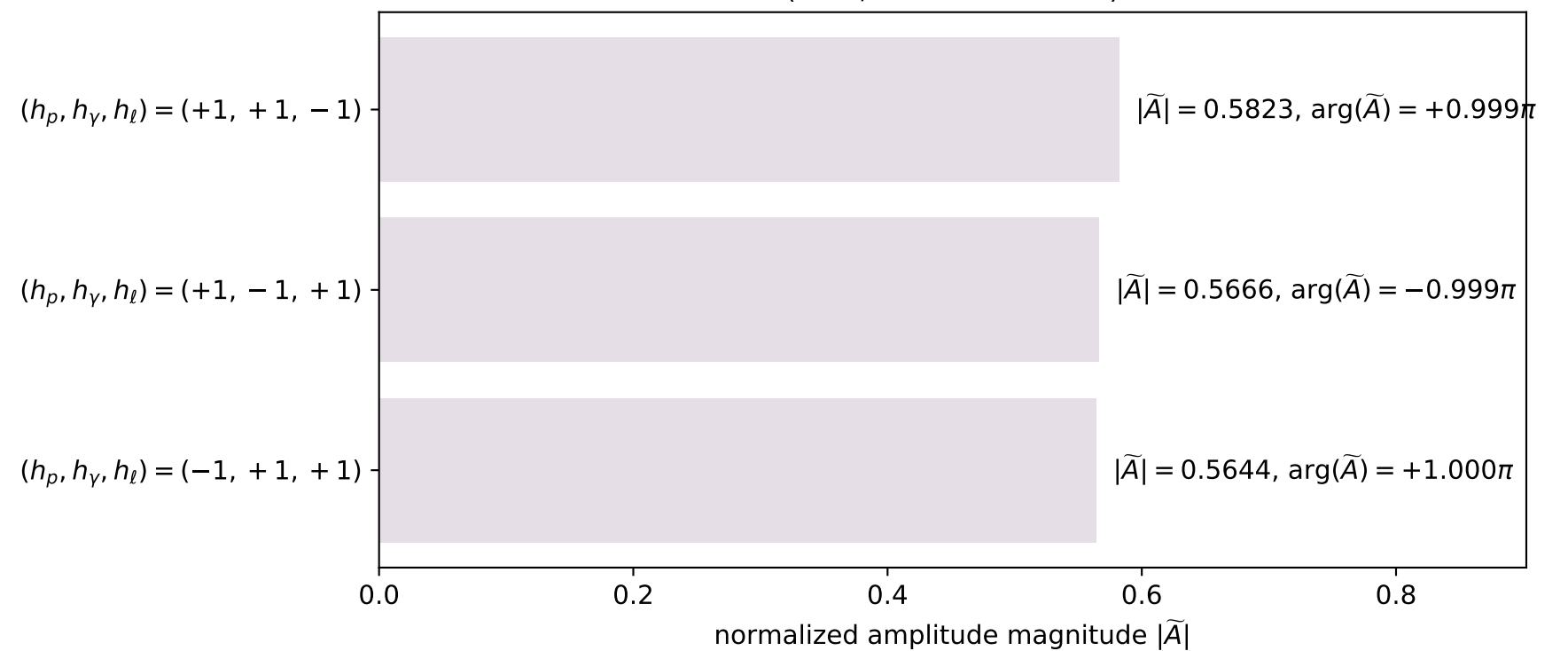
P $E= 0.9635$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.2201$

l' $E= 0.1219$ $p_x= -0.1191$ $p_y= 0.0003$ $p_z= -0.0258$

P' $E= 0.9395$ $p_x= -0.0001$ $p_y= 0.0000$ $p_z= 0.0527$

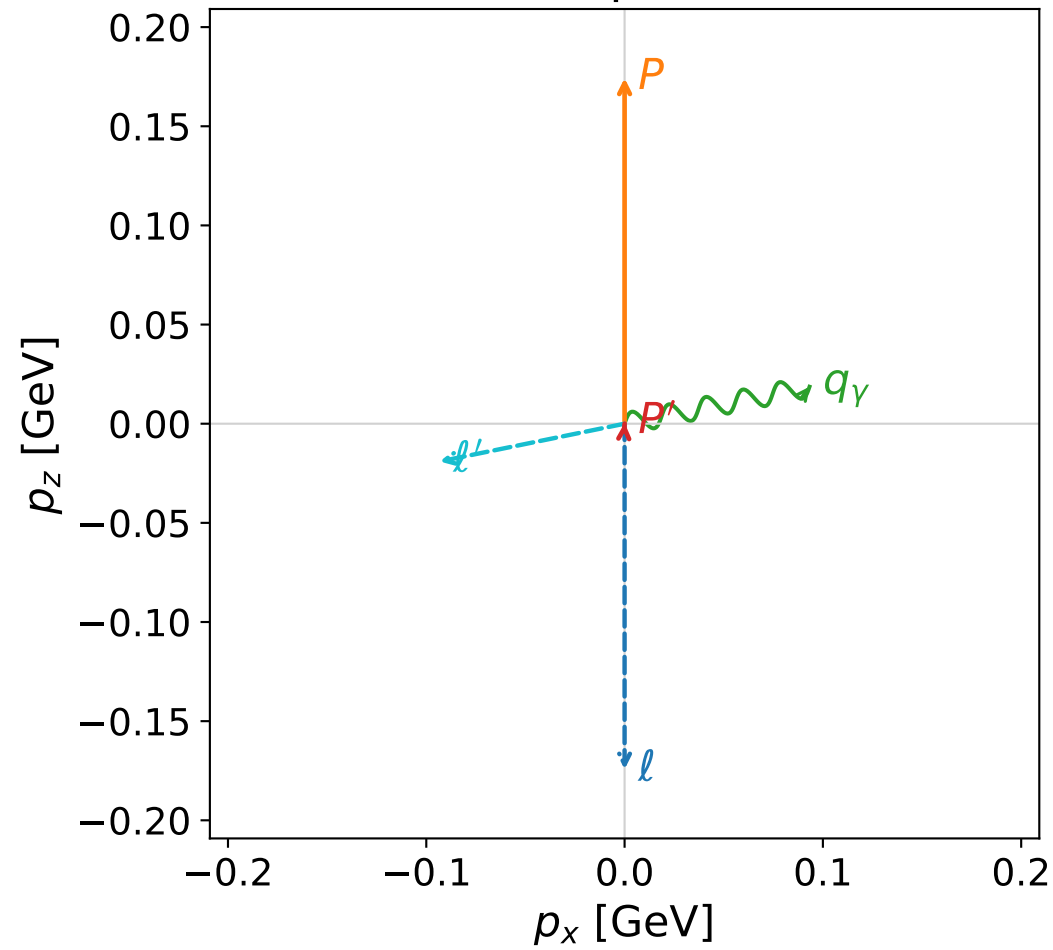
q_γ $E= 0.1223$ $p_x= 0.1193$ $p_y= -0.0003$ $p_z= -0.0269$

Leading normalized final-state amplitudes
(N=3, retained=97.9%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

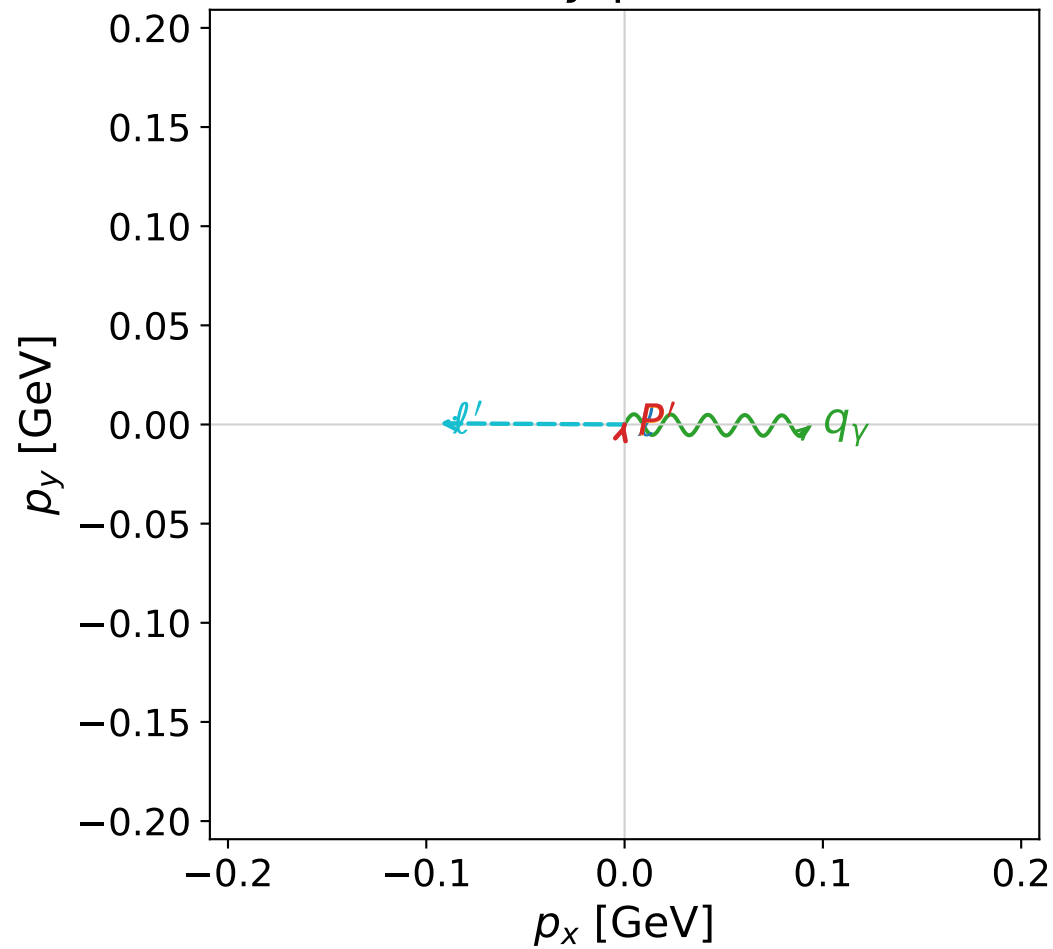


dw_mixing_angles_polarization_03_local_minimum_888
 $D_W=0.0132357$
 region: polarization_cluster_03_local_minimum_888
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0888
 lepton mixing angle: $\alpha_e=0.81471988$ rad
 incoming lepton: $0.686072|+\rangle + 0.727533|-\rangle$
 proton mixing angle: $\alpha_p=2.9588101$ rad
 incoming proton: $-0.983342|+\rangle + 0.181767|-\rangle$

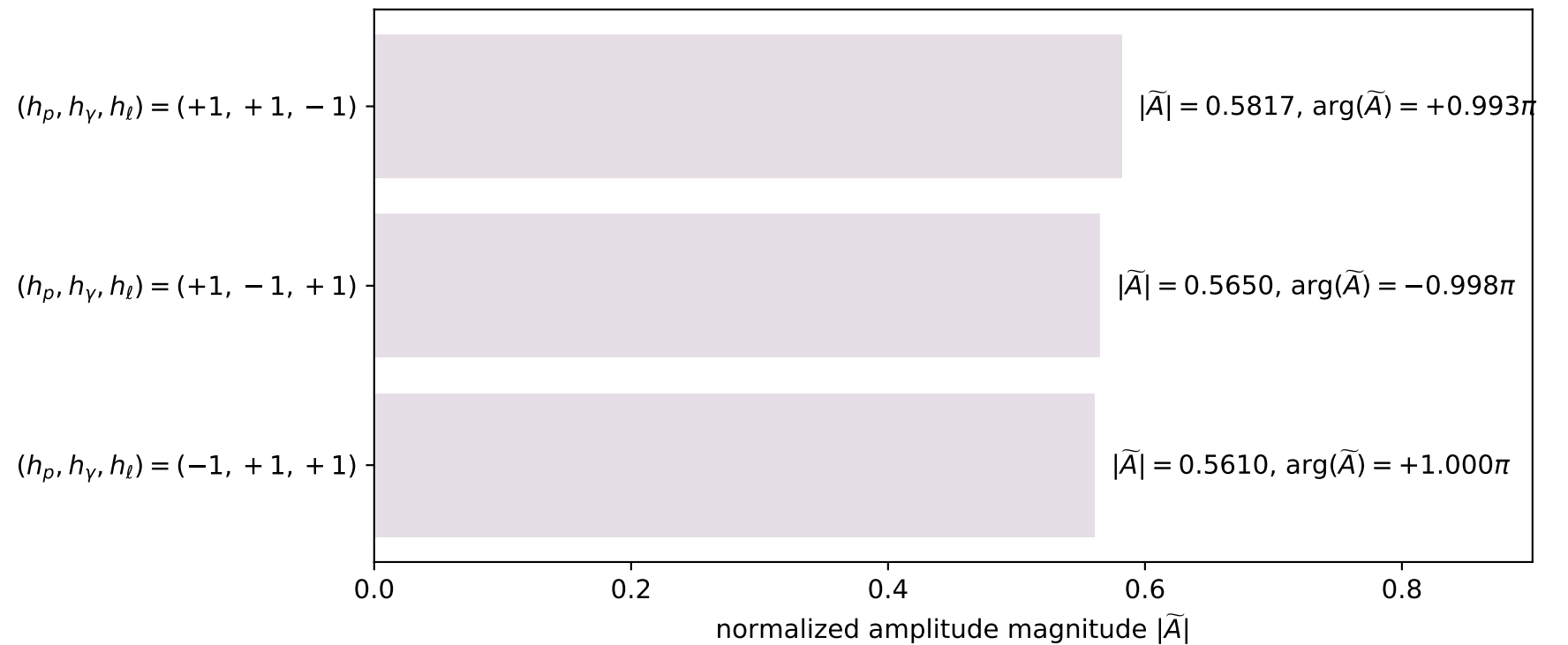
$s=1.27313$, $\sqrt{s}=1.12833$, $|\vec{P}|=0.174277$, $|\vec{P}'|=0.000416078$
 $\theta_{p'}=0.200961$, $\theta_\gamma=1.37346$, $\phi_{p'}=1.30137$, $\phi_\gamma=6.27608$
 $E_\gamma=0.0951137$, $\theta_{l'}=1.77228$, $\phi_{l'}=3.13535$
 $Q^2=0.0265461$, $x_B=0.129518$, $t=-0.029973$, $W^2=1.05826$, $y=0.521148$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E=0.1743$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.1743$
 P $E=0.9541$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1743$
 l' $E=0.0952$ $p_x=-0.0933$ $p_y=0.0006$ $p_z=-0.0191$
 P' $E=0.9380$ $p_x=0.0000$ $p_y=0.0001$ $p_z=0.0004$
 q_γ $E=0.0951$ $p_x=0.0933$ $p_y=-0.0007$ $p_z=0.0186$

x--y plane

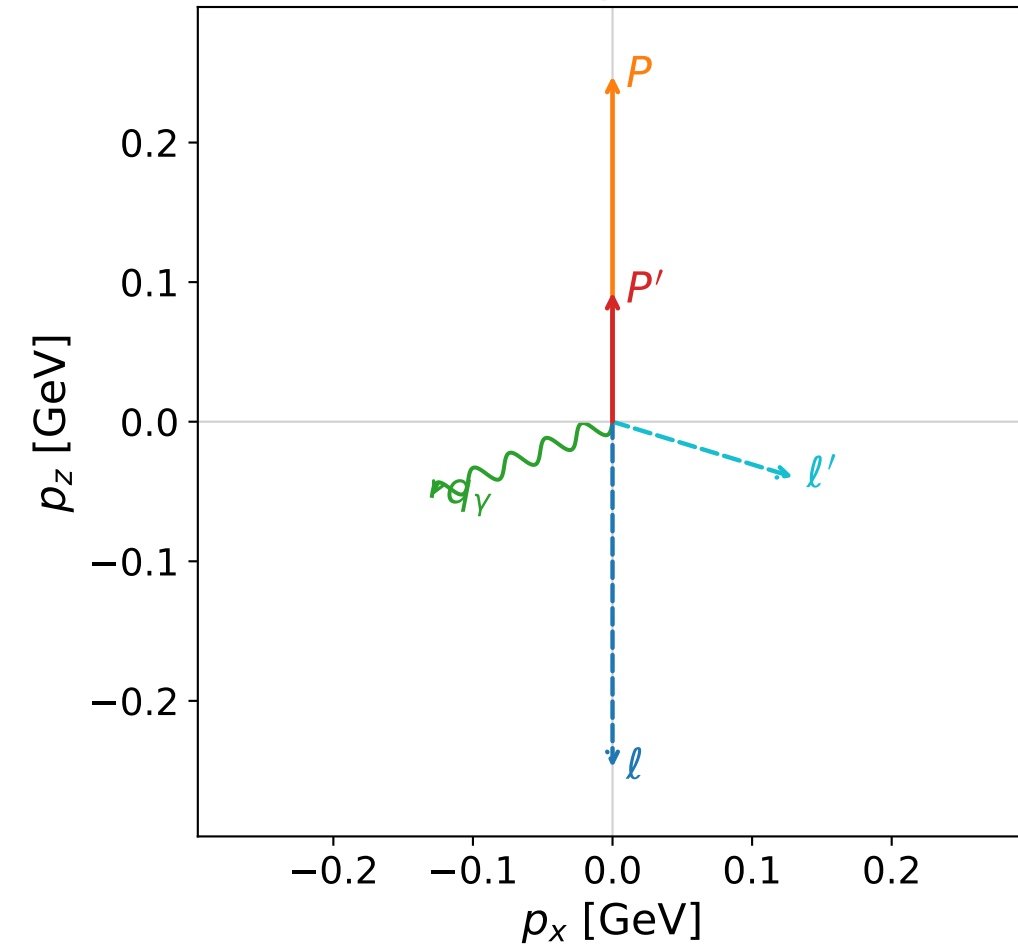


Leading normalized final-state amplitudes
 (N=3, retained=97.2%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

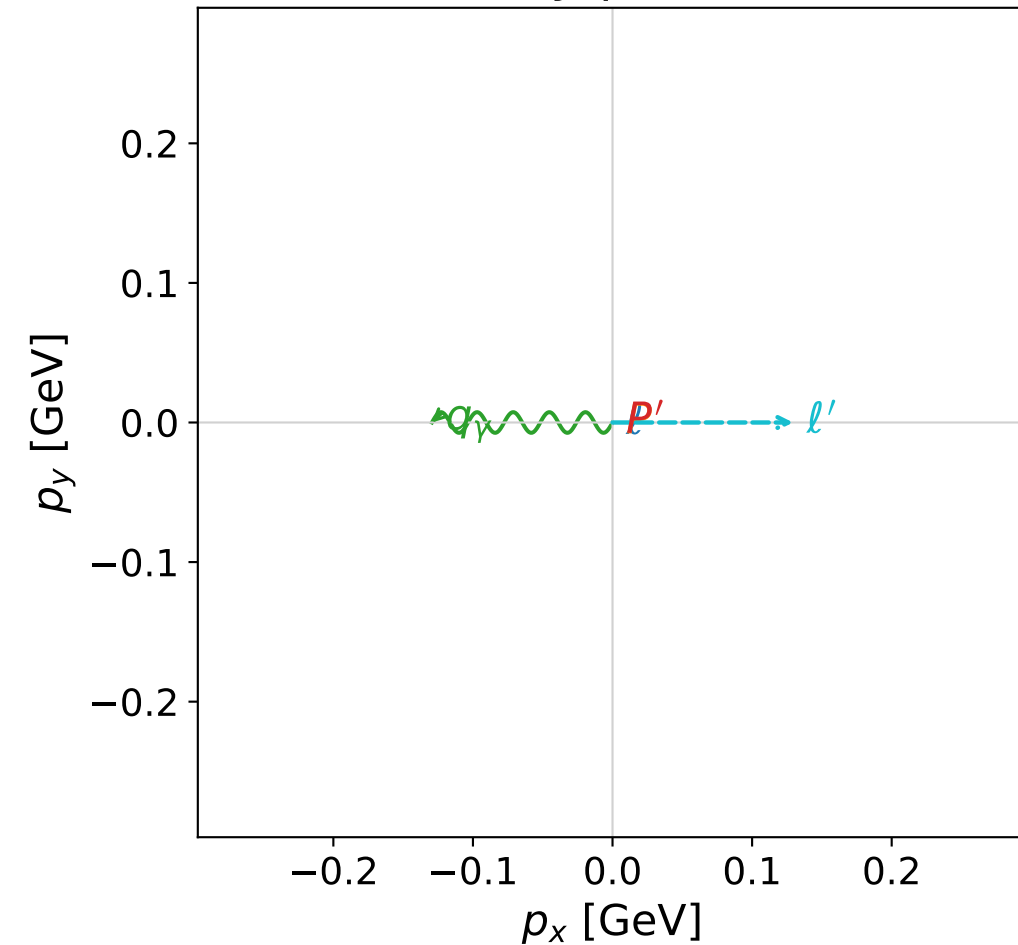


dw_mixing_angles_polarization_03_local_minimum_910
 $D_W=0.0140542$
 region: polarization_cluster_03_local_minimum_910
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0910
 lepton mixing angle: $\alpha_e=0.81021801$ rad
 incoming lepton: $0.689341|+\rangle + 0.724437|-\rangle$
 proton mixing angle: $\alpha_p=0.13099092$ rad
 incoming proton: $0.991433|+\rangle + 0.130617|-\rangle$

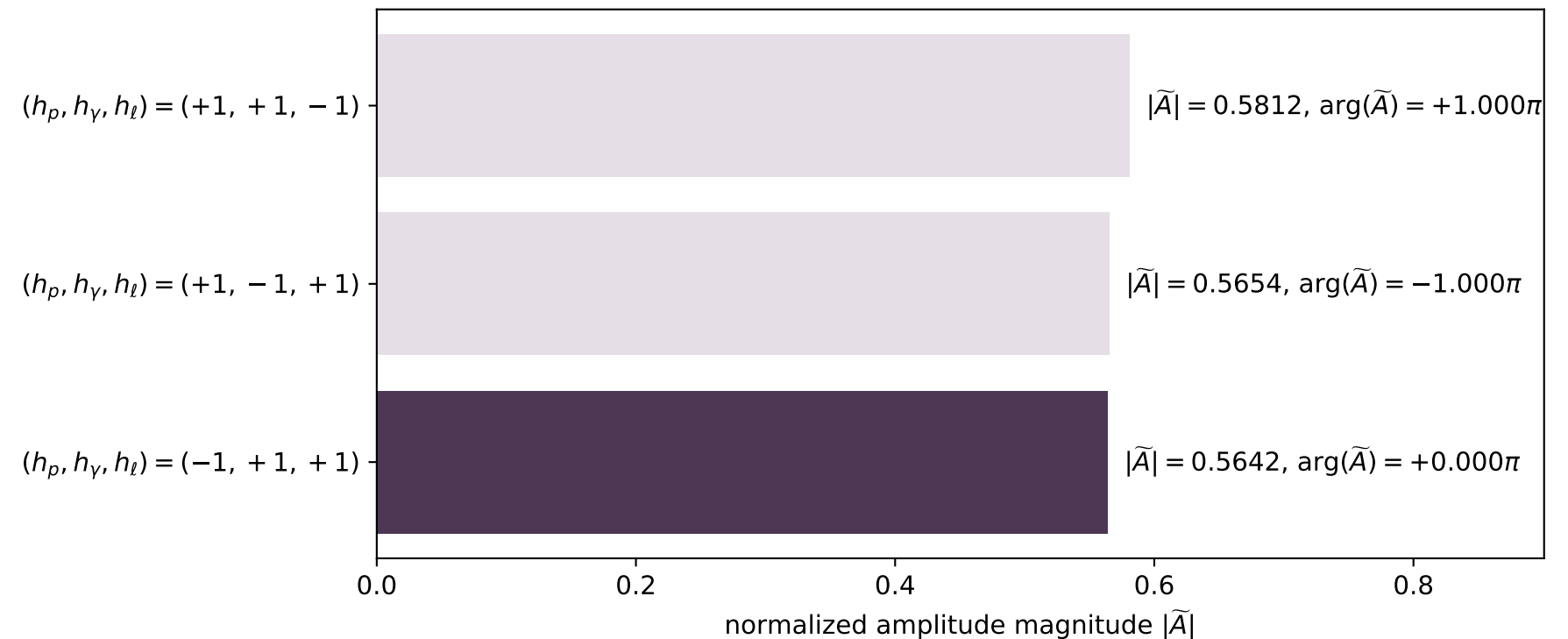
$s=1.48284$, $\sqrt{s}=1.21772$, $|\vec{P}|=0.247591$, $|\vec{P}'|=0.0926797$
 $\theta_{p'}=0$, $\theta_\gamma=1.96038$, $\phi_{p'}=3.1836$, $\phi_\gamma=3.14158$
 $E_\gamma=0.139858$, $\theta_{l'}=1.86755$, $\phi_{l'}=6.28318$
 $Q^2=0.047404$, $x_B=0.147721$, $t=-0.0232381$, $W^2=1.15334$, $y=0.53218$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.2476$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2476$
 P $E= 0.9701$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.2476$
 l' $E= 0.1353$ $p_x= 0.1294$ $p_y= -0.0000$ $p_z= -0.0396$
 P' $E= 0.9426$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.0927$
 q_γ $E= 0.1399$ $p_x= -0.1294$ $p_y= 0.0000$ $p_z= -0.0531$

x--y plane

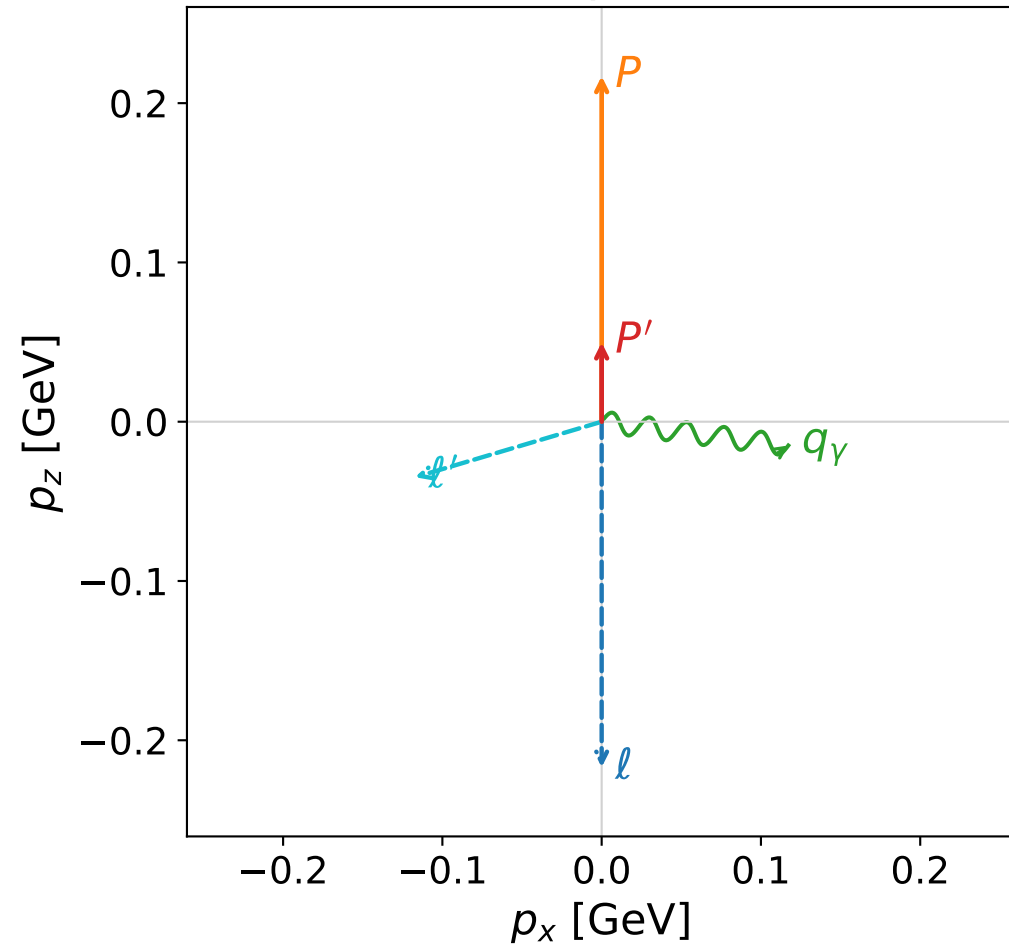


Leading normalized final-state amplitudes
 (N=3, retained=97.6%)

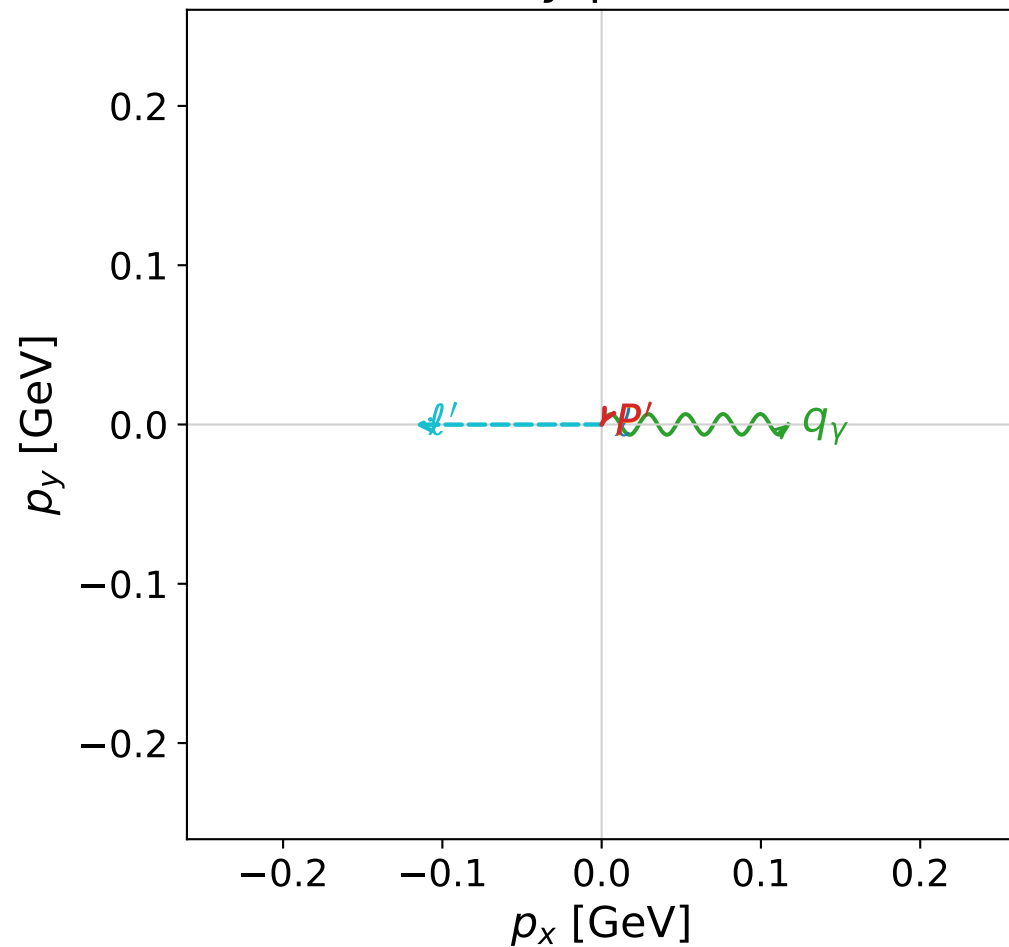


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



x--y plane

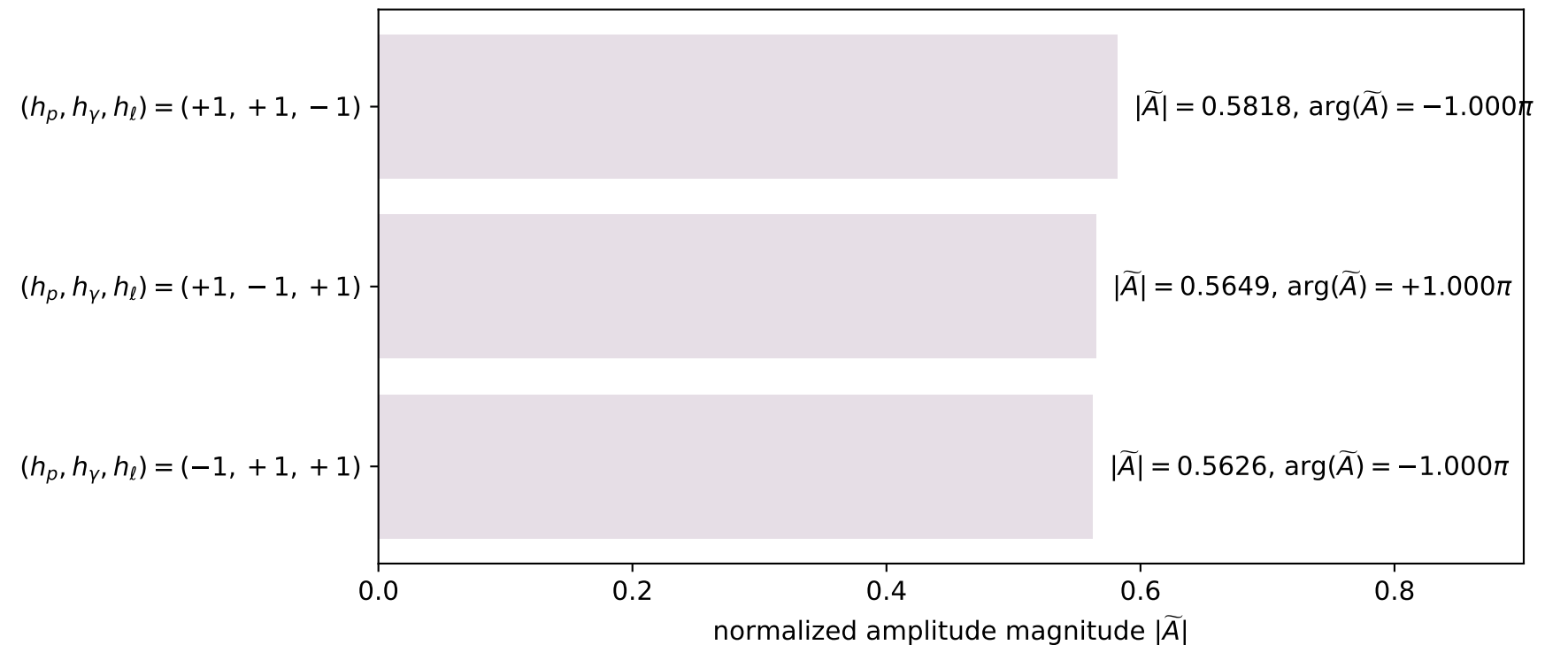


dw_mixing_angles_polarization_03_local_minimum_950
 $D_W=0.01601$
 region: polarization_cluster_03_local_minimum_950
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0950
 lepton mixing angle: $\alpha_e=0.81877057$ rad
 incoming lepton: $0.68312|+\rangle + 0.730307|-\rangle$
 proton mixing angle: $\alpha_p=2.9968052$ rad
 incoming proton: $-0.989537|+\rangle + 0.144282|-\rangle$

$s=1.39179$, $\sqrt{s}=1.17974$, $|\vec{P}|=0.216973$, $|\vec{P}'|=0.0497768$
 $\theta_{p'}=0.000110249$, $\theta_\gamma=1.69747$, $\phi_{p'}=4.21159$, $\phi_\gamma=0.00102048$
 $E_\gamma=0.118149$, $\theta_{\ell'}=1.85983$, $\phi_{\ell'}=3.14257$
 $Q^2=0.0379363$, $x_B=0.145137$, $t=-0.0274049$, $W^2=1.10329$, $y=0.510567$

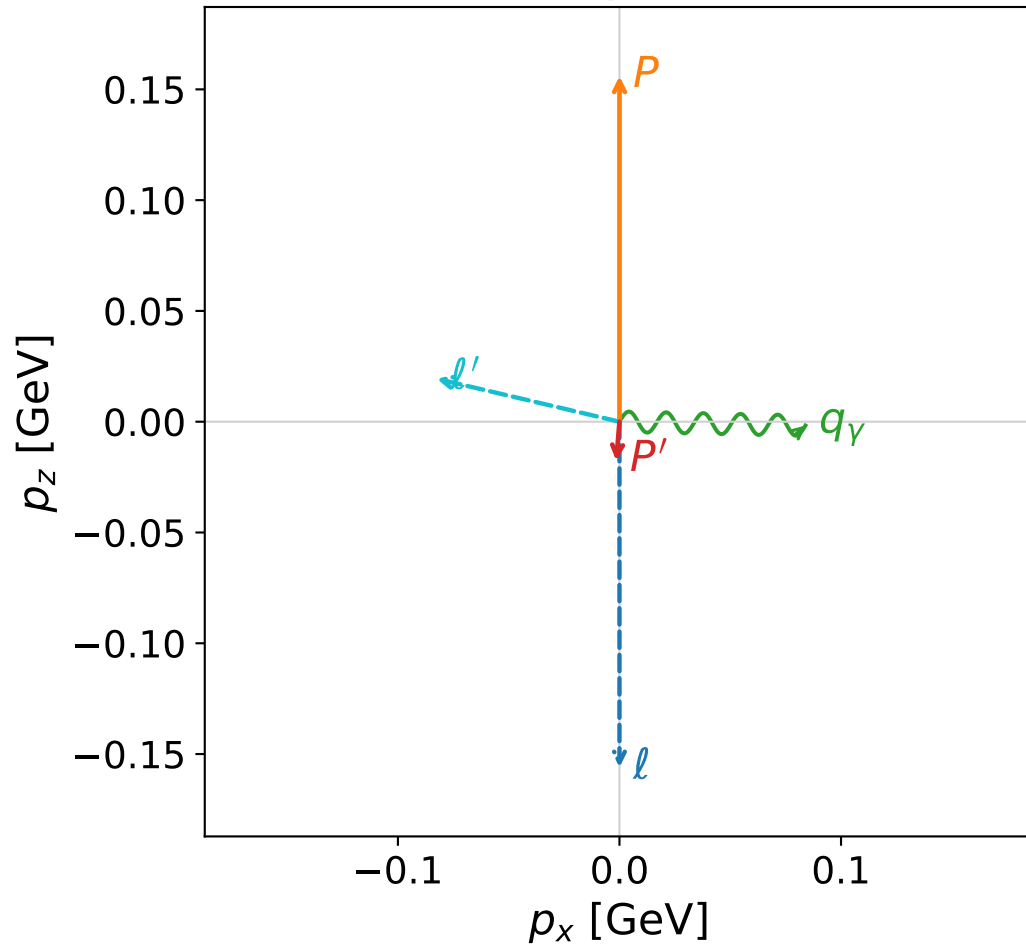
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E=0.2170$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.2170$
 P $E=0.9628$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.2170$
 ℓ' $E=0.1223$ $p_x=-0.1172$ $p_y=-0.0001$ $p_z=-0.0349$
 P' $E=0.9393$ $p_x=-0.0000$ $p_y=-0.0000$ $p_z=0.0498$
 q_γ $E=0.1181$ $p_x=0.1172$ $p_y=0.0001$ $p_z=-0.0149$

Leading normalized final-state amplitudes
 (N=3, retained=97.4%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

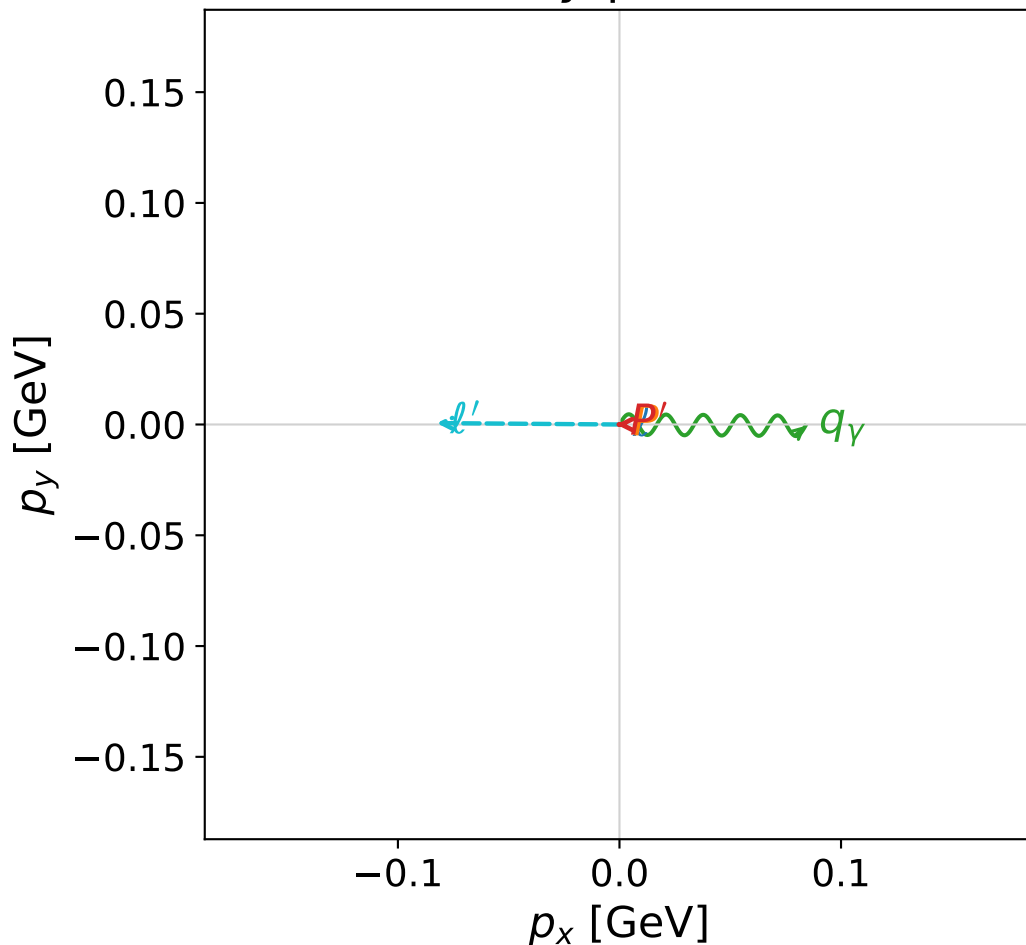


dw_mixing_angles_polarization_03_local_minimum_991
 $D_W=0.0179842$
 region: polarization_cluster_03_local_minimum_991
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0991
 lepton mixing angle: $\alpha_e=0.78916052$ rad
 incoming lepton: $0.704441|+\rangle + 0.709762|-\rangle$
 proton mixing angle: $\alpha_p=3.112534$ rad
 incoming proton: $-0.999578|+\rangle + 0.0290546|-\rangle$

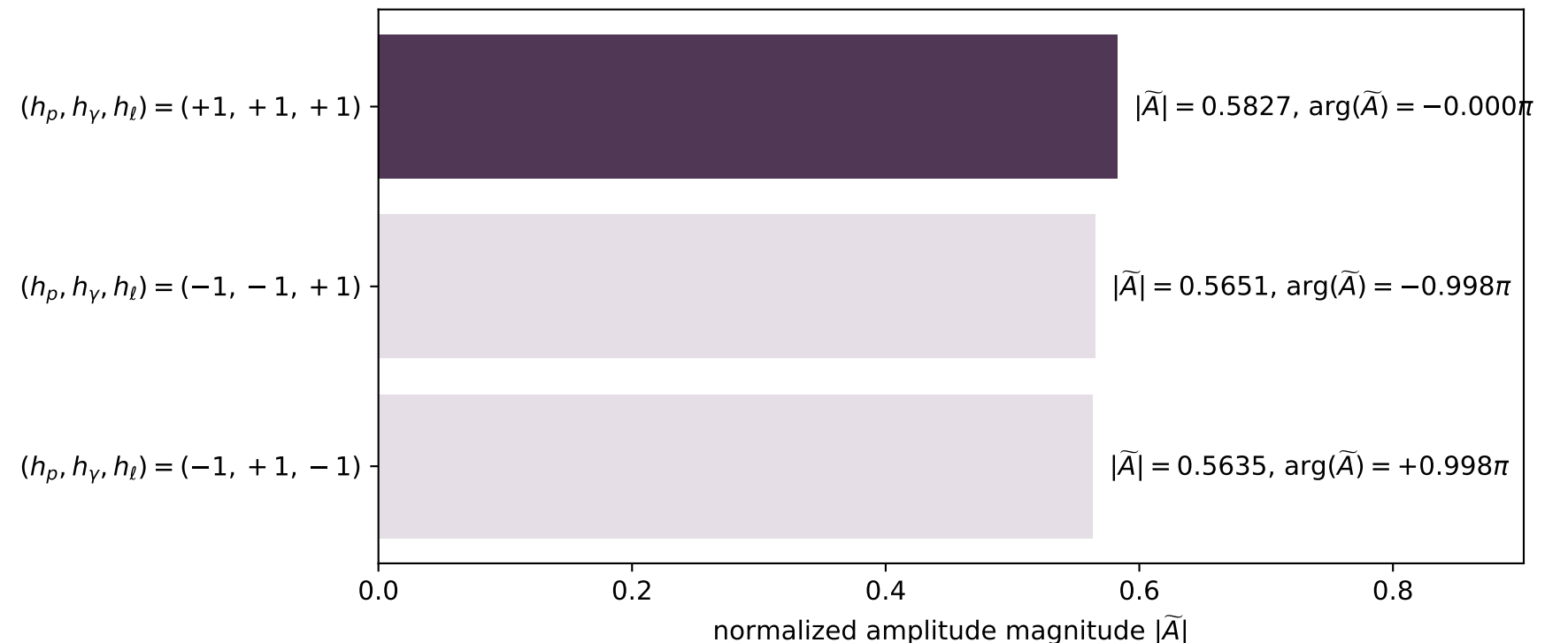
$s=1.22514$, $\sqrt{s}=1.10686$, $|\vec{P}|=0.155981$, $|\vec{P}'|=0.0177058$
 $\theta_{p'}=3.06042$, $\theta_\gamma=1.59143$, $\phi_{p'}=3.14178$, $\phi_\gamma=6.27569$
 $E_\gamma=0.0839506$, $\theta_{\ell'}=1.34006$, $\phi_{\ell'}=3.13397$
 $Q^2=0.0324823$, $x_B=0.170796$, $t=-0.0299871$, $W^2=1.03754$, $y=0.550773$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1560$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1560$
 P $E= 0.9509$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1560$
 ℓ' $E= 0.0847$ $p_x= -0.0825$ $p_y= 0.0006$ $p_z= 0.0194$
 P' $E= 0.9382$ $p_x= -0.0014$ $p_y= -0.0000$ $p_z= -0.0176$
 q_γ $E= 0.0840$ $p_x= 0.0839$ $p_y= -0.0006$ $p_z= -0.0017$

x--y plane

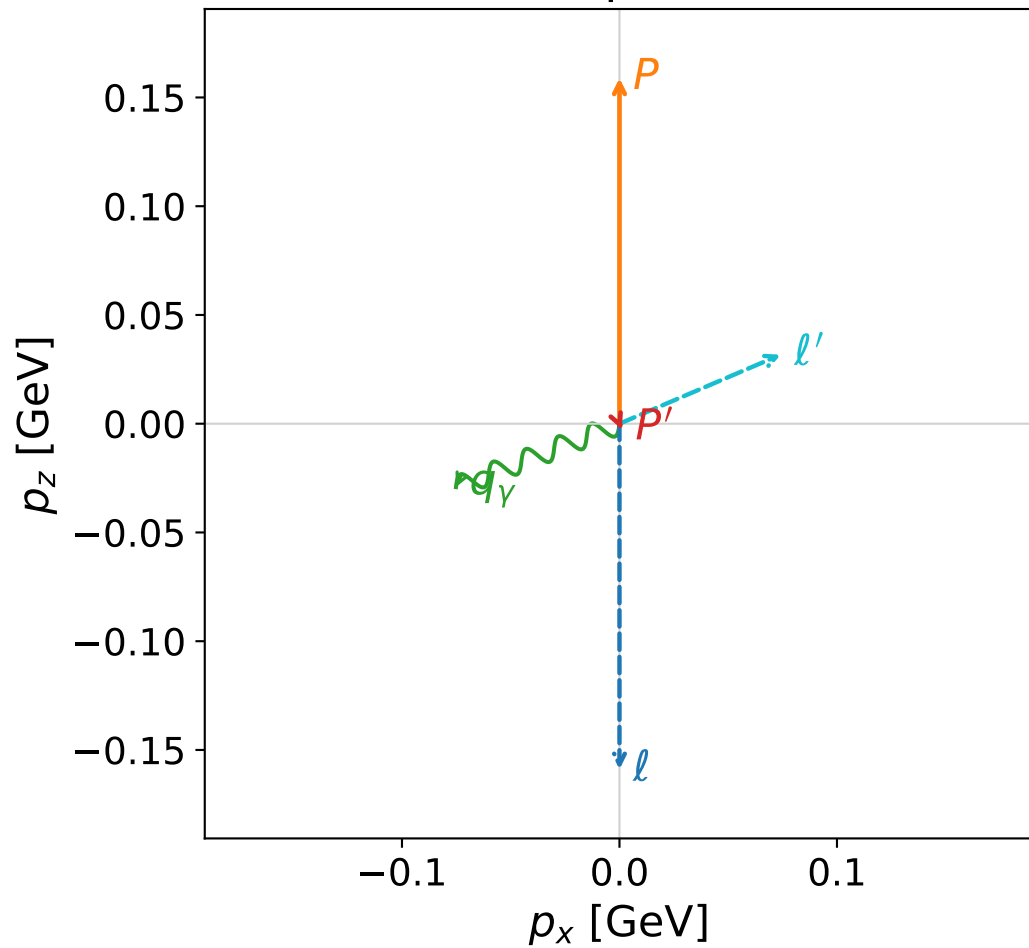


Leading normalized final-state amplitudes
 (N=3, retained=97.6%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

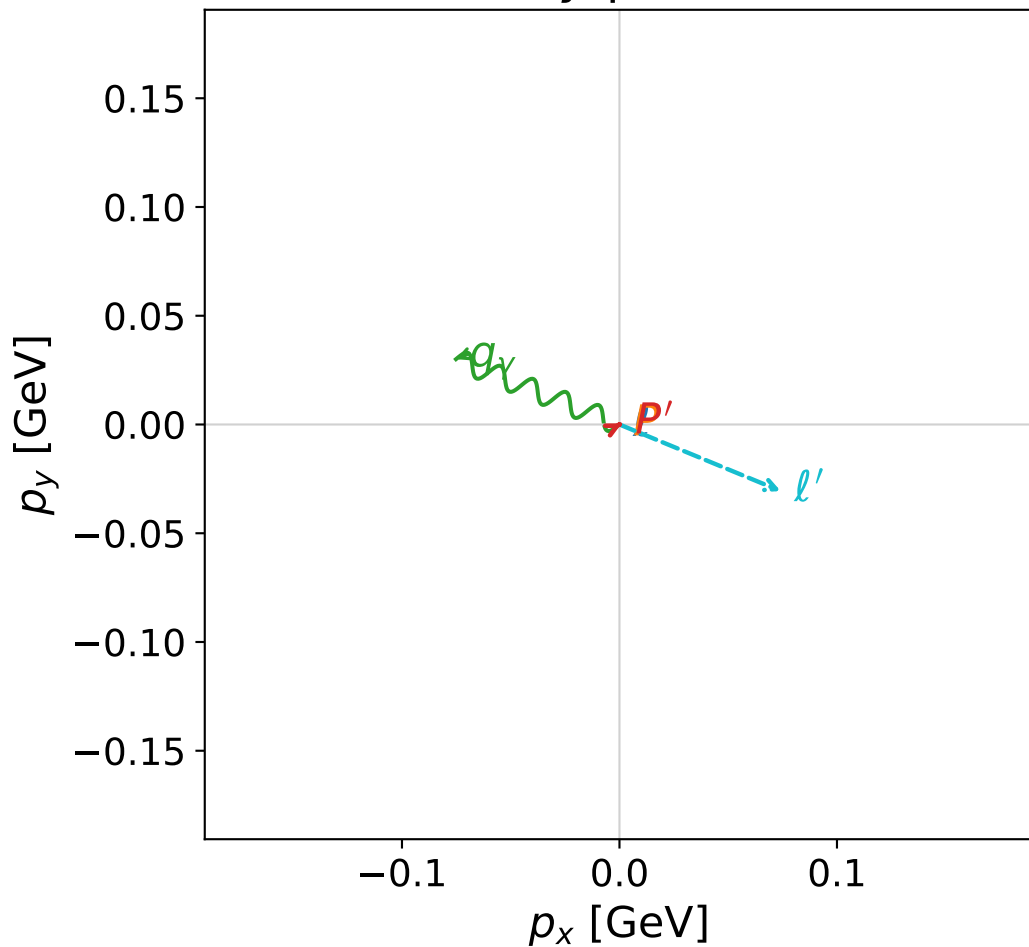


dw_mixing_angles_polarization_03_local_minimum_994
 $D_W=0.0180986$
 region: polarization_cluster_03_local_minimum_994
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0994
 lepton mixing angle: $\alpha_e=0.77750627$ rad
 incoming lepton: $0.712665|+\rangle + 0.701504|-\rangle$
 proton mixing angle: $\alpha_p=0.028423775$ rad
 incoming proton: $0.999596|+\rangle + 0.0284199|-\rangle$

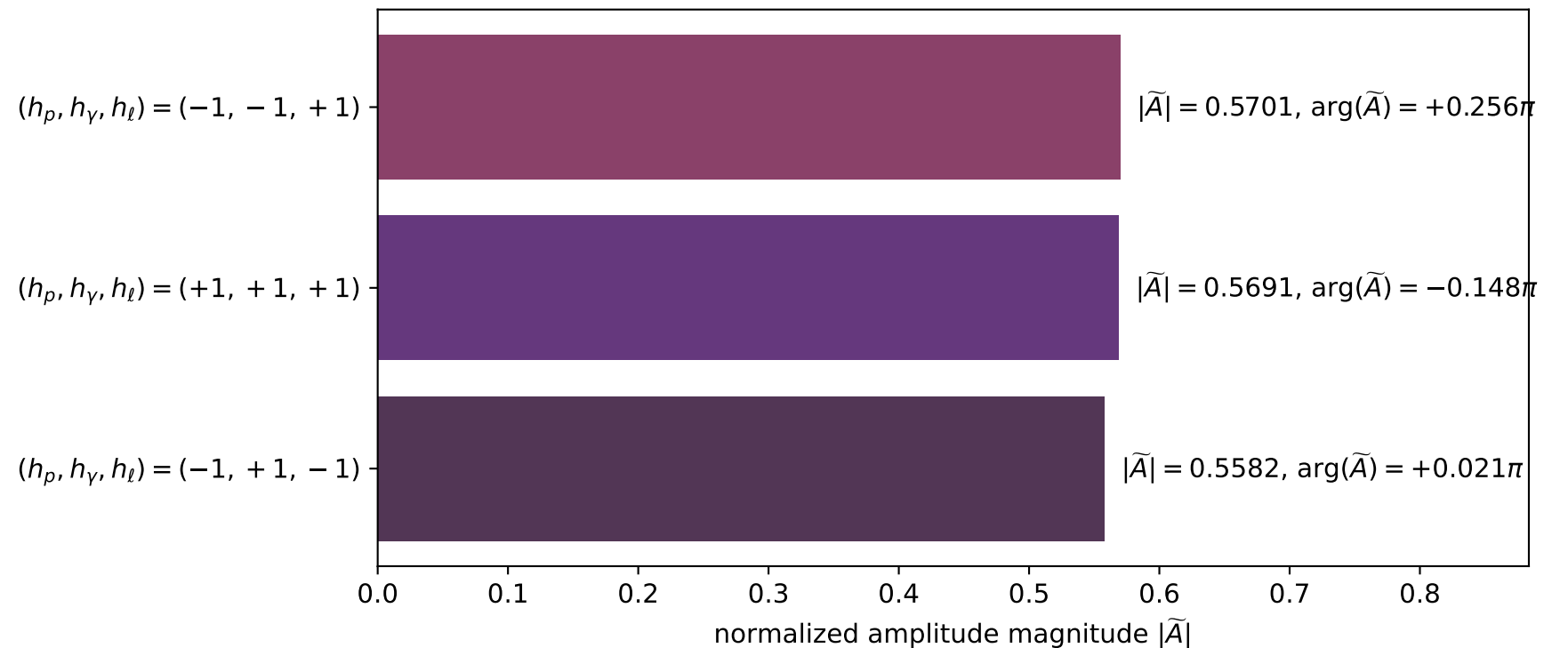
$s=1.23267$, $\sqrt{s}=1.11026$, $|\vec{P}|=0.158894$, $|\vec{P}'|=0.00281842$
 $\theta_{p'}=2.70197$, $\theta_Y=1.91653$, $\phi_{p'}=0.467859$, $\phi_Y=2.76154$
 $E_Y=0.0860442$, $\theta_{\ell'}=1.19413$, $\phi_{\ell'}=5.89191$
 $Q^2=0.0374729$, $x_B=0.188427$, $t=-0.0258873$, $W^2=1.04124$, $y=0.563652$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1589$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1589$
 P $E= 0.9514$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1589$
 ℓ' $E= 0.0862$ $p_x= 0.0741$ $p_y= -0.0306$ $p_z= 0.0317$
 P' $E= 0.9380$ $p_x= 0.0011$ $p_y= 0.0005$ $p_z= -0.0026$
 q_Y $E= 0.0860$ $p_x= -0.0752$ $p_y= 0.0300$ $p_z= -0.0292$

x--y plane

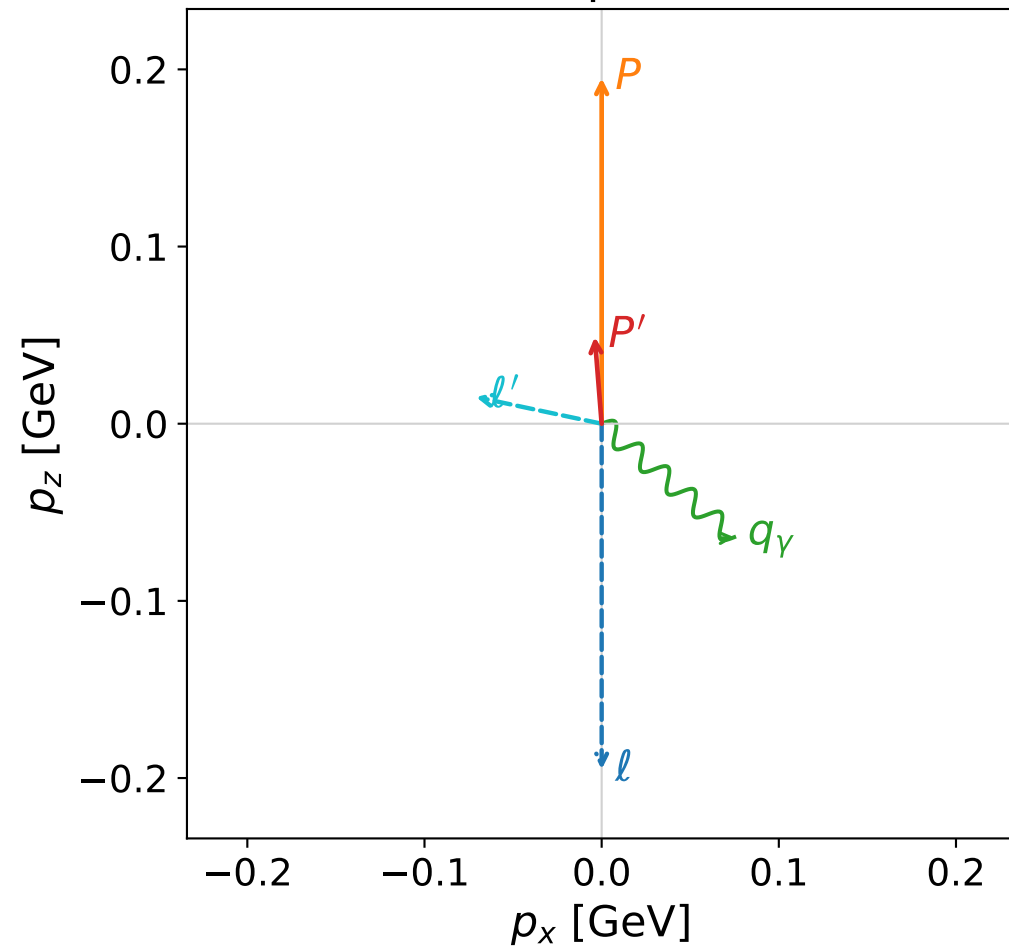


Leading normalized final-state amplitudes
 (N=3, retained=96.0%)

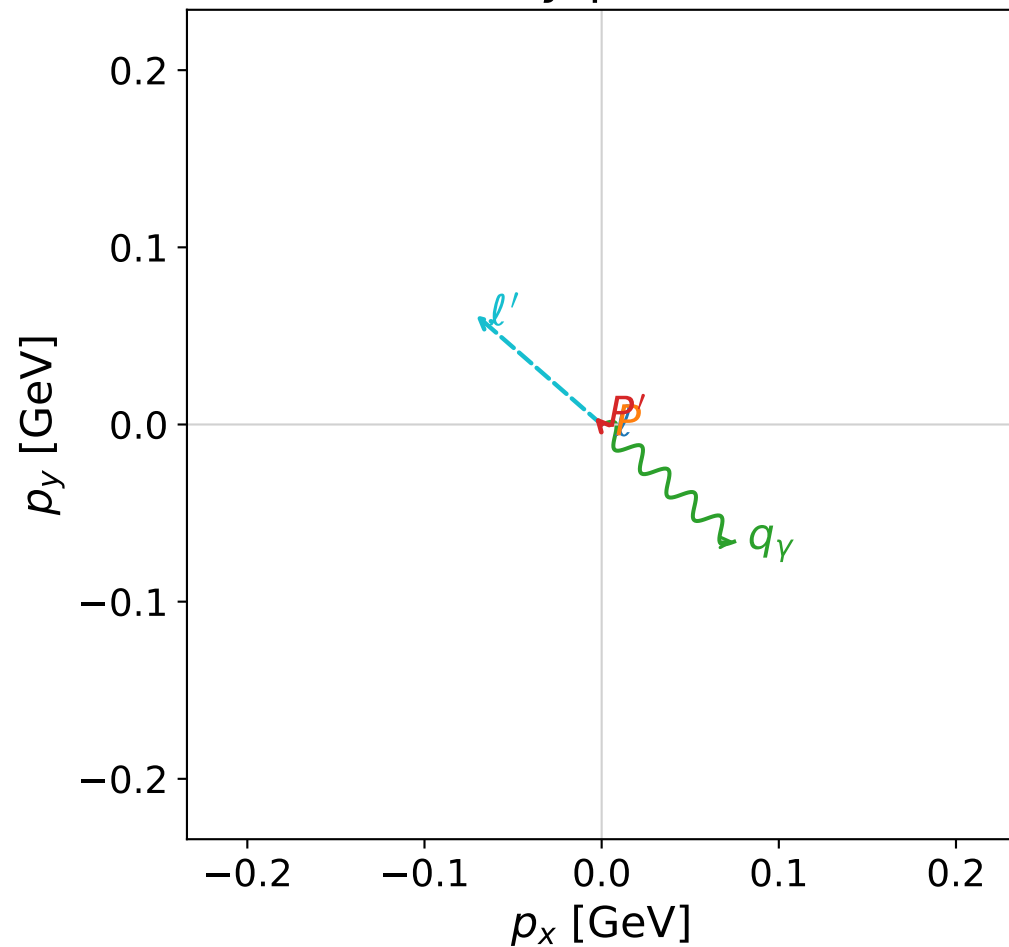


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



x--y plane



dw_mixing_angles_polarization_05_local_minimum_113

$D_W=0.00028775$

region: polarization_cluster_05_local_minimum_113

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0113

lepton mixing angle: $\alpha_e=2.3605141$ rad

incoming lepton: $-0.710155|+> +0.704046|->$

proton mixing angle: $\alpha_p=3.009653$ rad

incoming proton: $-0.991309|+> +0.131557|->$

$s=1.32973$, $\sqrt{s}=1.15314$, $|\vec{P}|=0.195069$, $|\vec{P}'|=0.0493626$

$\theta_{p'}=0.118924$, $\theta_\gamma=2.1413$, $\phi_{p'}=2.29736$, $\phi_\gamma=5.55919$

$E_\gamma=0.11862$, $\theta_{l'}=1.4121$, $\phi_{l'}=2.42506$

$Q^2=0.0430194$, $x_B=0.157407$, $t=-0.0210141$, $W^2=1.11012$, $y=0.607488$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

l $E= 0.1951$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1951$

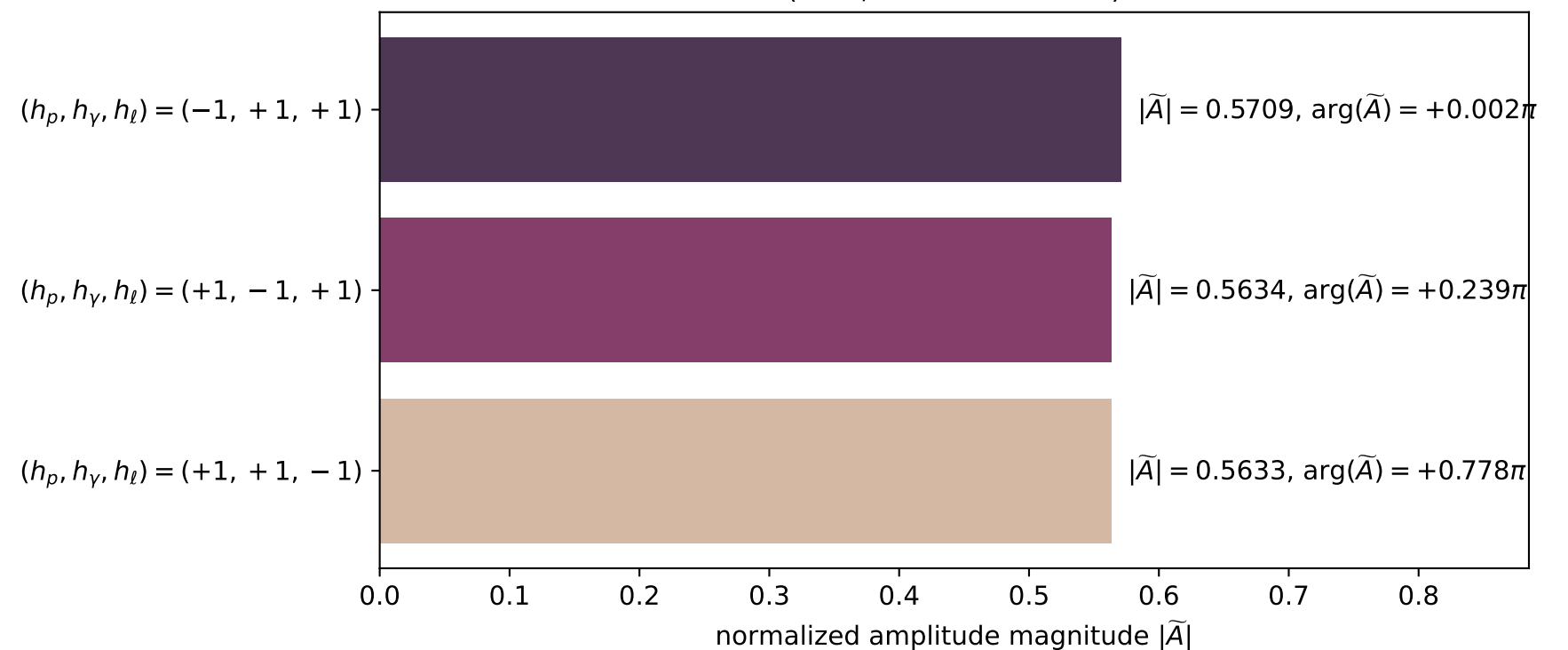
P $E= 0.9581$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1951$

l' $E= 0.0952$ $p_x= -0.0709$ $p_y= 0.0618$ $p_z= 0.0150$

P' $E= 0.9393$ $p_x= -0.0039$ $p_y= 0.0044$ $p_z= 0.0490$

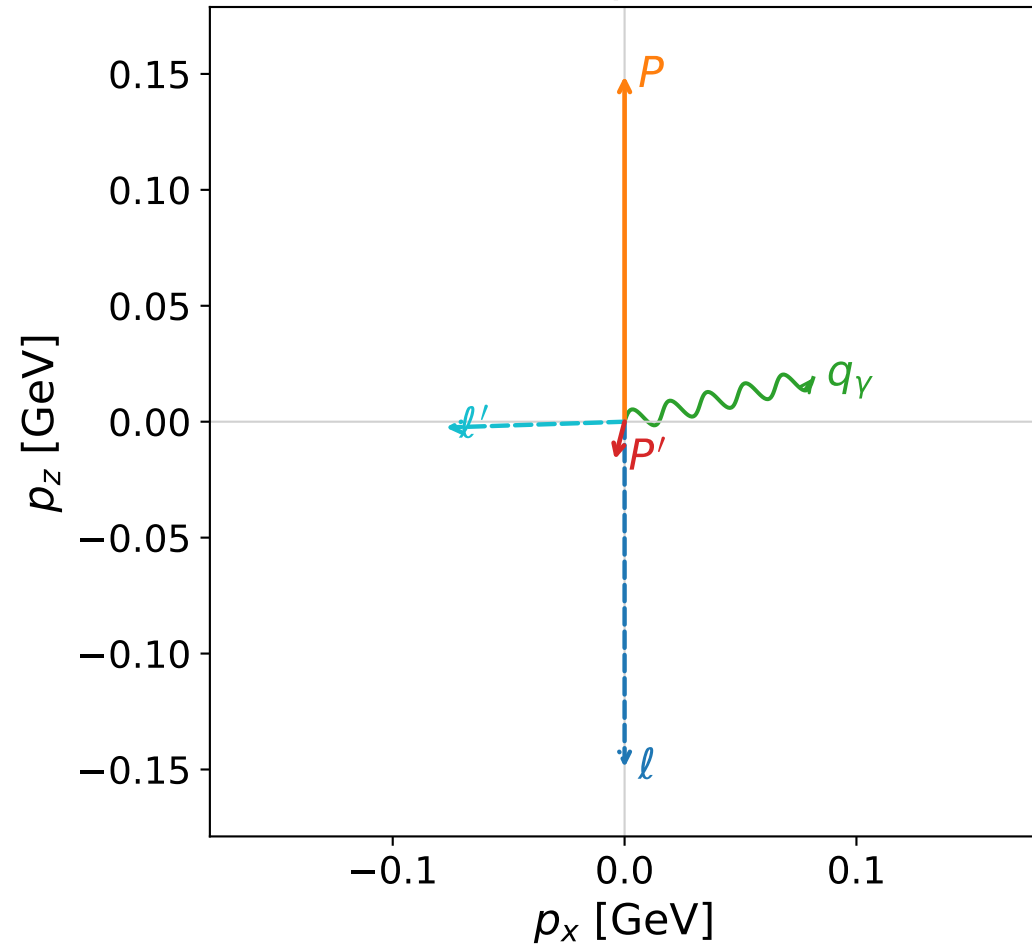
q_γ $E= 0.1186$ $p_x= 0.0748$ $p_y= -0.0661$ $p_z= -0.0641$

Leading normalized final-state amplitudes
(N=3, retained=96.1%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

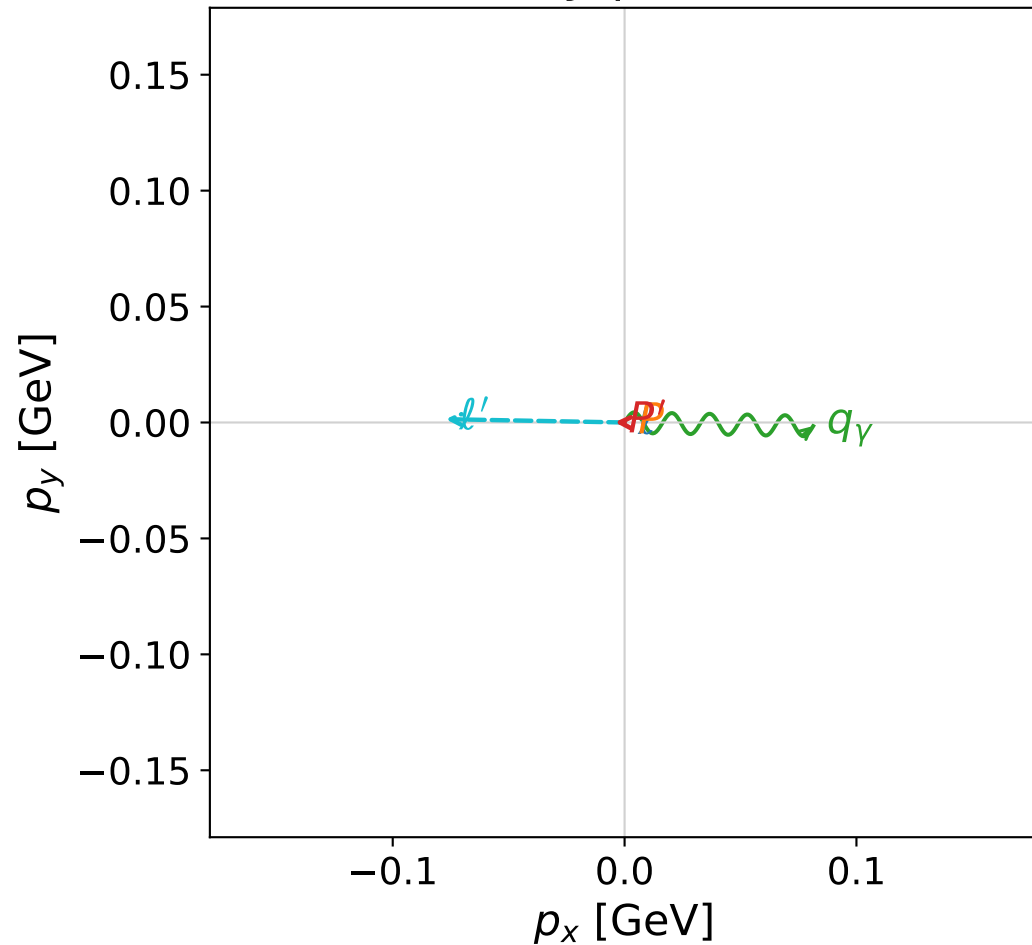


dw_mixing_angles_polarization_05_local_minimum_559
 $D_W=0.00332951$
 region: polarization_cluster_05_local_minimum_559
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0559
 lepton mixing angle: $\alpha_e=2.3556829$ rad
 incoming lepton: $-0.706745|+\rangle +0.707468|-\rangle$
 proton mixing angle: $\alpha_p=3.0233463$ rad
 incoming proton: $-0.993017|+\rangle +0.117971|-\rangle$

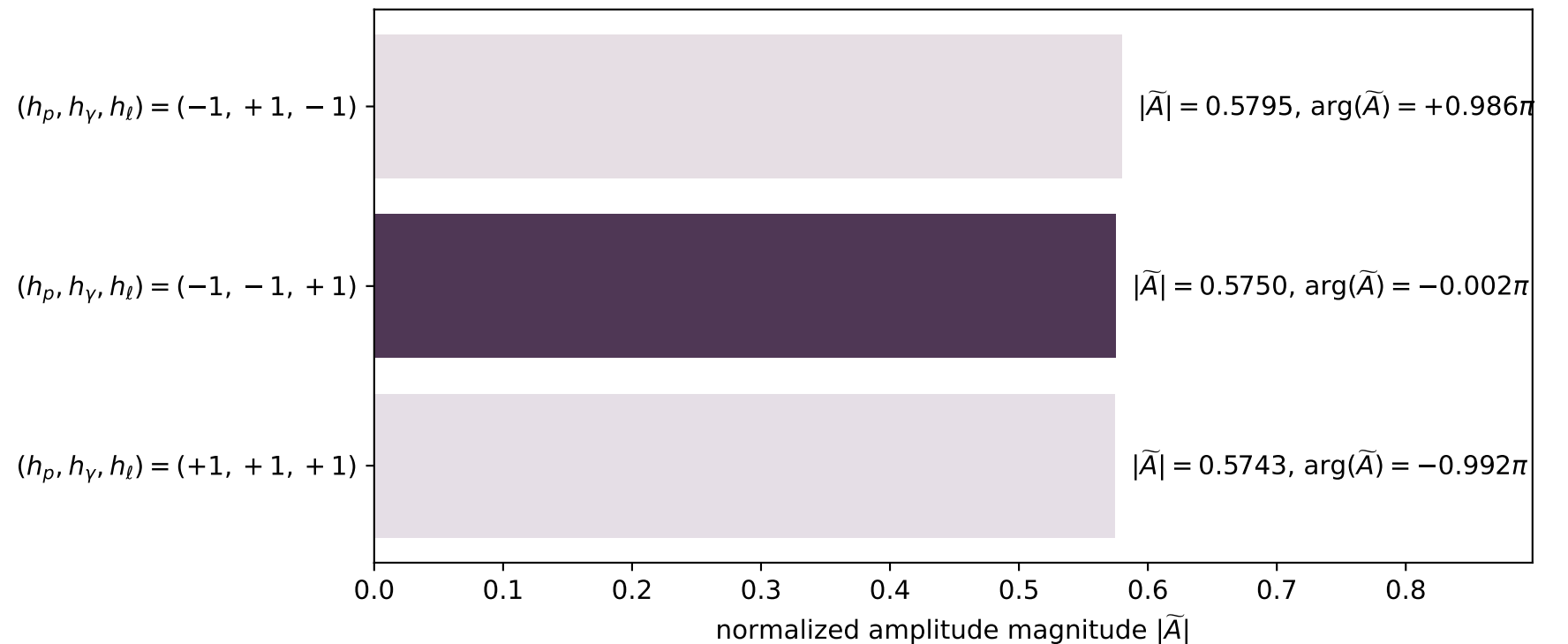
$s=1.20742$, $\sqrt{s}=1.09883$, $|\vec{P}|=0.149056$, $|\vec{P}'|=0.01683$
 $\theta_{p'}=2.88847$, $\theta_Y=1.34375$, $\phi_{p'}=3.11512$, $\phi_Y=6.26422$
 $E_Y=0.083495$, $\theta_{l'}=1.60321$, $\phi_{l'}=3.12304$
 $Q^2=0.0222623$, $x_B=0.123527$, $t=-0.0272232$, $W^2=1.0378$, $y=0.550173$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.1491$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1491$
 P $E= 0.9498$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1491$
 l' $E= 0.0772$ $p_x= -0.0771$ $p_y= 0.0014$ $p_z= -0.0025$
 P' $E= 0.9382$ $p_x= -0.0042$ $p_y= 0.0001$ $p_z= -0.0163$
 q_Y $E= 0.0835$ $p_x= 0.0813$ $p_y= -0.0015$ $p_z= 0.0188$

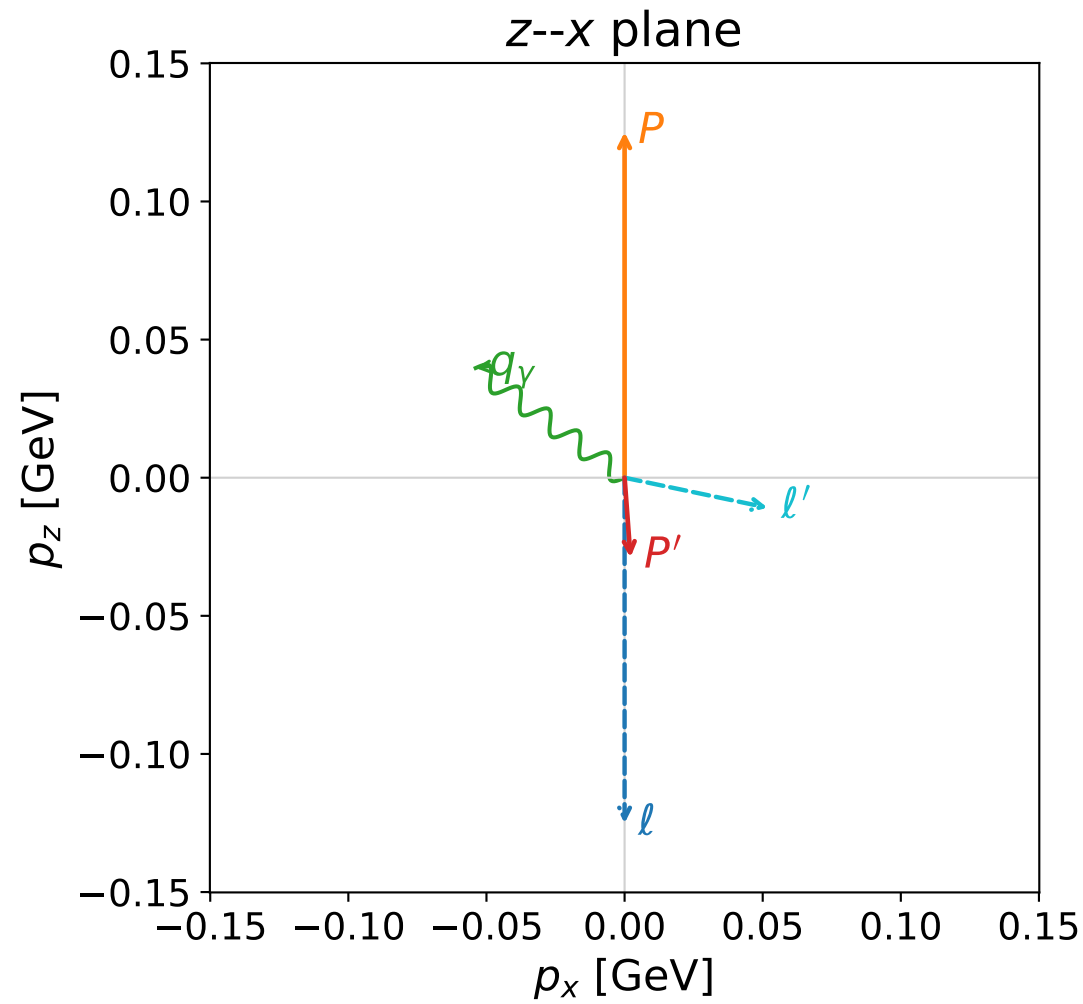
x--y plane



Leading normalized final-state amplitudes
 (N=3, retained=99.6%)



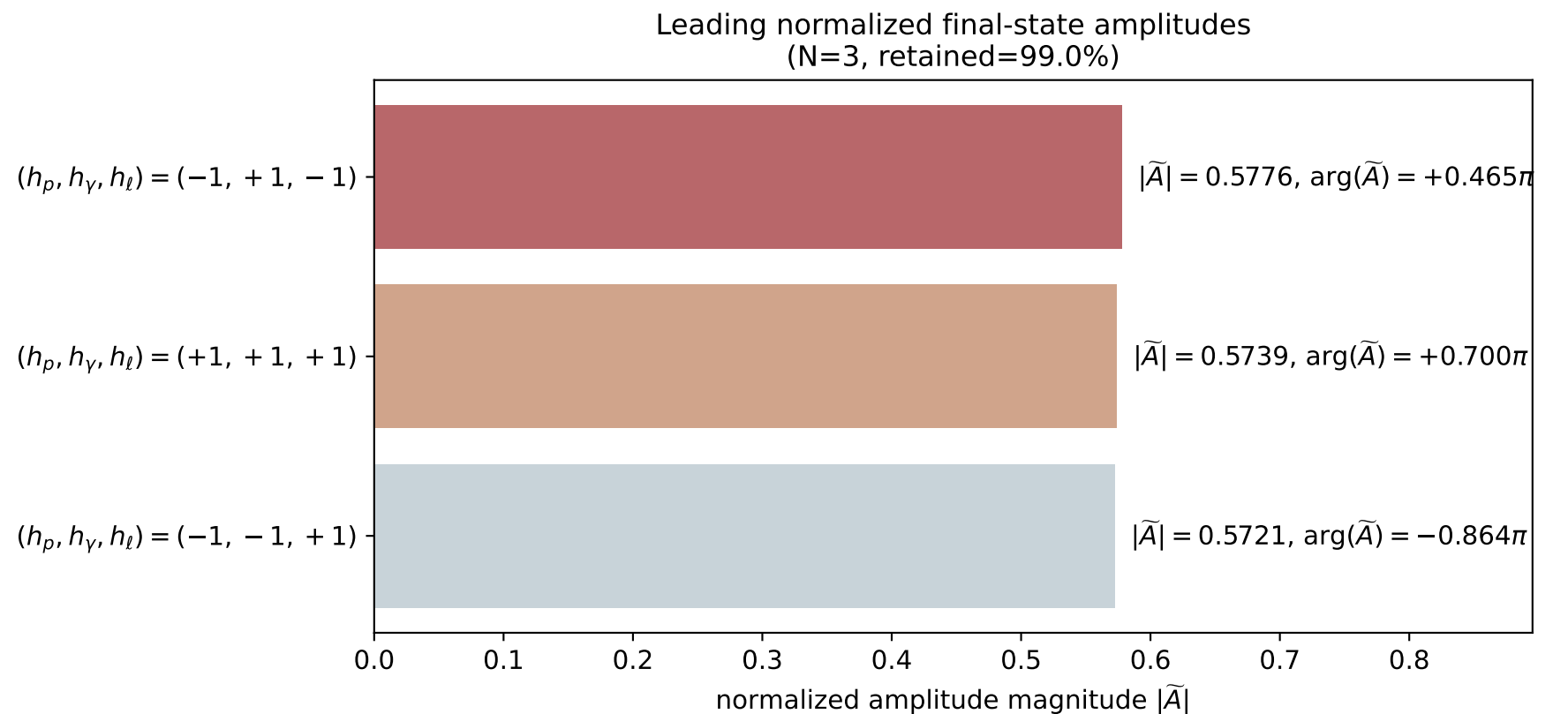
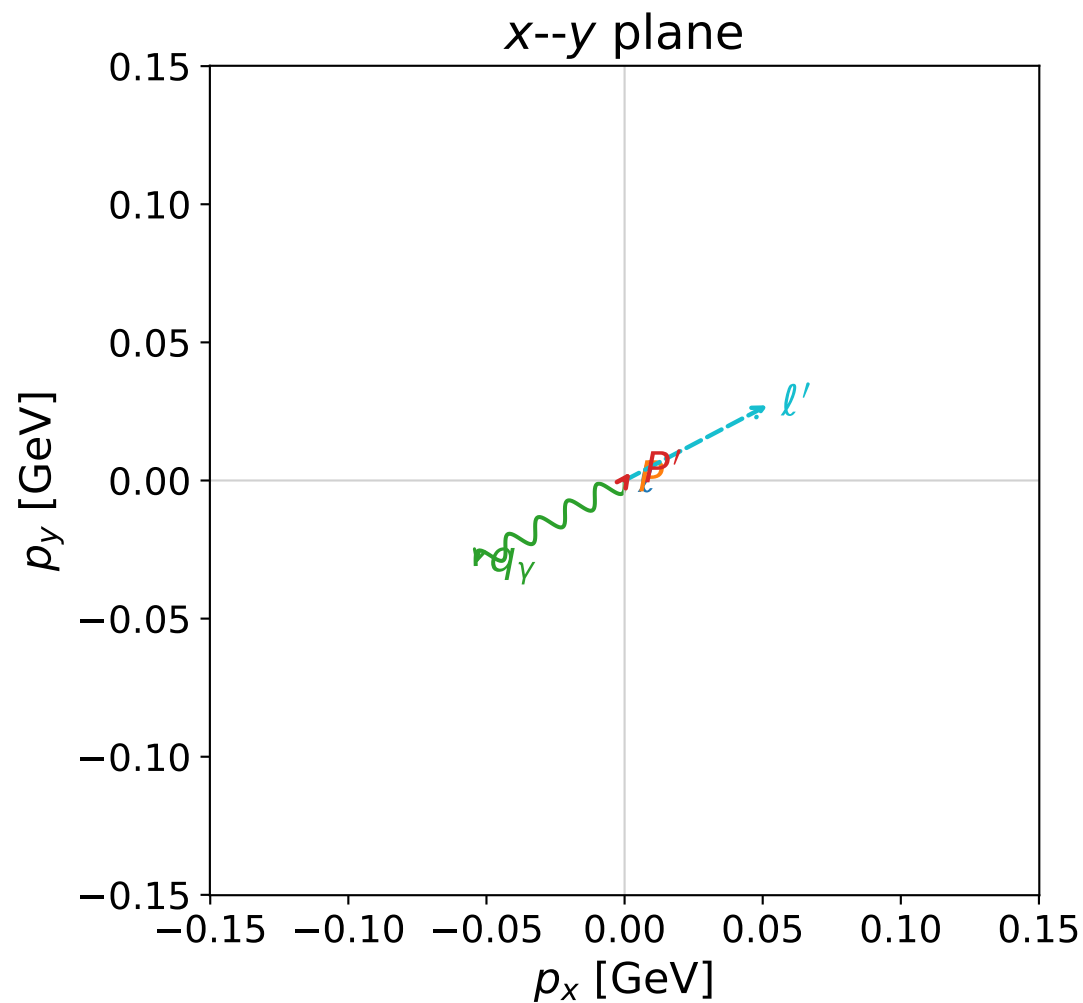
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)



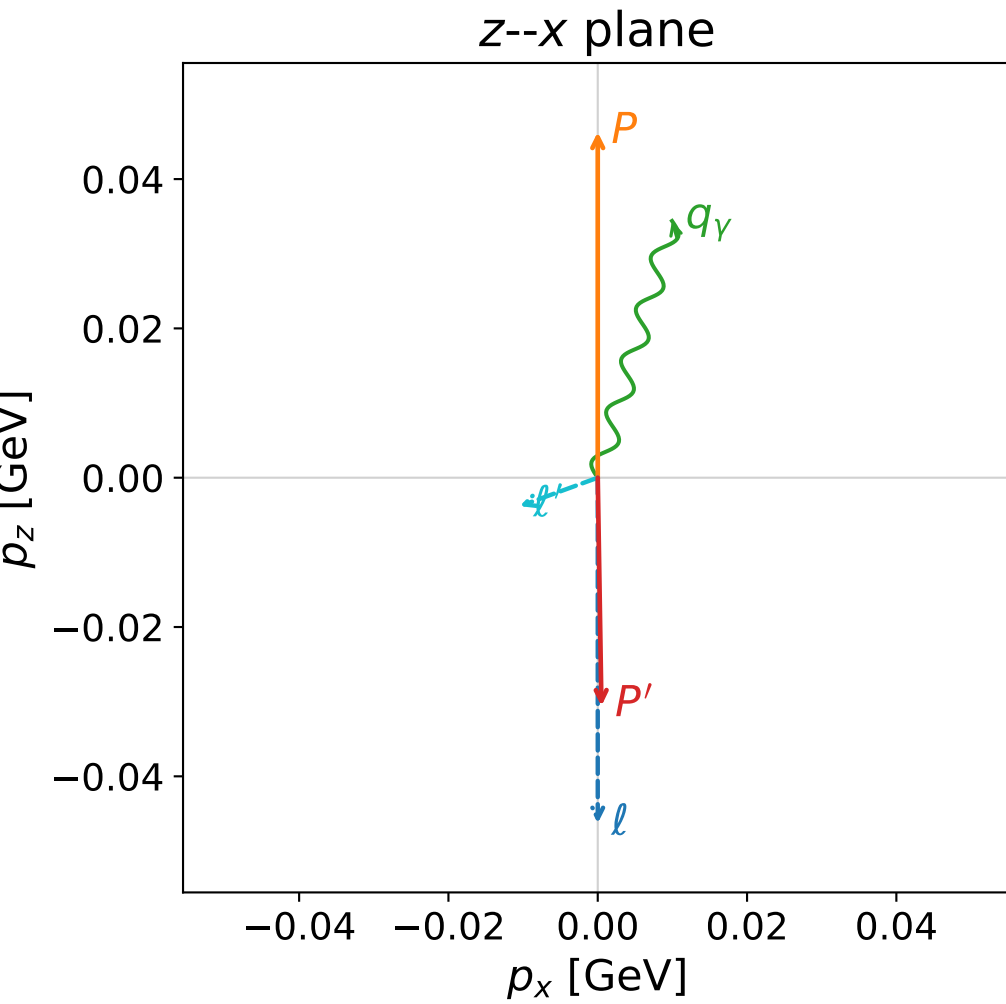
```
dw_mixing_angles_polarization_05_local_minimum_569
D_W=0.00349548
region: polarization_cluster_05_local_minimum_569
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0569
lepton mixing angle: alpha_e=2.3431432 rad
incoming lepton: -0.697818|+> +0.716275|->
proton mixing angle: alpha_p=0.13845223 rad
incoming proton: 0.990431|+> +0.13801|->
```

```
s=1.14793, sqrt(s)=1.07141, |P|=0.125106, |P'|=0.0290871
theta_p=3.01447, theta_gamma=0.99976, phi_p=0.954904, phi_gamma=3.65282
E_gamma=0.0734631, theta_l=1.75427, phi_l=0.484158
Q^2=0.012171, x_B=0.0796754, t=-0.023655, W^2=1.02043, y=0.569815
```

```
four-momenta [E, p_x, p_y, p_z] GeV:
l      E= 0.1251 p_x= 0.0000 p_y= 0.0000 p_z= -0.1251
P      E= 0.9463 p_x= 0.0000 p_y= 0.0000 p_z= 0.1251
l'     E= 0.0595 p_x= 0.0518 p_y= 0.0272 p_z= -0.0109
P'     E= 0.9385 p_x= 0.0021 p_y= 0.0030 p_z= -0.0289
q_gamma E= 0.0735 p_x= -0.0539 p_y= -0.0302 p_z= 0.0397
```



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)



dw_mixing_angles_polarization_05_local_minimum_606

$D_W=0.00424053$

region: polarization_cluster_05_local_minimum_606

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0606

lepton mixing angle: $\alpha_e=2.3490452$ rad

incoming lepton: $-0.702033|+> +0.712144|->$

proton mixing angle: $\alpha_p=3.0187472$ rad

incoming proton: $-0.992464|+> +0.122537|->$

$s=0.971085$, $\sqrt{s}=0.985437$, $|\vec{P}|=0.0462921$, $|\vec{P}'|=0.0305854$

$\theta_{p'}=3.12485$, $\theta_\gamma=0.281828$, $\phi_{p'}=0.254548$, $\phi_\gamma=6.24302$

$E_\gamma=0.0358045$, $\theta_{\ell'}=1.92053$, $\phi_{\ell'}=3.11566$

$Q^2=0.000677509$, $x_B=0.009682$, $t=-0.00590934$, $W^2=0.949143$, $y=0.766938$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

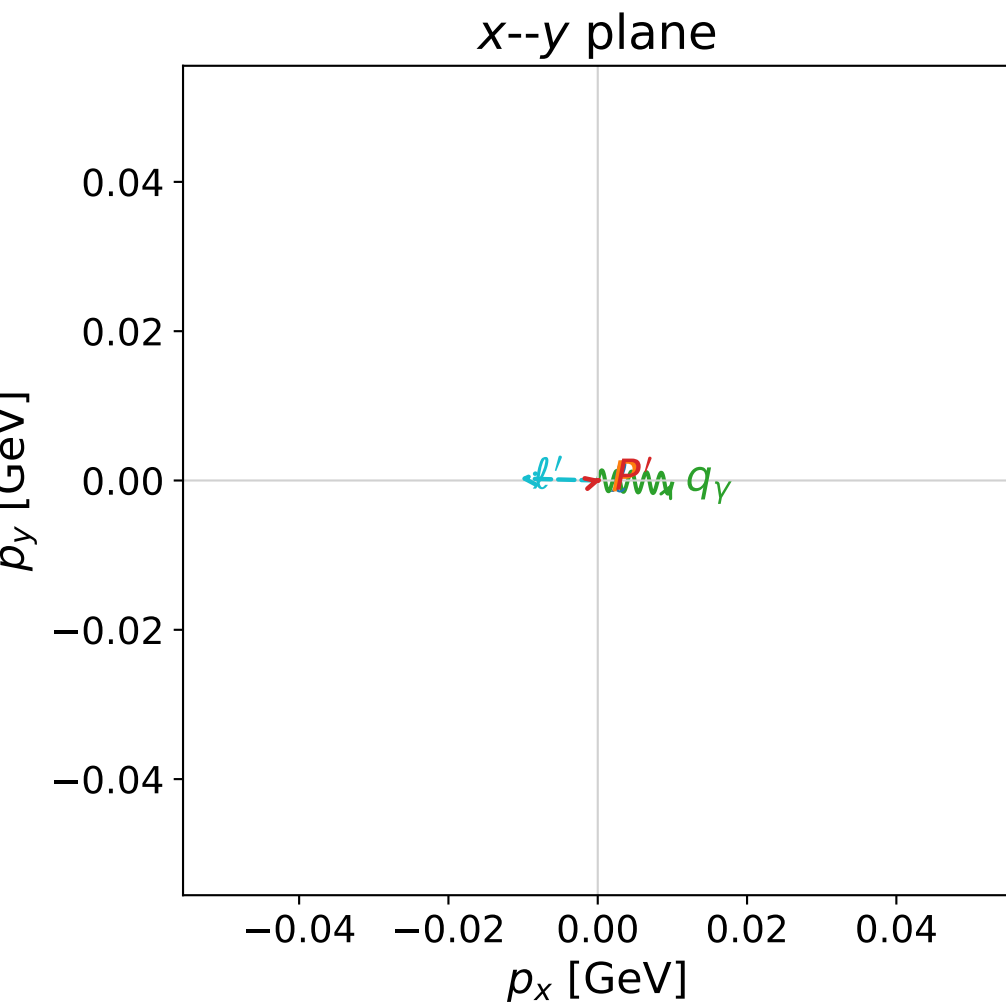
ℓ $E= 0.0463$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0463$

P $E= 0.9391$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0463$

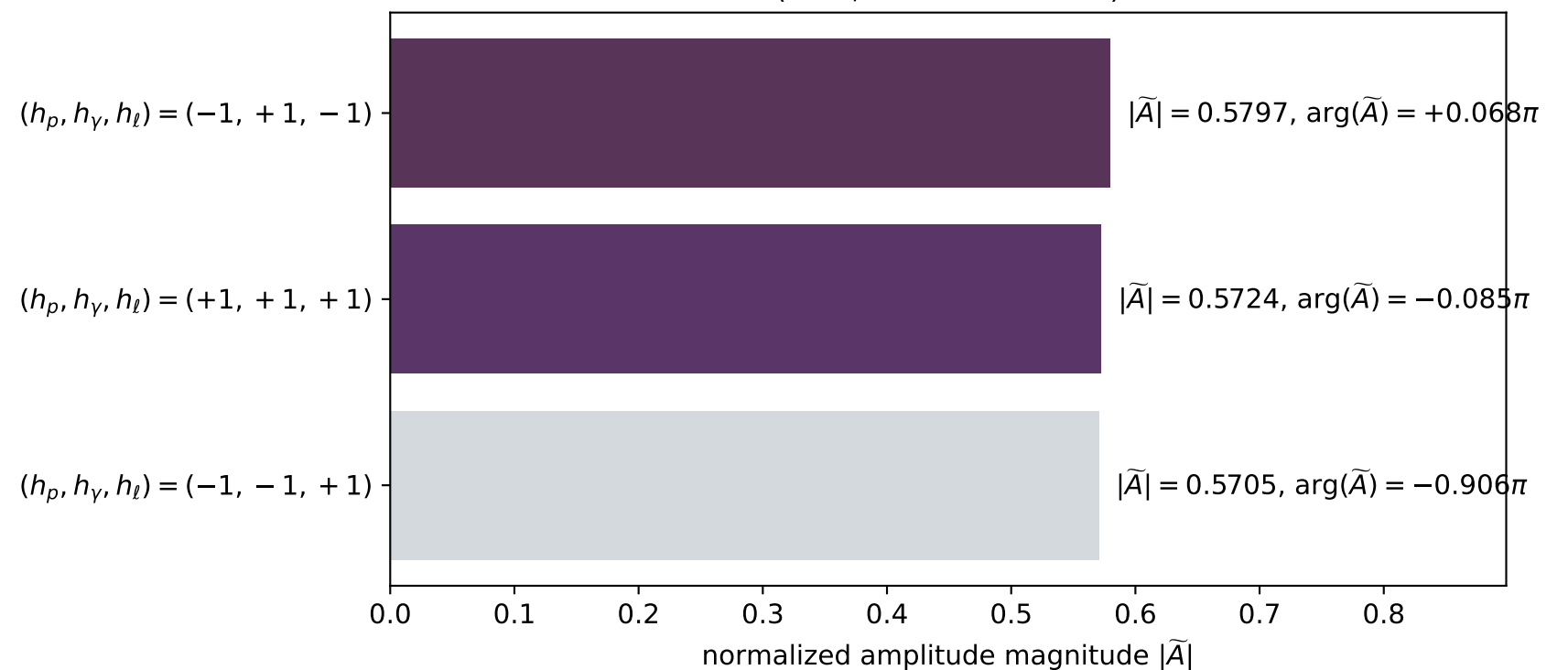
ℓ' $E= 0.0111$ $p_x= -0.0104$ $p_y= 0.0003$ $p_z= -0.0038$

P' $E= 0.9385$ $p_x= 0.0005$ $p_y= 0.0001$ $p_z= -0.0306$

q_γ $E= 0.0358$ $p_x= 0.0099$ $p_y= -0.0004$ $p_z= 0.0344$

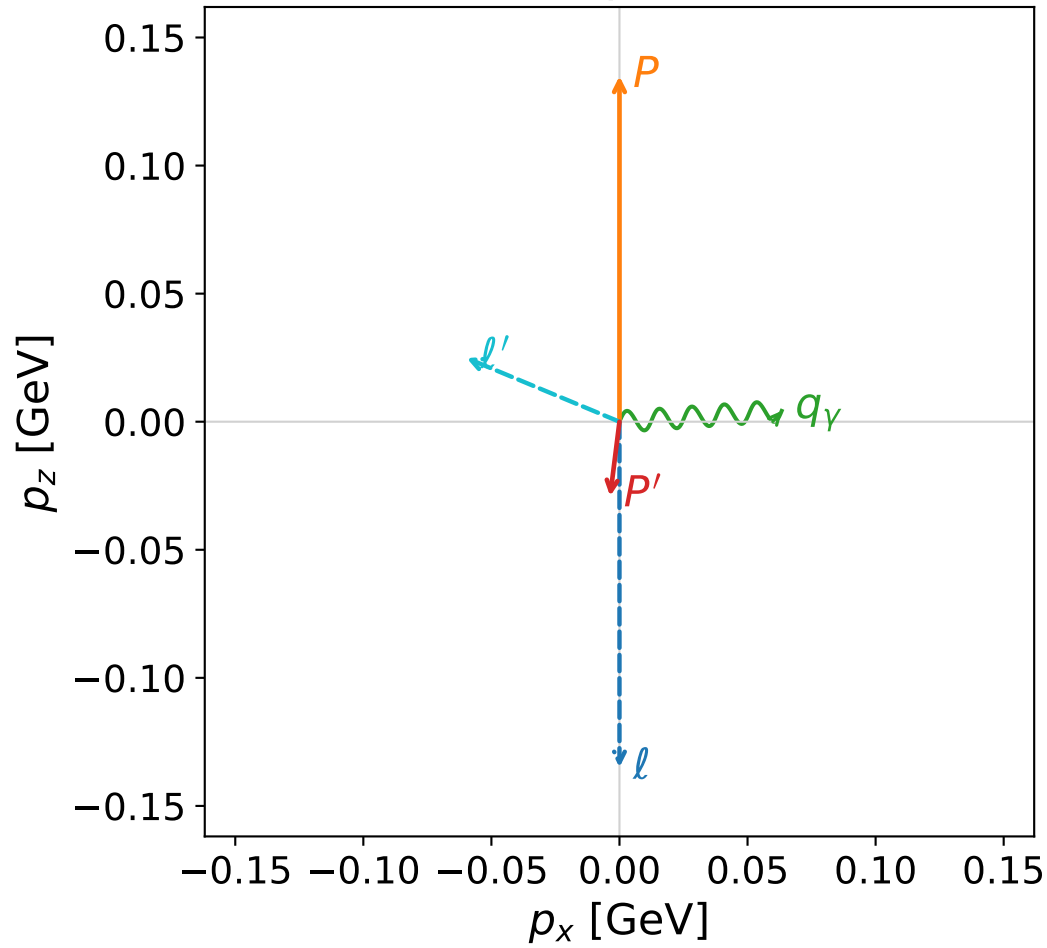


Leading normalized final-state amplitudes
(N=3, retained=98.9%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



dw_mixing_angles_polarization_05_local_minimum_728

$D_W=0.00832553$

region: polarization_cluster_05_local_minimum_728

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0728

lepton mixing angle: $\alpha_e=2.3656503$ rad

incoming lepton: $-0.713761|+\rangle + 0.700389|-\rangle$

proton mixing angle: $\alpha_p=3.0654637$ rad

incoming proton: $-0.997104|+\rangle + 0.0760554|-\rangle$

$s=1.17197$, $\sqrt{s}=1.08258$, $|\vec{P}|=0.134923$, $|\vec{P}'|=0.0293$

$\theta_{p'}=3.01563$, $\theta_\gamma=1.51191$, $\phi_{p'}=3.29583$, $\phi_\gamma=0.483699$

$E_\gamma=0.0716602$, $\theta_{\ell'}=1.22072$, $\phi_{\ell'}=3.64279$

$Q^2=0.0262585$, $x_B=0.162589$, $t=-0.0268219$, $W^2=1.01509$, $y=0.552843$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ $E= 0.1349$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1349$

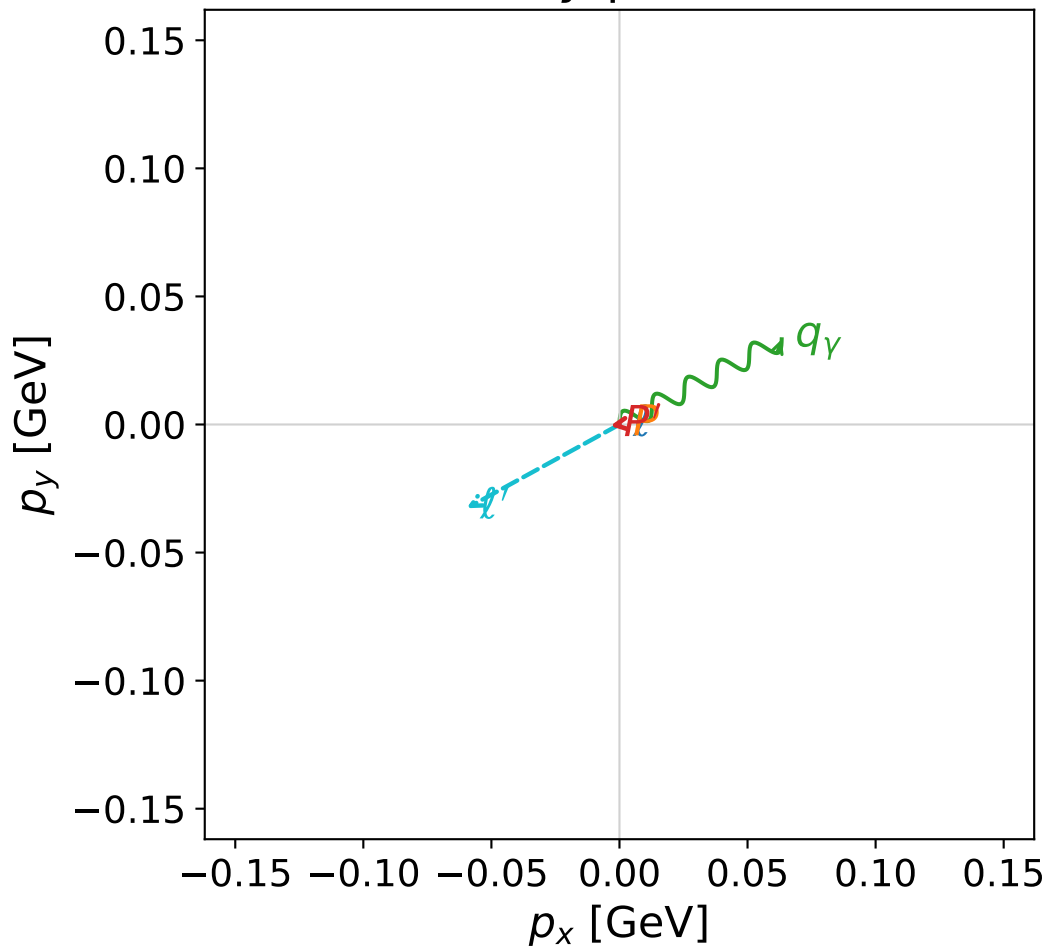
P $E= 0.9477$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1349$

ℓ' $E= 0.0725$ $p_x= -0.0597$ $p_y= -0.0327$ $p_z= 0.0249$

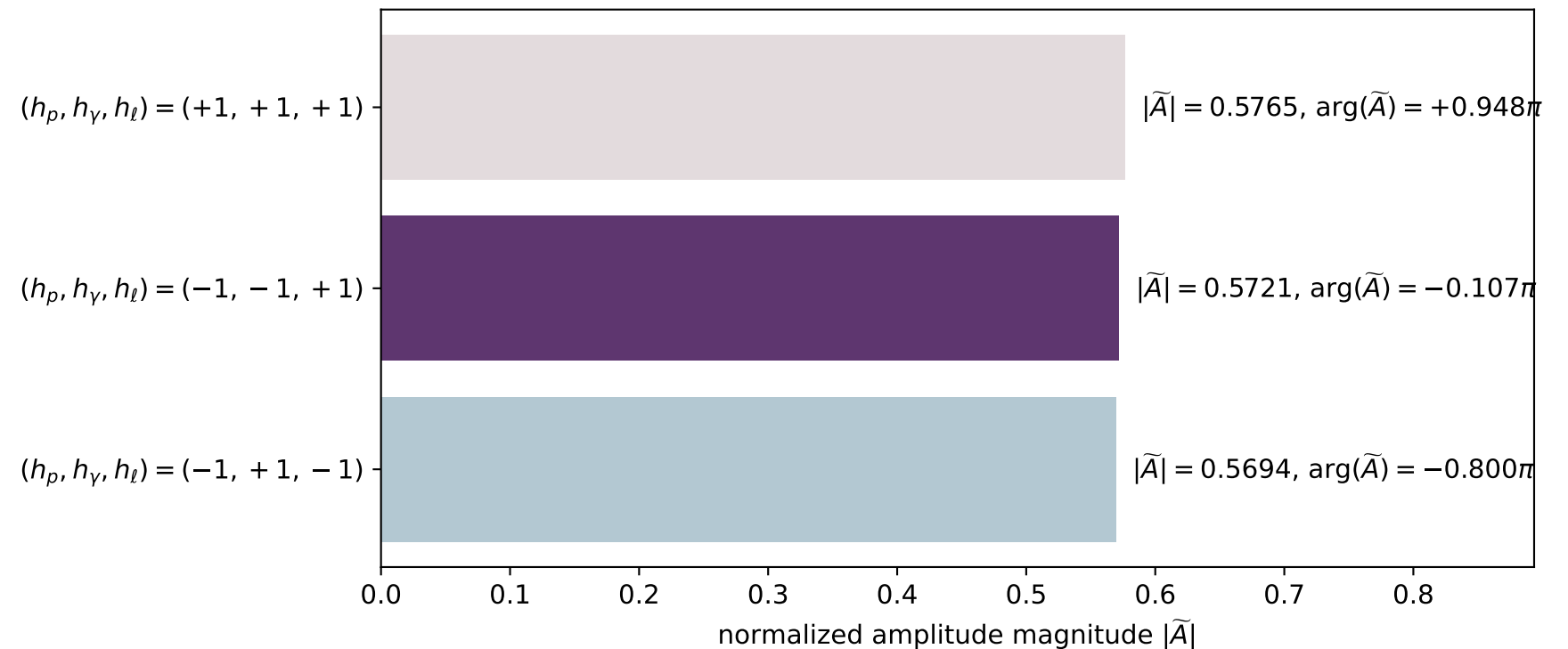
P' $E= 0.9385$ $p_x= -0.0036$ $p_y= -0.0006$ $p_z= -0.0291$

q_γ $E= 0.0717$ $p_x= 0.0633$ $p_y= 0.0333$ $p_z= 0.0042$

x--y plane

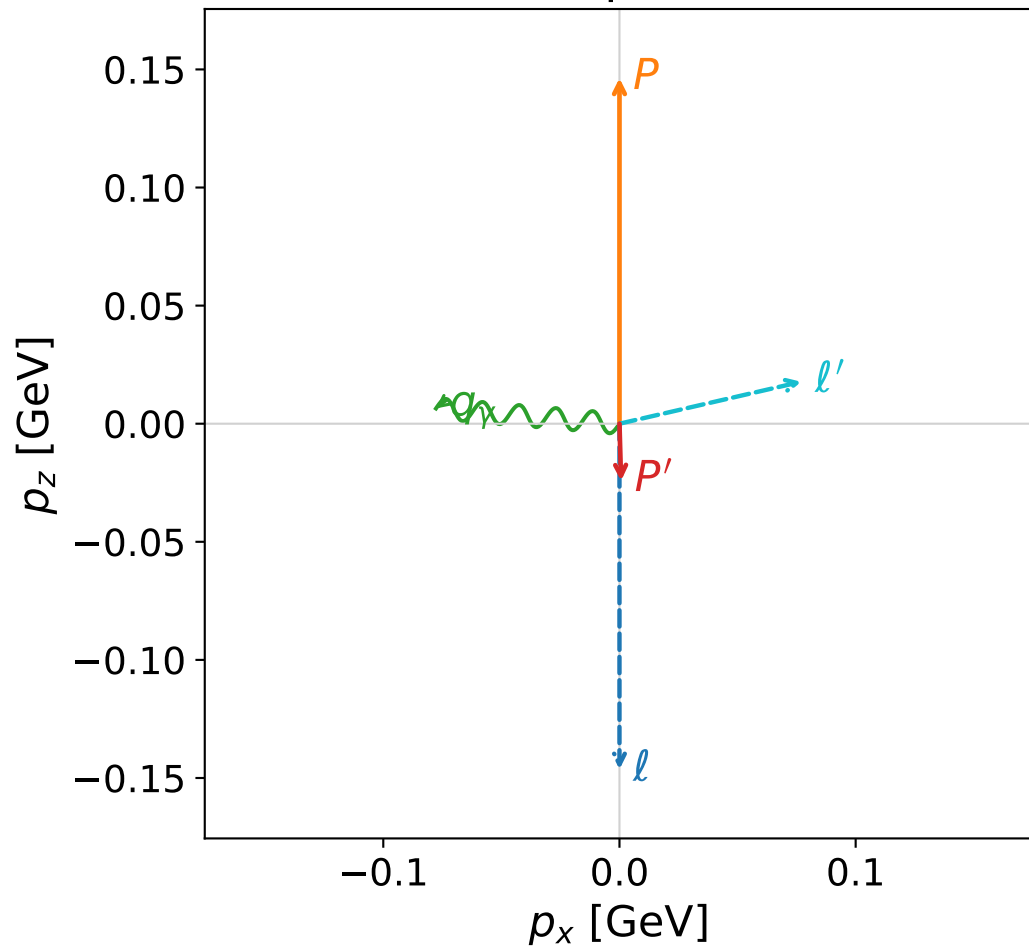


Leading normalized final-state amplitudes
(N=3, retained=98.4%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



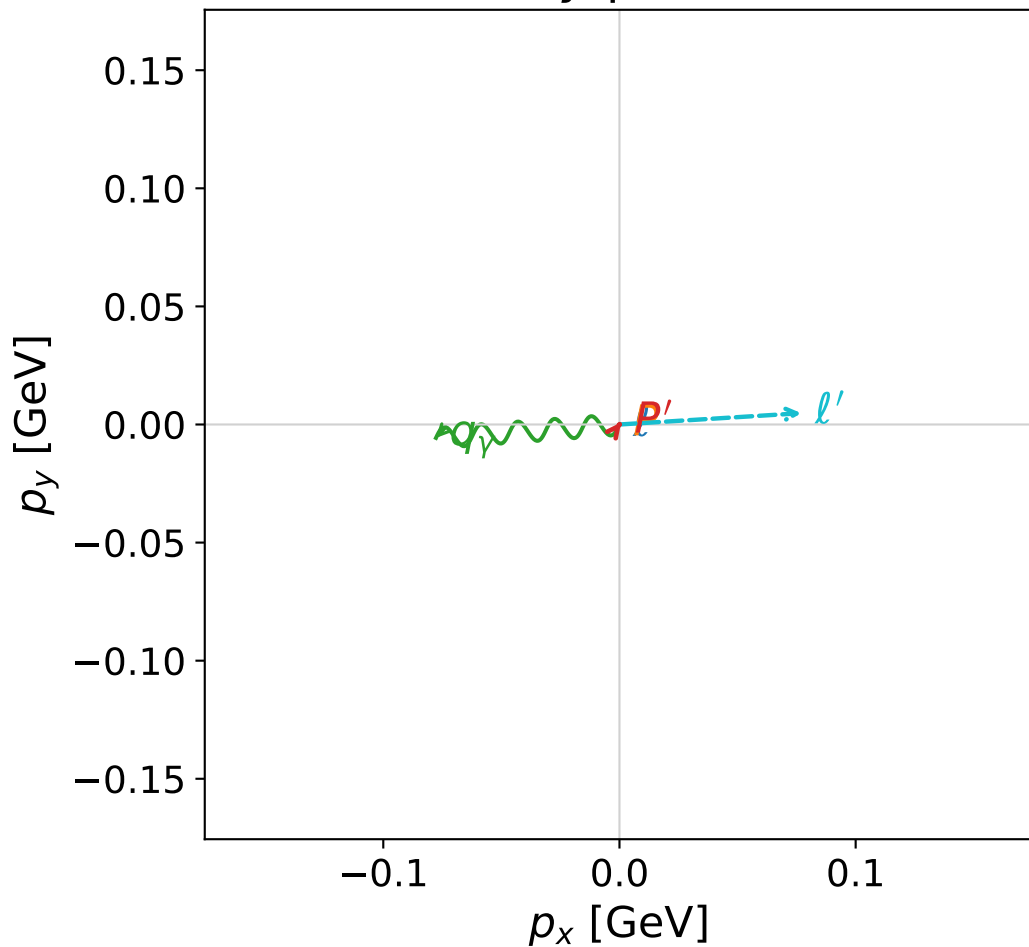
```
dw_mixing_angles_polarization_05_local_minimum_731
D_W=0.00839324
region: polarization_cluster_05_local_minimum_731
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0731
lepton mixing angle: alpha_e=2.3650174 rad
incoming lepton: -0.713318|+> +0.700841|->
proton mixing angle: alpha_p=0.18228078 rad
incoming proton: 0.983433|+> +0.181273|->
```

$s=1.20048$, $\sqrt{s}=1.09567$, $|\vec{P}|=0.146321$, $|\vec{P}'|=0.024235$
 $\theta_{p'}=3.0953$, $\theta_\gamma=1.48825$, $\phi_{p'}=0.920045$, $\phi_\gamma=3.21463$
 $E_\gamma=0.0781683$, $\theta_{\ell'}=1.34453$, $\phi_{\ell'}=0.0621503$
 $Q^2=0.0283707$, $x_B=0.161665$, $t=-0.0289599$, $W^2=1.02696$, $y=0.547318$

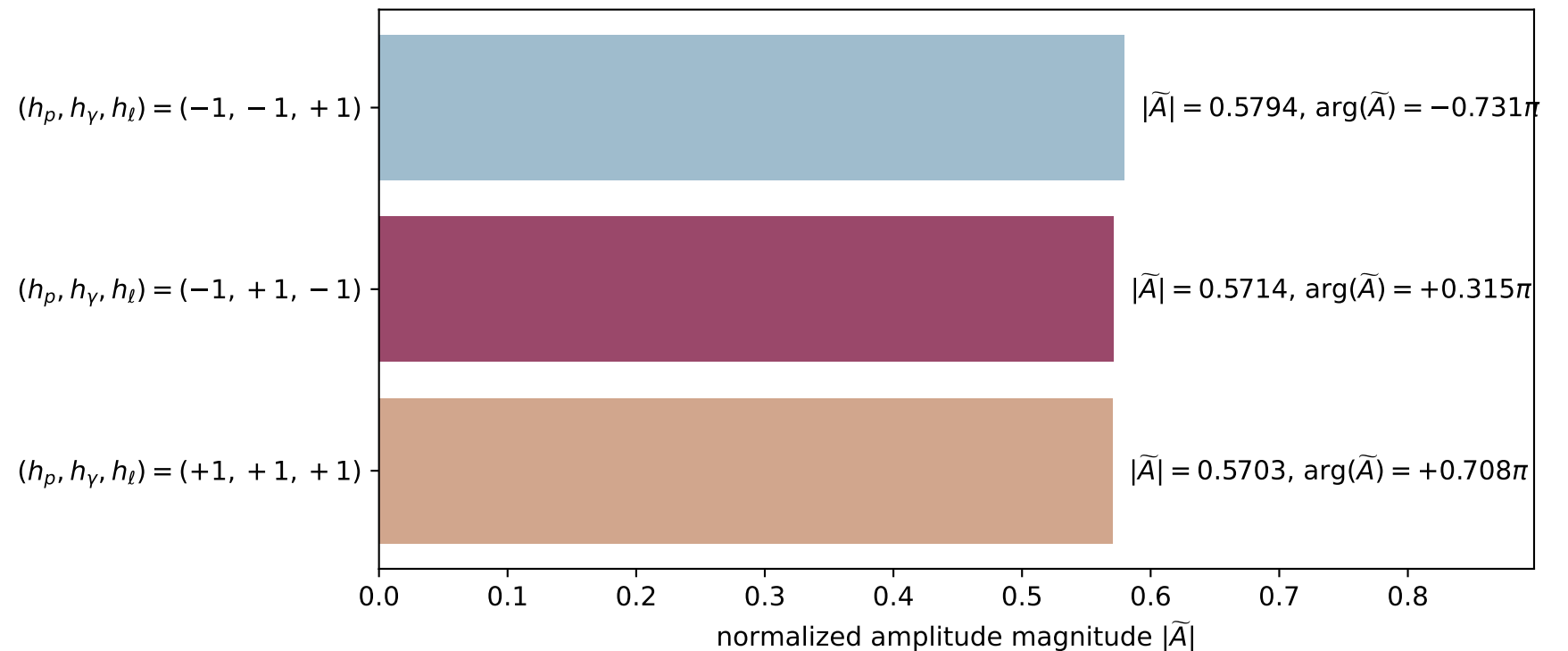
four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ	$E=$	0.1463	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	-0.1463
P	$E=$	0.9493	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	0.1463
ℓ'	$E=$	0.0792	$p_x=$	0.0770	$p_y=$	0.0048	$p_z=$	0.0178
P'	$E=$	0.9383	$p_x=$	0.0007	$p_y=$	0.0009	$p_z=$	-0.0242
q_γ	$E=$	0.0782	$p_x=$	-0.0777	$p_y=$	-0.0057	$p_z=$	0.0064

x--y plane

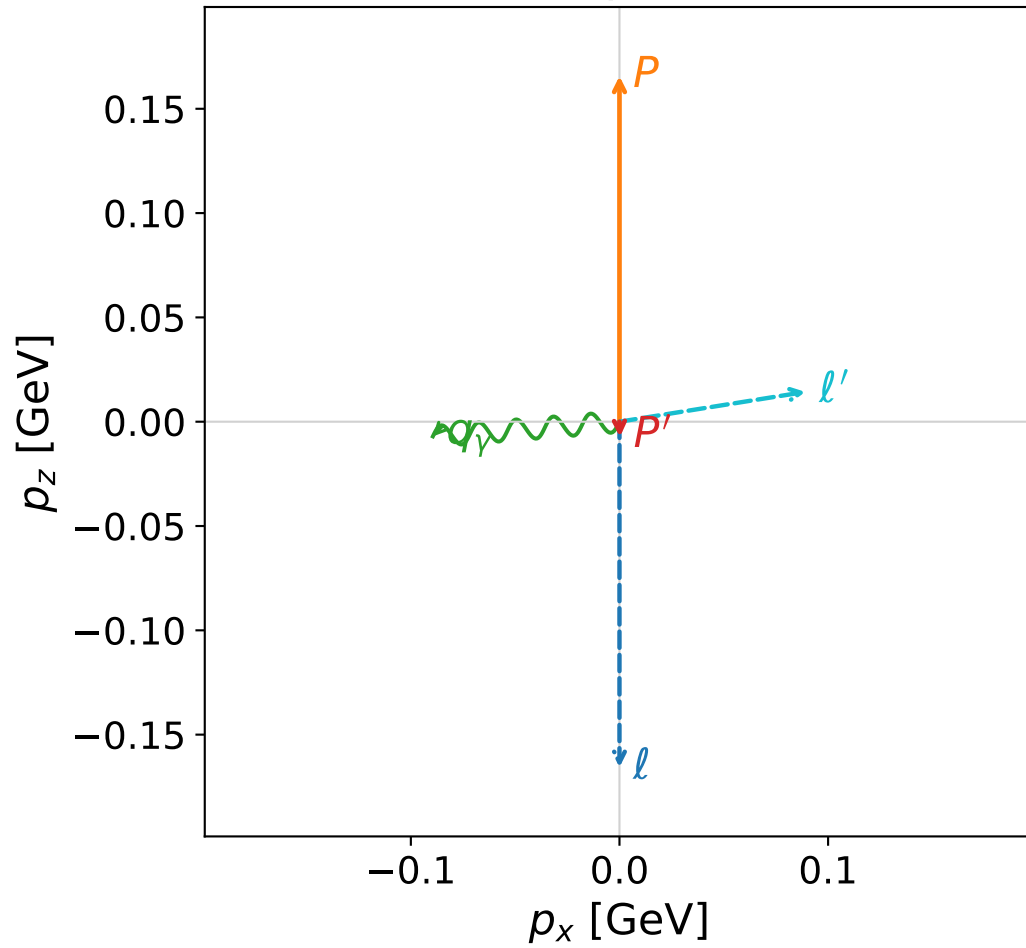


Leading normalized final-state amplitudes
(N=3, retained=98.8%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

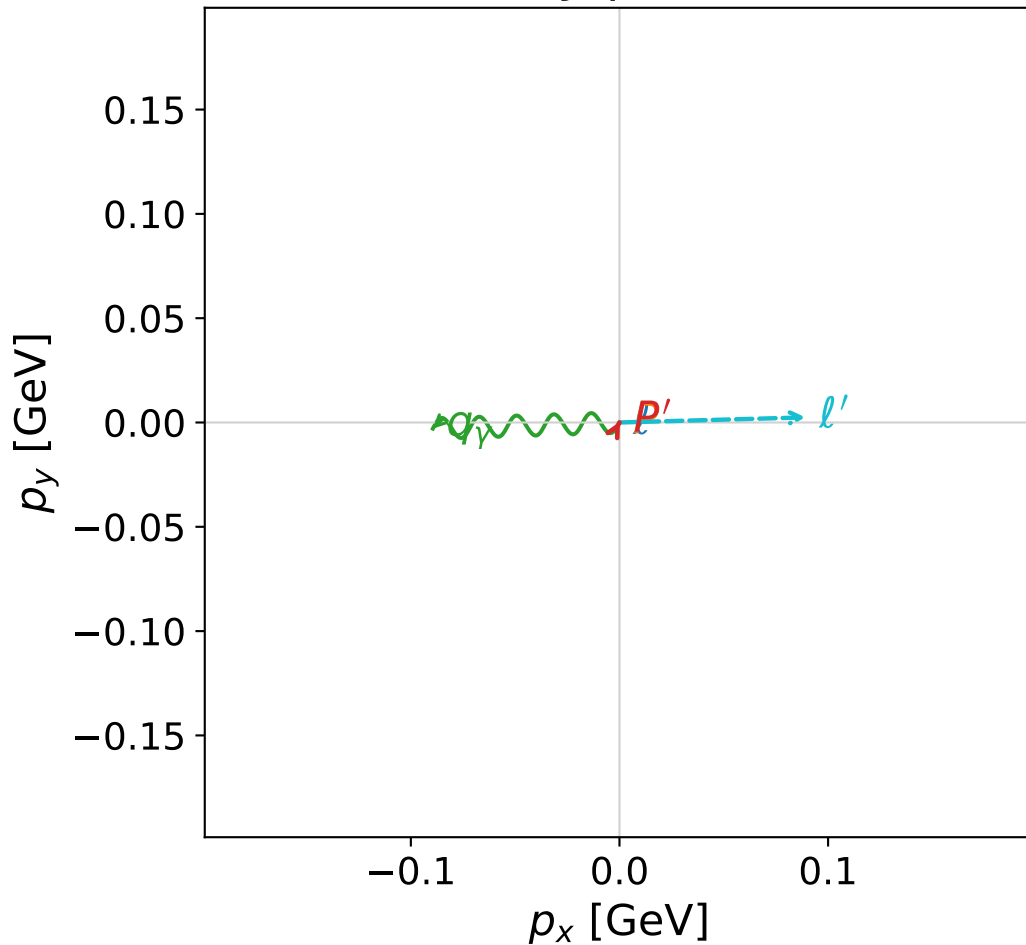


dw_mixing_angles_polarization_05_local_minimum_748
 $D_W=0.00890618$
 region: polarization_cluster_05_local_minimum_748
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0748
 lepton mixing angle: $\alpha_e=2.358533$ rad
 incoming lepton: $-0.708758|+\rangle +0.705451|-\rangle$
 proton mixing angle: $\alpha_p=0.19379091$ rad
 incoming proton: $0.981281|+\rangle +0.19258|-\rangle$

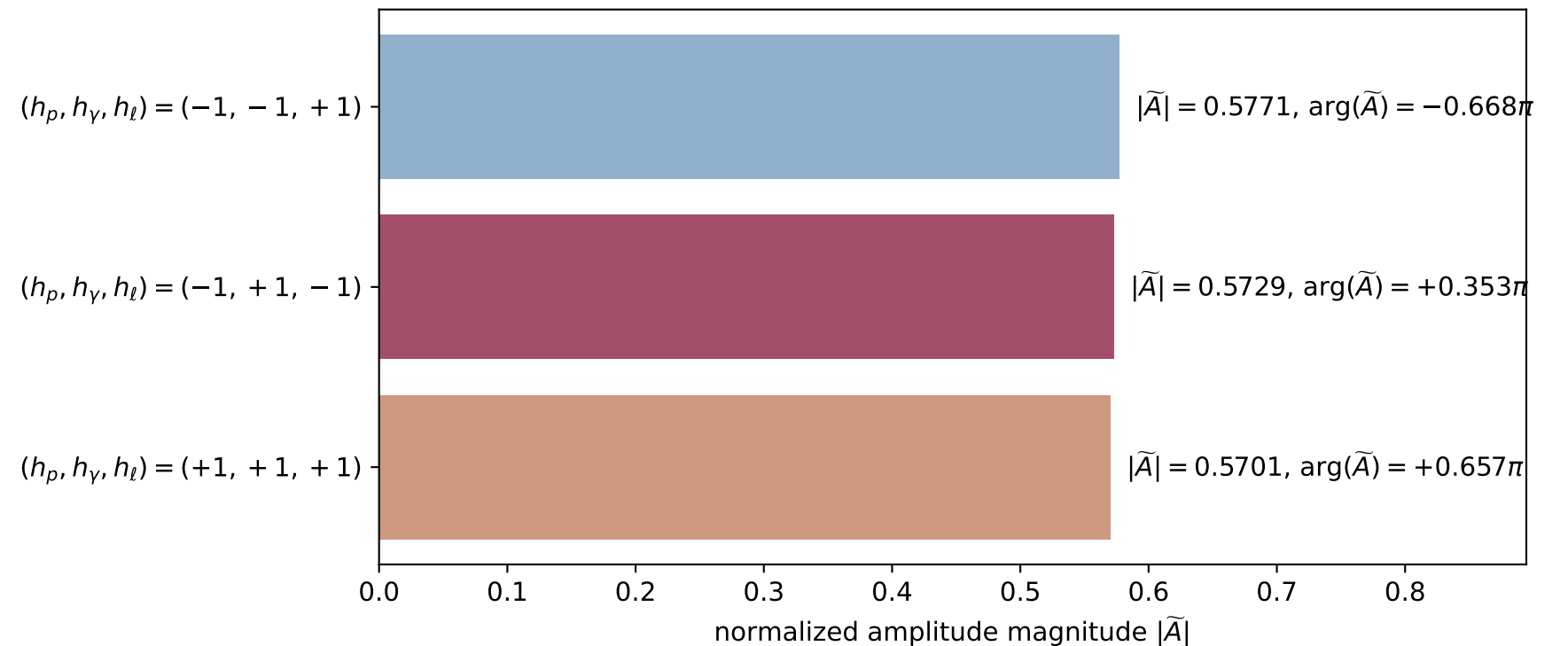
$s=1.25031$, $\sqrt{s}=1.11817$, $|\vec{P}|=0.165657$, $|\vec{P}'|=0.00726959$
 $\theta_{p'}=3.06736$, $\theta_\gamma=1.64926$, $\phi_{p'}=1.07968$, $\phi_\gamma=3.17433$
 $E_\gamma=0.0897761$, $\theta_{\ell'}=1.41203$, $\phi_{\ell'}=0.0275043$
 $Q^2=0.0346737$, $x_B=0.170769$, $t=-0.0296871$, $W^2=1.04822$, $y=0.548077$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E=0.1657$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.1657$
 P $E=0.9525$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1657$
 ℓ' $E=0.0904$ $p_x=0.0892$ $p_y=0.0025$ $p_z=0.0143$
 P' $E=0.9380$ $p_x=0.0003$ $p_y=0.0005$ $p_z=-0.0072$
 q_γ $E=0.0898$ $p_x=-0.0895$ $p_y=-0.0029$ $p_z=-0.0070$

x--y plane

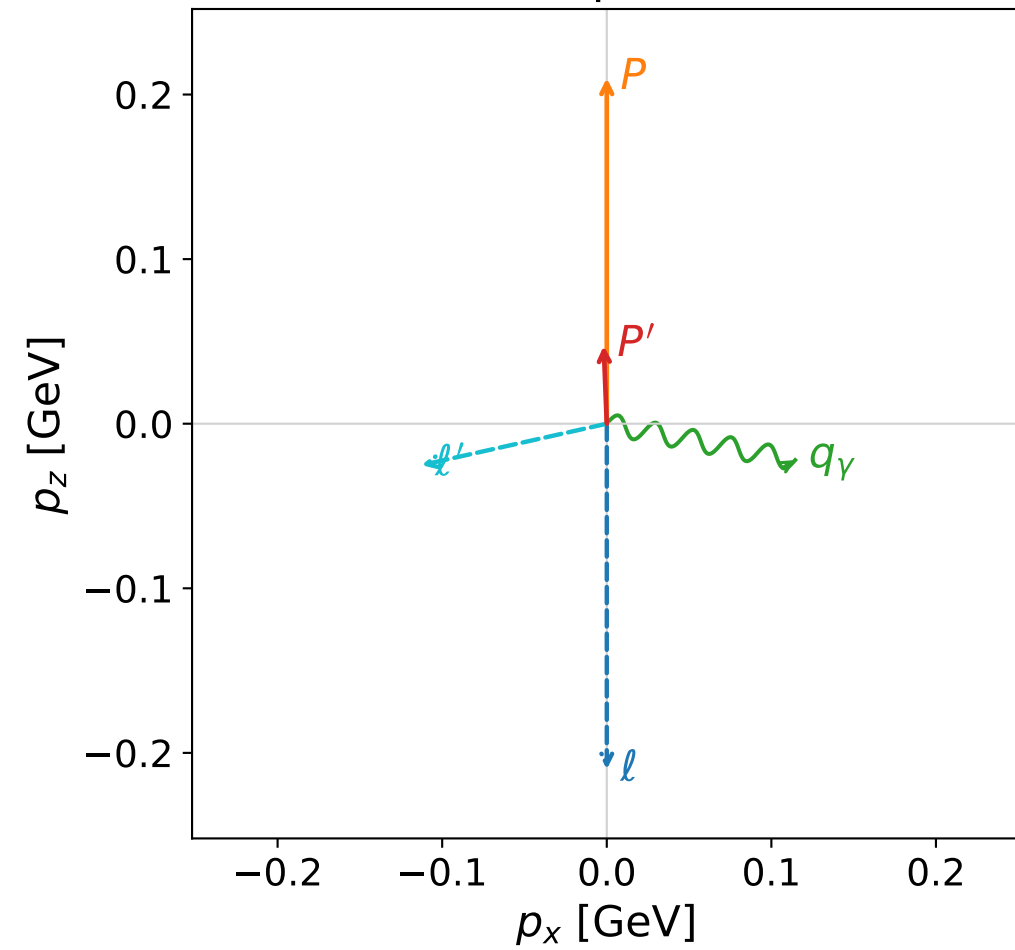


Leading normalized final-state amplitudes
(N=3, retained=98.6%)

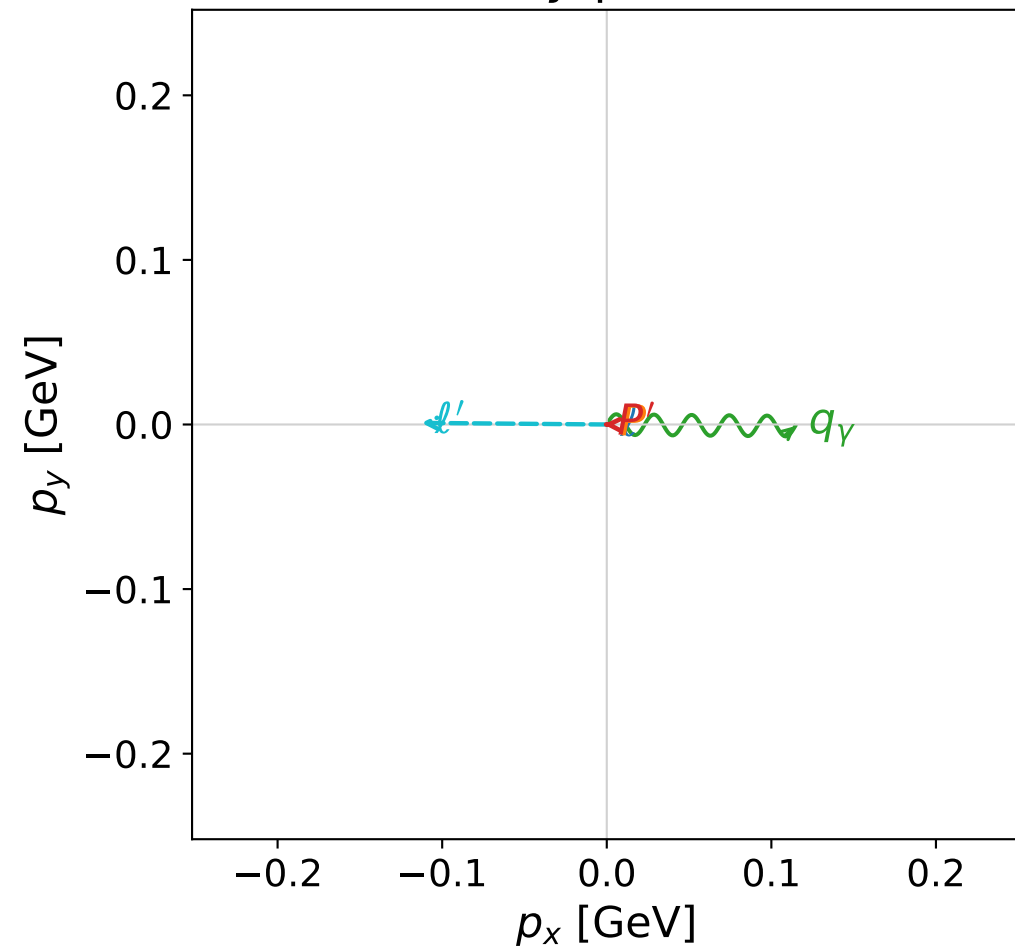


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



x--y plane

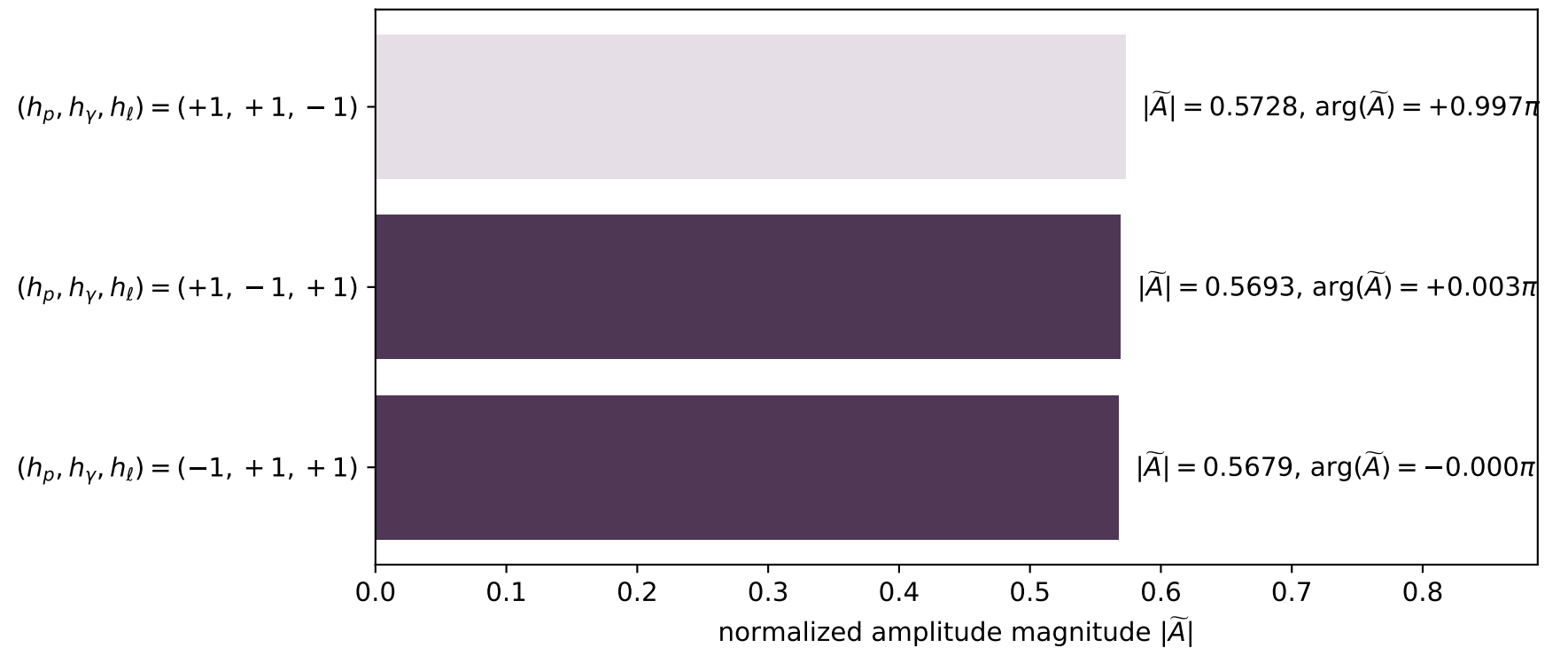


dw_mixing_angles_polarization_05_local_minimum_752
 $D_W=0.00917587$
 region: polarization_cluster_05_local_minimum_752
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0752
 lepton mixing angle: $\alpha_e=2.3432894$ rad
 incoming lepton: $-0.697923|+\rangle +0.716173|-\rangle$
 proton mixing angle: $\alpha_p=3.0217874$ rad
 incoming proton: $-0.992832|+\rangle +0.119519|-\rangle$

$s=1.37168$, $\sqrt{s}=1.17119$, $|\vec{P}|=0.209973$, $|\vec{P}'|=0.0473169$
 $\theta_{p'}=0.0392699$, $\theta_\gamma=1.76154$, $\phi_{p'}=3.12521$, $\phi_\gamma=6.27439$
 $E_\gamma=0.116593$, $\theta_{l'}=1.79072$, $\phi_{l'}=3.13292$
 $Q^2=0.0378896$, $x_B=0.14606$, $t=-0.0259872$, $W^2=1.10137$, $y=0.527436$

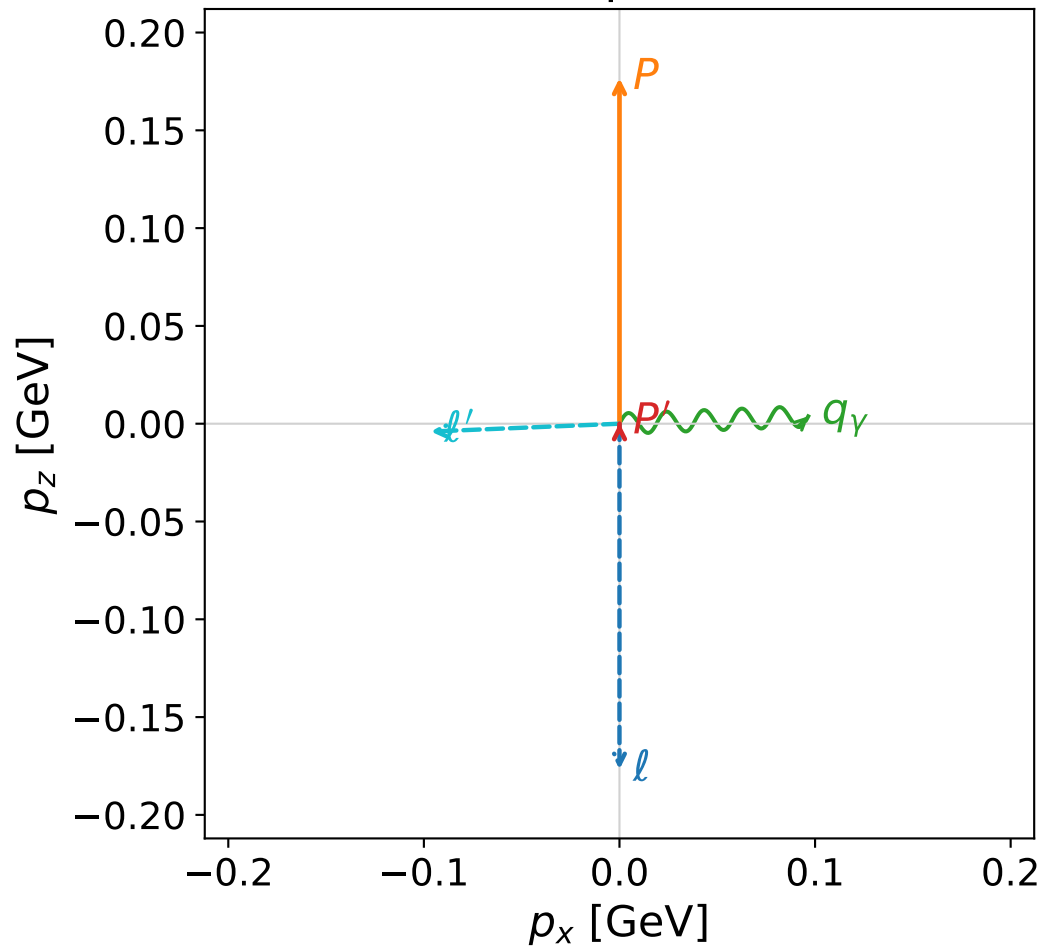
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E=0.2100$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.2100$
 P $E=0.9612$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.2100$
 l' $E=0.1154$ $p_x=-0.1126$ $p_y=0.0010$ $p_z=-0.0252$
 P' $E=0.9392$ $p_x=-0.0019$ $p_y=0.0000$ $p_z=0.0473$
 q_γ $E=0.1166$ $p_x=0.1145$ $p_y=-0.0010$ $p_z=-0.0221$

Leading normalized final-state amplitudes
 (N=3, retained=97.5%)

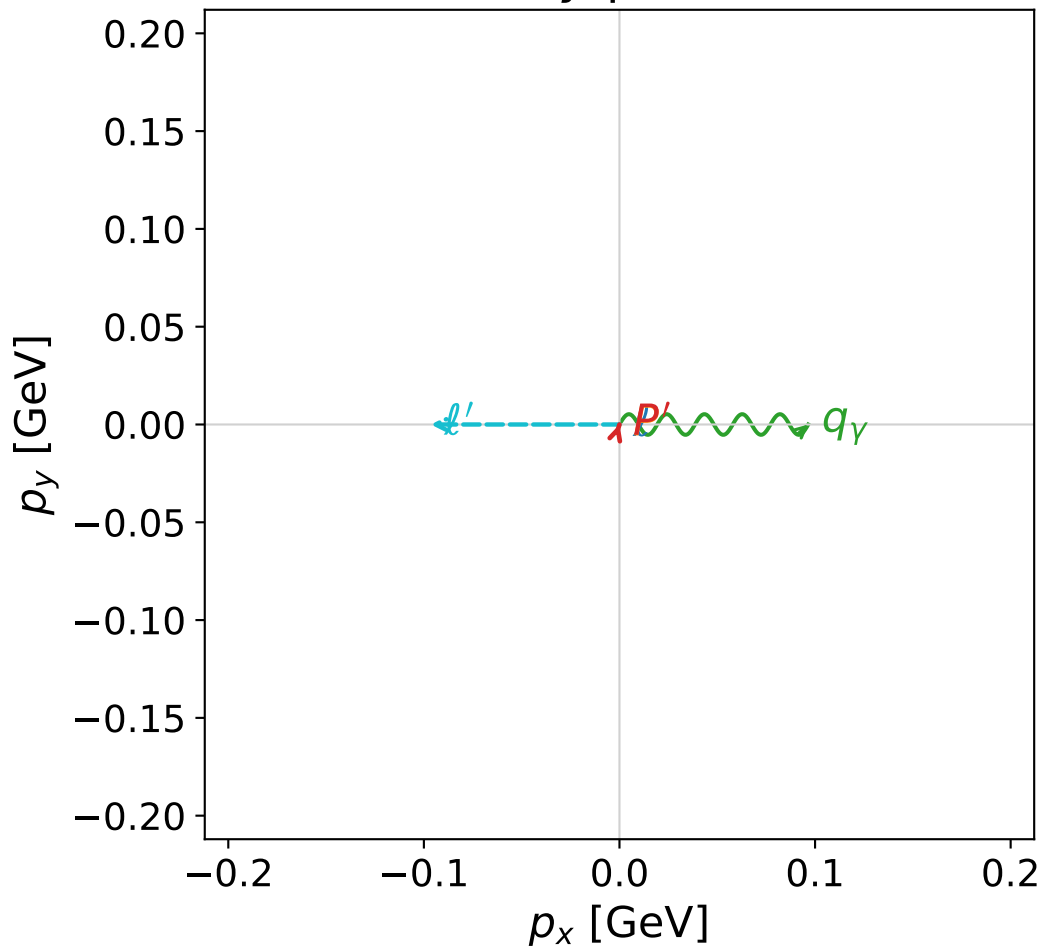


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



x--y plane



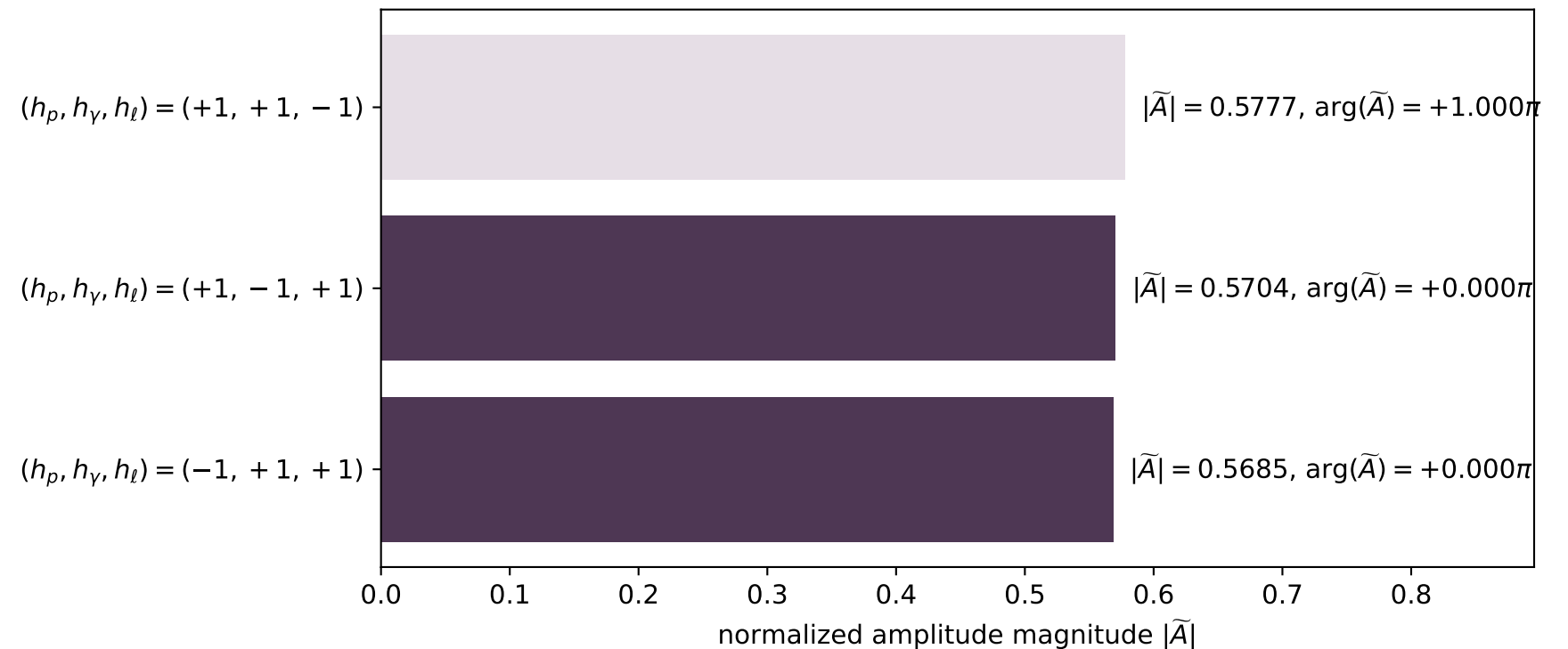
dw_mixing_angles_polarization_05_local_minimum_776
 $D_W=0.00968106$
 region: polarization_cluster_05_local_minimum_776
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0776
 lepton mixing angle: $\alpha_e=2.3464431$ rad
 incoming lepton: $-0.700178|+\rangle +0.713968|-\rangle$
 proton mixing angle: $\alpha_p=2.9353127$ rad
 incoming proton: $-0.9788|+\rangle +0.20482|-\rangle$

$s=1.27959$, $\sqrt{s}=1.13119$, $|\vec{P}|=0.176693$, $|\vec{P}'|=0.000242767$
 $\theta_{p'}=8.30818e-07$, $\theta_\gamma=1.53179$, $\phi_{p'}=1.27106$, $\phi_\gamma=6.28317$
 $E_\gamma=0.09659$, $\theta_{\ell'}=1.61231$, $\phi_{\ell'}=3.14157$
 $Q^2=0.0327203$, $x_B=0.152955$, $t=-0.0308627$, $W^2=1.06104$, $y=0.535139$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

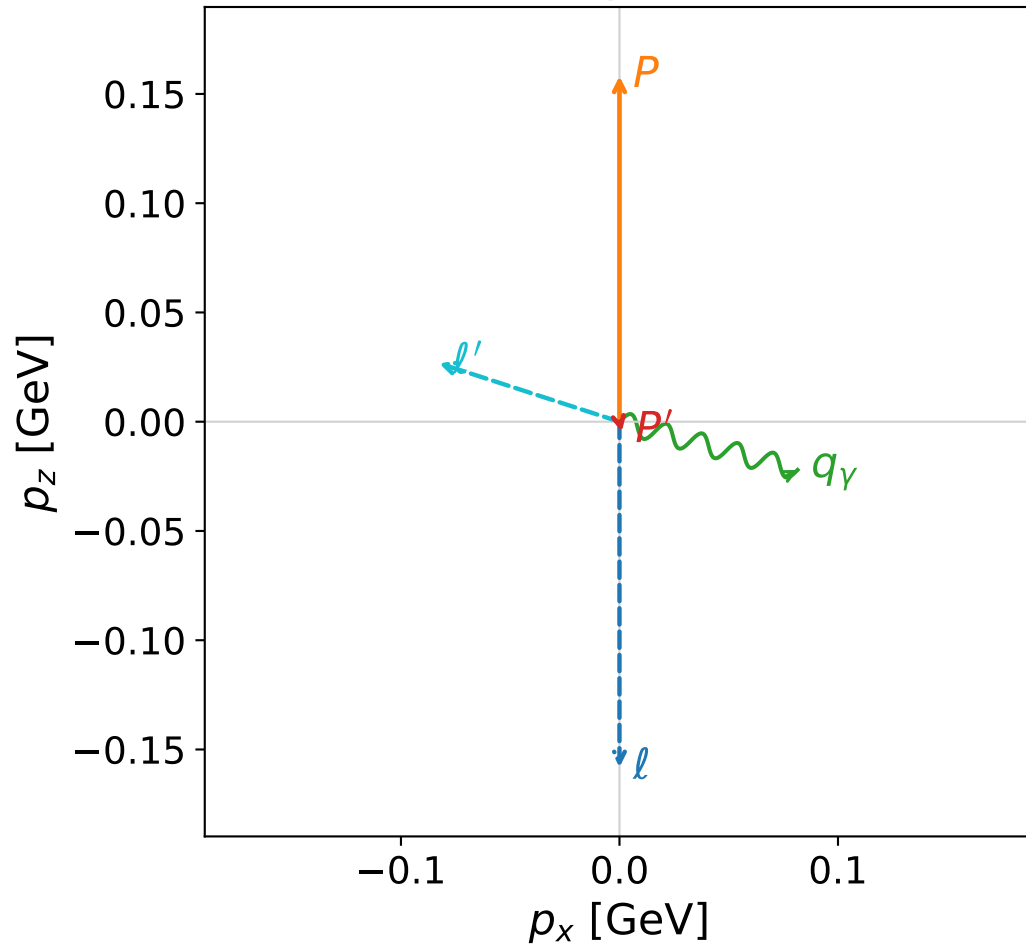
ℓ	$E=$	0.1767	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	-0.1767
P	$E=$	0.9545	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	0.1767
ℓ'	$E=$	0.0966	$p_x=$	-0.0965	$p_y=$	0.0000	$p_z=$	-0.0040
P'	$E=$	0.9380	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	0.0002
q_γ	$E=$	0.0966	$p_x=$	0.0965	$p_y=$	-0.0000	$p_z=$	0.0038

Leading normalized final-state amplitudes
 (N=3, retained=98.2%)

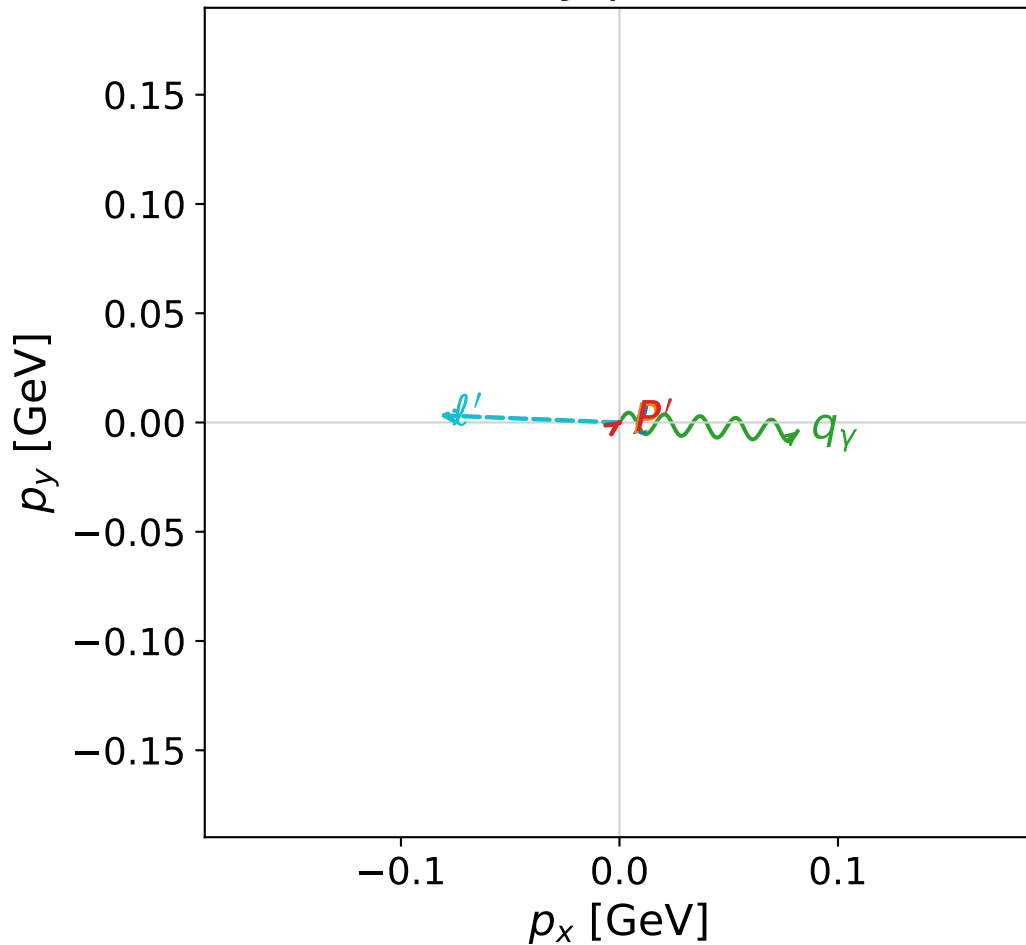


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



x--y plane



dw_mixing_angles_polarization_05_local_minimum_793

D_W=0.0102042

region: polarization_cluster_05_local_minimum_793

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0793

lepton mixing angle: $\alpha_e=2.3866647$ rad

incoming lepton: $-0.728321|+\rangle + 0.685236|-\rangle$

proton mixing angle: $\alpha_p=2.9399128$ rad

incoming proton: $-0.979731|+\rangle + 0.200315|-\rangle$

$s=1.23076$, $\sqrt{s}=1.1094$, $|\vec{P}|=0.158156$, $|\vec{P}'|=0.00484159$

$\theta_{p'}=2.88114$, $\theta_Y=1.83482$, $\phi_{p'}=0.579384$, $\phi_Y=6.23421$

$E_Y=0.0845404$, $\theta_{\ell'}=1.25779$, $\phi_{\ell'}=3.10149$

$Q^2=0.0359278$, $x_B=0.185045$, $t=-0.0263416$, $W^2=1.03807$, $y=0.553284$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ $E= 0.1582$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1582$

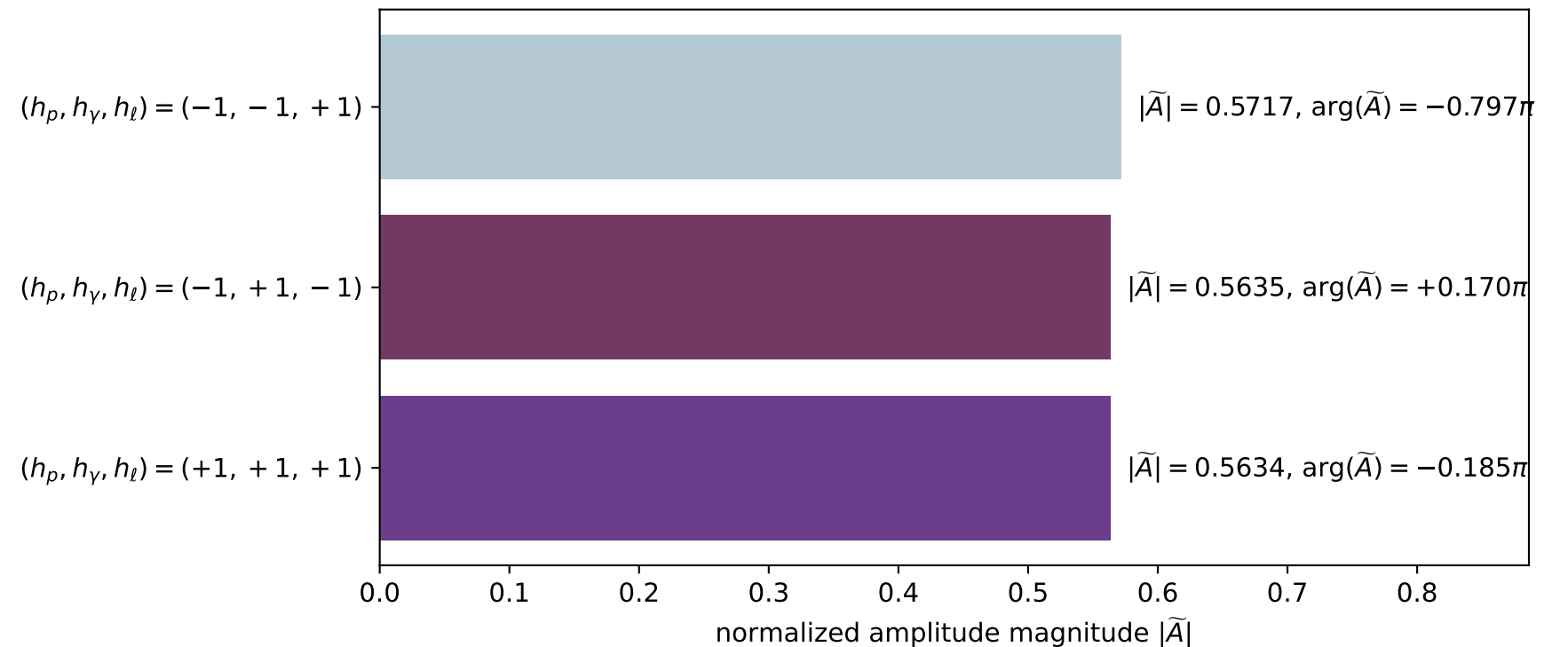
P $E= 0.9512$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1582$

ℓ' $E= 0.0868$ $p_x= -0.0826$ $p_y= 0.0033$ $p_z= 0.0267$

P' $E= 0.9380$ $p_x= 0.0010$ $p_y= 0.0007$ $p_z= -0.0047$

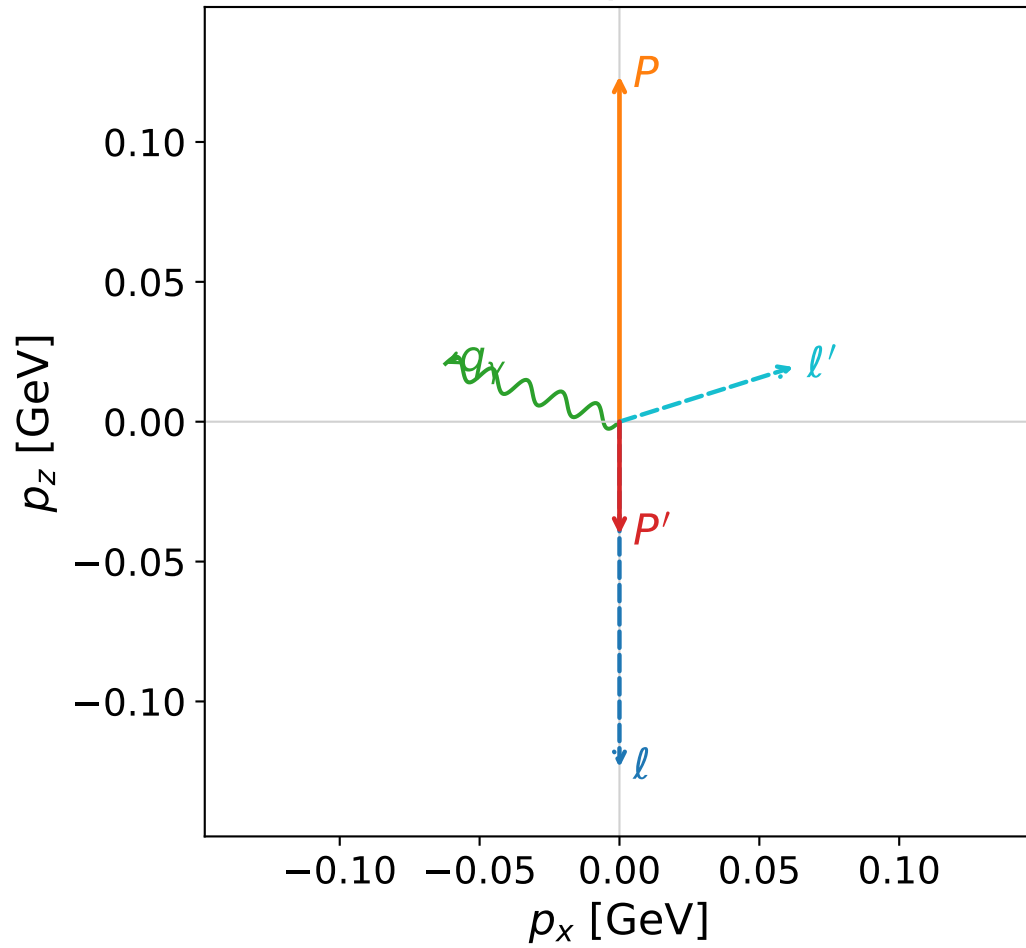
q_Y $E= 0.0845$ $p_x= 0.0815$ $p_y= -0.0040$ $p_z= -0.0221$

Leading normalized final-state amplitudes
(N=3, retained=96.2%)

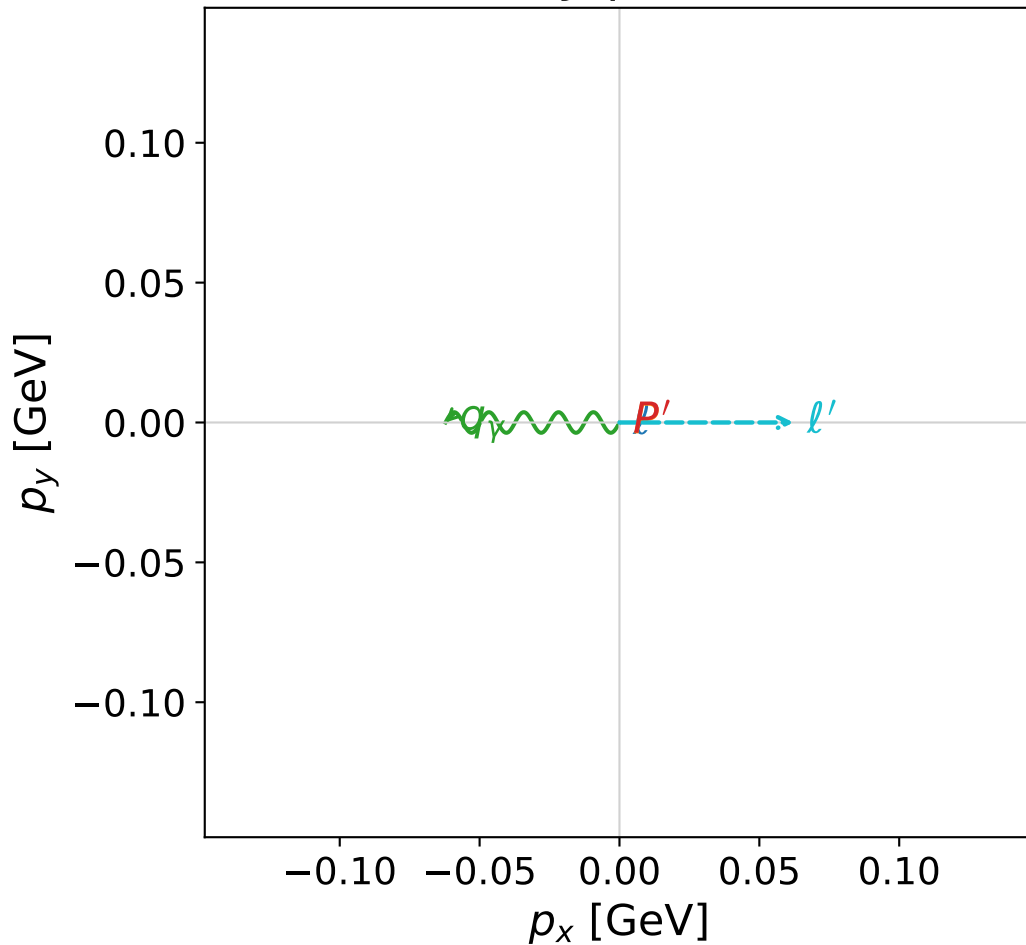


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



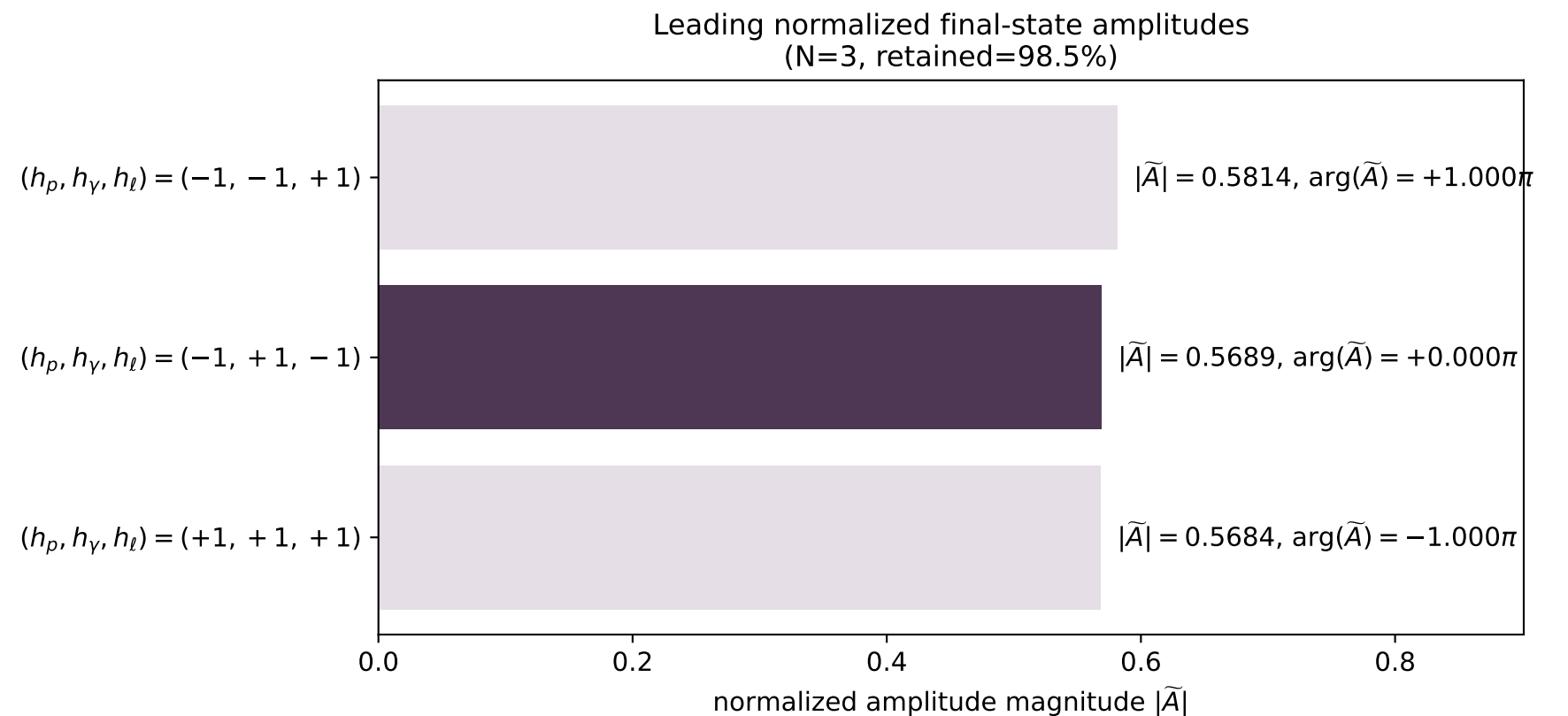
x--y plane



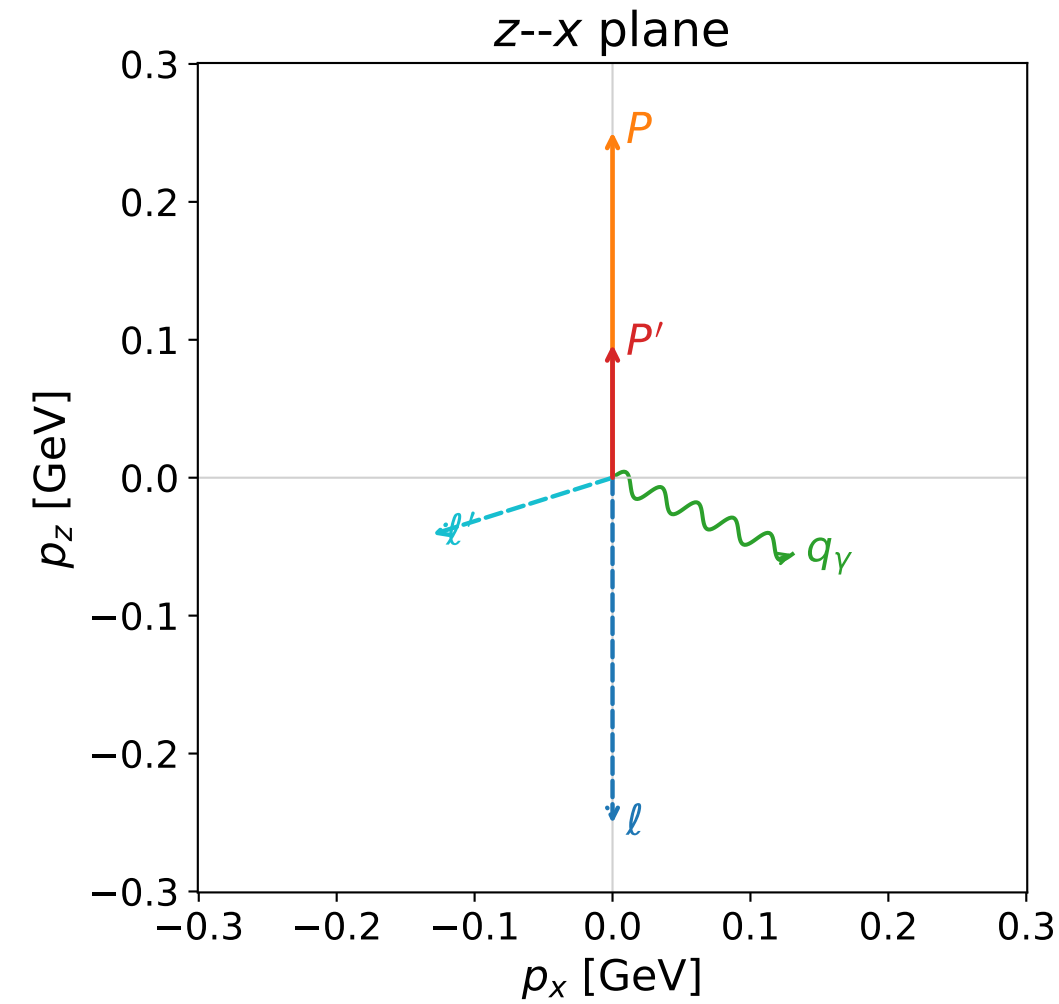
dw_mixing_angles_polarization_05_local_minimum_794
 $D_W=0.0102103$
 region: polarization_cluster_05_local_minimum_794
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0794
 lepton mixing angle: $\alpha_e=2.372709$ rad
 incoming lepton: $-0.718687|+\rangle + 0.695333|-\rangle$
 proton mixing angle: $\alpha_p=0.17283578$ rad
 incoming proton: $0.985101|+\rangle + 0.171977|-\rangle$

$s=1.14414$, $\sqrt{s}=1.06964$, $|\vec{P}|=0.123543$, $|\vec{P}'|=0.0402749$
 $\theta_{p'}=3.14159$, $\theta_\gamma=1.24932$, $\phi_{p'}=2.93467$, $\phi_\gamma=3.14162$
 $E_\gamma=0.0655678$, $\theta_{\ell'}=1.26619$, $\phi_{\ell'}=2.75644e-05$
 $Q^2=0.0209452$, $x_B=0.143724$, $t=-0.0267839$, $W^2=1.00463$, $y=0.551399$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1235$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1235$
 P $E= 0.9461$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1235$
 ℓ' $E= 0.0652$ $p_x= 0.0622$ $p_y= 0.0000$ $p_z= 0.0196$
 P' $E= 0.9389$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.0403$
 q_γ $E= 0.0656$ $p_x= -0.0622$ $p_y= -0.0000$ $p_z= 0.0207$



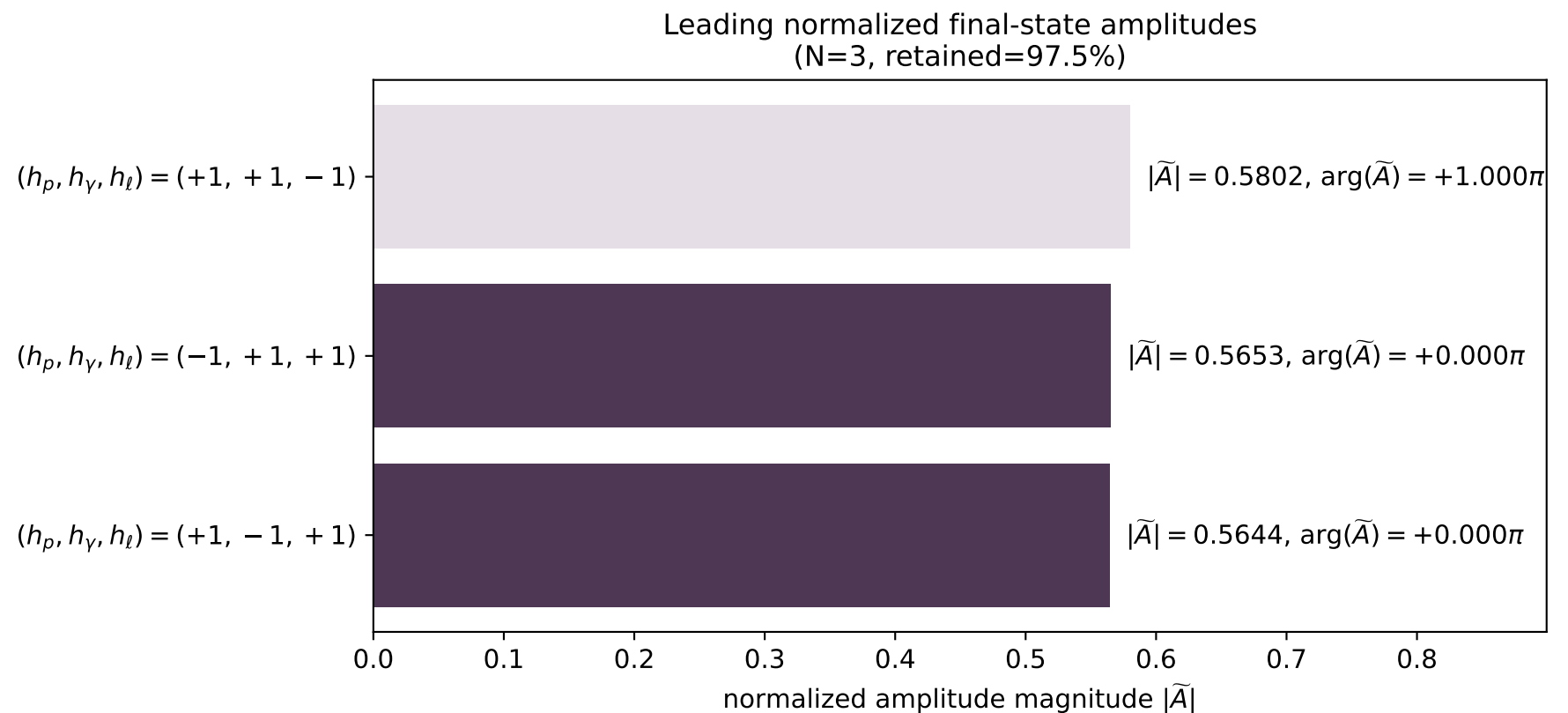
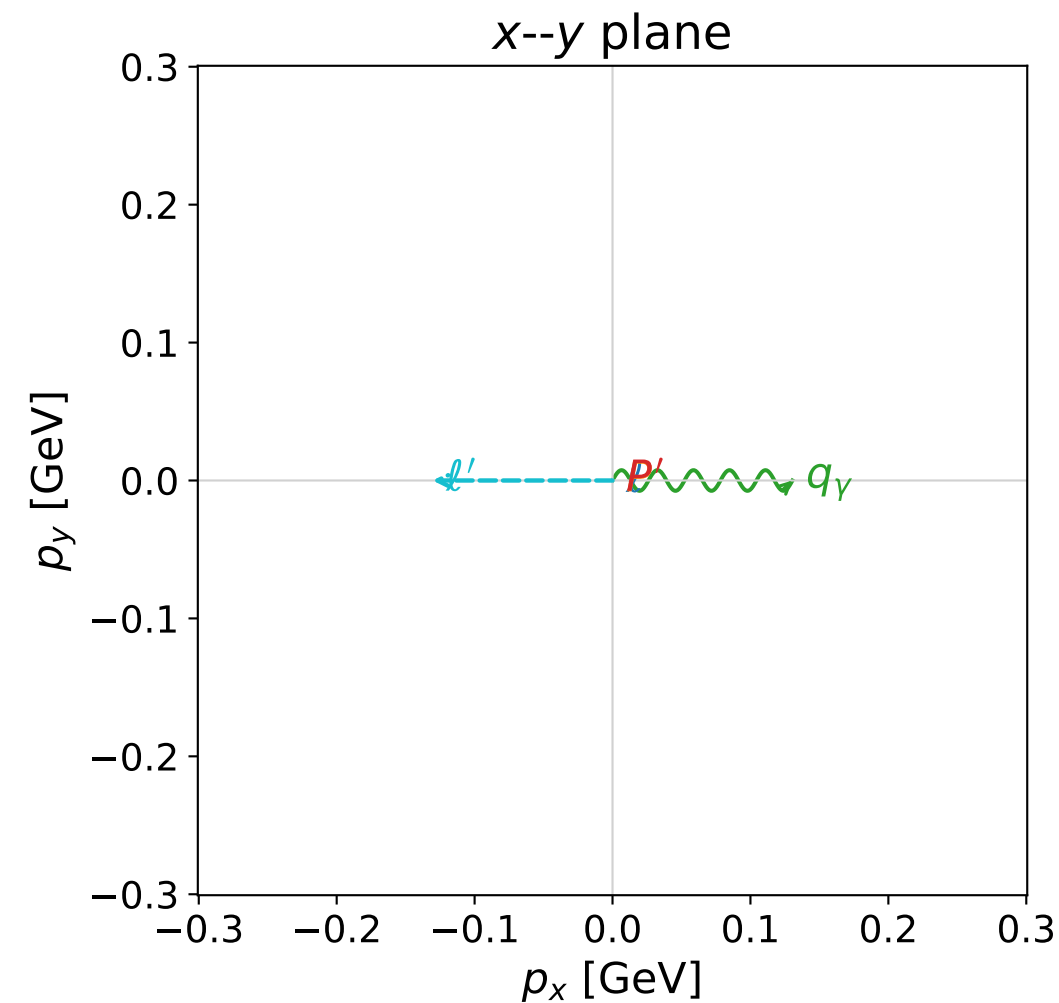
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)



dw_mixing_angles_polarization_05_local_minimum_921
 $D_W=0.0145517$
 region: polarization_cluster_05_local_minimum_921
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0921
 lepton mixing angle: $\alpha_e=2.3318716$ rad
 incoming lepton: $-0.6897|+\rangle +0.724095|-\rangle$
 proton mixing angle: $\alpha_p=3.0185833$ rad
 incoming proton: $-0.992444|+\rangle +0.122699|-\rangle$

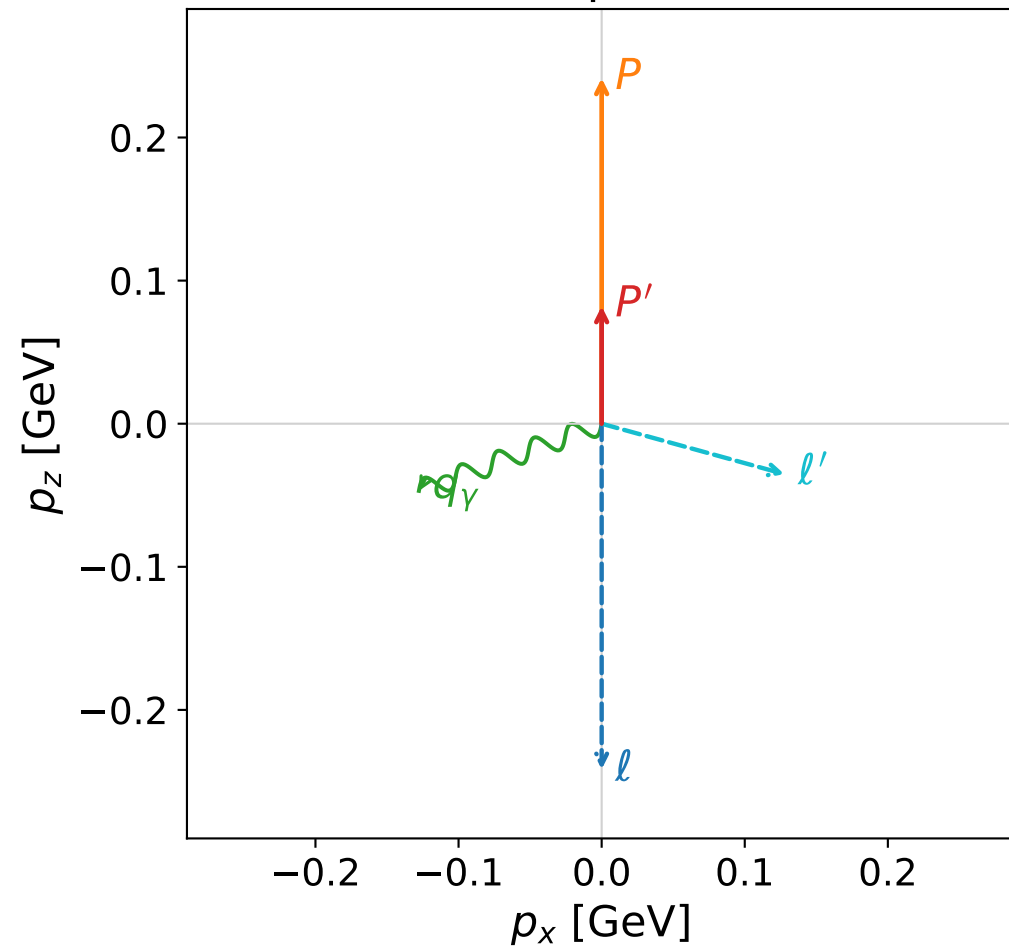
$s=1.49199$, $\sqrt{s}=1.22147$, $|\vec{P}|=0.250577$, $|\vec{P}'|=0.0965147$
 $\theta_{p'}=0$, $\theta_\gamma=1.97239$, $\phi_{p'}=4.7718$, $\phi_\gamma=6.28317$
 $E_\gamma=0.141734$, $\theta_{\ell'}=1.87609$, $\phi_{\ell'}=3.14157$
 $Q^2=0.0479455$, $x_B=0.1471$, $t=-0.0229545$, $W^2=1.15784$, $y=0.532451$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.2506$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2506$
 P $E= 0.9709$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.2506$
 ℓ' $E= 0.1368$ $p_x= -0.1305$ $p_y= 0.0000$ $p_z= -0.0411$
 P' $E= 0.9430$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0965$
 q_γ $E= 0.1417$ $p_x= 0.1305$ $p_y= -0.0000$ $p_z= -0.0554$

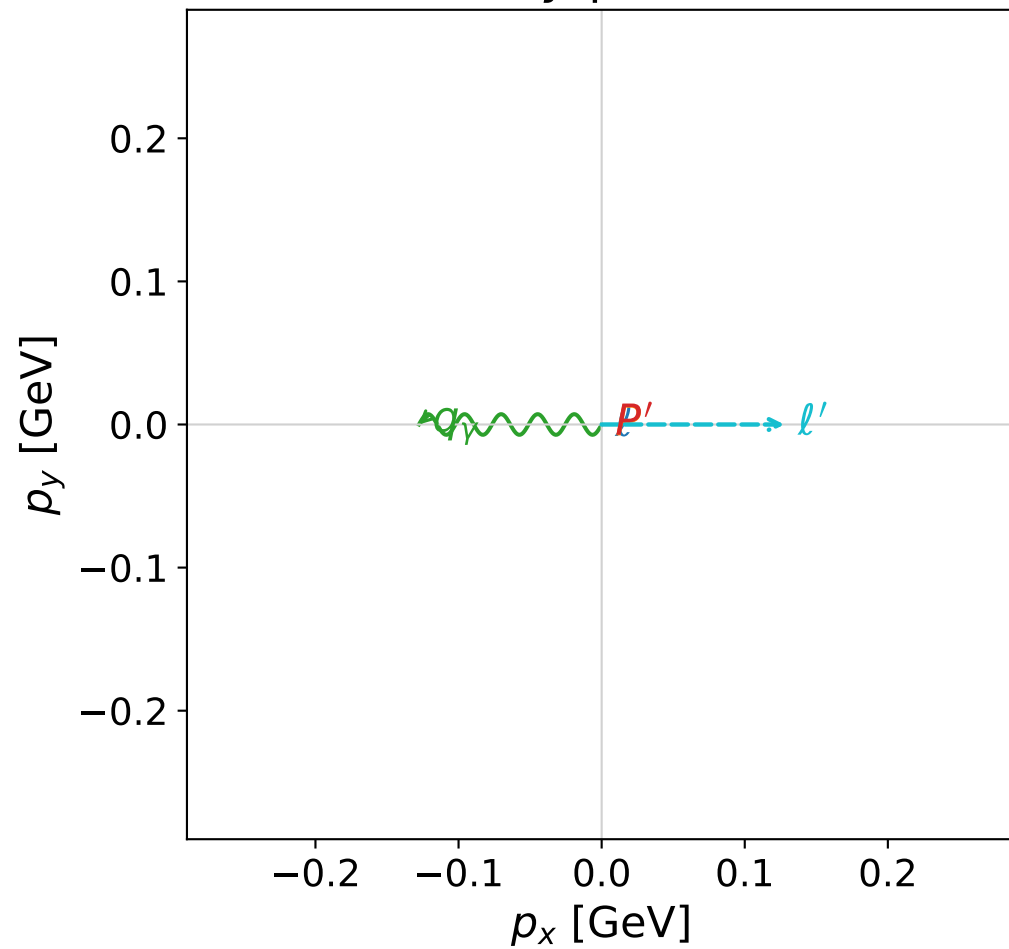


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



x--y plane

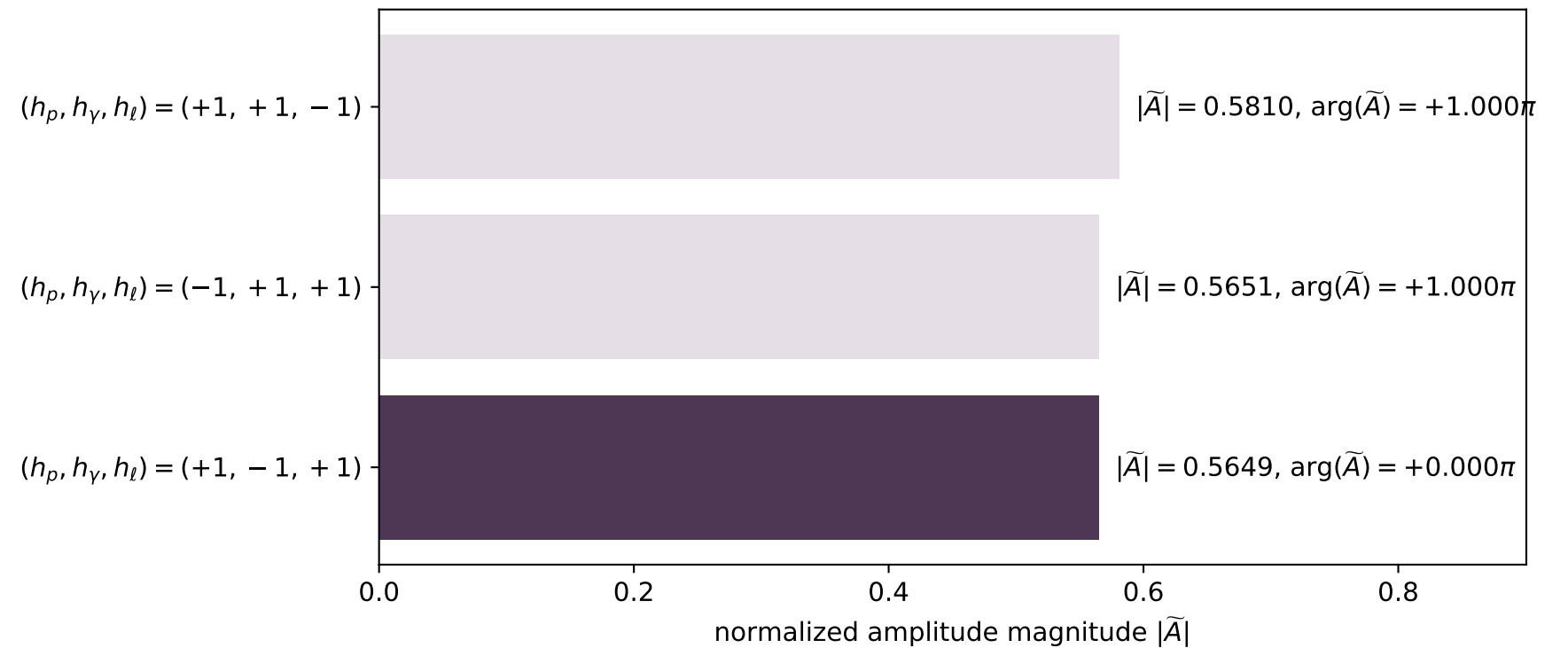


dw_mixing_angles_polarization_05_local_minimum_923
 $D_W=0.0146924$
 region: polarization_cluster_05_local_minimum_923
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0923
 lepton mixing angle: $\alpha_e=2.324786$ rad
 incoming lepton: $-0.684552|+\rangle +0.728964|-\rangle$
 proton mixing angle: $\alpha_p=0.13355488$ rad
 incoming proton: $0.991095|+\rangle +0.133158|-\rangle$

$s=1.46432$, $\sqrt{s}=1.21009$, $|\vec{P}|=0.241502$, $|\vec{P}'|=0.0821138$
 $\theta_{P'}=0$, $\theta_\gamma=1.92292$, $\phi_{P'}=3.23812$, $\phi_\gamma=3.1415$
 $E_\gamma=0.136046$, $\theta_{\ell'}=1.83971$, $\phi_{\ell'}=6.28309$
 $Q^2=0.0469803$, $x_B=0.151118$, $t=-0.0246754$, $W^2=1.14375$, $y=0.531898$

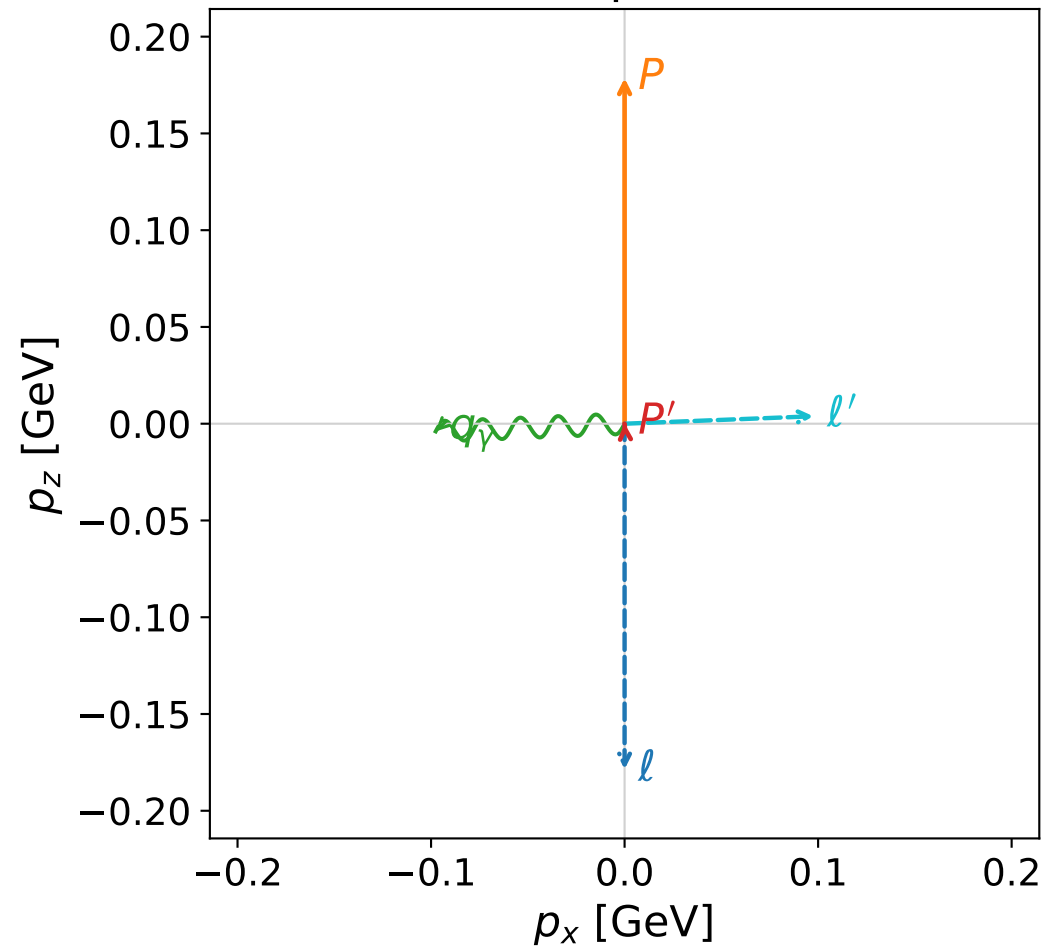
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E=0.2415$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.2415$
 P $E=0.9686$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.2415$
 ℓ' $E=0.1325$ $p_x=0.1277$ $p_y=-0.0000$ $p_z=-0.0352$
 P' $E=0.9416$ $p_x=-0.0000$ $p_y=-0.0000$ $p_z=0.0821$
 q_γ $E=0.1360$ $p_x=-0.1277$ $p_y=0.0000$ $p_z=-0.0469$

Leading normalized final-state amplitudes
 (N=3, retained=97.6%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

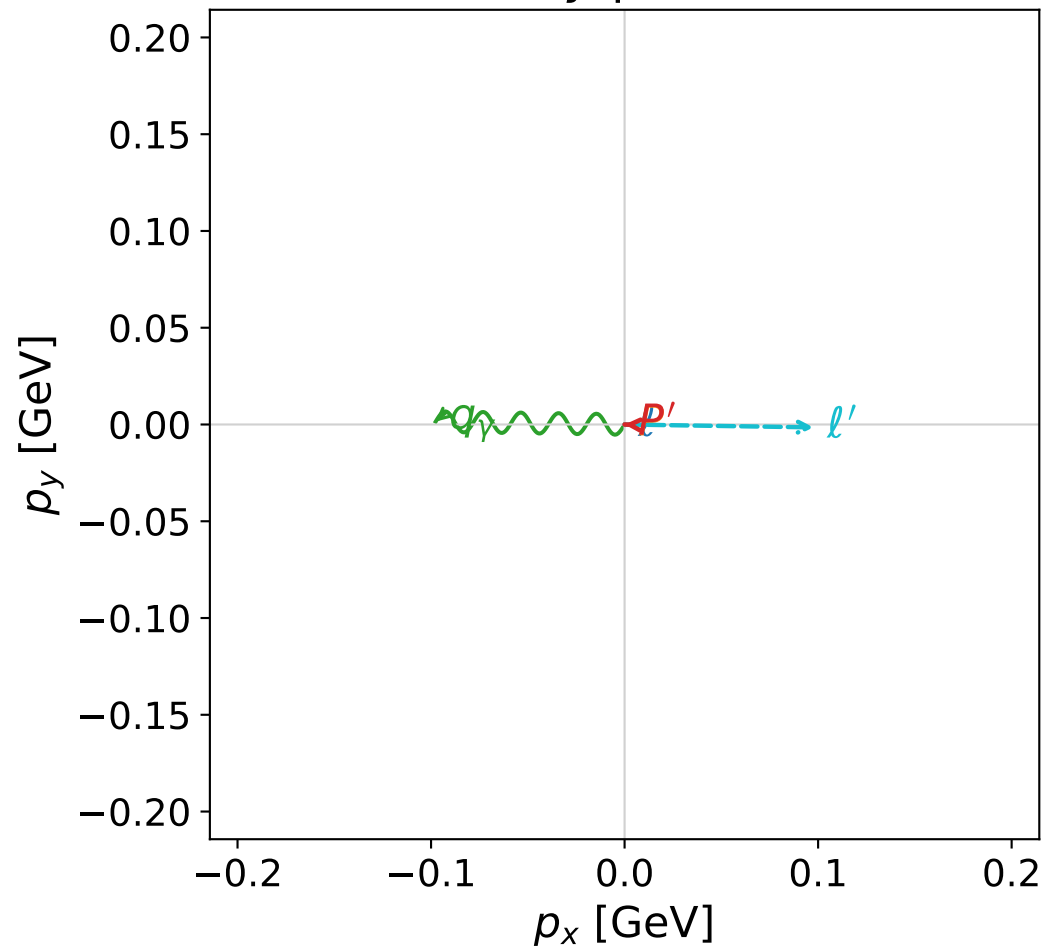


dw_mixing_angles_polarization_05_local_minimum_1091
 $D_W=0.0262184$
 region: polarization_cluster_05_local_minimum_1091
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_1091
 lepton mixing angle: $\alpha_e=2.3526069$ rad
 incoming lepton: $-0.704565|+\rangle +0.709639|-\rangle$
 proton mixing angle: $\alpha_p=0.0056745158$ rad
 incoming proton: $0.999984|+\rangle +0.00567449|-\rangle$

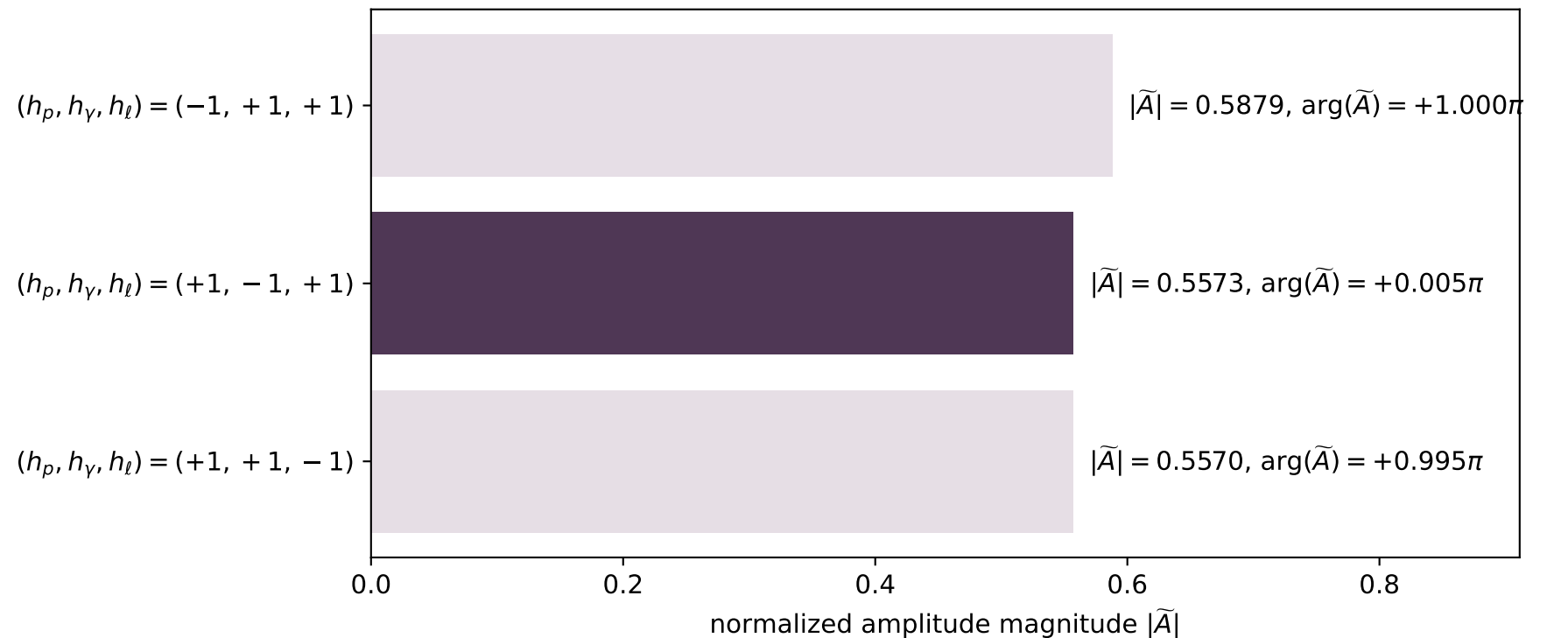
$s=1.28465$, $\sqrt{s}=1.13342$, $|\vec{P}|=0.178576$, $|\vec{P}'|=0.000129391$
 $\theta_{p'}=0.0541711$, $\theta_\gamma=1.61158$, $\phi_{p'}=3.10435$, $\phi_\gamma=3.12631$
 $E_\gamma=0.0977106$, $\theta_{\ell'}=1.53134$, $\phi_{\ell'}=6.2679$
 $Q^2=0.036275$, $x_B=0.165202$, $t=-0.0315595$, $W^2=1.06315$, $y=0.542431$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1786$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1786$
 P $E= 0.9548$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1786$
 ℓ' $E= 0.0977$ $p_x= 0.0976$ $p_y= -0.0015$ $p_z= 0.0039$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.0001$
 q_γ $E= 0.0977$ $p_x= -0.0976$ $p_y= 0.0015$ $p_z= -0.0040$

x--y plane

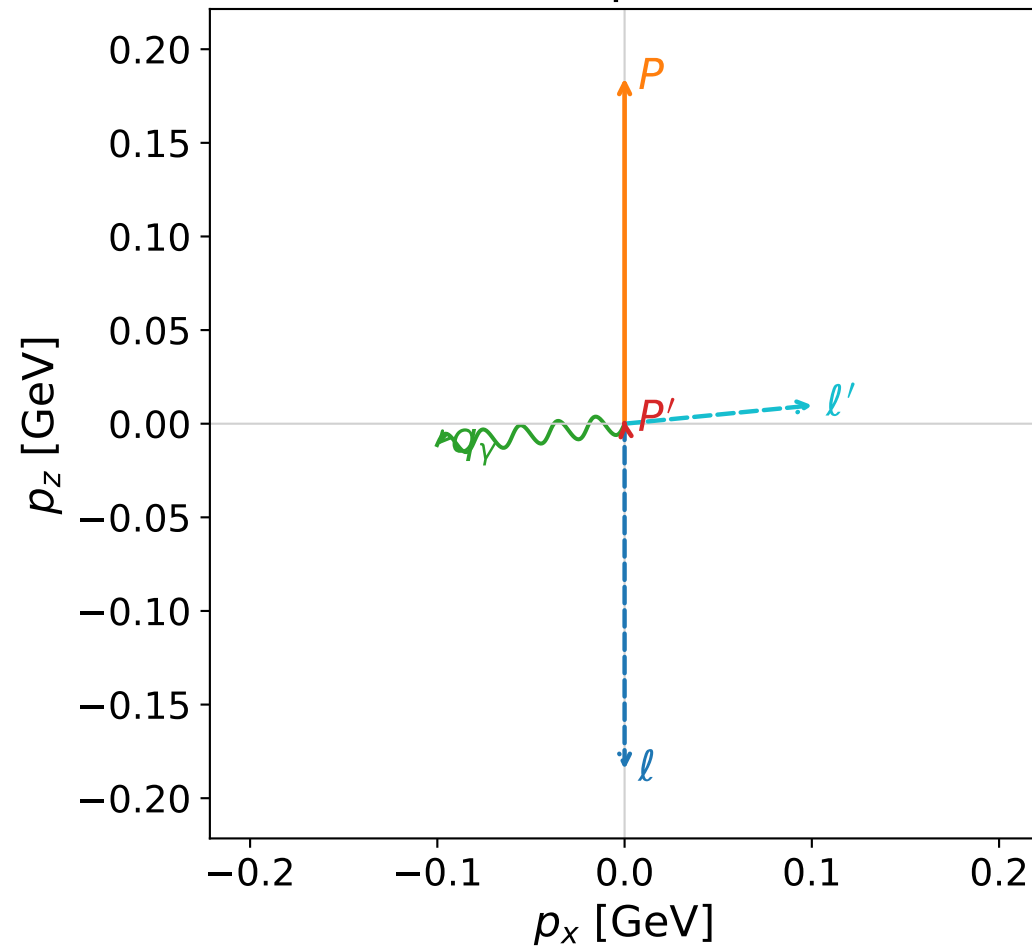


Leading normalized final-state amplitudes
(N=3, retained=96.6%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

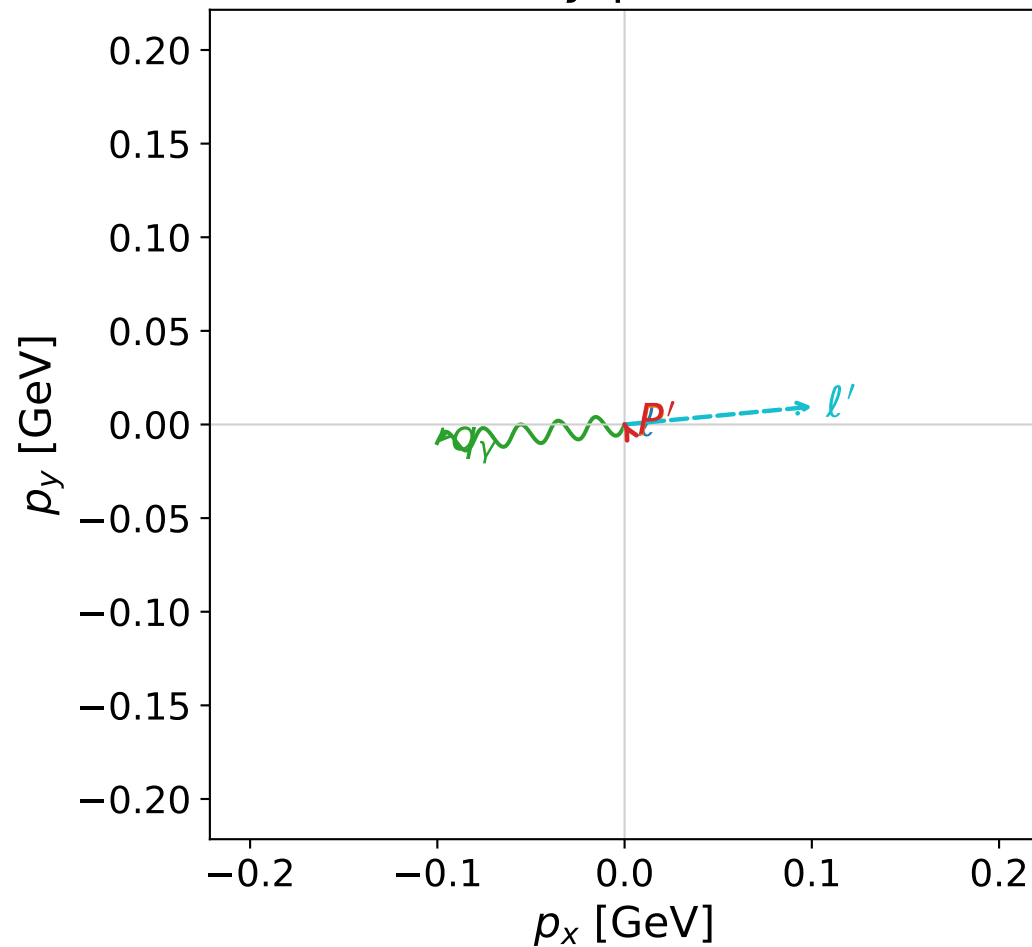


dw_mixing_angles_polarization_05_local_minimum_1123
 $D_W=0.031282$
 region: polarization_cluster_05_local_minimum_1123
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_1123
 lepton mixing angle: $\alpha_e=2.3460623$ rad
 incoming lepton: $-0.699906|+\rangle +0.714235|-\rangle$
 proton mixing angle: $\alpha_p=0.0032211657$ rad
 incoming proton: $0.999995|+\rangle +0.00322116|-\rangle$

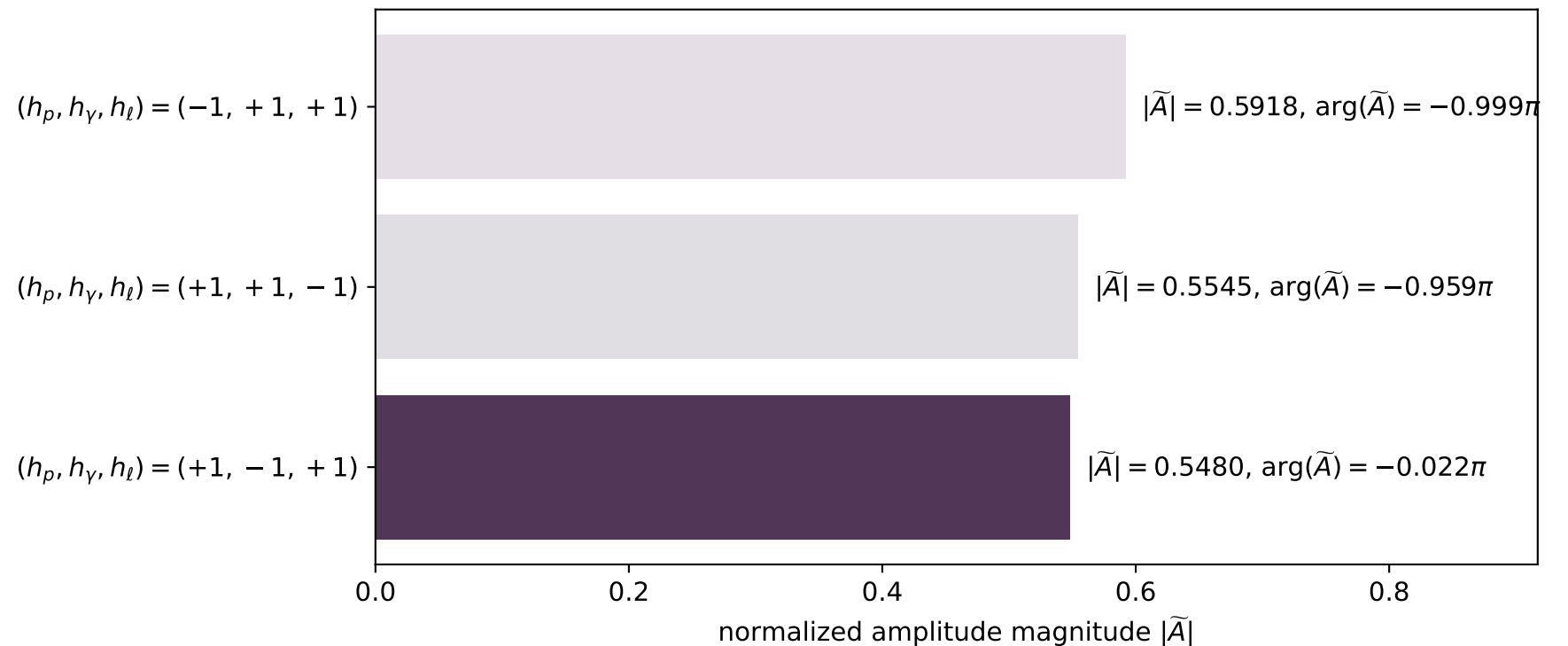
$s=1.30086$, $\sqrt{s}=1.14055$, $|\vec{P}|=0.184565$, $|\vec{P}'|=0.0016487$
 $\theta_{P'}=0.239026$, $\theta_\gamma=1.68355$, $\phi_{P'}=2.06697$, $\phi_\gamma=3.23994$
 $E_\gamma=0.101283$, $\theta_{l'}=1.47393$, $\phi_{l'}=0.094778$
 $Q^2=0.040996$, $x_B=0.177465$, $t=-0.0331524$, $W^2=1.06986$, $y=0.548696$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E=0.1846$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.1846$
 P $E=0.9560$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1846$
 l' $E=0.1013$ $p_x=0.1003$ $p_y=0.0095$ $p_z=0.0098$
 P' $E=0.9380$ $p_x=-0.0002$ $p_y=0.0003$ $p_z=0.0016$
 q_γ $E=0.1013$ $p_x=-0.1002$ $p_y=-0.0099$ $p_z=-0.0114$

x--y plane

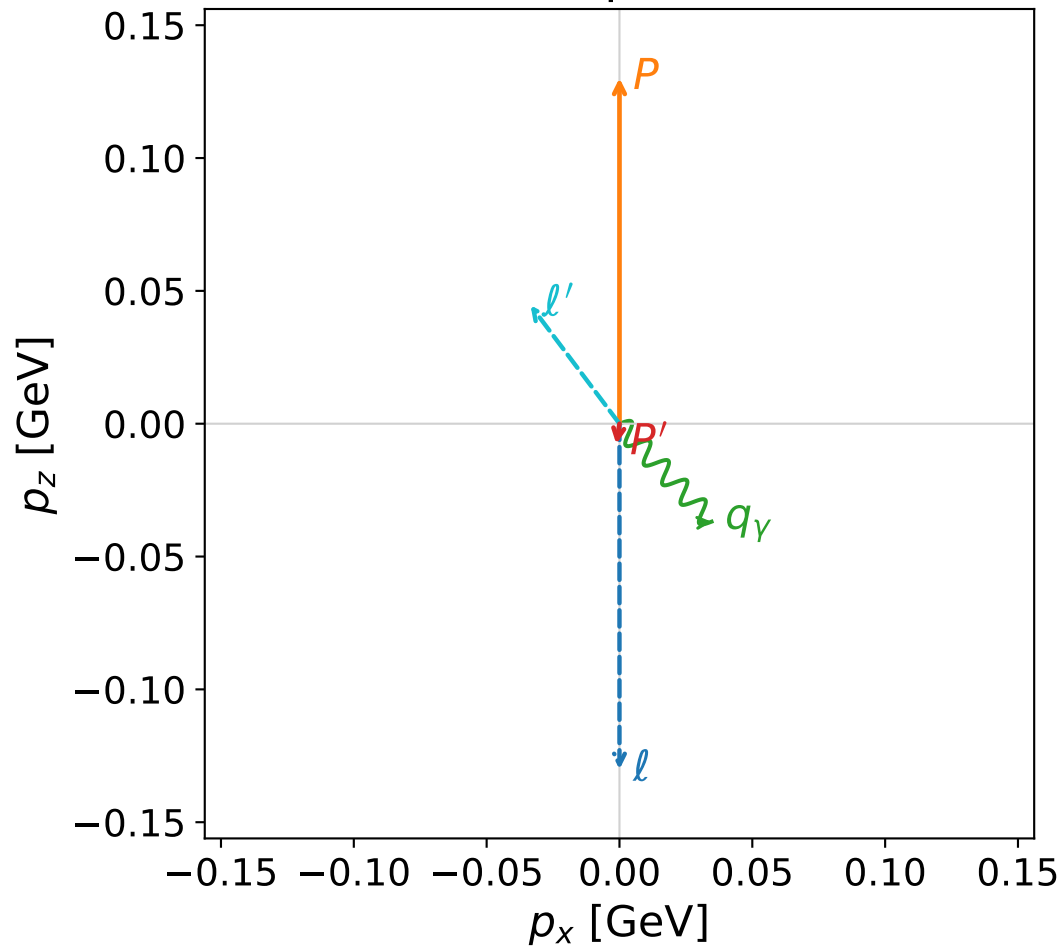


Leading normalized final-state amplitudes
 (N=3, retained=95.8%)

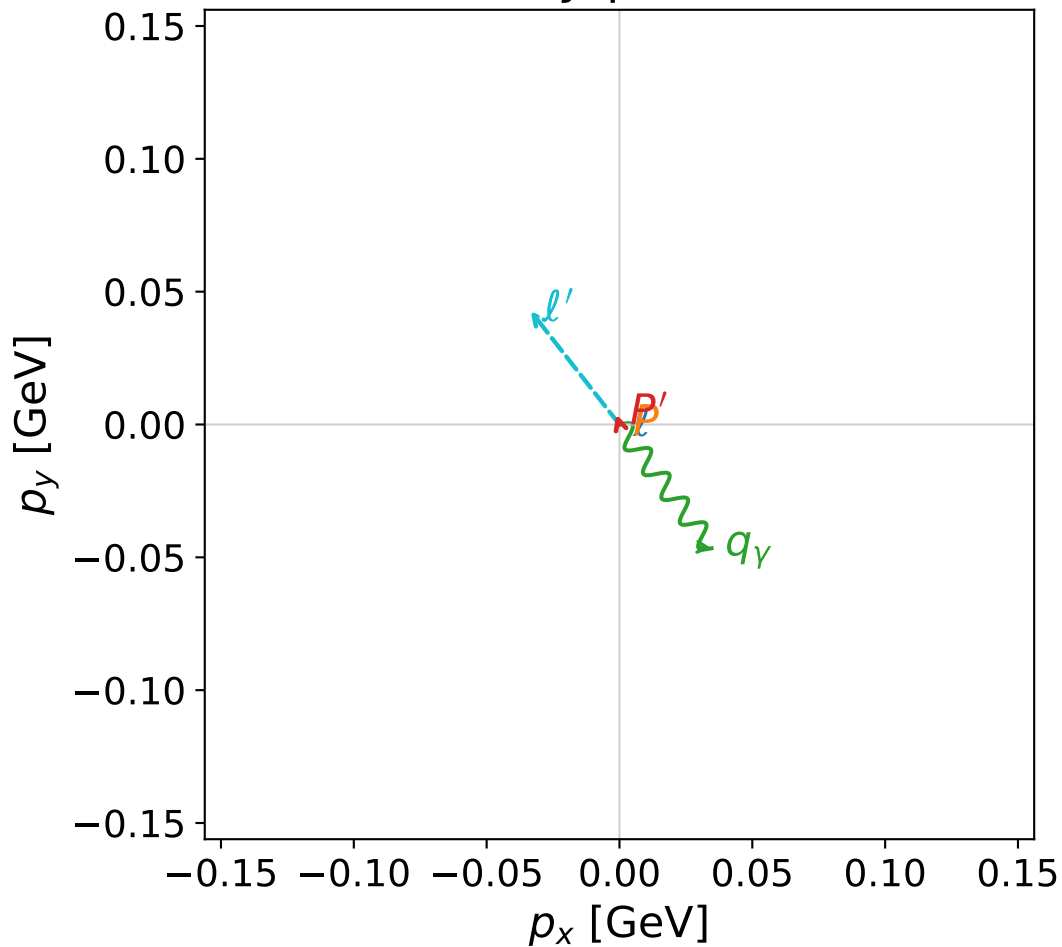


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



x--y plane

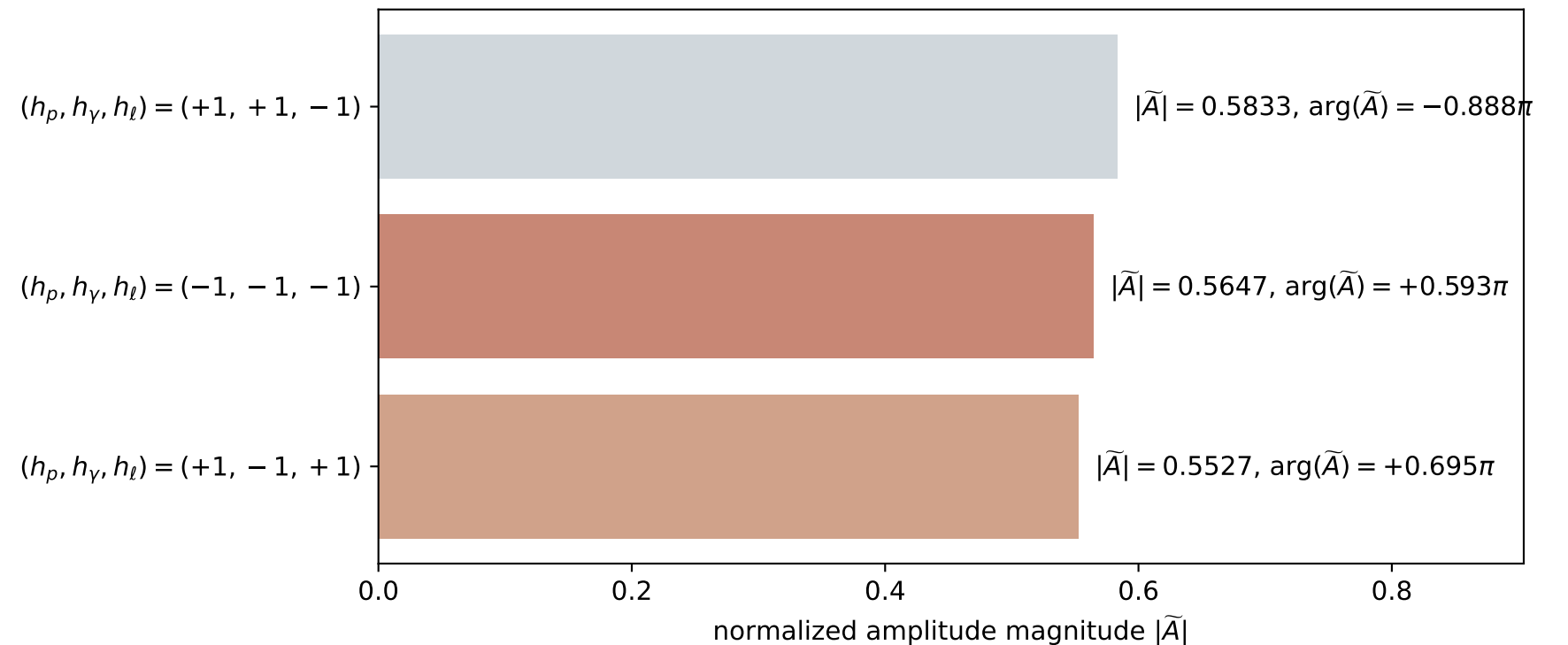


dw_mixing_angles_polarization_06_local_minimum_509
 $D_W=0.00259542$
 region: polarization_cluster_06_local_minimum_509
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0509
 lepton mixing angle: $\alpha_e=2.3474721$ rad
 incoming lepton: $-0.700912|+\rangle +0.713247|-\rangle$
 proton mixing angle: $\alpha_p=1.3918247$ rad
 incoming proton: $0.178018|+\rangle +0.984027|-\rangle$

$s=1.16007$, $\sqrt{s}=1.07706$, $|\vec{P}|=0.130085$, $|\vec{P}'|=0.00864853$
 $\theta_{p'}=2.66104$, $\theta_Y=2.13648$, $\phi_{p'}=1.87001$, $\phi_Y=5.35198$
 $E_Y=0.0687822$, $\theta_{\ell'}=0.884012$, $\phi_{\ell'}=2.23496$
 $Q^2=0.0298612$, $x_B=0.188073$, $t=-0.0189123$, $W^2=1.00876$, $y=0.566603$

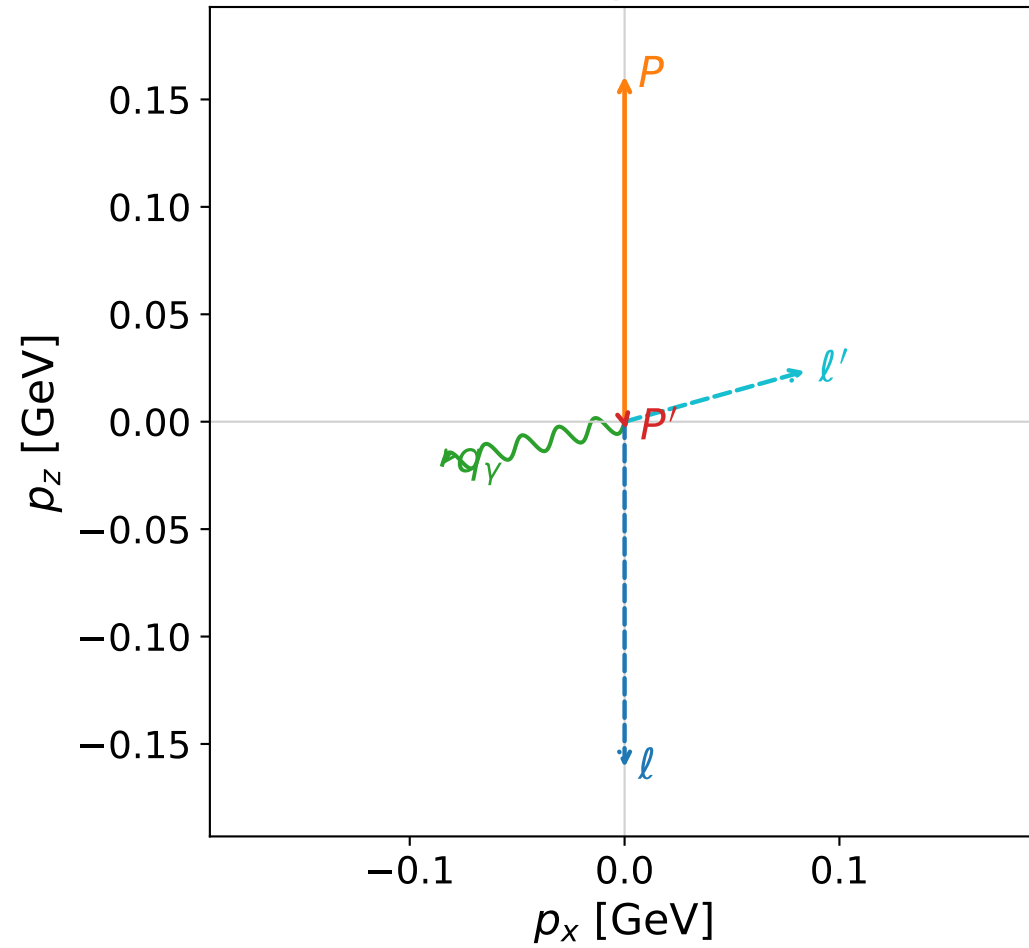
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1301$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1301$
 P $E= 0.9470$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1301$
 ℓ' $E= 0.0702$ $p_x= -0.0335$ $p_y= 0.0428$ $p_z= 0.0445$
 P' $E= 0.9380$ $p_x= -0.0012$ $p_y= 0.0038$ $p_z= -0.0077$
 q_Y $E= 0.0688$ $p_x= 0.0347$ $p_y= -0.0466$ $p_z= -0.0369$

Leading normalized final-state amplitudes
 (N=3, retained=96.5%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

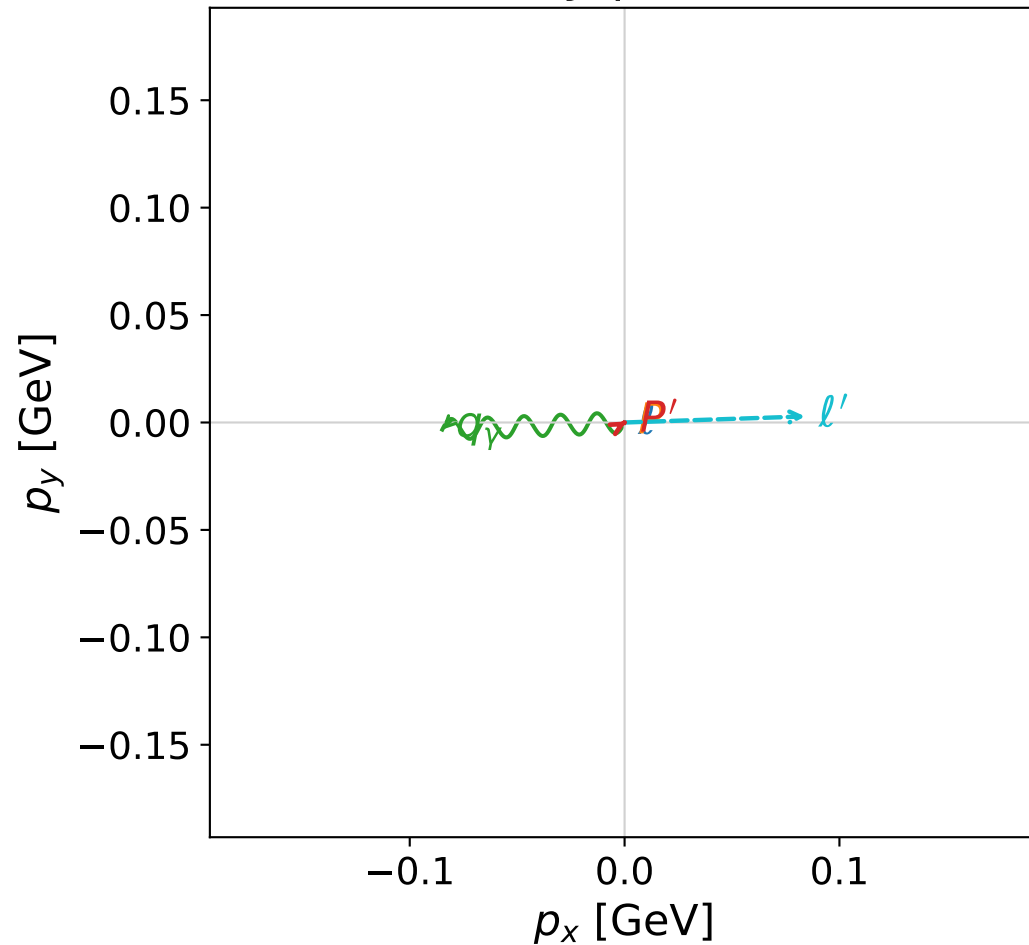


dw_mixing_angles_polarization_06_local_minimum_687
 $D_W=0.00662348$
 region: polarization_cluster_06_local_minimum_687
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0687
 lepton mixing angle: $\alpha_e=2.3355152$ rad
 incoming lepton: $-0.692334|+\rangle +0.721577|-\rangle$
 proton mixing angle: $\alpha_p=1.7539808$ rad
 incoming proton: $-0.182162|+\rangle +0.983269|-\rangle$

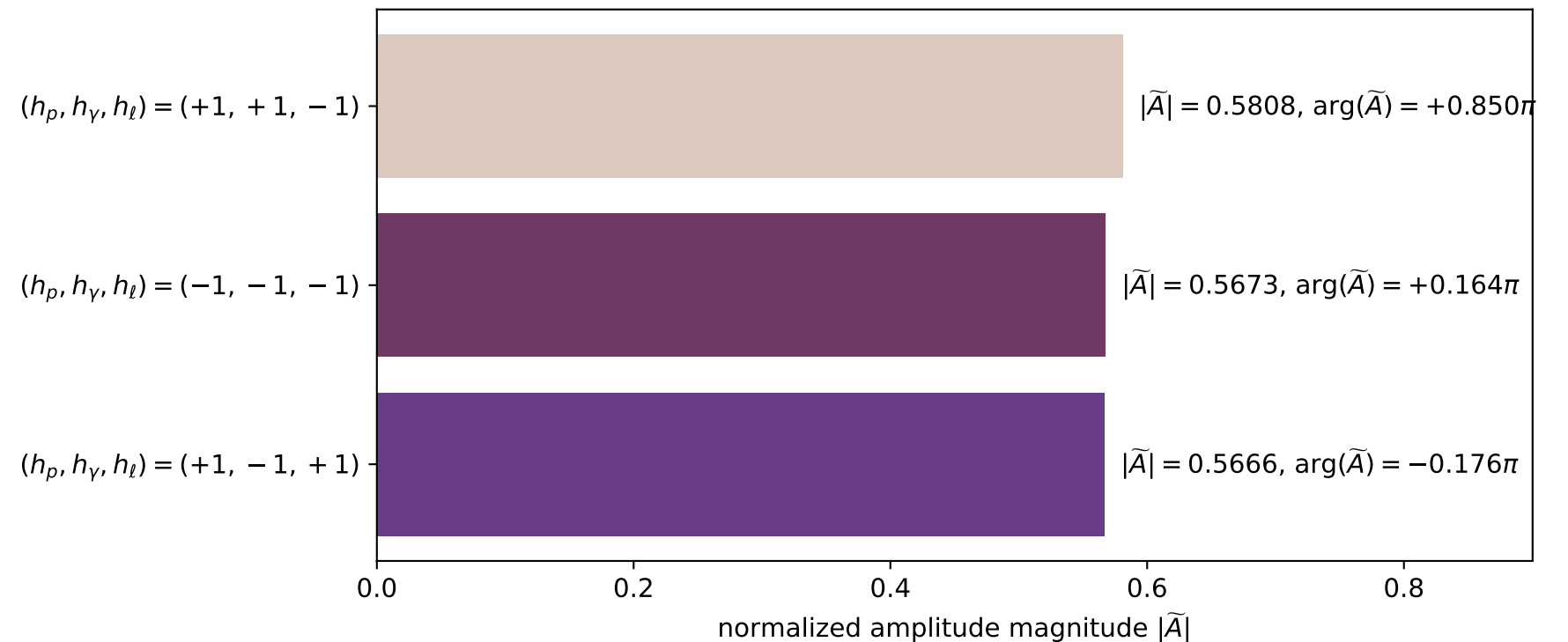
$s=1.23777$, $\sqrt{s}=1.11255$, $|\vec{P}|=0.160858$, $|\vec{P}'|=0.003618$
 $\theta_{p'}=2.87163$, $\theta_\gamma=1.80235$, $\phi_{p'}=0.51582$, $\phi_\gamma=3.18054$
 $E_\gamma=0.0872511$, $\theta_{l'}=1.2981$, $\phi_{l'}=0.0336836$
 $Q^2=0.035647$, $x_B=0.178828$, $t=-0.0268229$, $W^2=1.04353$, $y=0.556922$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.1609$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1609$
 P $E= 0.9517$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1609$
 l' $E= 0.0873$ $p_x= 0.0840$ $p_y= 0.0028$ $p_z= 0.0235$
 P' $E= 0.9380$ $p_x= 0.0008$ $p_y= 0.0005$ $p_z= -0.0035$
 q_γ $E= 0.0873$ $p_x= -0.0849$ $p_y= -0.0033$ $p_z= -0.0200$

x--y plane

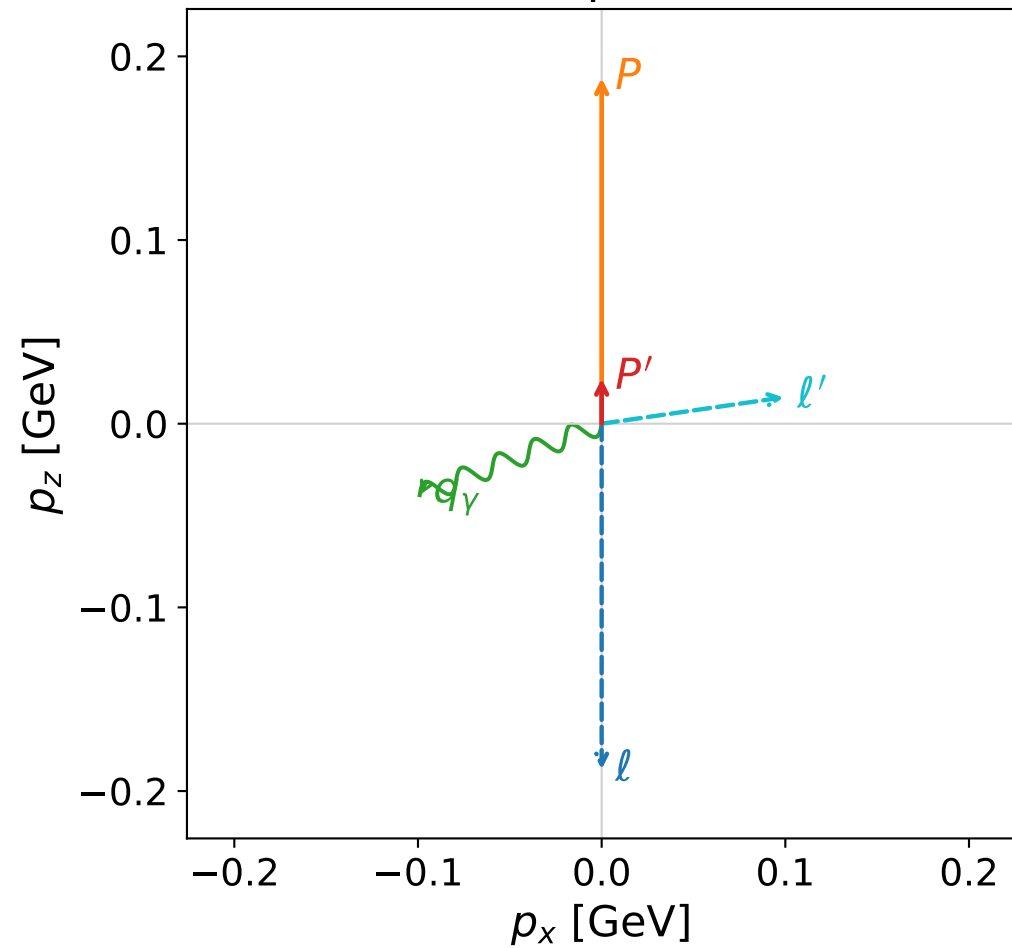


Leading normalized final-state amplitudes
 (N=3, retained=98.0%)

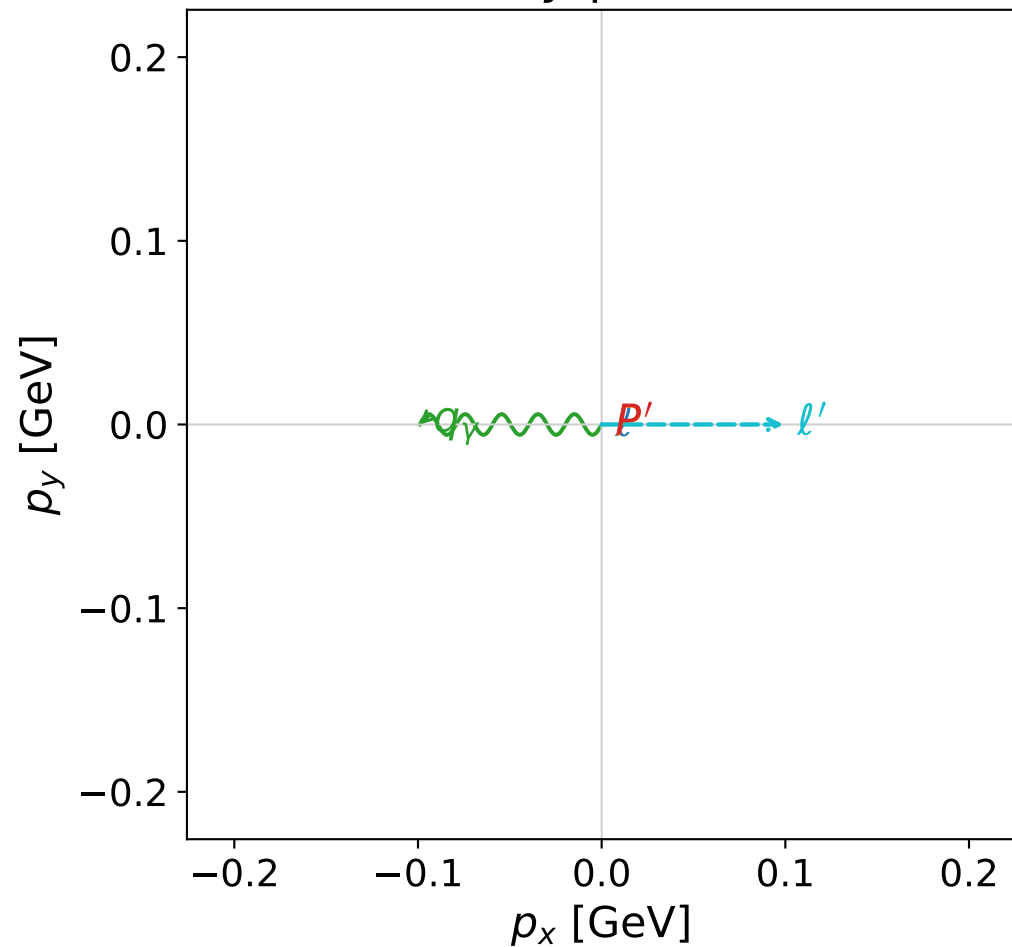


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



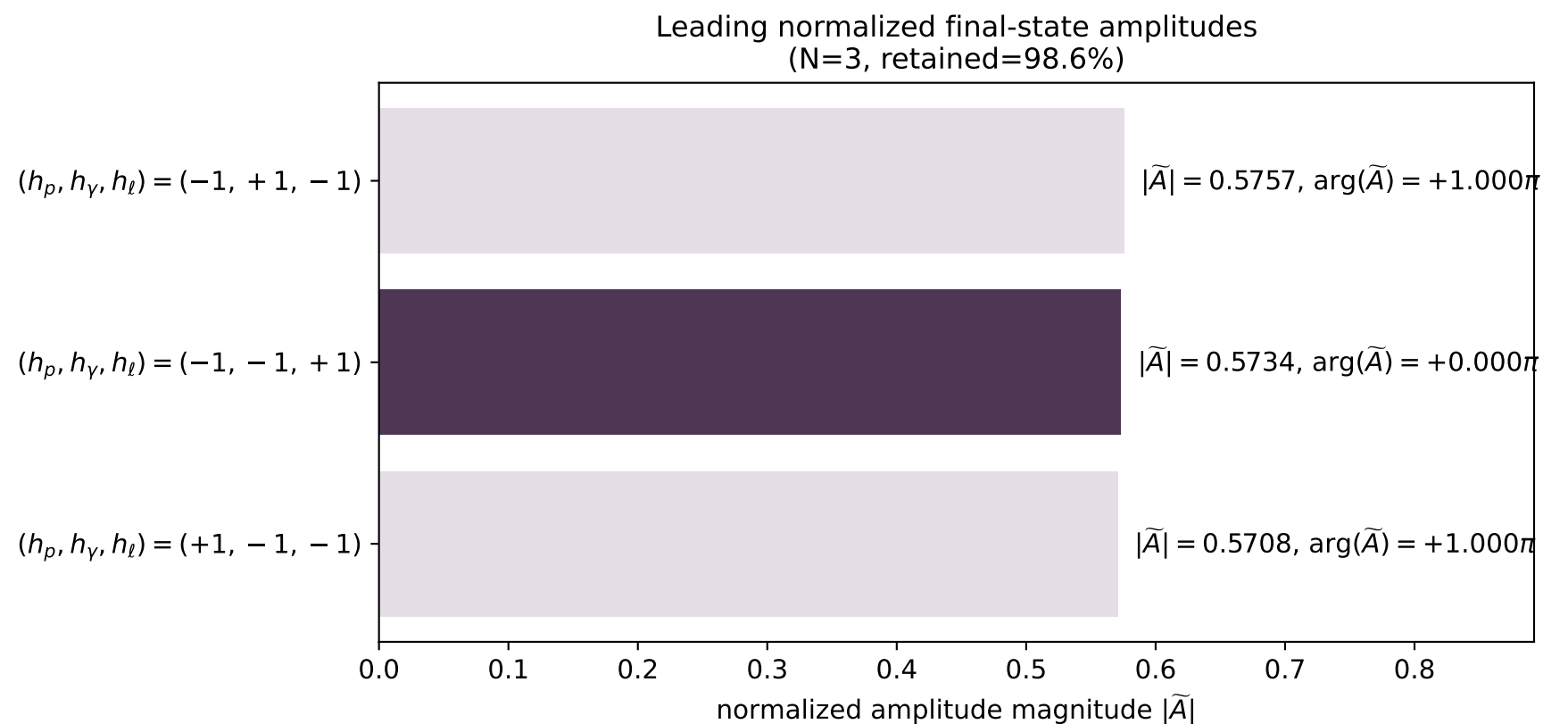
x--y plane



dw_mixing_angles_polarization_06_local_minimum_705
 $D_W=0.00745435$
 region: polarization_cluster_06_local_minimum_705
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0705
 lepton mixing angle: $\alpha_e=2.3477982$ rad
 incoming lepton: $-0.701145|+> +0.713019|->$
 proton mixing angle: $\alpha_p=1.7659151$ rad
 incoming proton: $-0.193883|+> +0.981025|->$

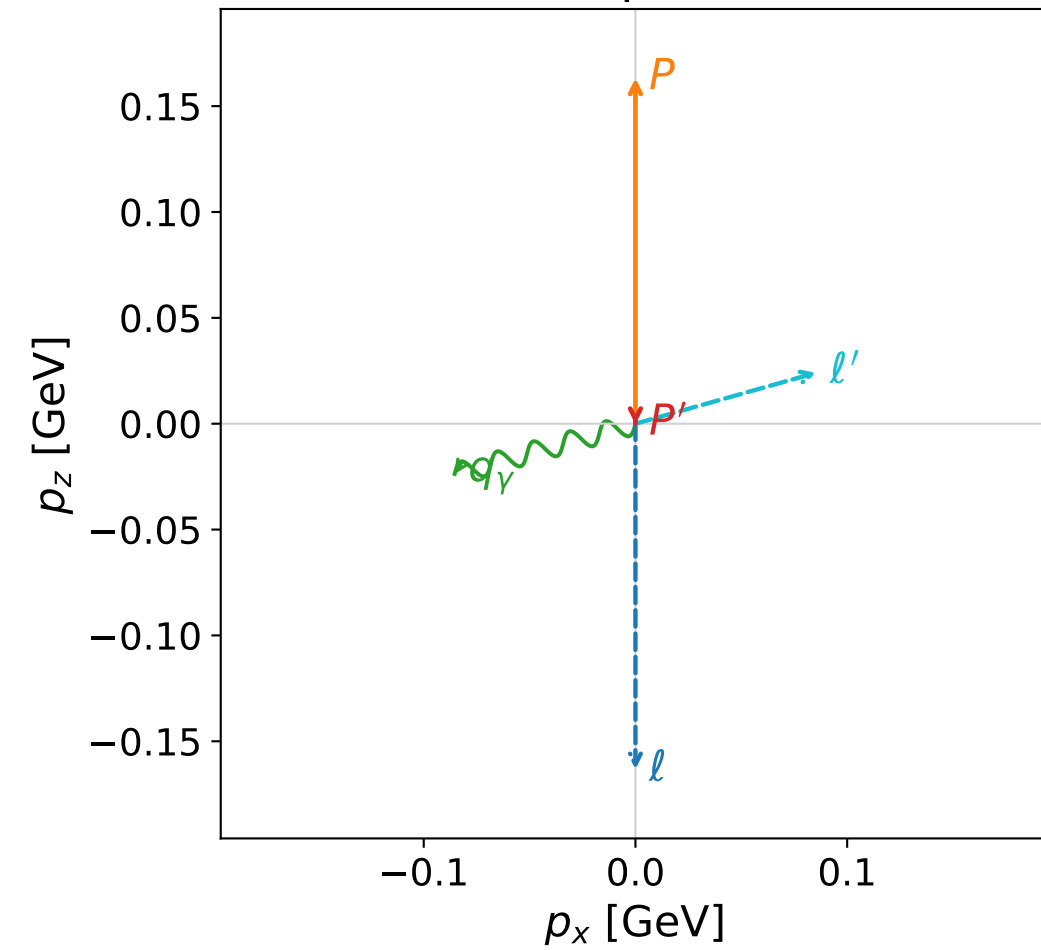
$s=1.31067$, $\sqrt{s}=1.14485$, $|\vec{P}|=0.18816$, $|\vec{P}'|=0.0245489$
 $\theta_{p'}=0$, $\theta_\gamma=1.94491$, $\phi_{p'}=3.06061$, $\phi_\gamma=3.14158$
 $E_\gamma=0.106426$, $\theta_{\ell'}=1.427$, $\phi_{\ell'}=6.28317$
 $Q^2=0.0430666$, $x_B=0.175998$, $t=-0.0264312$, $W^2=1.08148$, $y=0.567974$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1882$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1882$
 P $E= 0.9567$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1882$
 ℓ' $E= 0.1001$ $p_x= 0.0991$ $p_y= -0.0000$ $p_z= 0.0143$
 P' $E= 0.9383$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.0245$
 q_γ $E= 0.1064$ $p_x= -0.0991$ $p_y= 0.0000$ $p_z= -0.0389$



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



dw_mixing_angles_polarization_06_local_minimum_715

$D_W=0.00790999$

region: polarization_cluster_06_local_minimum_715

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0715

lepton mixing angle: $\alpha_e=2.3328713$ rad

incoming lepton: $-0.690424|+\rangle +0.723405|-\rangle$

proton mixing angle: $\alpha_p=1.763399$ rad

incoming proton: $-0.191414|+\rangle +0.981509|-\rangle$

$s=1.24396$, $\sqrt{s}=1.11533$, $|\vec{P}|=0.163233$, $|\vec{P}'|=0.000348371$

$\theta_{p'}=3.14148$, $\theta_\gamma=1.8417$, $\phi_{p'}=3.24294$, $\phi_\gamma=3.14161$

$E_\gamma=0.0886176$, $\theta_{\ell'}=1.29611$, $\phi_{\ell'}=1.5997e-05$

$Q^2=0.0368169$, $x_B=0.181322$, $t=-0.02656$, $W^2=1.04607$, $y=0.55764$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ $E= 0.1632$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1632$

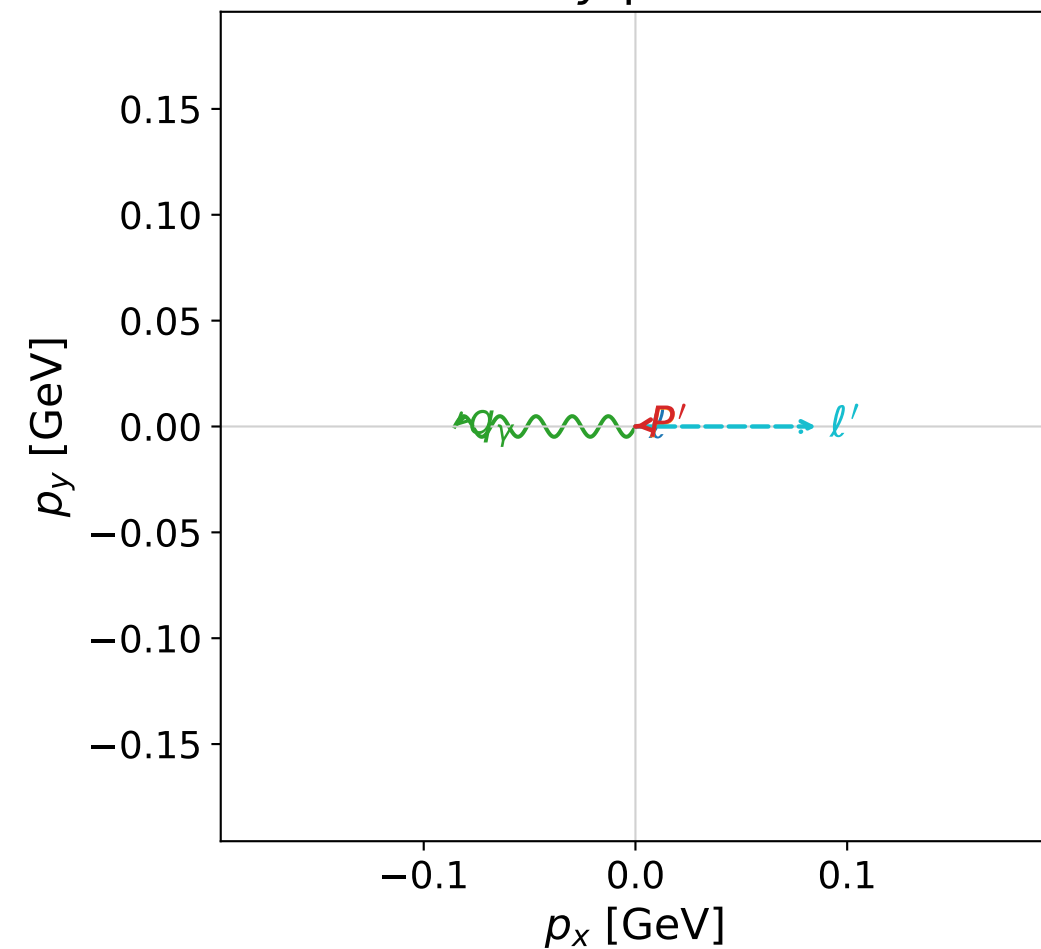
P $E= 0.9521$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1632$

ℓ' $E= 0.0887$ $p_x= 0.0854$ $p_y= 0.0000$ $p_z= 0.0241$

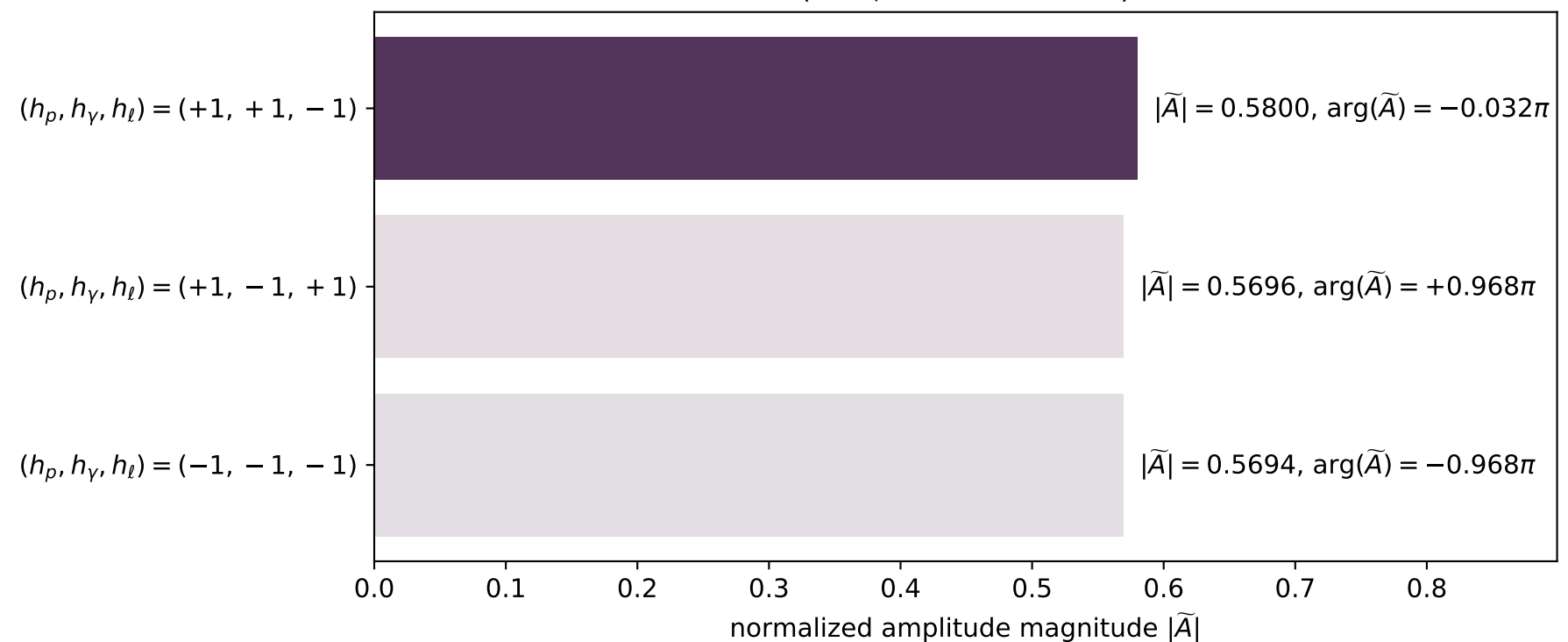
P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.0003$

q_γ $E= 0.0886$ $p_x= -0.0854$ $p_y= -0.0000$ $p_z= -0.0237$

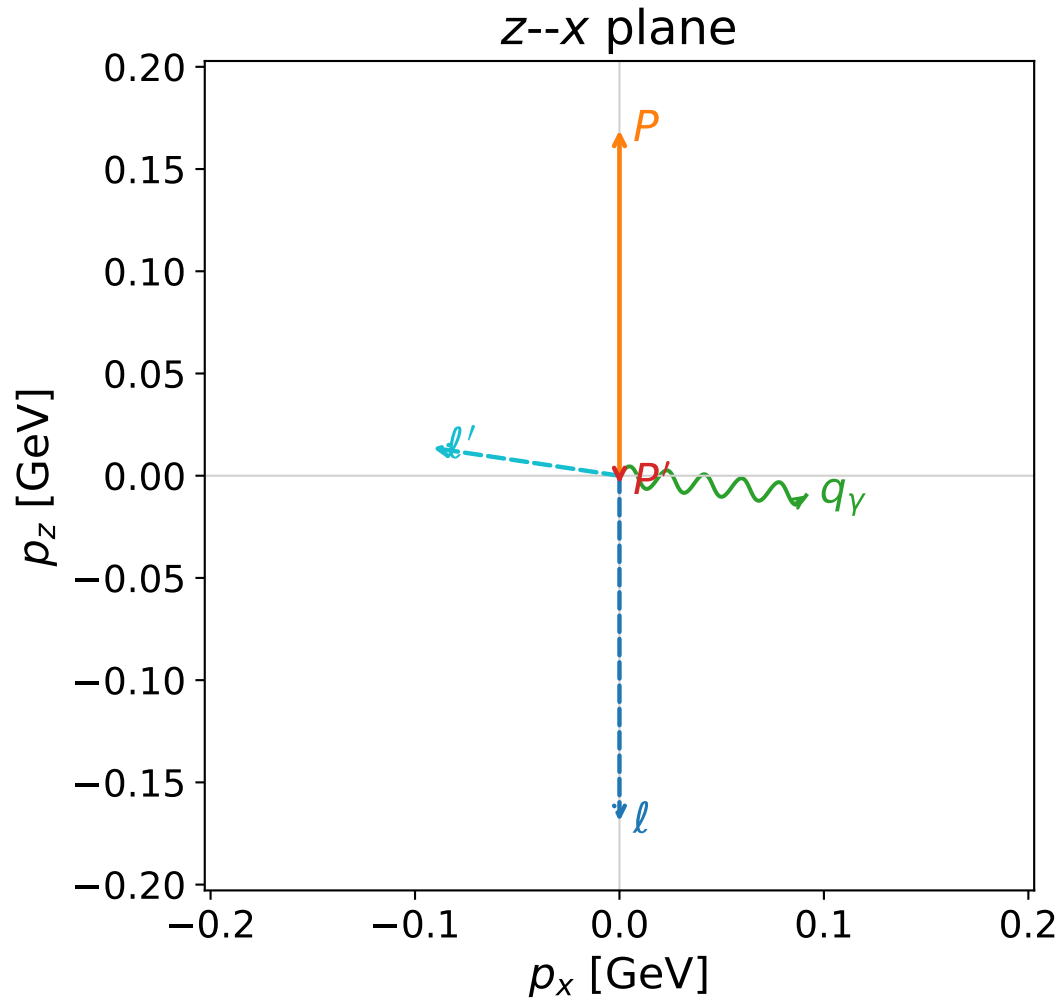
x--y plane



Leading normalized final-state amplitudes
(N=3, retained=98.5%)



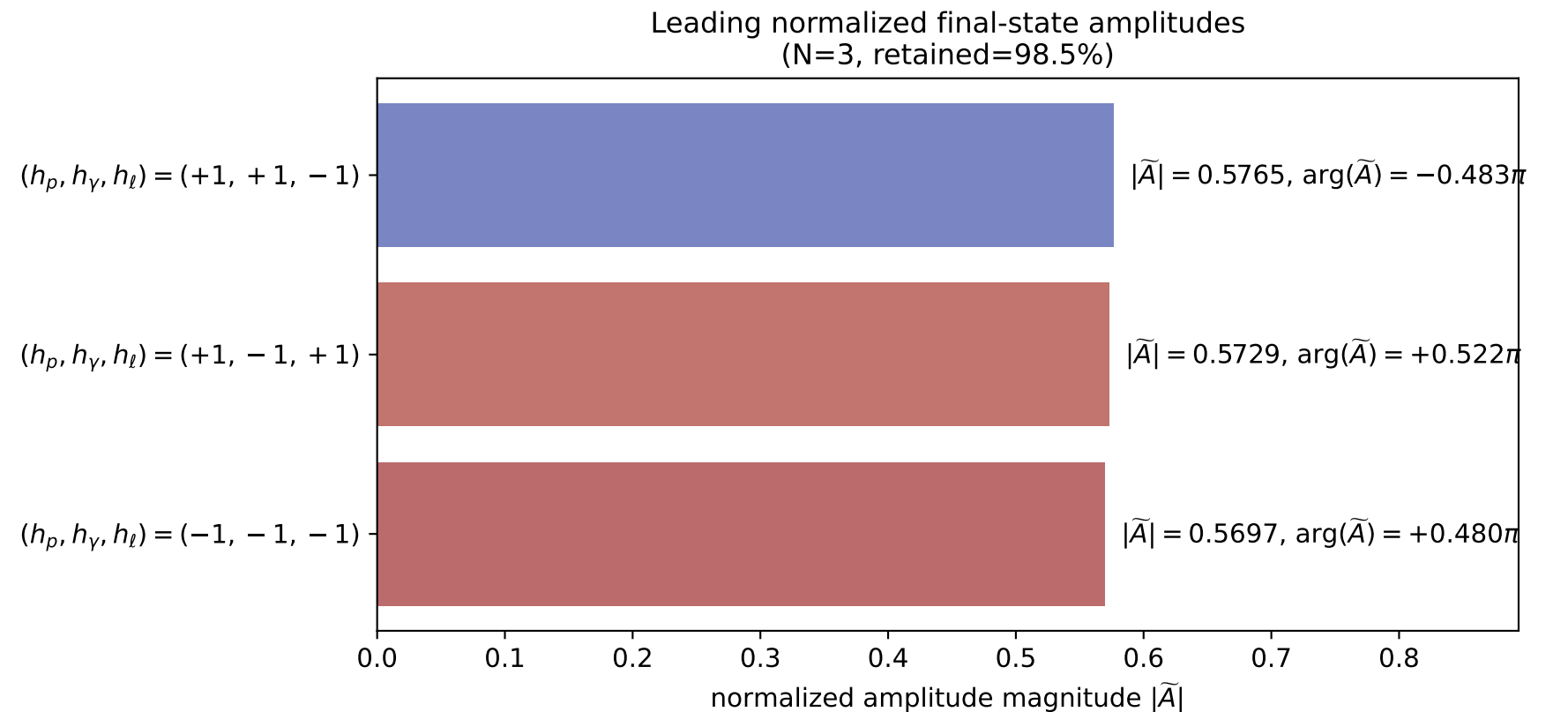
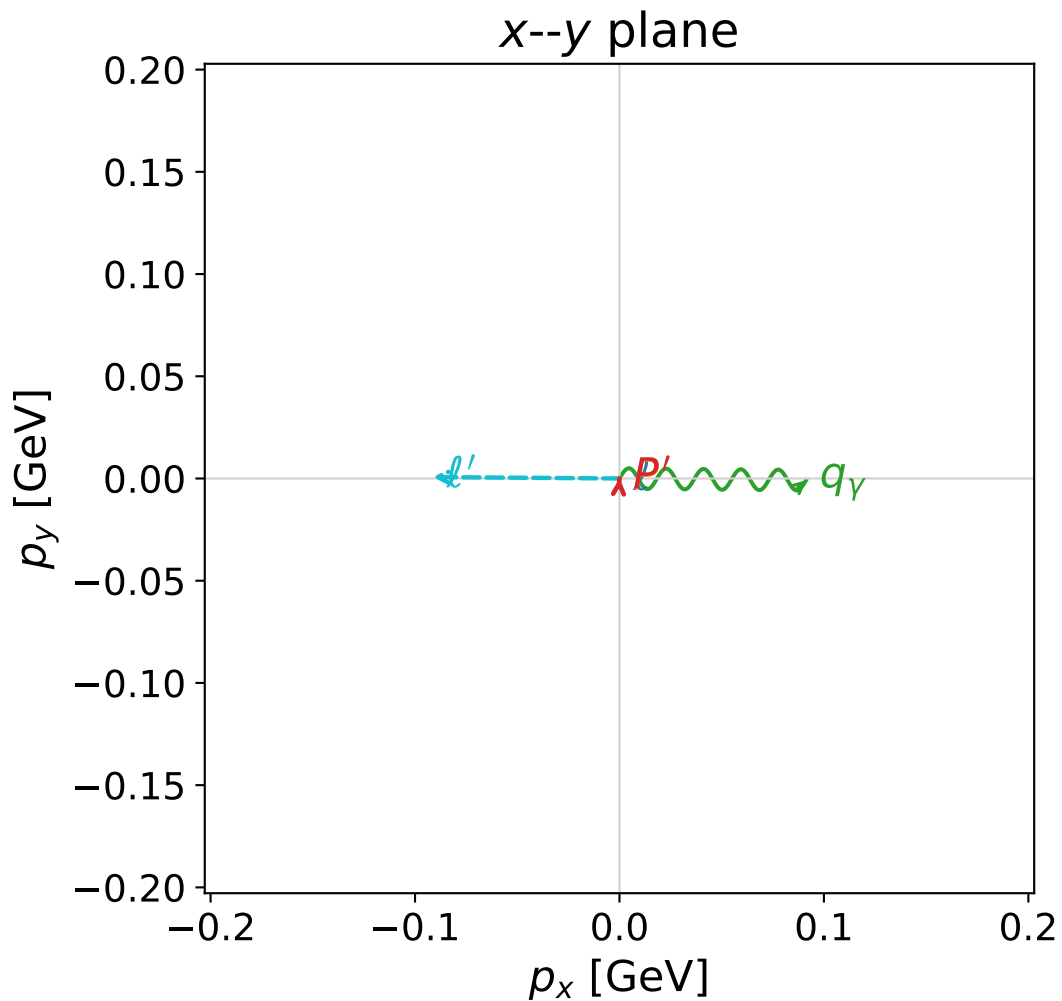
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)



```
dw_mixing_angles_polarization_06_local_minimum_759
D_W=0.00935482
region: polarization_cluster_06_local_minimum_759
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0759
lepton mixing angle: alpha_e=2.3545656 rad
incoming lepton: -0.705954|+> +0.708258|->
proton mixing angle: alpha_p=1.3737025 rad
incoming proton: 0.19582|+> +0.98064|->
```

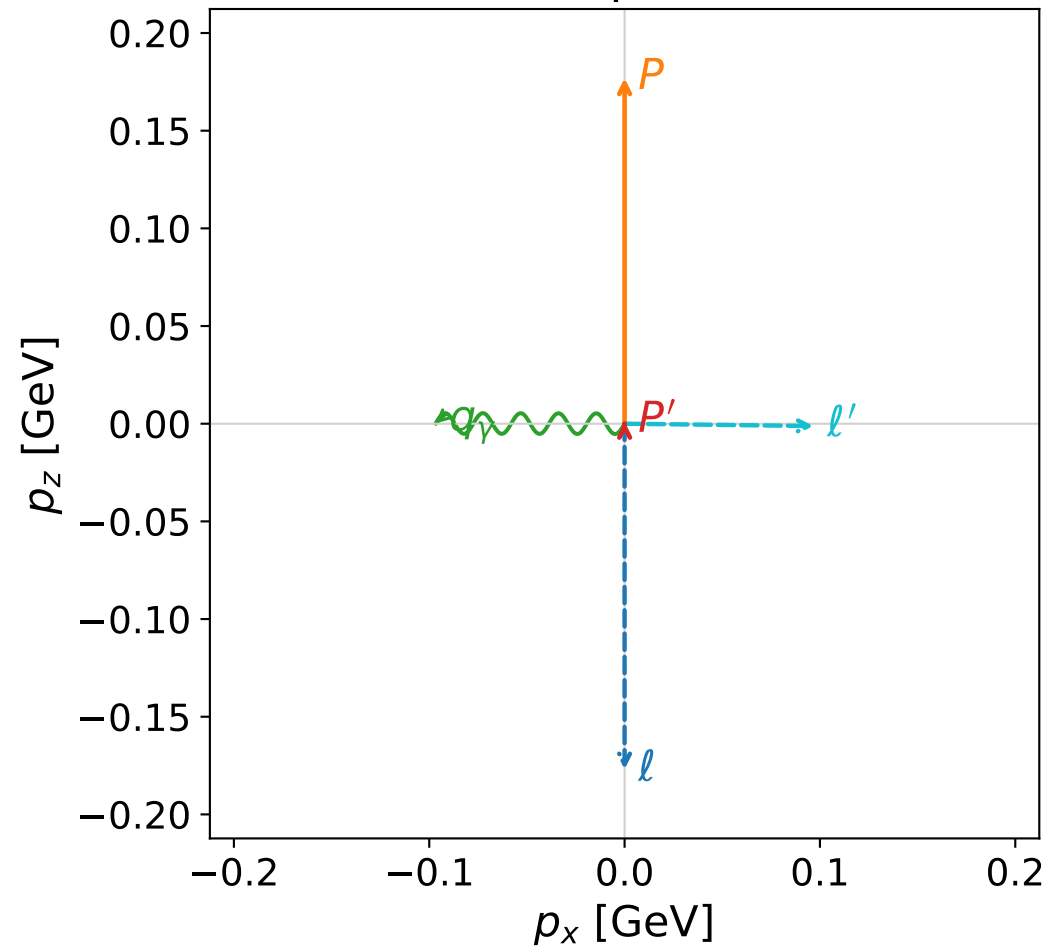
```
s=1.25922, sqrt(s)=1.12215, |P|=0.169039, |P'|=0.00397886
theta_p=3.10698, theta_gamma=1.67588, phi_p'=1.50837, phi_gamma=6.27405
E_gamma=0.091815, theta_l'=1.42287, phi_l'=3.13397
Q^2=0.0358132, x_B=0.172194, t=-0.0297062, W^2=1.05201, y=0.548223
```

```
four-momenta [E, p_x, p_y, p_z] GeV:
l      E=  0.1690 p_x=  0.0000 p_y=  0.0000 p_z= -0.1690
P      E=  0.9531 p_x=  0.0000 p_y=  0.0000 p_z=  0.1690
l'     E=  0.0923 p_x= -0.0913 p_y=  0.0007 p_z=  0.0136
P'     E=  0.9380 p_x=  0.0000 p_y=  0.0001 p_z= -0.0040
q_gamma E=  0.0918 p_x=  0.0913 p_y= -0.0008 p_z= -0.0096
```



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

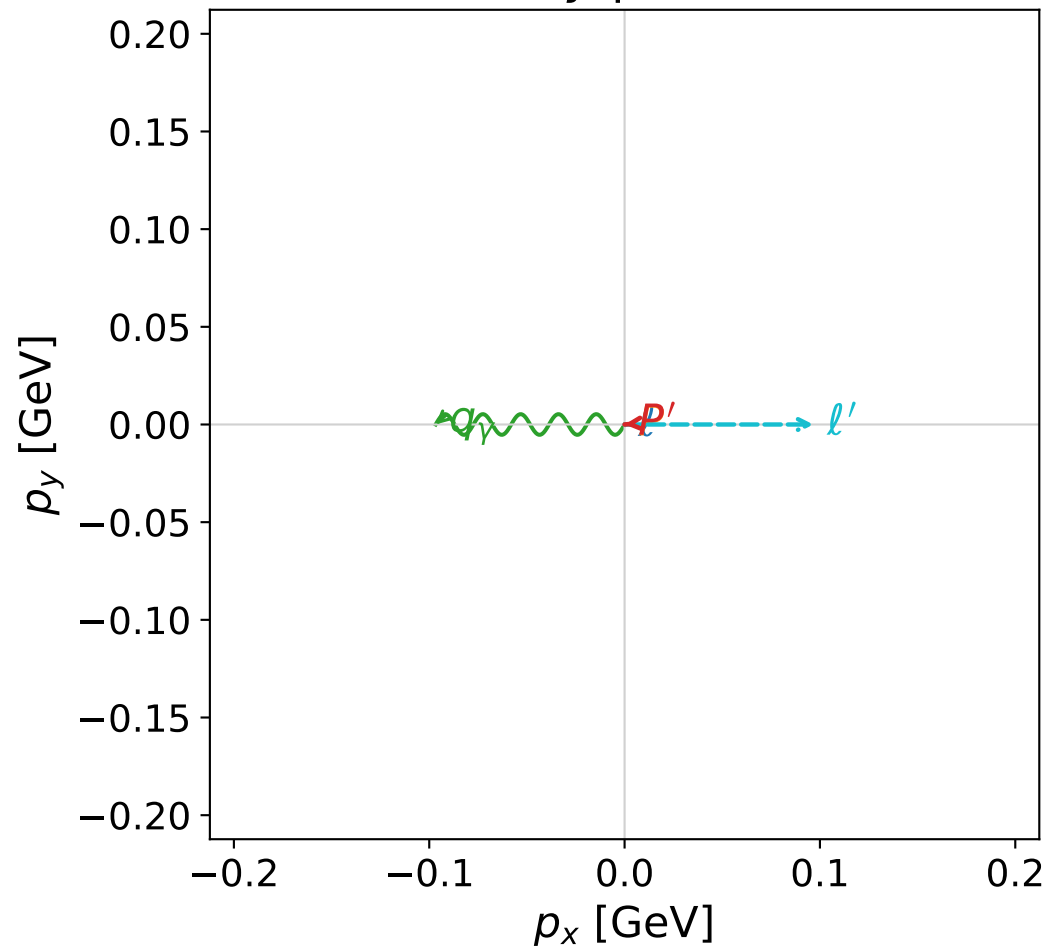


dw_mixing_angles_polarization_06_local_minimum_762
 $D_W=0.00938309$
 region: polarization_cluster_06_local_minimum_762
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0762
 lepton mixing angle: $\alpha_e=2.362513$ rad
 incoming lepton: $-0.71156|+\rangle +0.702625|-\rangle$
 proton mixing angle: $\alpha_p=1.7776891$ rad
 incoming proton: $-0.20542|+\rangle +0.978674|-\rangle$

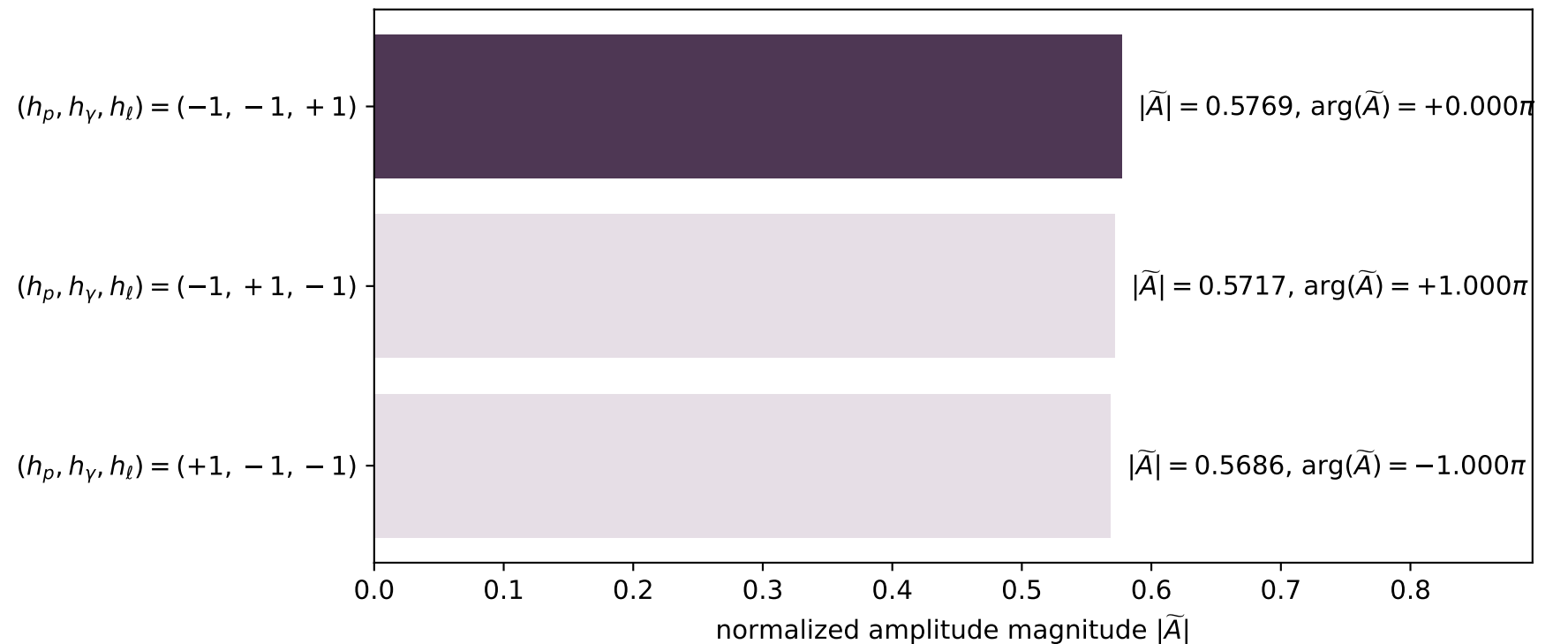
$s=1.28018$, $\sqrt{s}=1.13145$, $|\vec{P}|=0.176913$, $|\vec{P}'|=0.00103867$
 $\theta_{p'}=0.00343516$, $\theta_\gamma=1.56915$, $\phi_{p'}=3.22195$, $\phi_\gamma=3.14155$
 $E_\gamma=0.0967195$, $\theta_{\ell'}=1.58318$, $\phi_{\ell'}=6.28314$
 $Q^2=0.033802$, $x_B=0.157038$, $t=-0.0306584$, $W^2=1.06129$, $y=0.537662$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E=0.1769$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.1769$
 P $E=0.9545$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1769$
 ℓ' $E=0.0967$ $p_x=0.0967$ $p_y=-0.0000$ $p_z=-0.0012$
 P' $E=0.9380$ $p_x=-0.0000$ $p_y=-0.0000$ $p_z=0.0010$
 q_γ $E=0.0967$ $p_x=-0.0967$ $p_y=0.0000$ $p_z=0.0002$

x--y plane

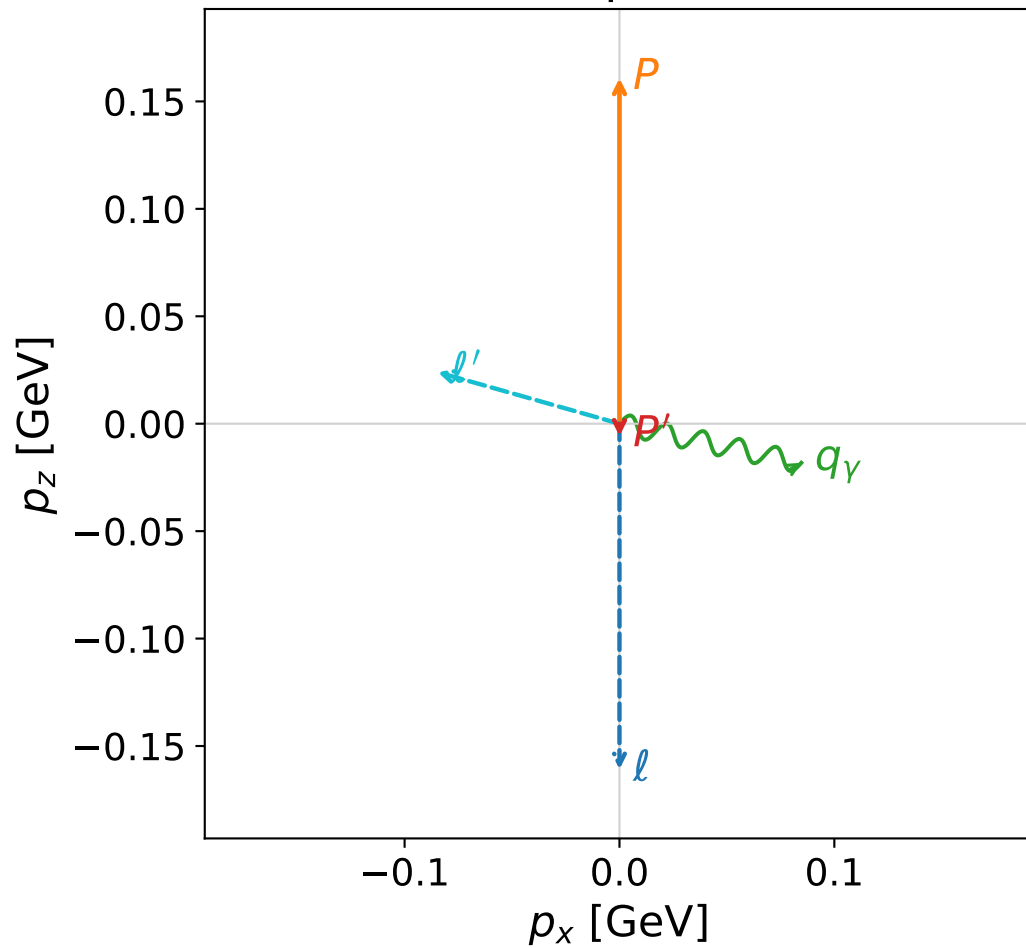


Leading normalized final-state amplitudes
 (N=3, retained=98.3%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

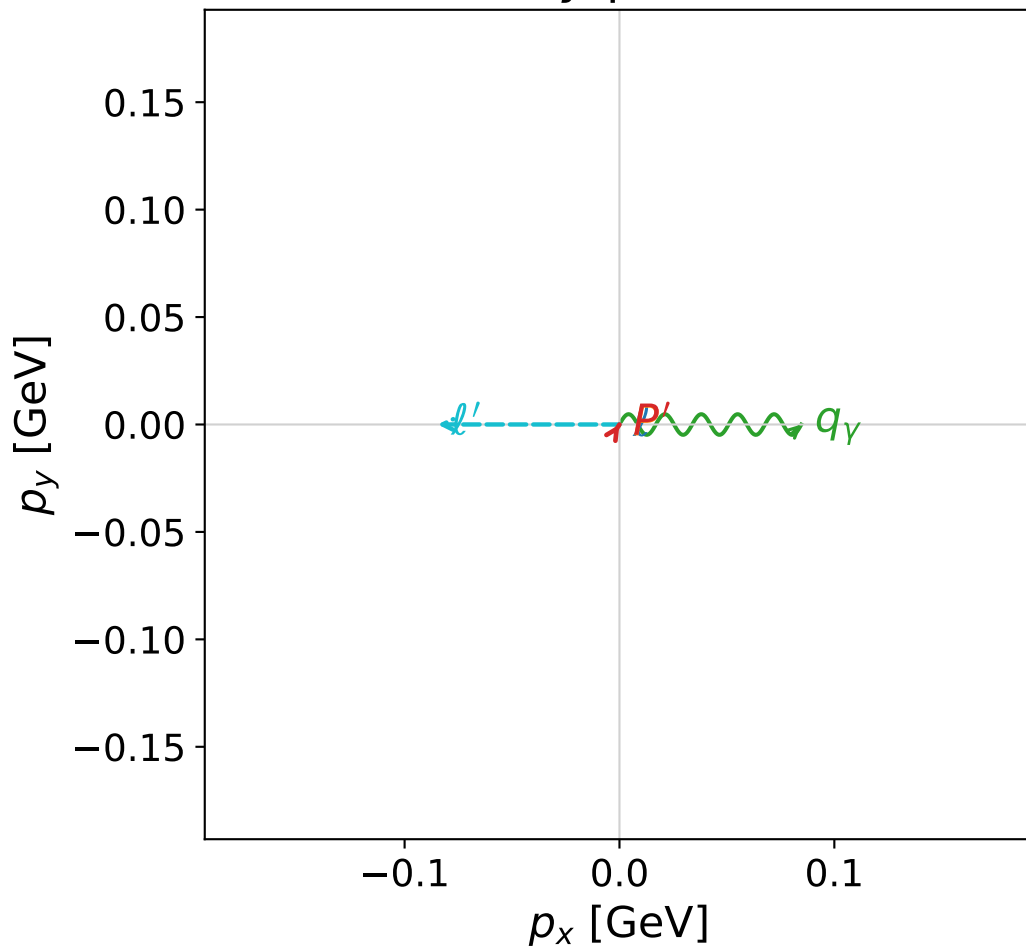


dw_mixing_angles_polarization_06_local_minimum_765
 $D_W=0.00940191$
 region: polarization_cluster_06_local_minimum_765
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0765
 lepton mixing angle: $\alpha_e=2.3440441$ rad
 incoming lepton: $-0.698463|+\rangle +0.715646|-\rangle$
 proton mixing angle: $\alpha_p=1.3862429$ rad
 incoming proton: $0.183508|+\rangle +0.983018|-\rangle$

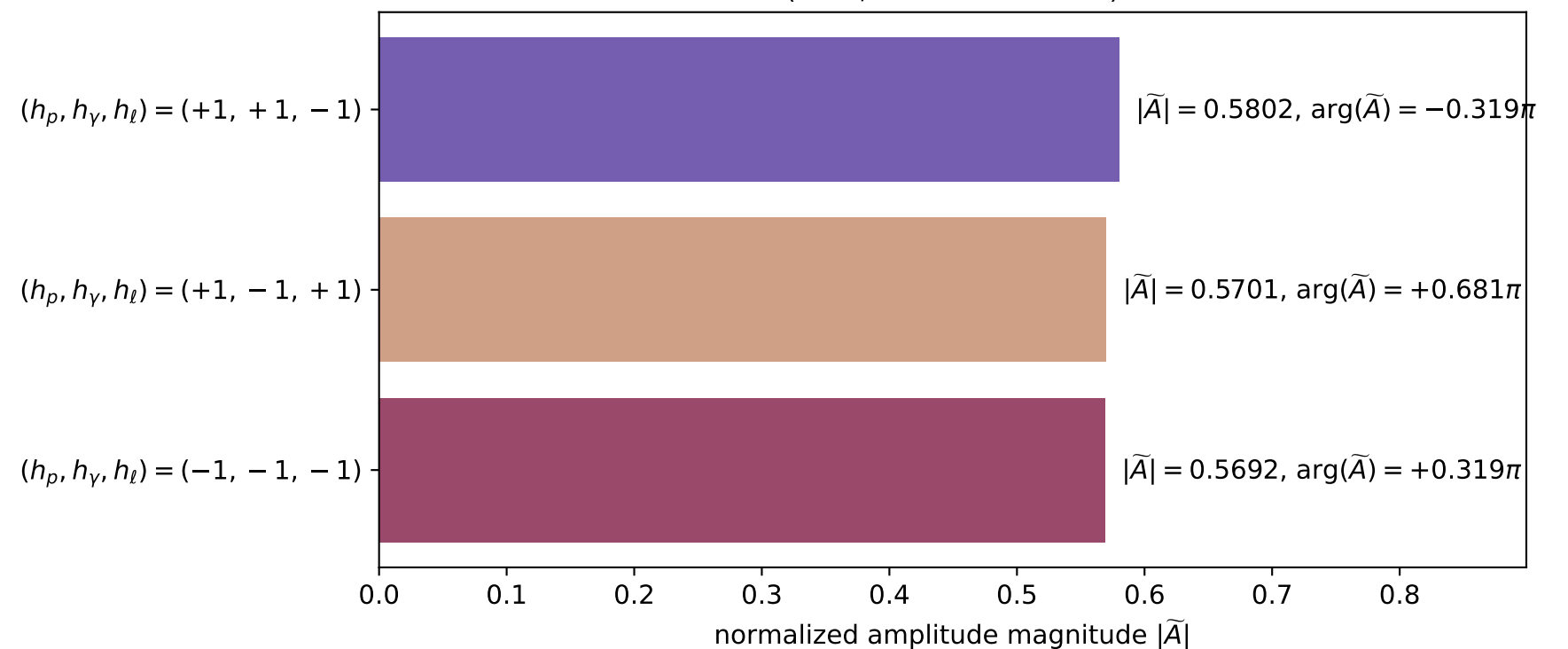
$s=1.23772$, $\sqrt{s}=1.11253$, $|\vec{P}|=0.16084$, $|\vec{P}'|=0.00576713$
 $\theta_{p'}=3.14098$, $\theta_\gamma=1.7826$, $\phi_{p'}=1.00097$, $\phi_\gamma=6.28294$
 $E_\gamma=0.086558$, $\theta_{\ell'}=1.29485$, $\phi_{\ell'}=3.14138$
 $Q^2=0.0360014$, $x_B=0.181663$, $t=-0.0275709$, $W^2=1.04202$, $y=0.553754$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E=0.1608$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.1608$
 P $E=0.9517$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1608$
 ℓ' $E=0.0880$ $p_x=-0.0846$ $p_y=0.0000$ $p_z=0.0240$
 P' $E=0.9380$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.0058$
 q_γ $E=0.0866$ $p_x=0.0846$ $p_y=-0.0000$ $p_z=-0.0182$

x--y plane

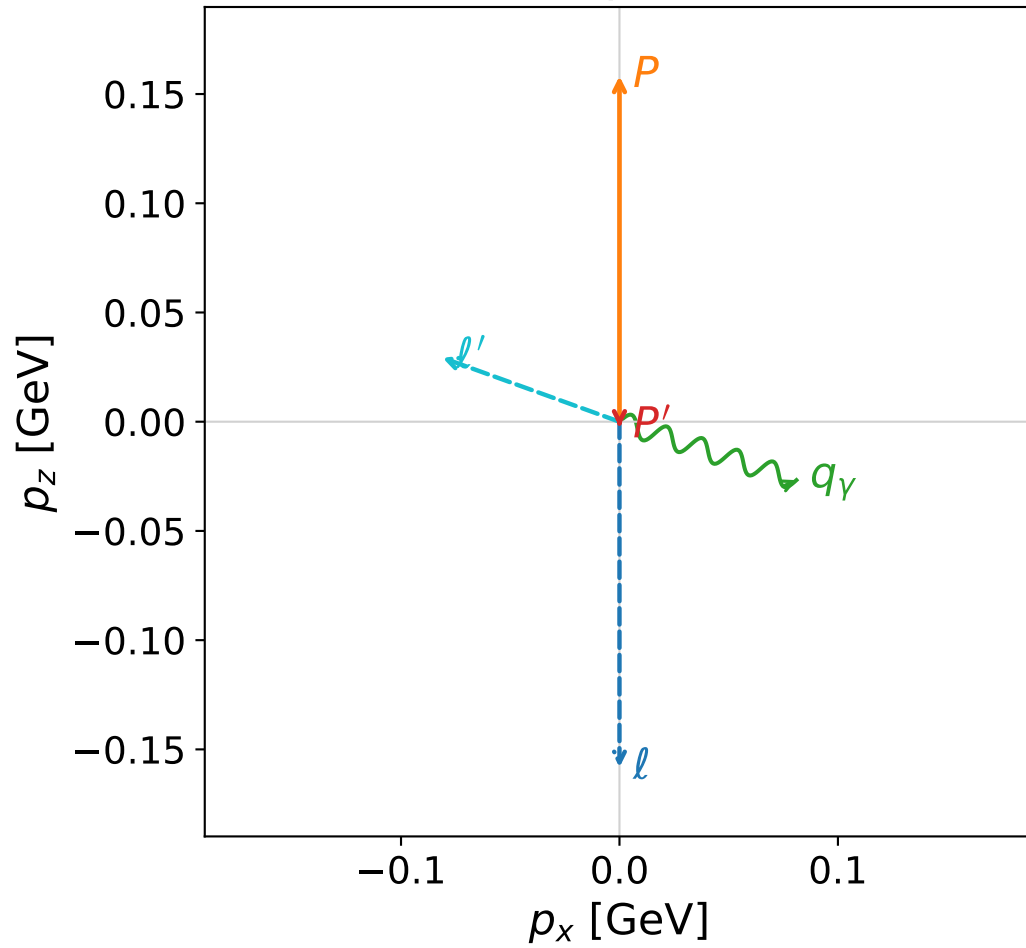


Leading normalized final-state amplitudes
 (N=3, retained=98.6%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

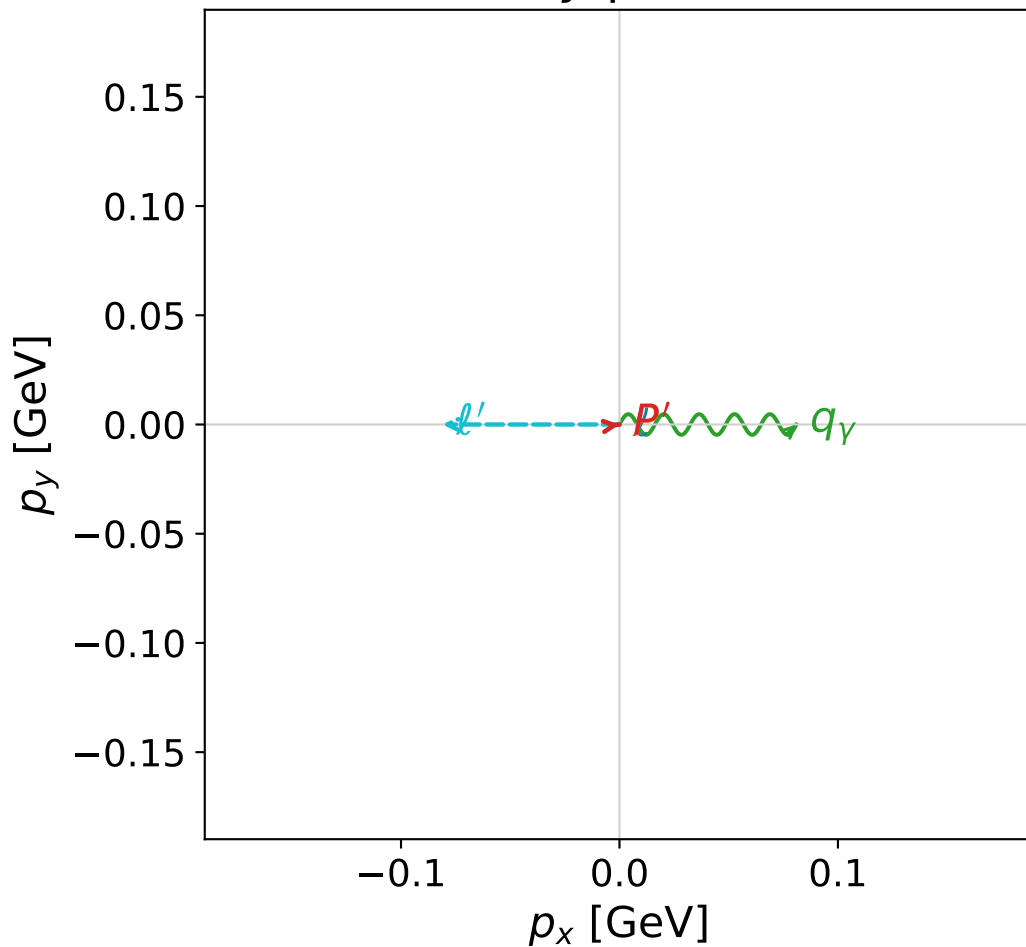


dw_mixing_angles_polarization_06_local_minimum_777
 $D_W=0.00968236$
 region: polarization_cluster_06_local_minimum_777
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0777
 lepton mixing angle: $\alpha_e=2.3407176$ rad
 incoming lepton: $-0.696079|+\rangle +0.717965|-\rangle$
 proton mixing angle: $\alpha_p=1.3995518$ rad
 incoming proton: $0.170409|+\rangle +0.985373|-\rangle$

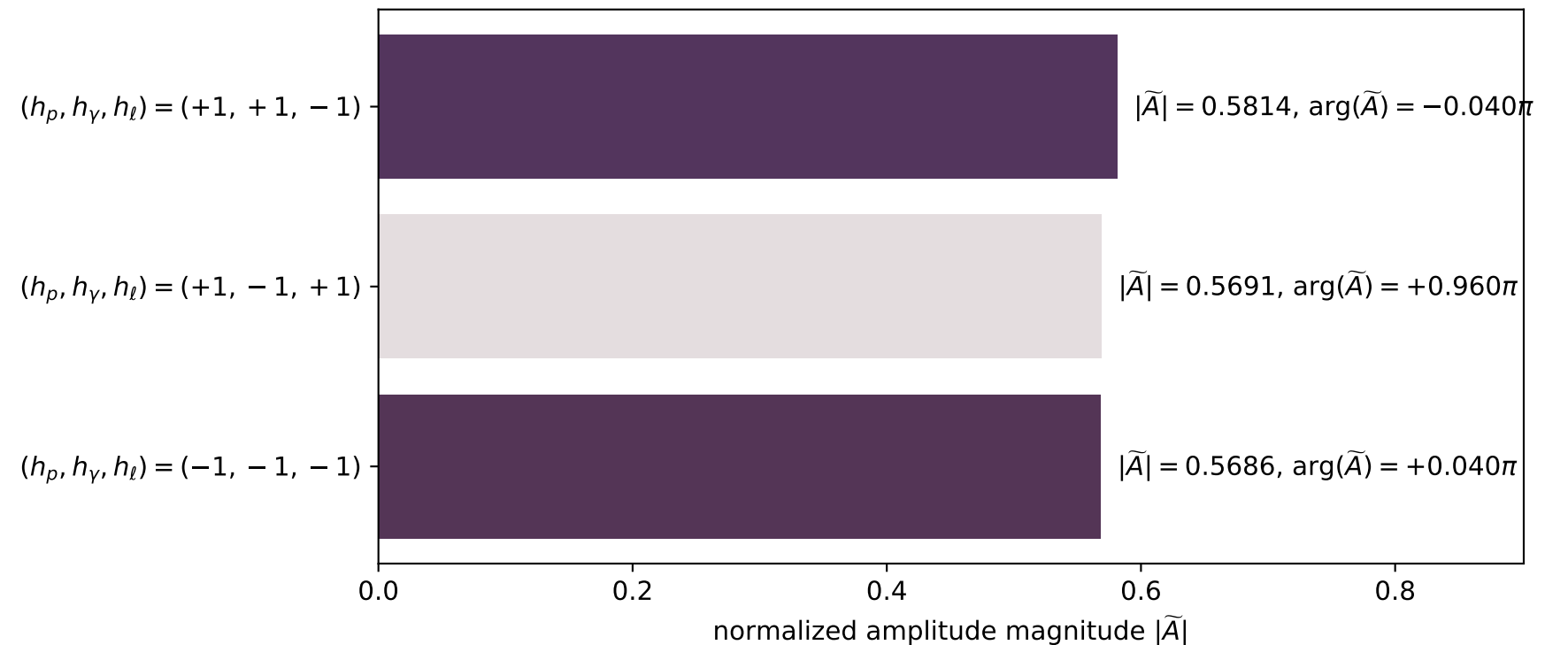
$s=1.23088$, $\sqrt{s}=1.10945$, $|\vec{P}|=0.158201$, $|\vec{P}'|=0.00251337$
 $\theta_{p'}=3.14152$, $\theta_\gamma=1.88965$, $\phi_{p'}=0.125414$, $\phi_\gamma=0$
 $E_\gamma=0.0853116$, $\theta_{\ell'}=1.22423$, $\phi_{\ell'}=3.14159$
 $Q^2=0.0365096$, $x_B=0.185875$, $t=-0.0256537$, $W^2=1.03975$, $y=0.559549$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1582$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1582$
 P $E= 0.9512$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1582$
 ℓ' $E= 0.0861$ $p_x= -0.0810$ $p_y= -0.0000$ $p_z= 0.0293$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0025$
 q_γ $E= 0.0853$ $p_x= 0.0810$ $p_y= 0.0000$ $p_z= -0.0267$

x--y plane

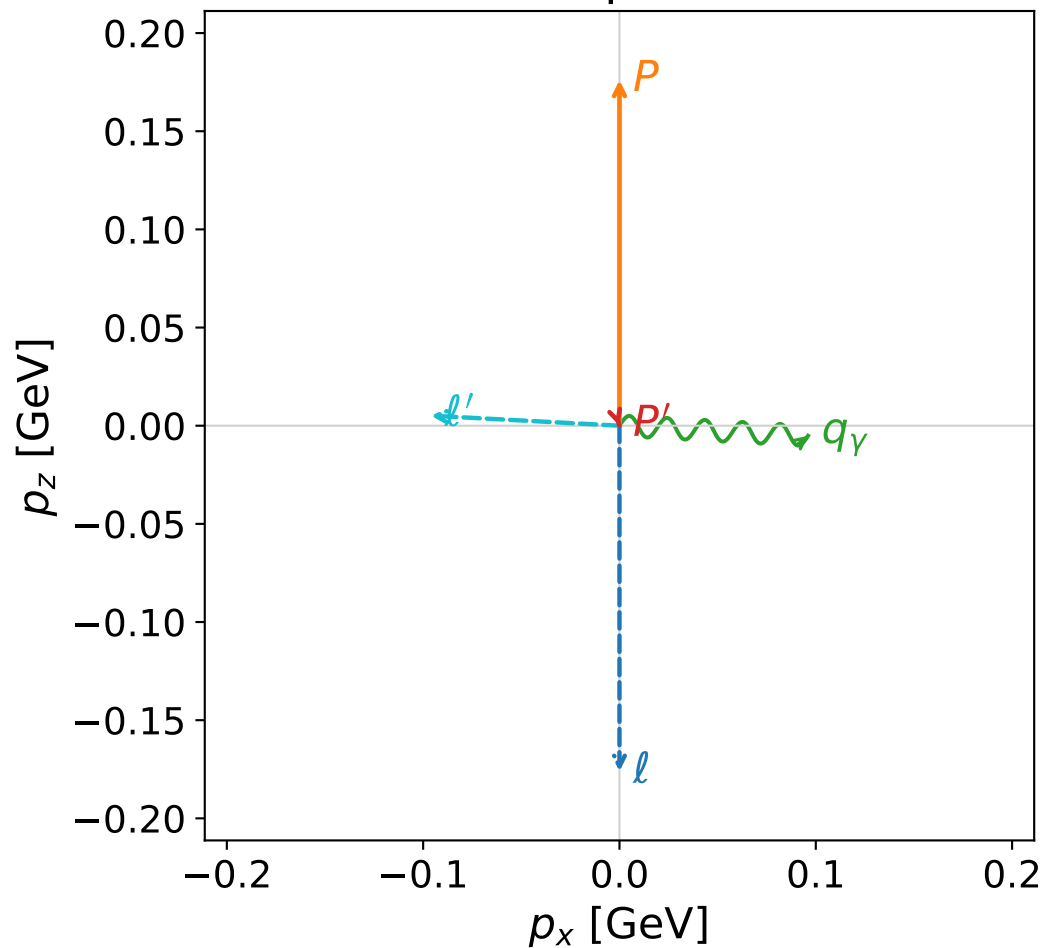


Leading normalized final-state amplitudes
 (N=3, retained=98.5%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



dw_mixing_angles_polarization_06_local_minimum_782

$D_W=0.00984539$

region: polarization_cluster_06_local_minimum_782

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0782

lepton mixing angle: $\alpha_e=2.3645136$ rad

incoming lepton: $-0.712965|+\rangle +0.7012|-\rangle$

proton mixing angle: $\alpha_p=1.3707723$ rad

incoming proton: $0.198693|+\rangle +0.980062|-\rangle$

$s=1.27783$, $\sqrt{s}=1.13041$, $|\vec{P}|=0.176036$, $|\vec{P}'|=4.9131e-08$

$\theta_{p'}=2.82852$, $\theta_\gamma=1.62411$, $\phi_{p'}=0.190647$, $\phi_\gamma=2.23335e-05$

$E_\gamma=0.0962054$, $\theta_{\ell'}=1.51749$, $\phi_{\ell'}=3.14162$

$Q^2=0.0356761$, $x_B=0.165047$, $t=-0.0307205$, $W^2=1.06033$, $y=0.543126$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ $E= 0.1760$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1760$

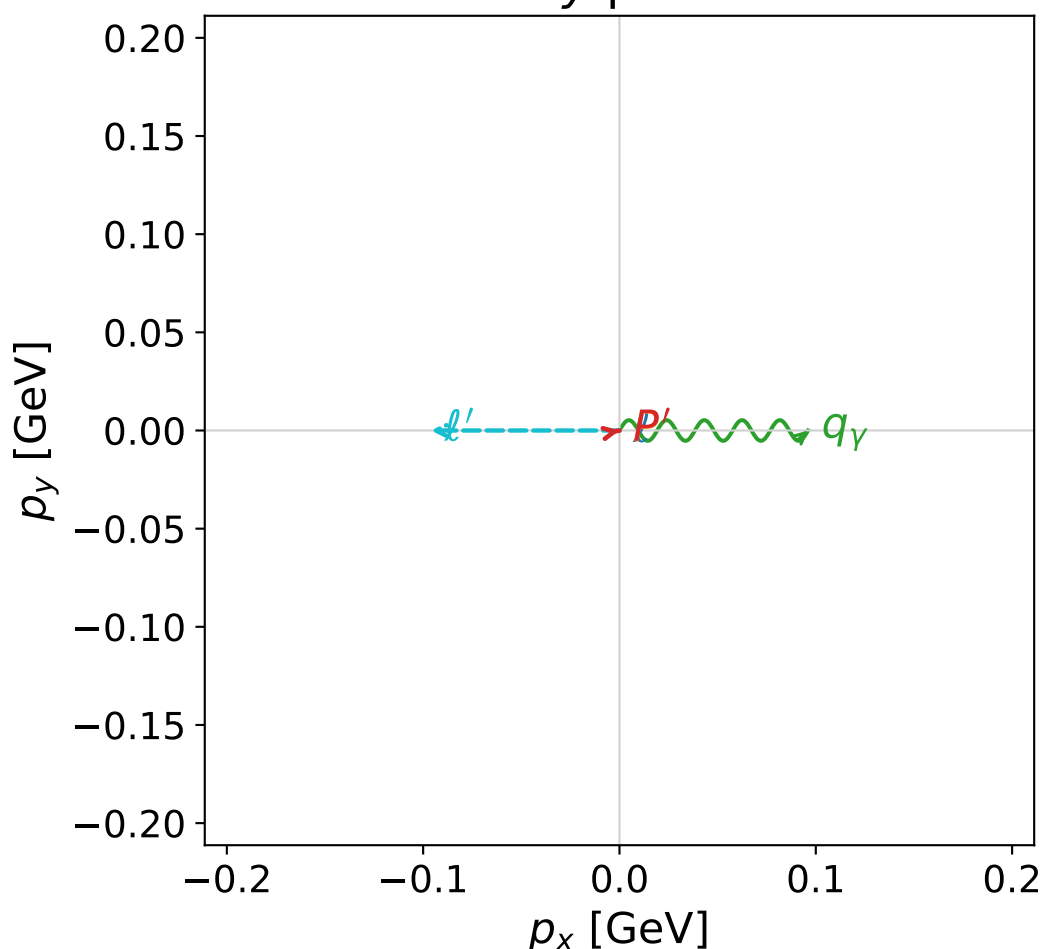
P $E= 0.9544$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1760$

ℓ' $E= 0.0962$ $p_x= -0.0961$ $p_y= -0.0000$ $p_z= 0.0051$

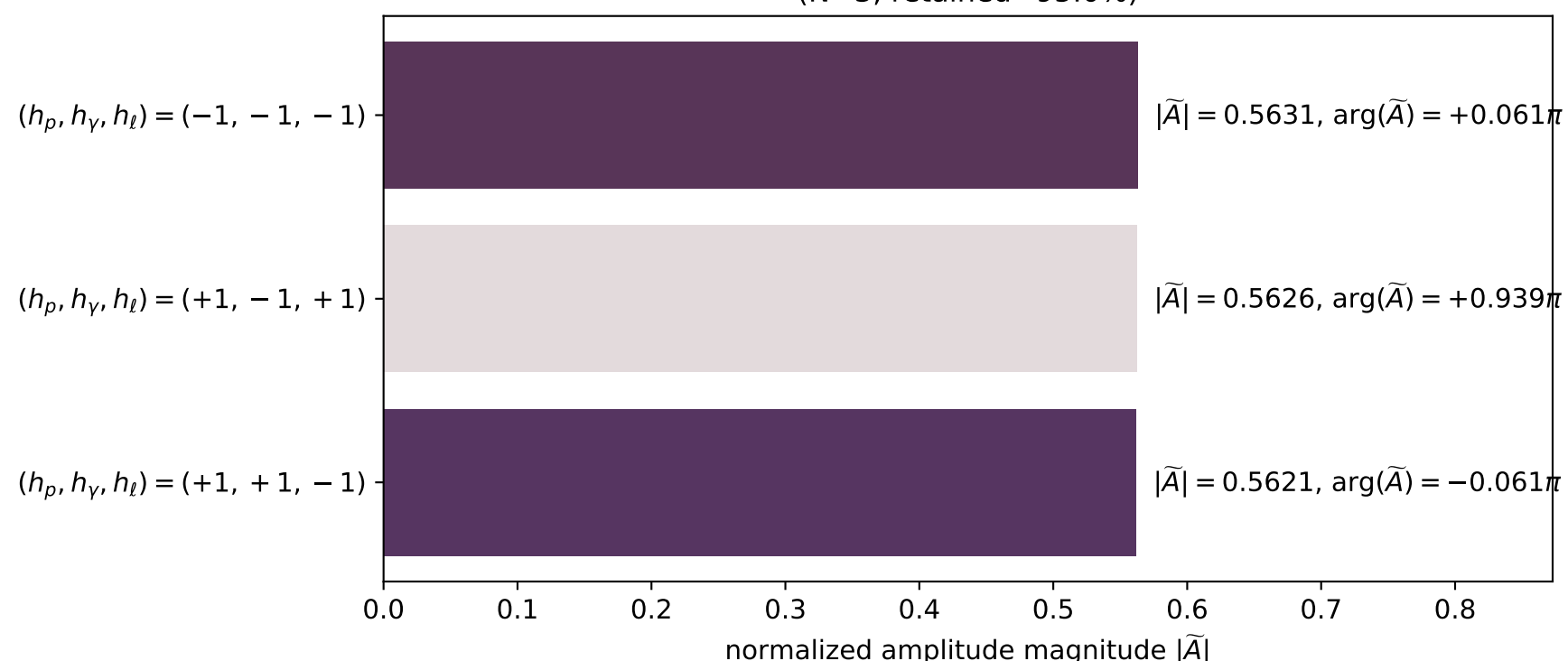
P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0000$

q_γ $E= 0.0962$ $p_x= 0.0961$ $p_y= 0.0000$ $p_z= -0.0051$

x--y plane

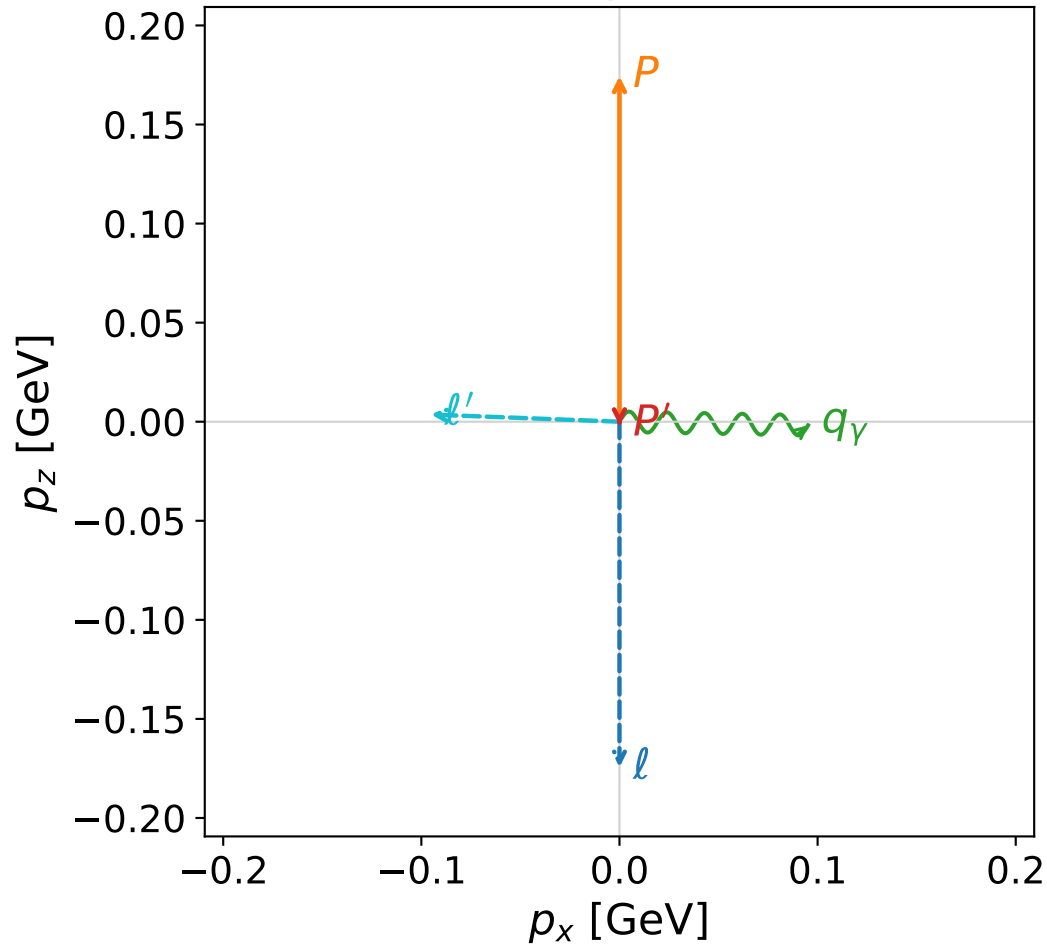


Leading normalized final-state amplitudes
(N=3, retained=95.0%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



dw_mixing_angles_polarization_06_local_minimum_786

$D_W=0.0099536$

region: polarization_cluster_06_local_minimum_786

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0786

lepton mixing angle: $\alpha_e=2.3653262$ rad

incoming lepton: $-0.713534|+\rangle +0.70062|-\rangle$

proton mixing angle: $\alpha_p=1.3705843$ rad

incoming proton: $0.198877|+\rangle +0.980024|-\rangle$

$s=1.27353$, $\sqrt{s}=1.12851$, $|\vec{P}|=0.174427$, $|\vec{P}'|=0.00184551$

$\theta_{p'}=3.05371$, $\theta_Y=1.58959$, $\phi_{p'}=1.53975$, $\phi_Y=6.27289$

$E_Y=0.0952242$, $\theta_{\ell'}=1.53271$, $\phi_{\ell'}=3.133$

$Q^2=0.0345044$, $x_B=0.161888$, $t=-0.0308108$, $W^2=1.05848$, $y=0.541393$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ $E= 0.1744$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1744$

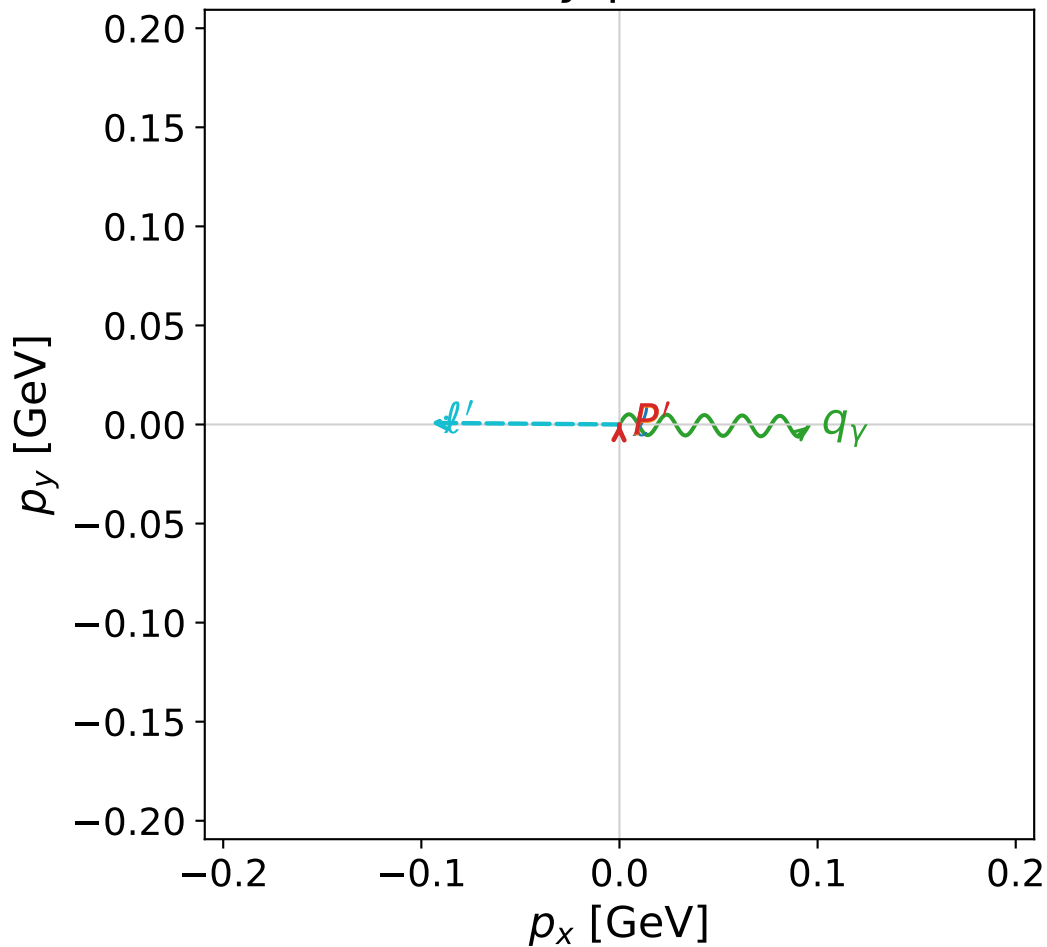
P $E= 0.9541$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1744$

ℓ' $E= 0.0953$ $p_x= -0.0952$ $p_y= 0.0008$ $p_z= 0.0036$

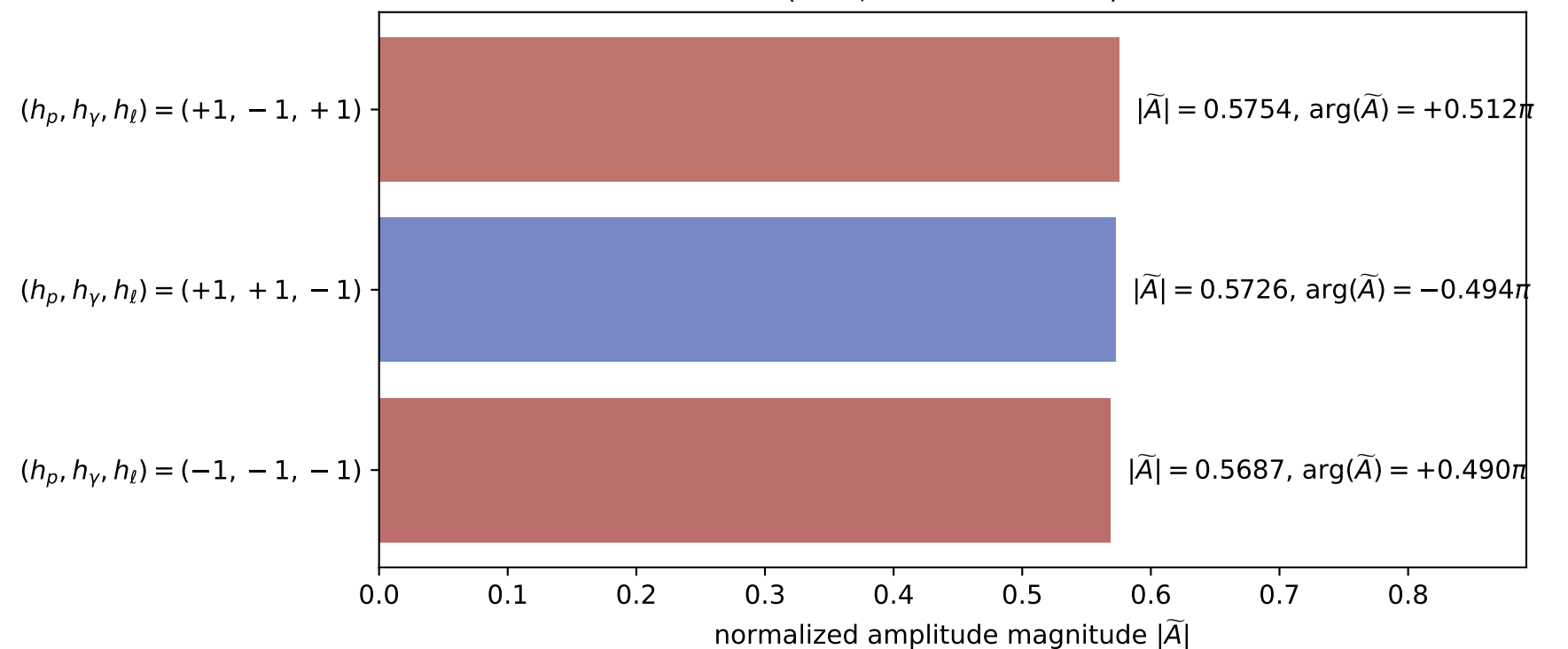
P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0002$ $p_z= -0.0018$

q_Y $E= 0.0952$ $p_x= 0.0952$ $p_y= -0.0010$ $p_z= -0.0018$

x--y plane

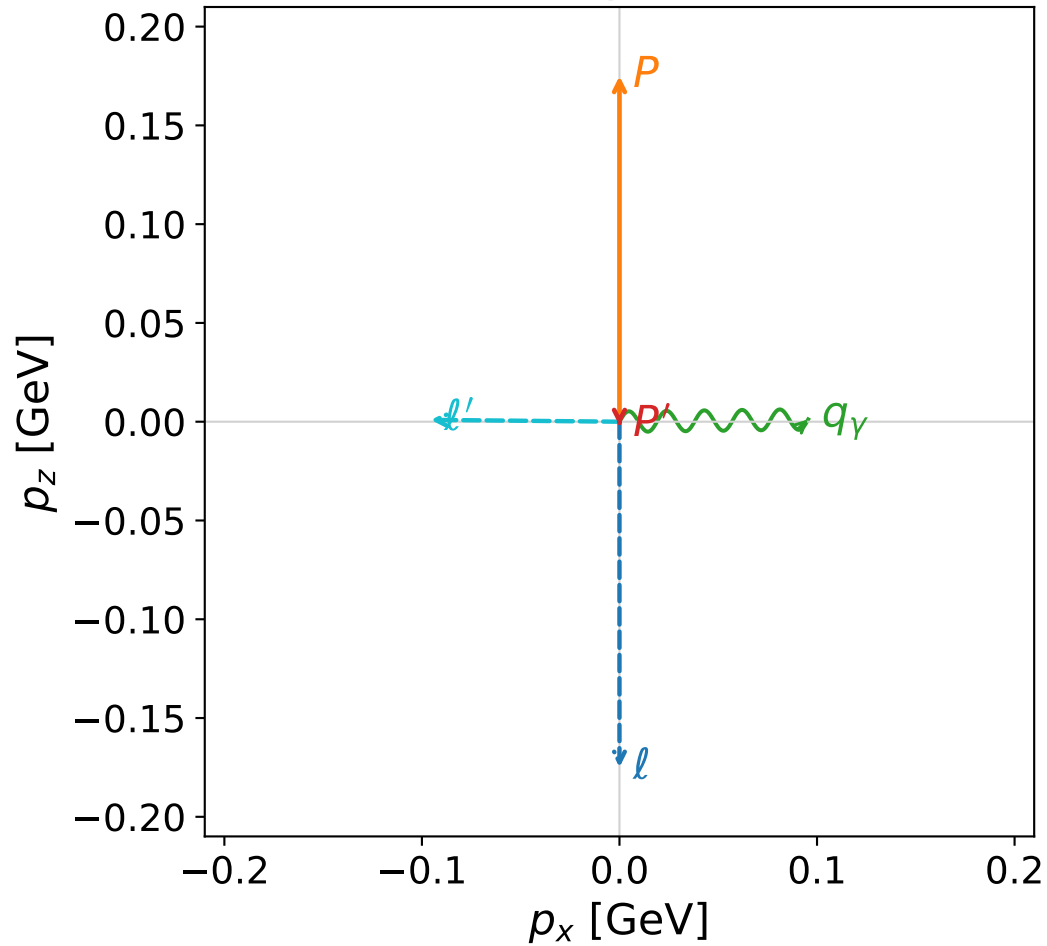


Leading normalized final-state amplitudes
(N=3, retained=98.2%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

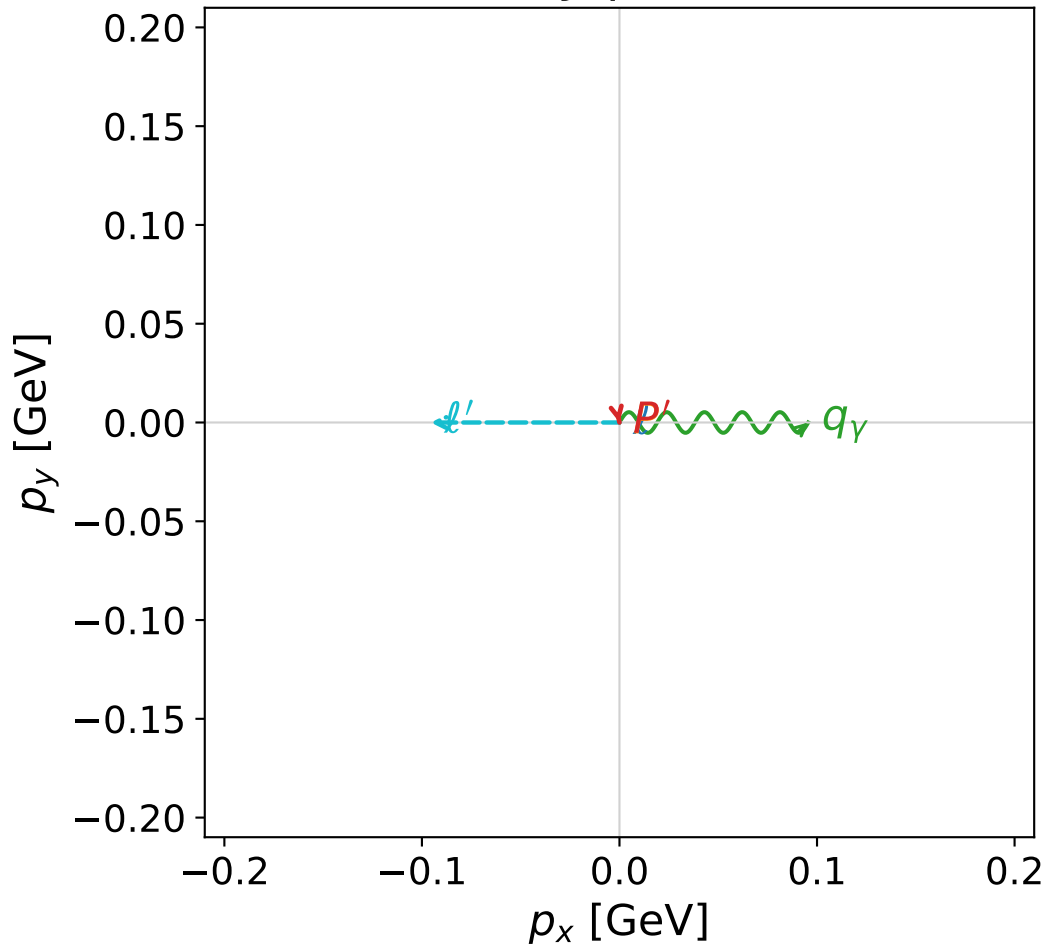


dw_mixing_angles_polarization_06_local_minimum_795
 $D_W=0.0102153$
 region: polarization_cluster_06_local_minimum_795
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0795
 lepton mixing angle: $\alpha_e=2.3684745$ rad
 incoming lepton: $-0.715737|+\rangle +0.69837|-\rangle$
 proton mixing angle: $\alpha_p=1.3705715$ rad
 incoming proton: $0.19889|+\rangle +0.980022|-\rangle$

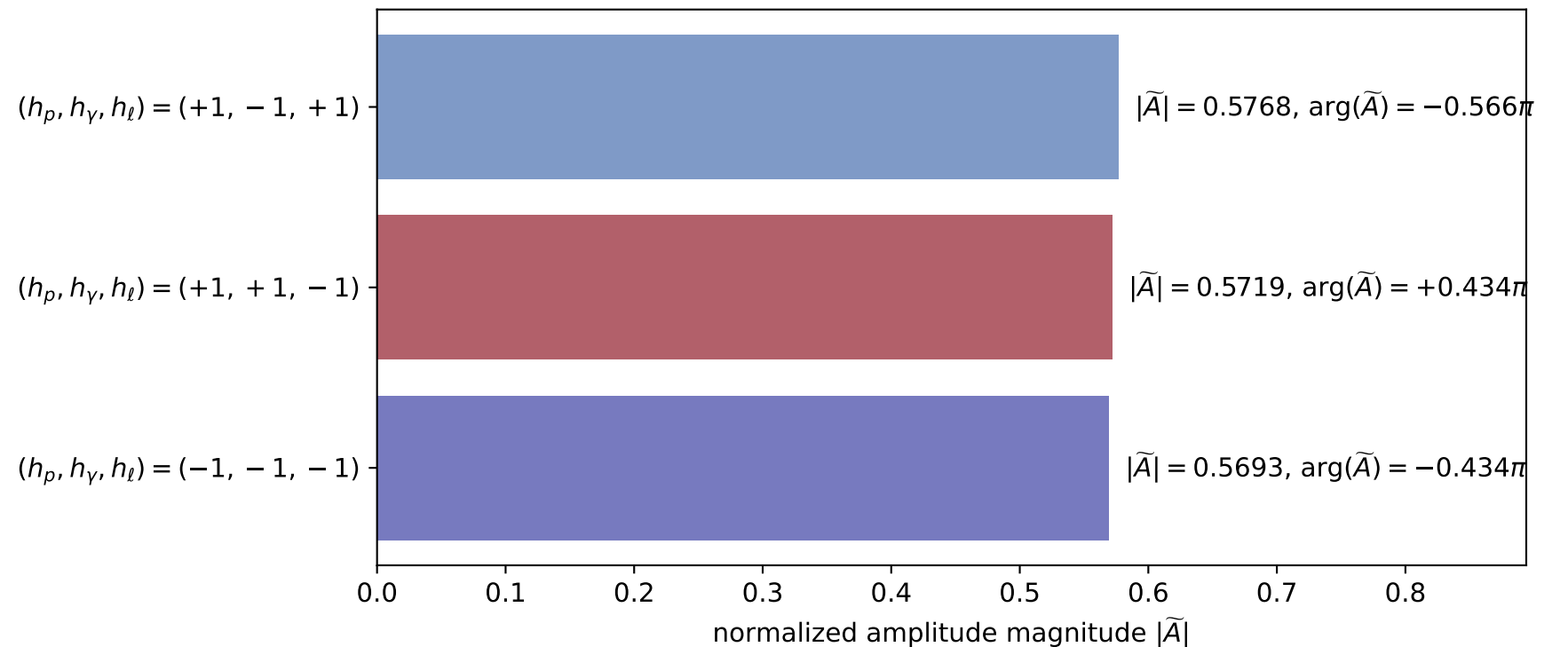
$s=1.2749$, $\sqrt{s}=1.12911$, $|\vec{P}|=0.174939$, $|\vec{P}'|=0.00210552$
 $\theta_{p'}=3.14096$, $\theta_\gamma=1.55855$, $\phi_{p'}=4.91864$, $\phi_\gamma=0.000108903$
 $E_\gamma=0.095556$, $\theta_{l'}=1.56101$, $\phi_{l'}=3.14169$
 $Q^2=0.0337594$, $x_B=0.158474$, $t=-0.0310832$, $W^2=1.05911$, $y=0.539239$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.1749$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1749$
 P $E= 0.9542$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1749$
 l' $E= 0.0956$ $p_x= -0.0955$ $p_y= -0.0000$ $p_z= 0.0009$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.0021$
 q_γ $E= 0.0956$ $p_x= 0.0955$ $p_y= 0.0000$ $p_z= 0.0012$

x--y plane

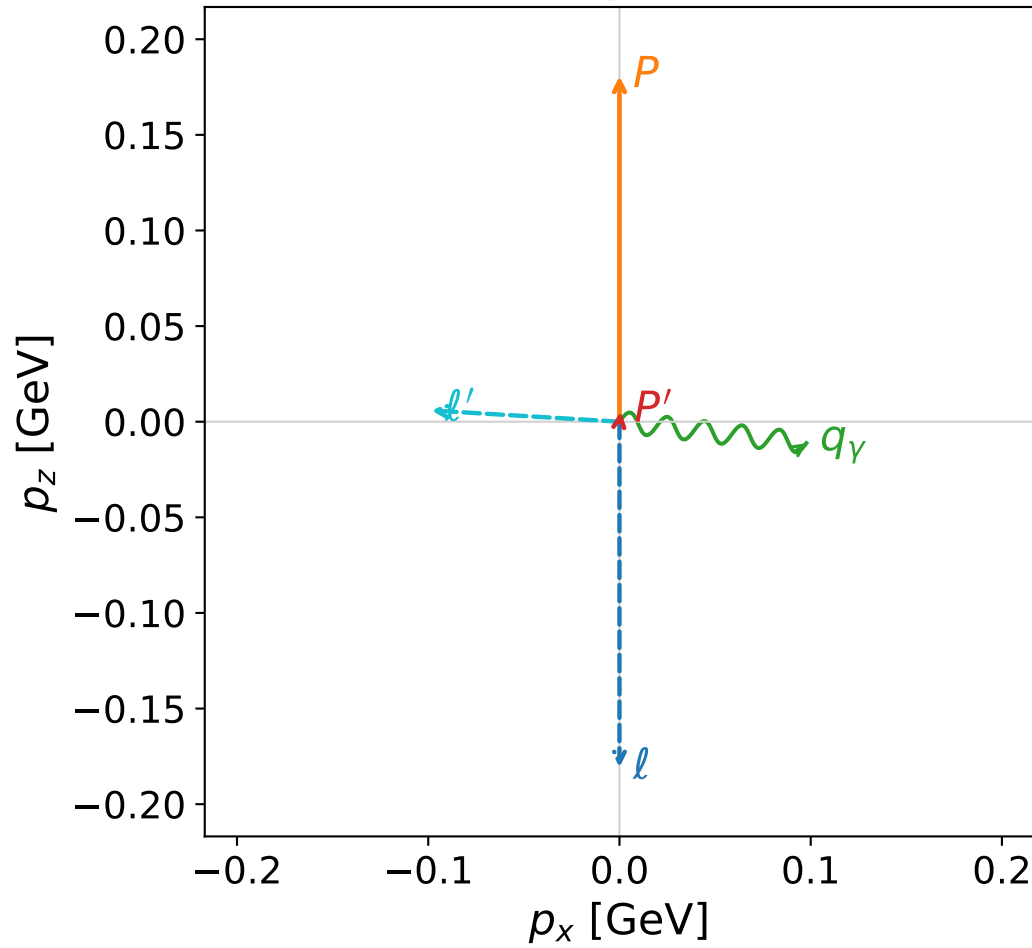


Leading normalized final-state amplitudes
 (N=3, retained=98.4%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

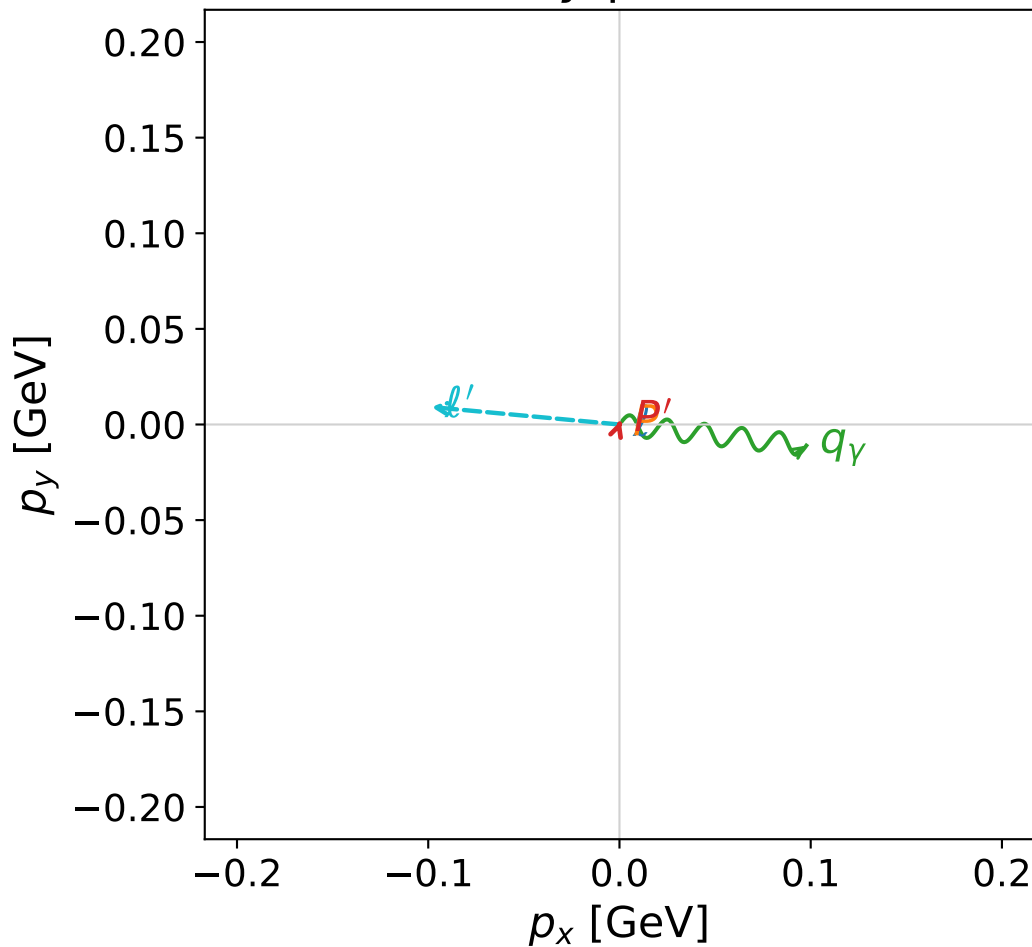


dw_mixing_angles_polarization_06_local_minimum_812
 $D_W=0.0107861$
 region: polarization_cluster_06_local_minimum_812
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0812
 lepton mixing angle: $\alpha_e=2.3638088$ rad
 incoming lepton: $-0.71247|+\rangle +0.701702|-\rangle$
 proton mixing angle: $\alpha_p=1.3635838$ rad
 incoming proton: $0.205733|+\rangle +0.978608|-\rangle$

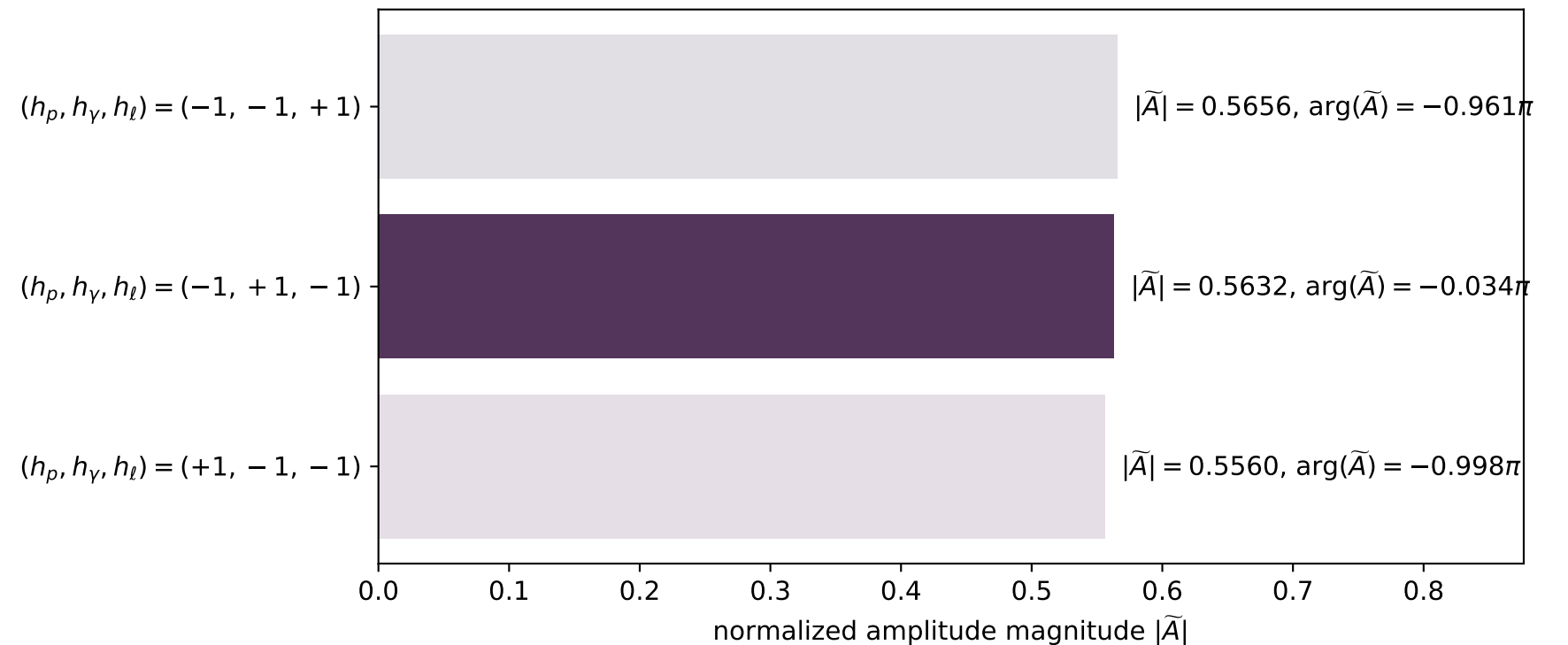
$s=1.29042$, $\sqrt{s}=1.13597$, $|\vec{P}|=0.180716$, $|\vec{P}'|=0.00566642$
 $\theta_{p'}=0.359605$, $\theta_\gamma=1.68413$, $\phi_{p'}=1.24282$, $\phi_\gamma=6.1708$
 $E_\gamma=0.0989809$, $\theta_{\ell'}=1.51125$, $\phi_{\ell'}=3.04893$
 $Q^2=0.0378986$, $x_B=0.169474$, $t=-0.0304764$, $W^2=1.06557$, $y=0.544662$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E=0.1807$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.1807$
 P $E=0.9552$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1807$
 ℓ' $E=0.0990$ $p_x=-0.0984$ $p_y=0.0091$ $p_z=0.0059$
 P' $E=0.9380$ $p_x=0.0006$ $p_y=0.0019$ $p_z=0.0053$
 q_γ $E=0.0990$ $p_x=0.0977$ $p_y=-0.0110$ $p_z=-0.0112$

x--y plane

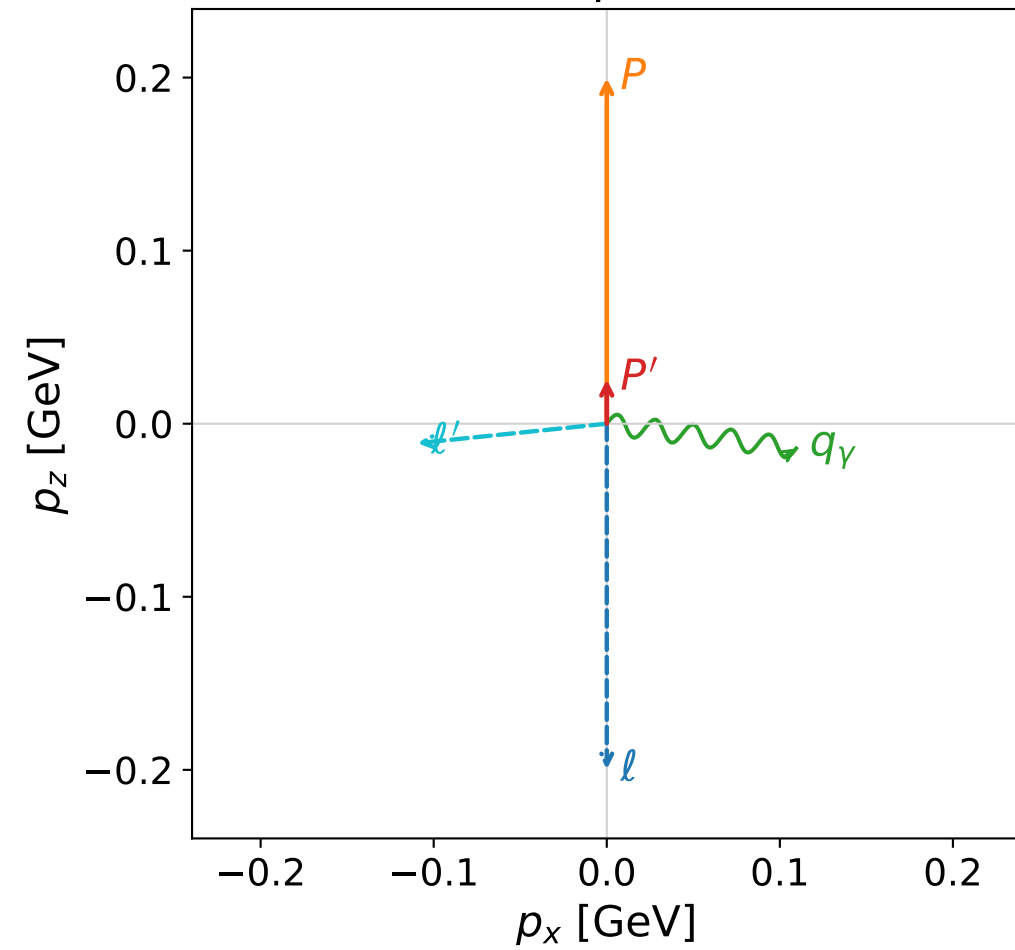


Leading normalized final-state amplitudes
 (N=3, retained=94.6%)

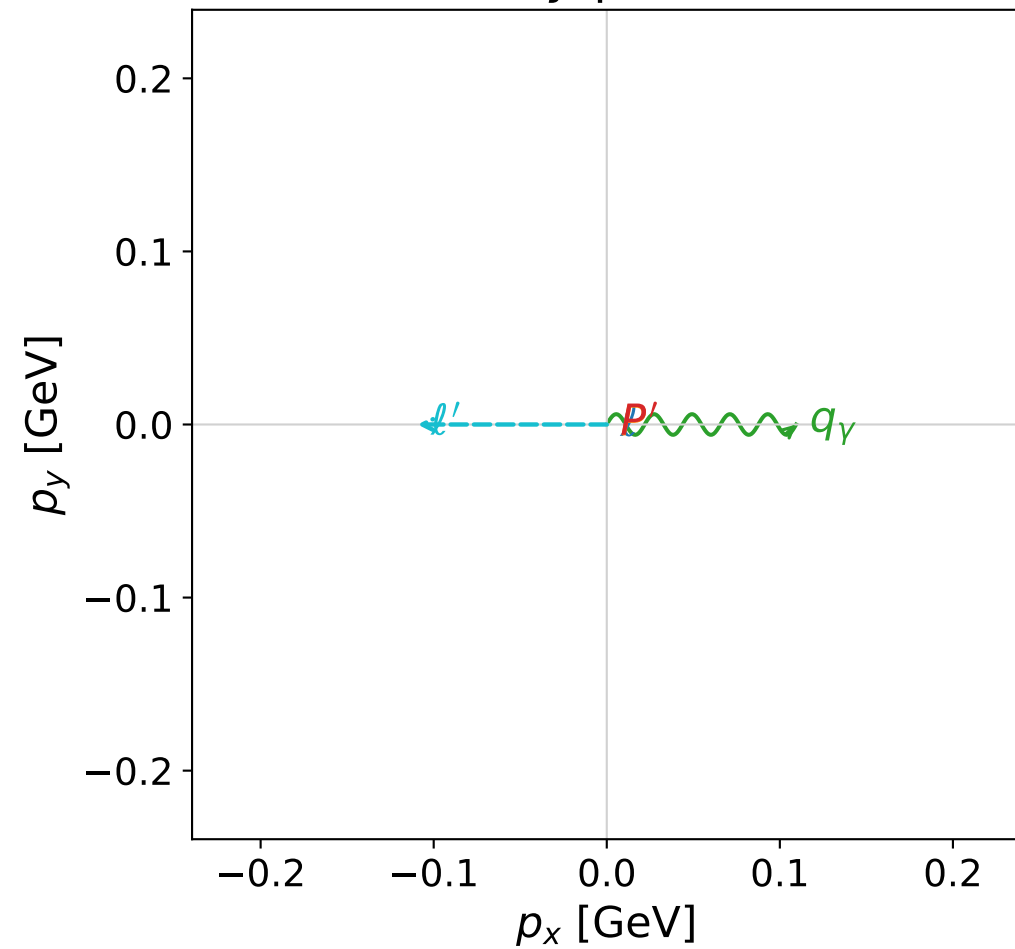


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



x--y plane

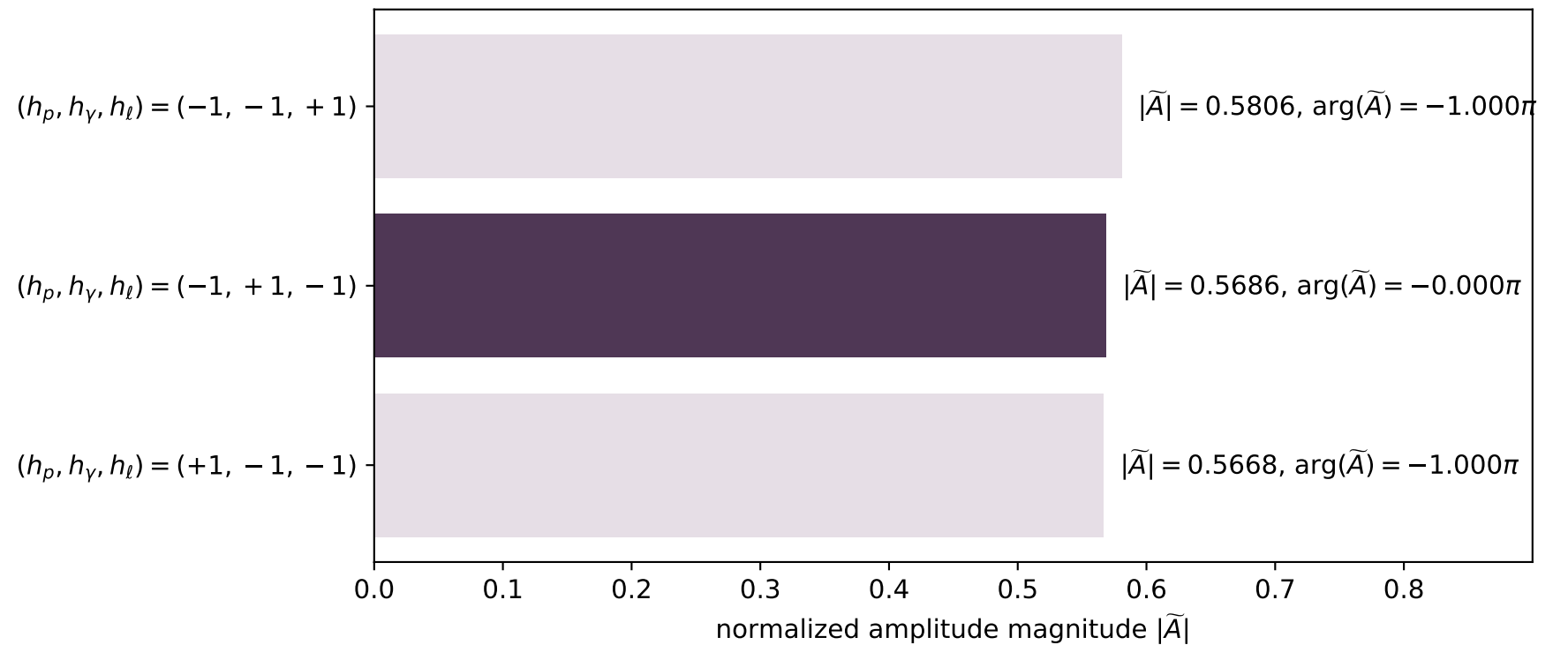


dw_mixing_angles_polarization_06_local_minimum_831
 $D_W=0.0114378$
 region: polarization_cluster_06_local_minimum_831
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0831
 lepton mixing angle: $\alpha_e=2.3807655$ rad
 incoming lepton: $-0.724266|+> +0.689521|->$
 proton mixing angle: $\alpha_p=1.3809873$ rad
 incoming proton: $0.188671|+> +0.98204|->$

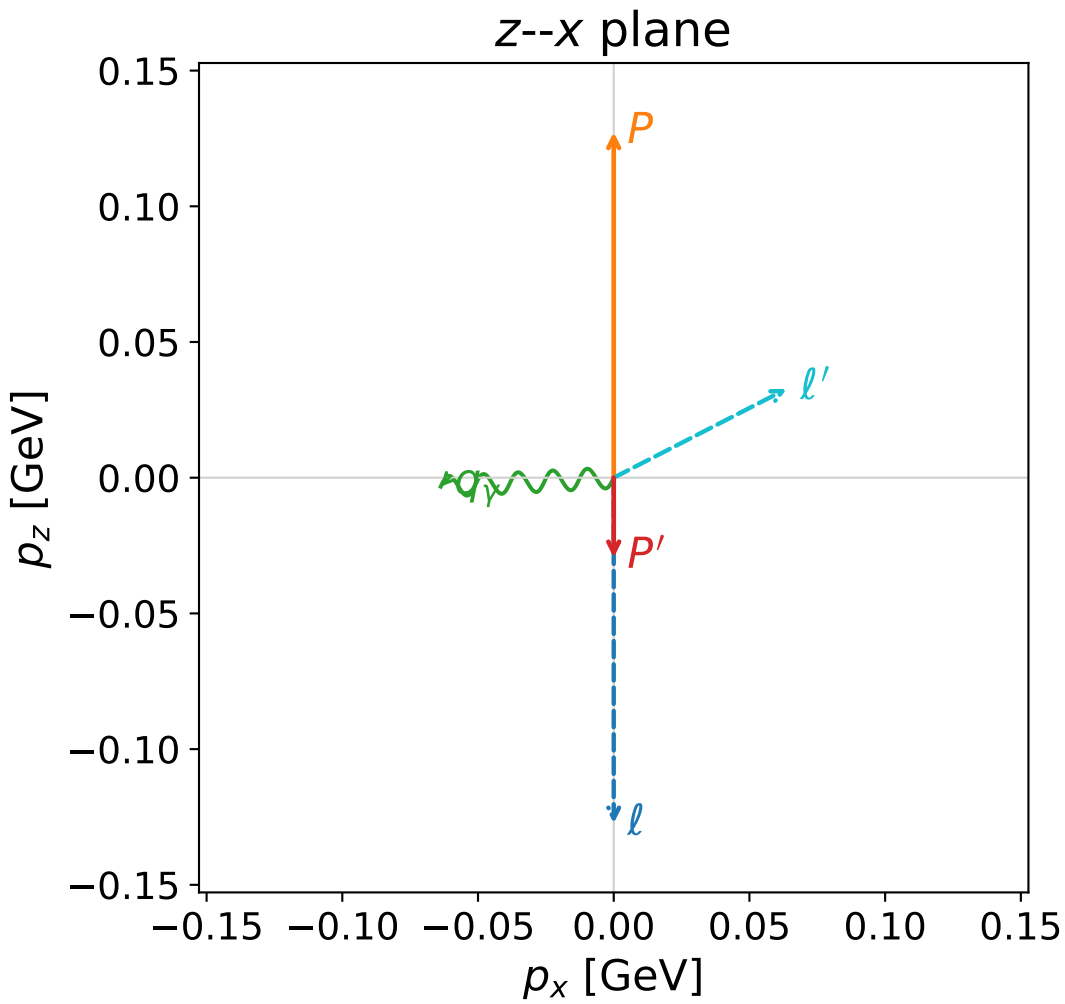
$s=1.34258$, $\sqrt{s}=1.1587$, $|\vec{P}|=0.199679$, $|\vec{P}'|=0.0255397$
 $\theta_{P'}=0$, $\theta_\gamma=1.70042$, $\phi_{P'}=0.802418$, $\phi_\gamma=6.28318$
 $E_\gamma=0.110347$, $\theta_{l'}=1.67348$, $\phi_{l'}=3.14158$
 $Q^2=0.0394268$, $x_B=0.159465$, $t=-0.0298971$, $W^2=1.08766$, $y=0.53431$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.1997$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1997$
 P $E= 0.9590$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1997$
 l' $E= 0.1100$ $p_x= -0.1094$ $p_y= 0.0000$ $p_z= -0.0113$
 P' $E= 0.9383$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0255$
 q_γ $E= 0.1103$ $p_x= 0.1094$ $p_y= -0.0000$ $p_z= -0.0143$

Leading normalized final-state amplitudes
 (N=3, retained=98.2%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

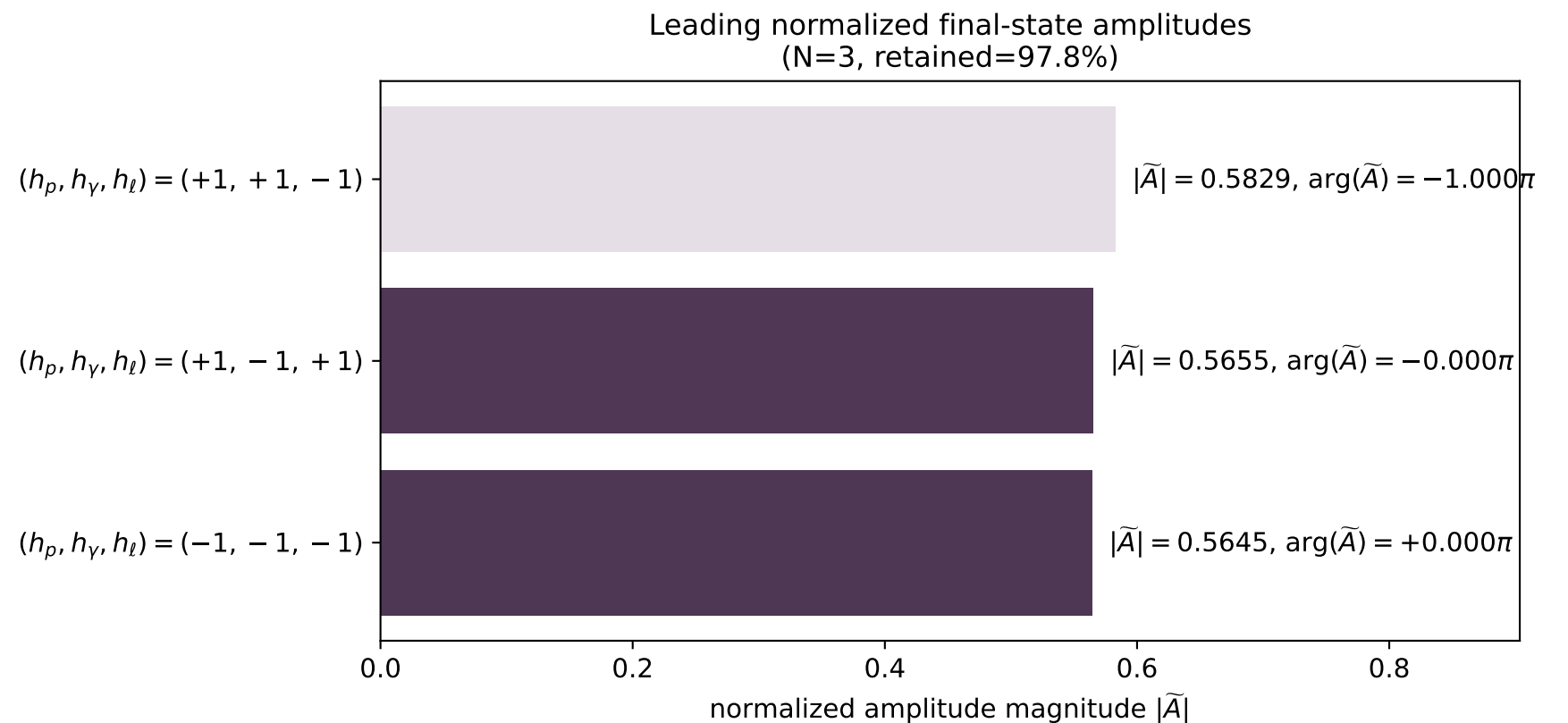
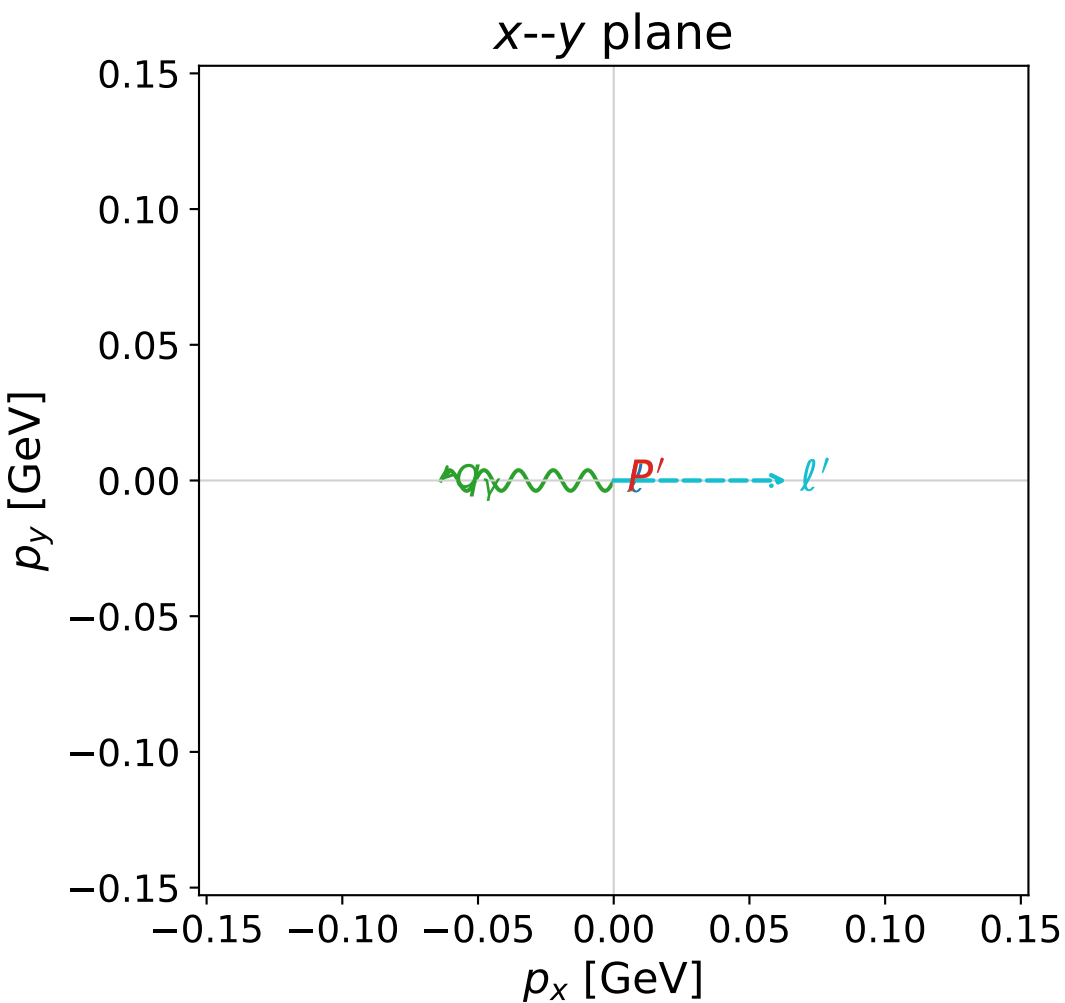


dw_mixing_angles_polarization_06_local_minimum_840
 $D_W=0.0116403$
 region: polarization_cluster_06_local_minimum_840
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0840
 lepton mixing angle: $\alpha_e=2.3185035$ rad
 incoming lepton: $-0.679959|+> +0.73325|->$
 proton mixing angle: $\alpha_p=1.7221857$ rad
 incoming proton: $-0.150812|+> +0.988562|->$

$s=1.15333$, $\sqrt{s}=1.07393$, $|\vec{P}|=0.12733$, $|\vec{P}'|=0.0294481$
 $\theta_{p'}=3.14159$, $\theta_\gamma=1.62221$, $\phi_{p'}=4.91412$, $\phi_\gamma=3.14161$
 $E_\gamma=0.0638214$, $\theta_{\ell'}=1.09642$, $\phi_{\ell'}=1.27813e-05$
 $Q^2=0.0265808$, $x_B=0.181842$, $t=-0.0245132$, $W^2=0.999439$, $y=0.534482$

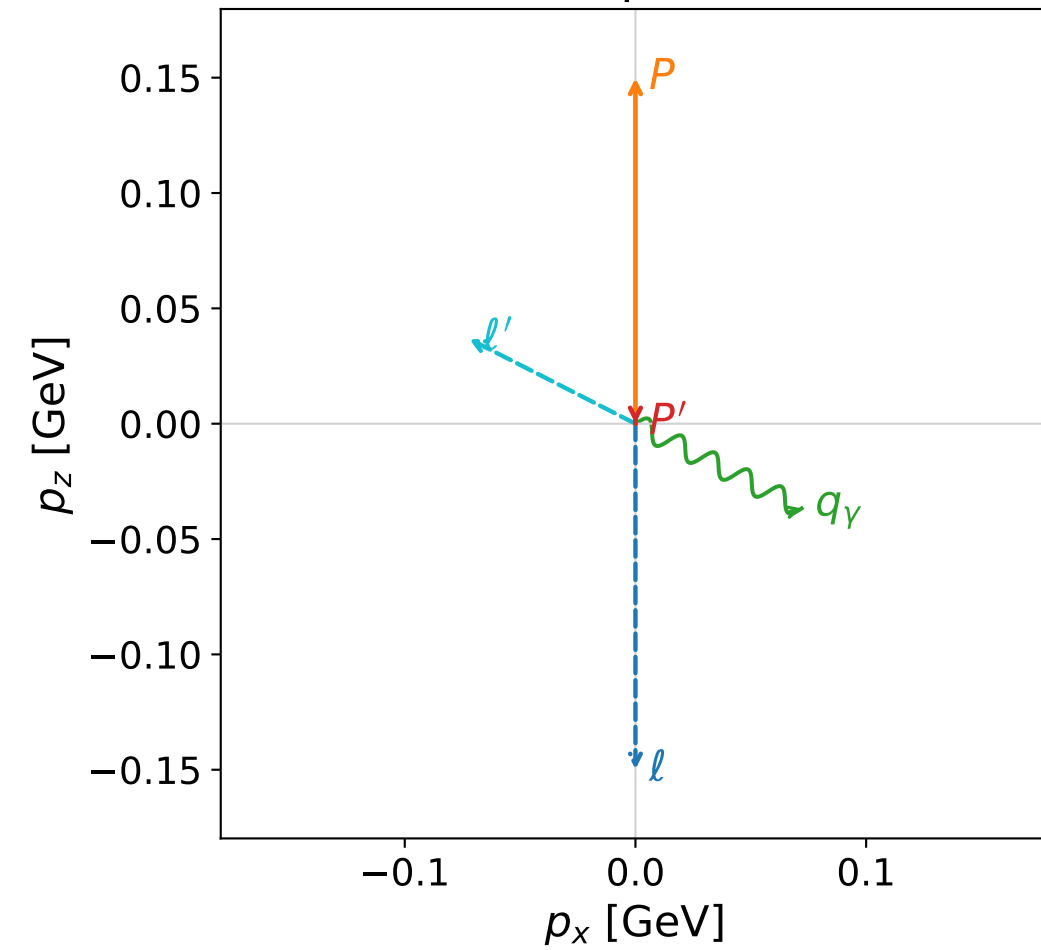
four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ	$E=$	0.1273	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	-0.1273
P	$E=$	0.9466	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	0.1273
ℓ'	$E=$	0.0717	$p_x=$	0.0637	$p_y=$	0.0000	$p_z=$	0.0327
P'	$E=$	0.9385	$p_x=$	0.0000	$p_y=$	-0.0000	$p_z=$	-0.0294
q_γ	$E=$	0.0638	$p_x=$	-0.0637	$p_y=$	-0.0000	$p_z=$	-0.0033



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



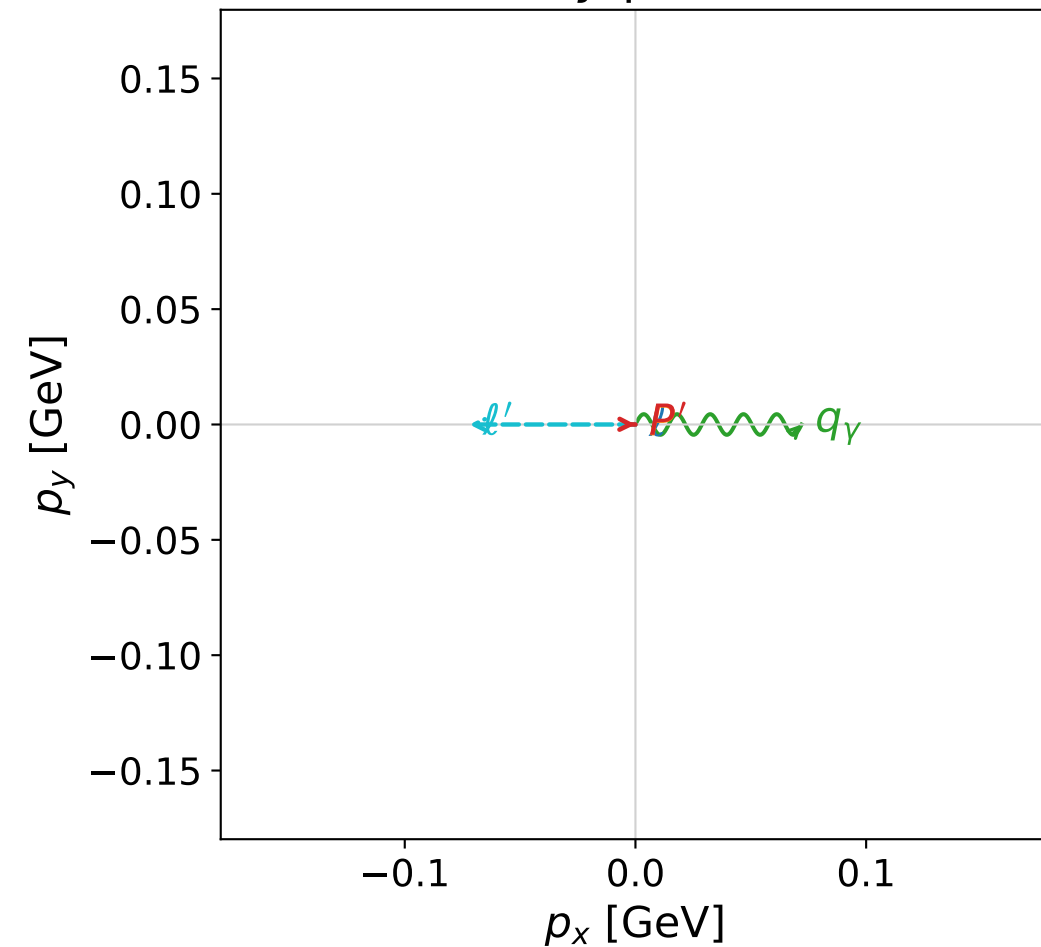
dw_mixing_angles_polarization_06_local_minimum_847
 $D_W=0.0118716$
 region: polarization_cluster_06_local_minimum_847
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0847
 lepton mixing angle: $\alpha_e=2.3406718$ rad
 incoming lepton: $-0.696046|+\rangle +0.717997|-\rangle$
 proton mixing angle: $\alpha_p=1.4421737$ rad
 incoming proton: $0.128268|+\rangle +0.99174|-\rangle$

$s=1.20934$, $\sqrt{s}=1.0997$, $|\vec{P}|=0.149809$, $|\vec{P}'|=8.16231e-06$
 $\theta_{p'}=3.14003$, $\theta_\gamma=2.04218$, $\phi_{p'}=6.26908$, $\phi_\gamma=8.62237e-06$
 $E_\gamma=0.0808463$, $\theta_{\ell'}=1.09932$, $\phi_{\ell'}=3.1416$
 $Q^2=0.035227$, $x_B=0.188486$, $t=-0.022304$, $W^2=1.03151$, $y=0.56722$

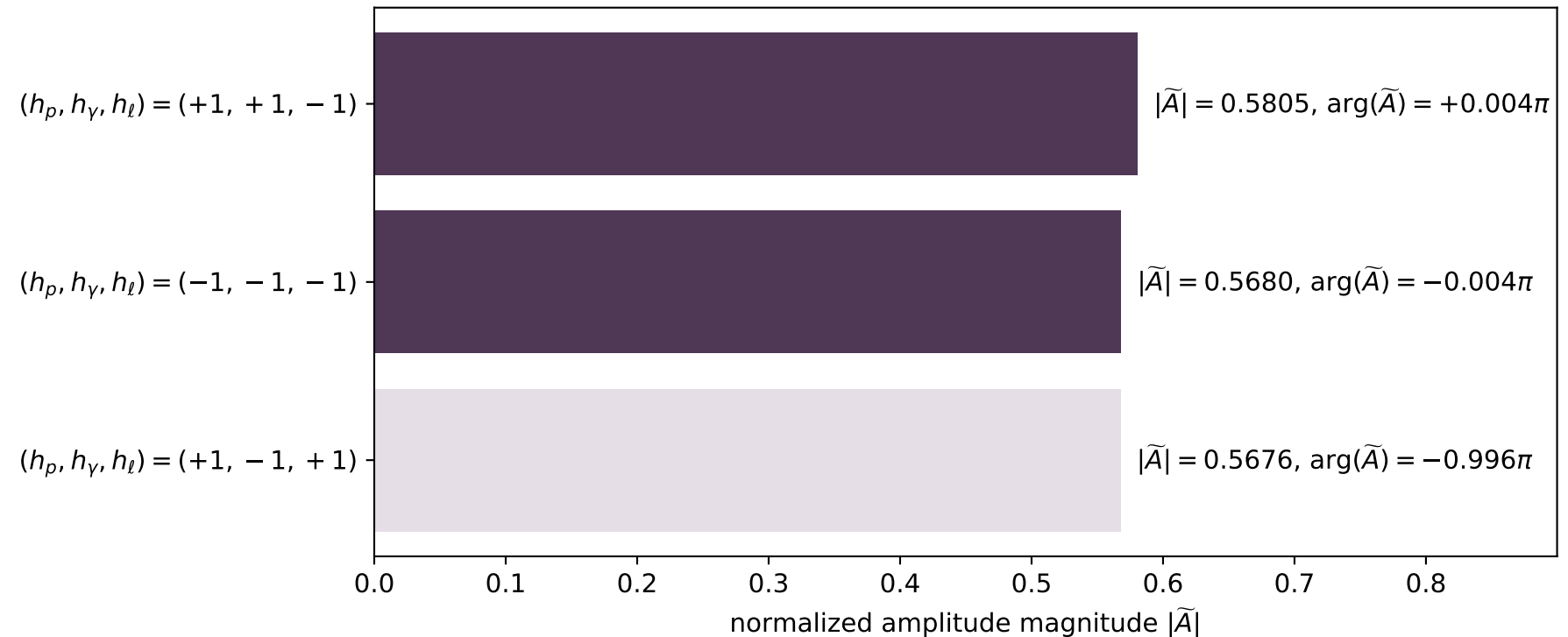
four-momenta $[E, p_x, p_y, p_z]$ GeV:

Particle	E	p_x	p_y	p_z
ℓ	0.1498	0.0000	0.0000	-0.1498
P	0.9499	0.0000	0.0000	0.1498
ℓ'	0.0809	-0.0720	-0.0000	0.0367
P'	0.9380	0.0000	-0.0000	-0.0000
q_γ	0.0808	0.0720	0.0000	-0.0367

x--y plane

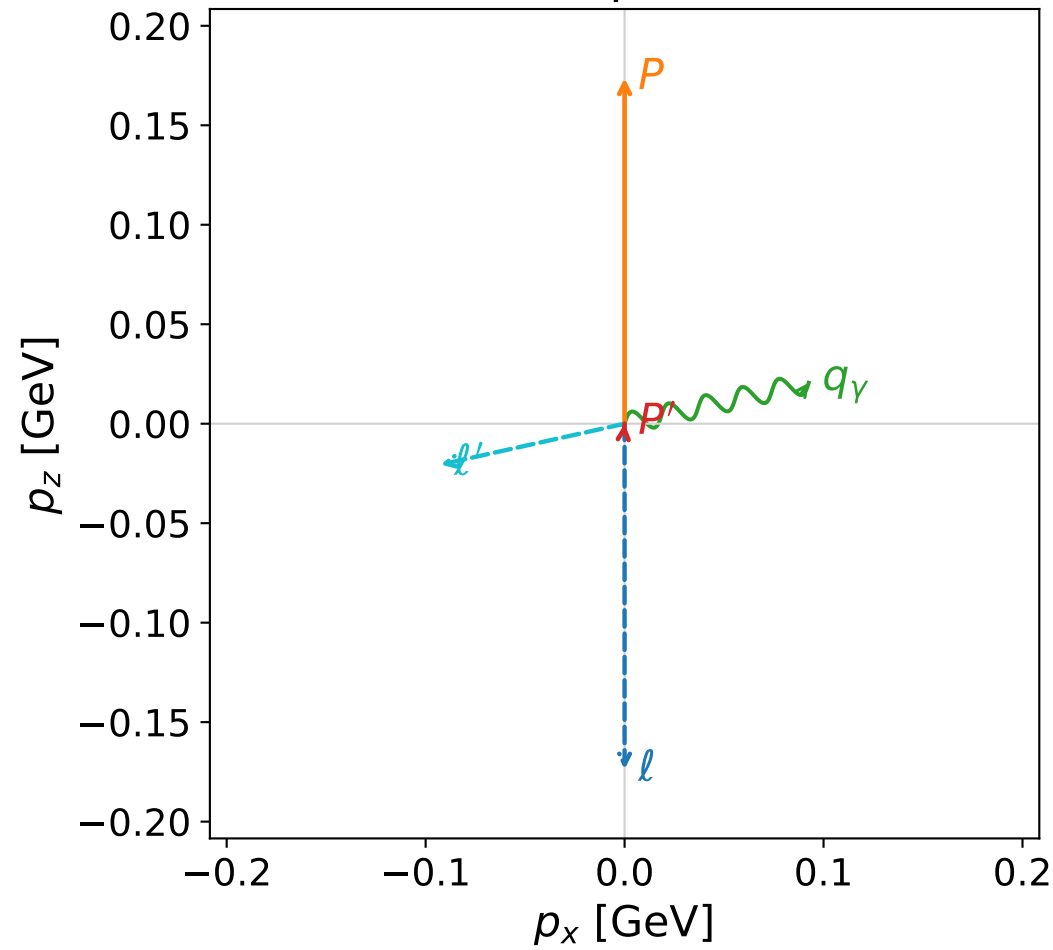


Leading normalized final-state amplitudes
(N=3, retained=98.2%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

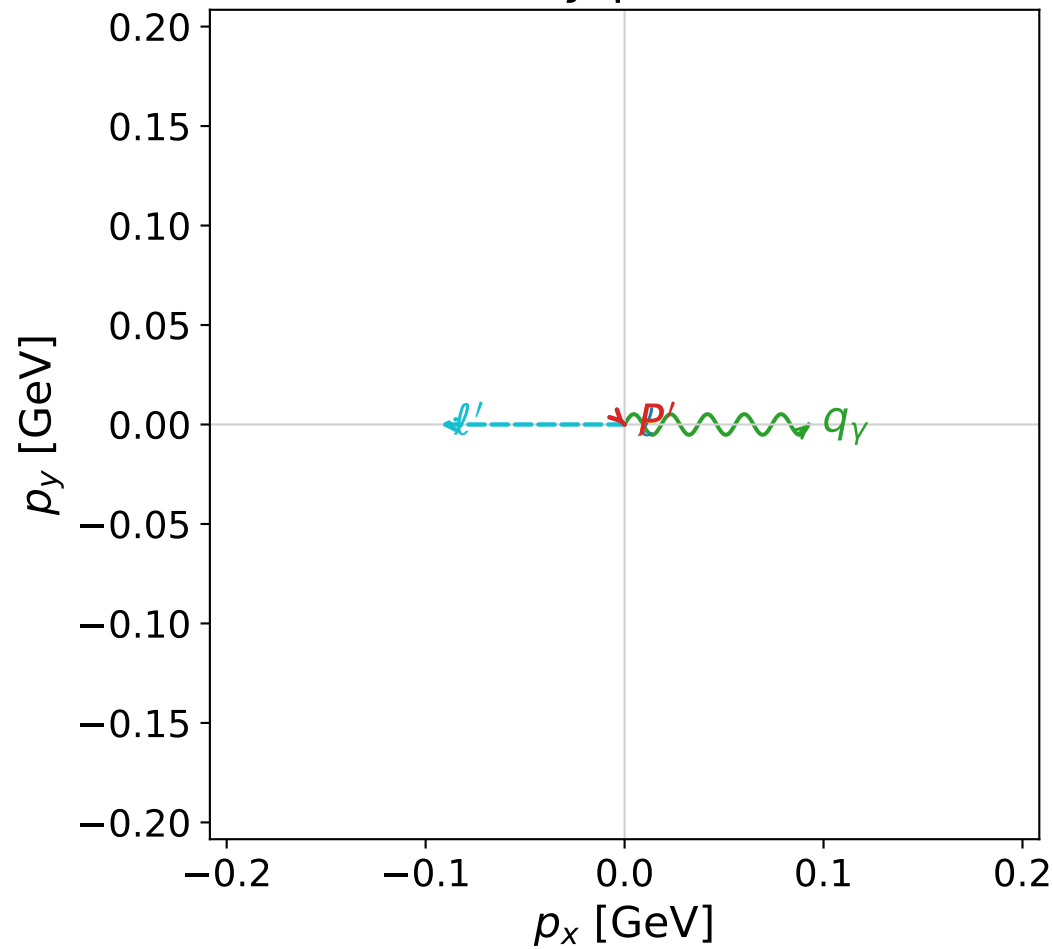


dw_mixing_angles_polarization_06_local_minimum_890
 $D_W=0.0133474$
 region: polarization_cluster_06_local_minimum_890
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0890
 lepton mixing angle: $\alpha_e=2.3874645$ rad
 incoming lepton: $-0.728869|+\rangle +0.684653|-\rangle$
 proton mixing angle: $\alpha_p=1.3908398$ rad
 incoming proton: $0.178987|+\rangle +0.983851|-\rangle$

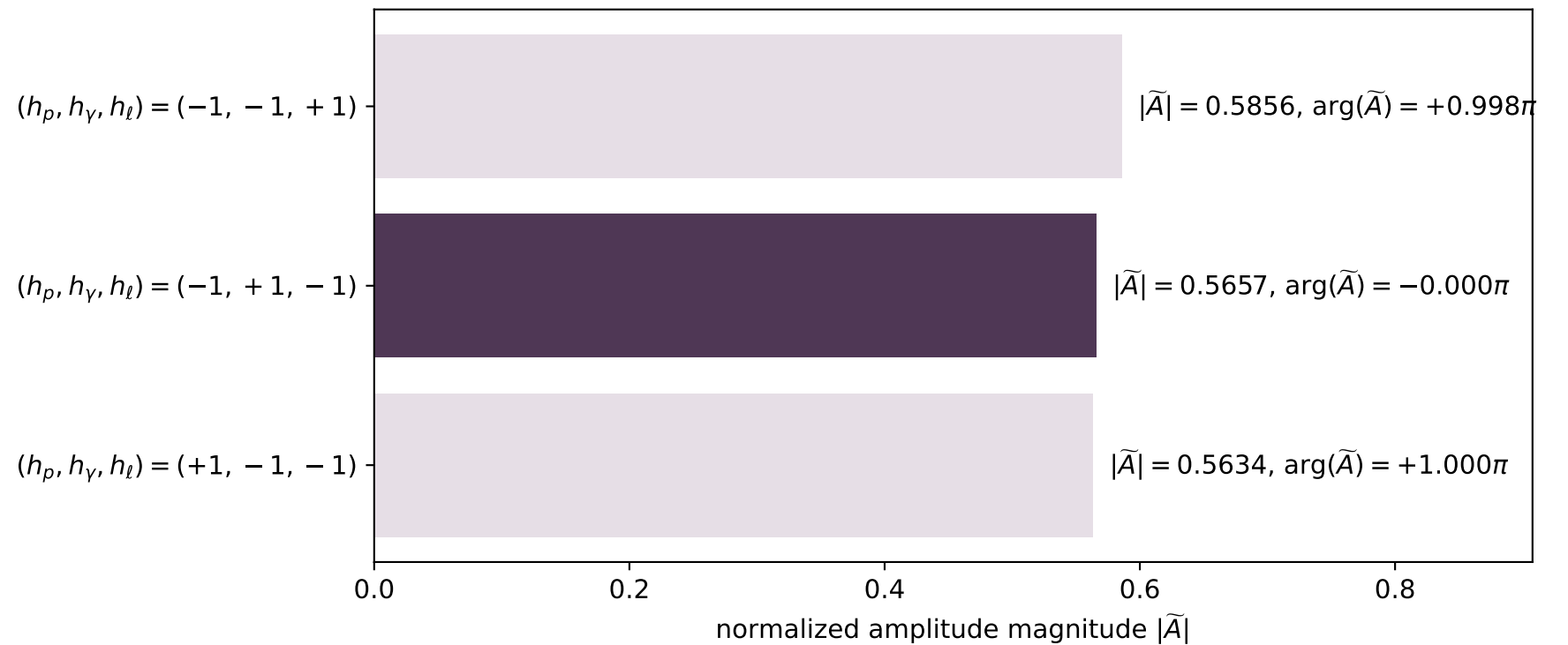
$s=1.2716$, $\sqrt{s}=1.12765$, $|\vec{P}|=0.173702$, $|\vec{P}'|=8.55738e-06$
 $\theta_{p'}=0.117307$, $\theta_\gamma=1.3521$, $\phi_{p'}=5.51155$, $\phi_\gamma=8.8004e-05$
 $E_\gamma=0.0948232$, $\theta_{l'}=1.78958$, $\phi_{l'}=3.14167$
 $Q^2=0.0257928$, $x_B=0.126634$, $t=-0.029915$, $W^2=1.05773$, $y=0.519924$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.1737$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1737$
 P $E= 0.9539$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1737$
 l' $E= 0.0948$ $p_x= -0.0926$ $p_y= -0.0000$ $p_z= -0.0206$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0000$
 q_γ $E= 0.0948$ $p_x= 0.0926$ $p_y= 0.0000$ $p_z= 0.0206$

x--y plane

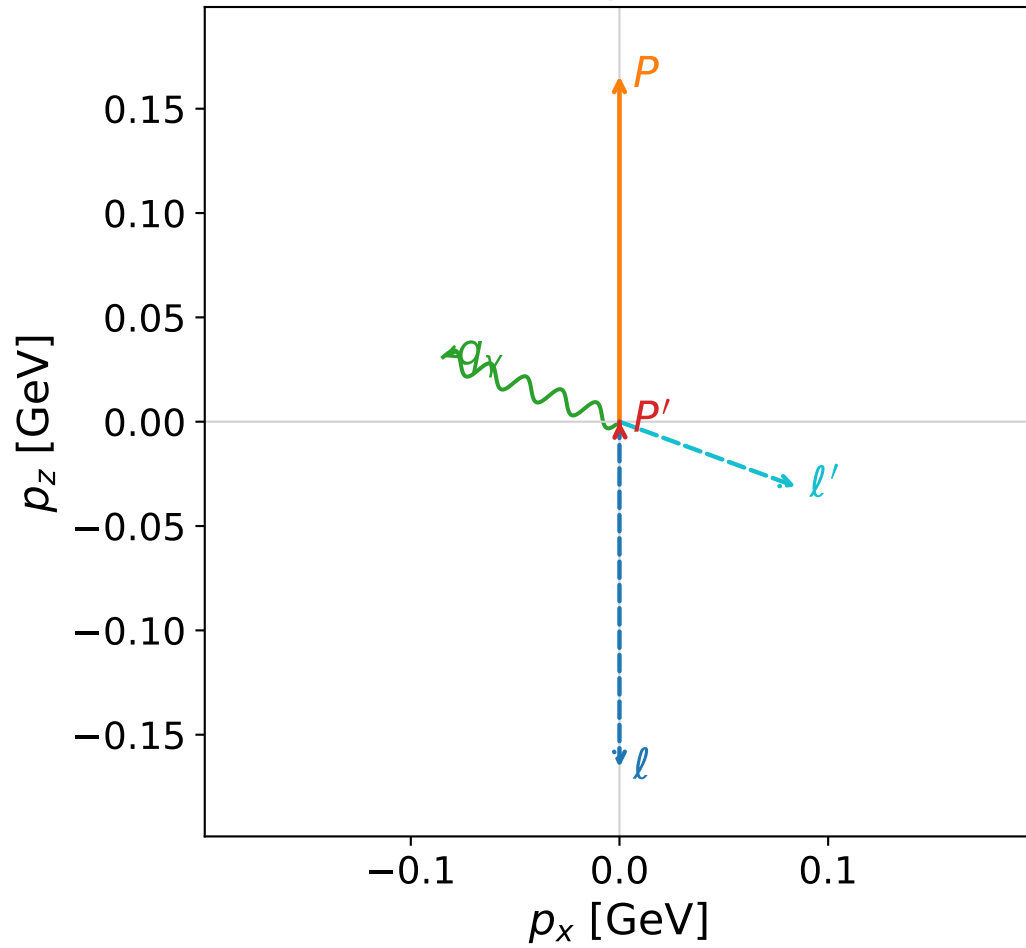


Leading normalized final-state amplitudes
 (N=3, retained=98.0%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

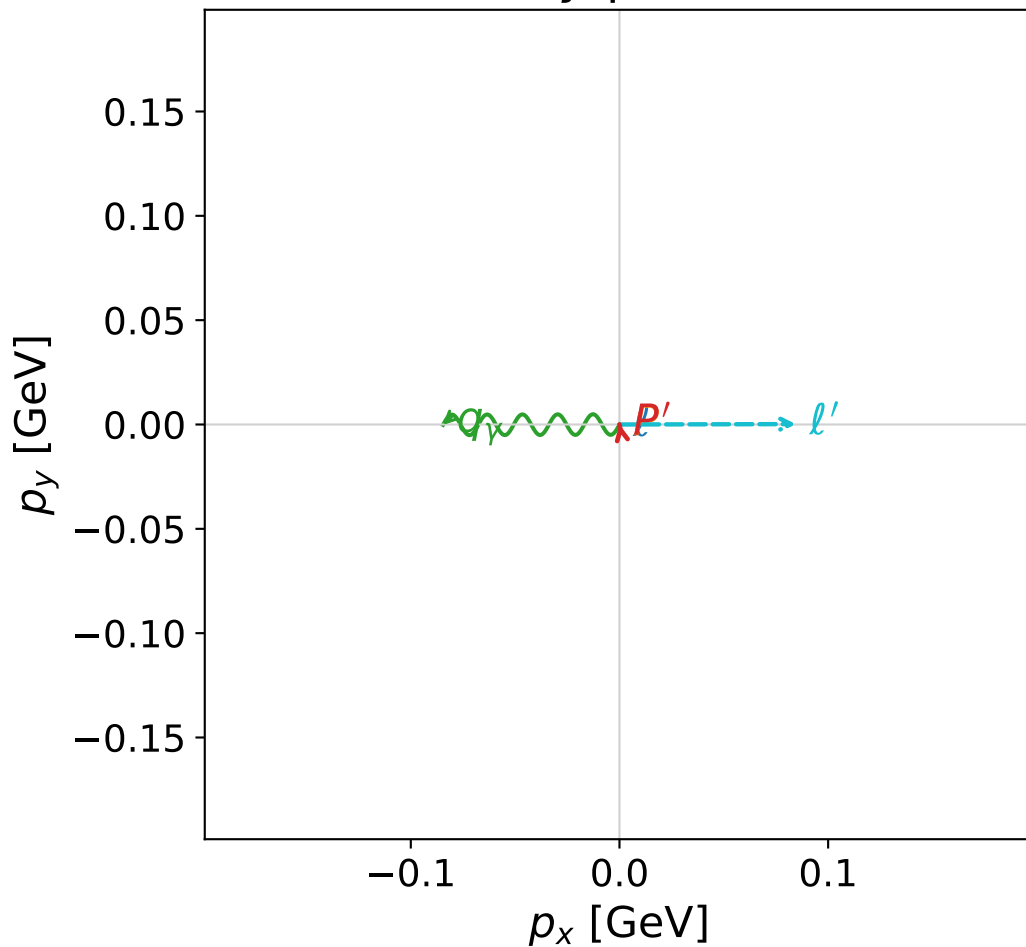


```
dw_mixing_angles_polarization_06_local_minimum_930
D_W=0.0148952
region: polarization_cluster_06_local_minimum_930
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0930
lepton mixing angle: alpha_e=2.3871059 rad
incoming lepton: -0.728623|+> +0.684915|->
proton mixing angle: alpha_p=1.728226 rad
incoming proton: -0.15678|+> +0.987634|->
```

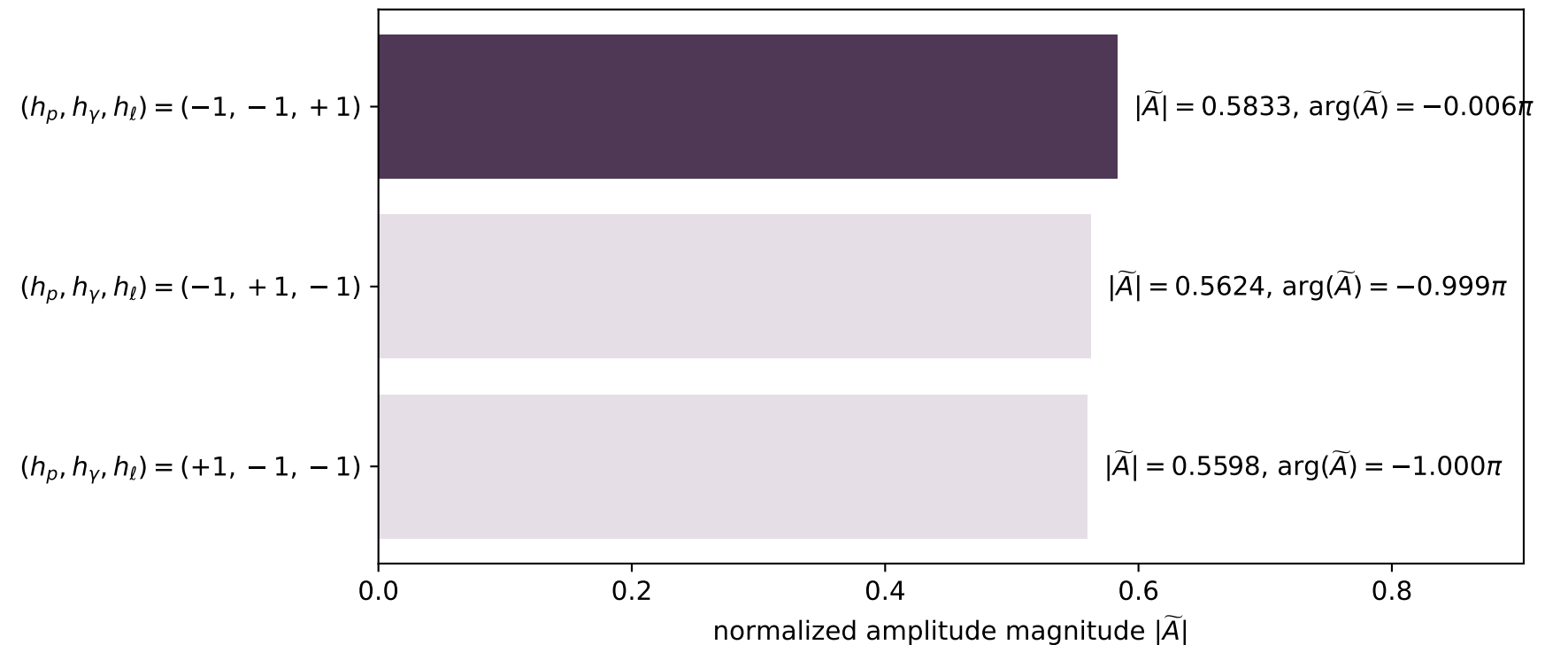
```
s=1.25021, sqrt(s)=1.11813, |P|=0.165617, |P'|=8.61277e-05
theta_p'=0.166897, theta_gamma=1.21964, phi_p'=1.75585, phi_gamma=3.14396
E_gamma=0.0900467, theta_l'=1.92283, phi_l'=0.00219716
Q^2=0.0195491, x_B=0.103724, t=-0.0271904, W^2=1.04877, y=0.508882
```

```
four-momenta [E, p_x, p_y, p_z] GeV:
l      E=  0.1656 p_x=  0.0000 p_y=  0.0000 p_z= -0.1656
P      E=  0.9525 p_x=  0.0000 p_y=  0.0000 p_z=  0.1656
l'     E=  0.0901 p_x=  0.0846 p_y=  0.0002 p_z= -0.0311
P'     E=  0.9380 p_x= -0.0000 p_y=  0.0000 p_z=  0.0001
q_gamma E=  0.0900 p_x= -0.0846 p_y= -0.0002 p_z=  0.0310
```

x--y plane

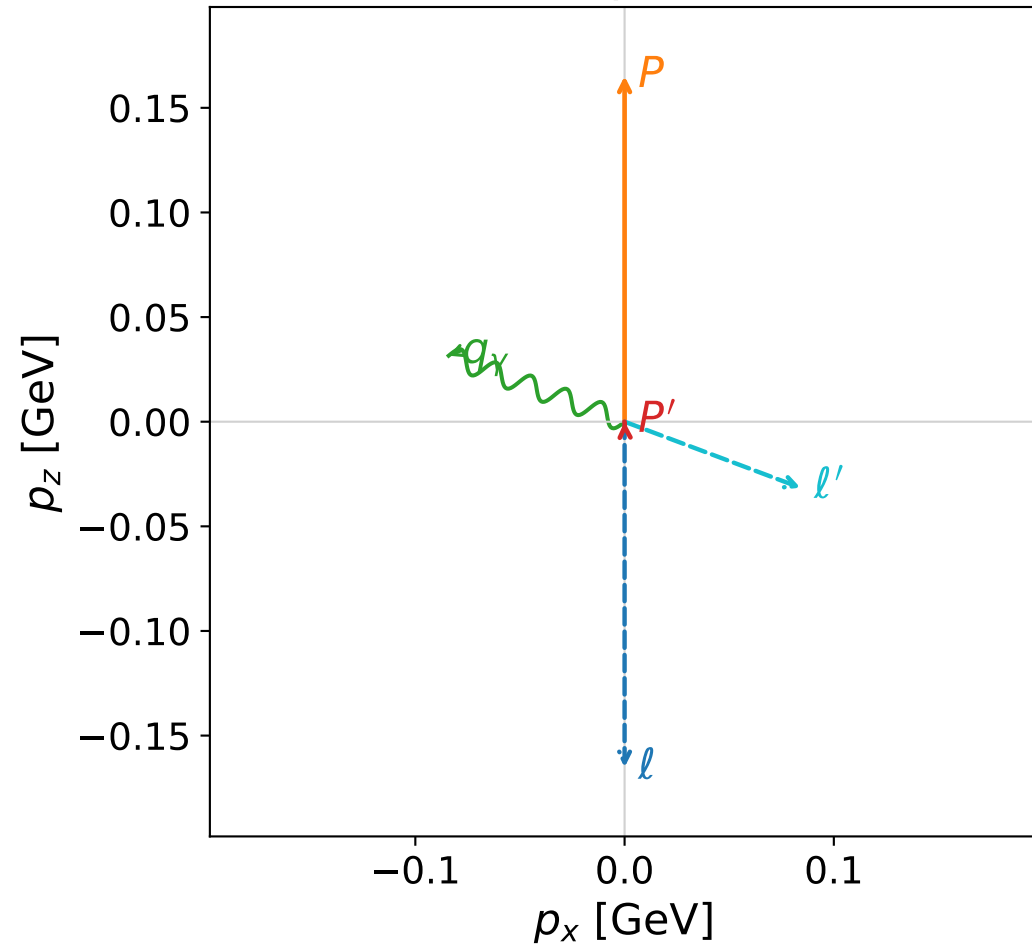


Leading normalized final-state amplitudes
(N=3, retained=97.0%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

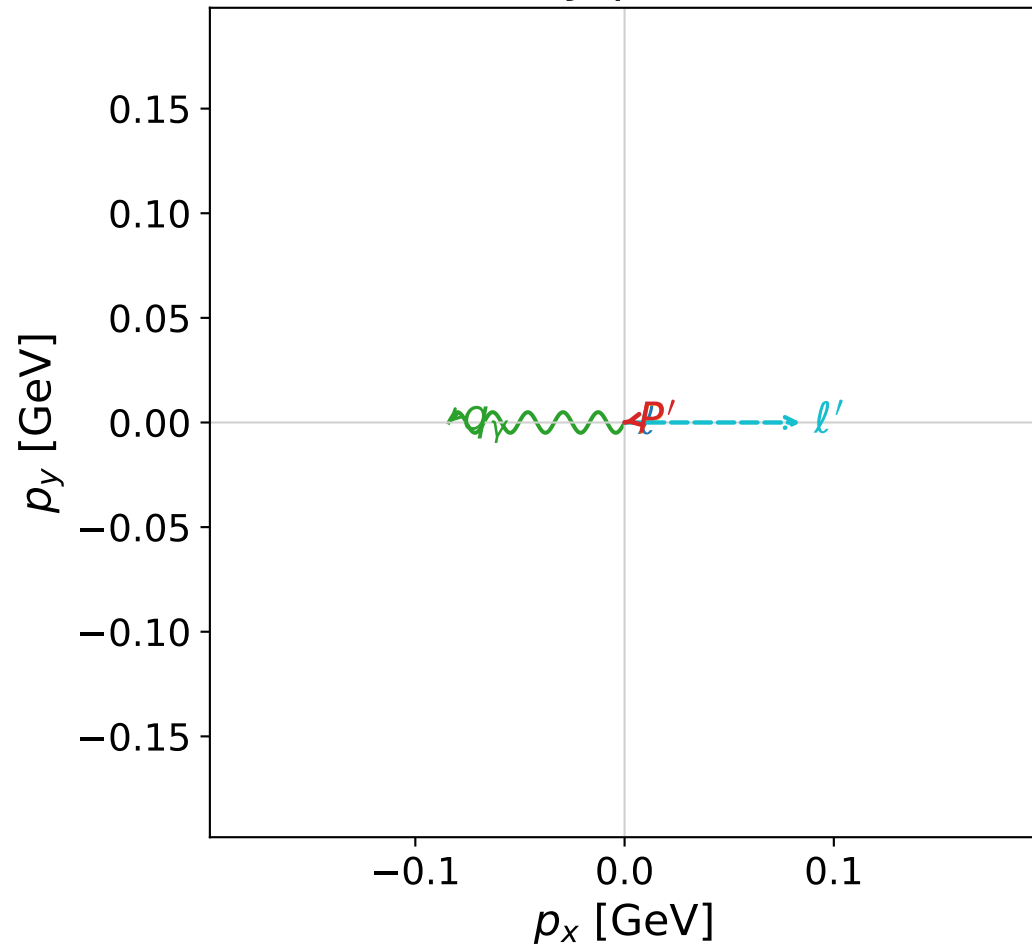


dw_mixing_angles_polarization_06_local_minimum_932
 $D_W=0.0150586$
 region: polarization_cluster_06_local_minimum_932
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0932
 lepton mixing angle: $\alpha_e=2.3873934$ rad
 incoming lepton: $-0.72882|+\rangle +0.684705|-\rangle$
 proton mixing angle: $\alpha_p=1.7256129$ rad
 incoming proton: $-0.154199|+\rangle +0.98804|-\rangle$

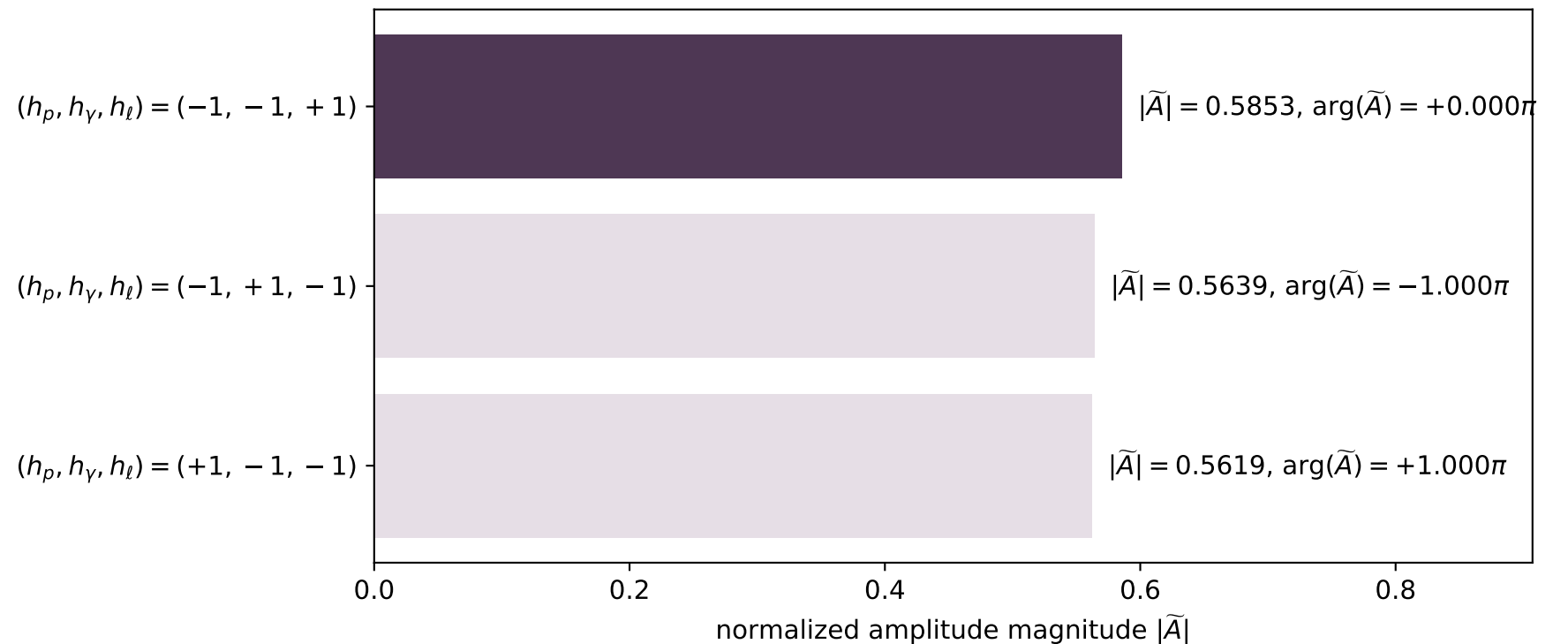
$s=1.249$, $\sqrt{s}=1.11758$, $|\vec{P}|=0.165155$, $|\vec{P}'|=1.2777e-07$
 $\theta_{P'}=0.0340212$, $\theta_\gamma=1.21186$, $\phi_{P'}=3.33858$, $\phi_\gamma=3.14161$
 $E_\gamma=0.0897916$, $\theta_{l'}=1.92973$, $\phi_{l'}=1.74269e-05$
 $Q^2=0.0192406$, $x_B=0.102513$, $t=-0.027068$, $W^2=1.04829$, $y=0.508435$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E=0.1652$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.1652$
 P $E=0.9524$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1652$
 l' $E=0.0898$ $p_x=0.0841$ $p_y=0.0000$ $p_z=-0.0315$
 P' $E=0.9380$ $p_x=-0.0000$ $p_y=-0.0000$ $p_z=0.0000$
 q_γ $E=0.0898$ $p_x=-0.0841$ $p_y=-0.0000$ $p_z=0.0315$

x--y plane

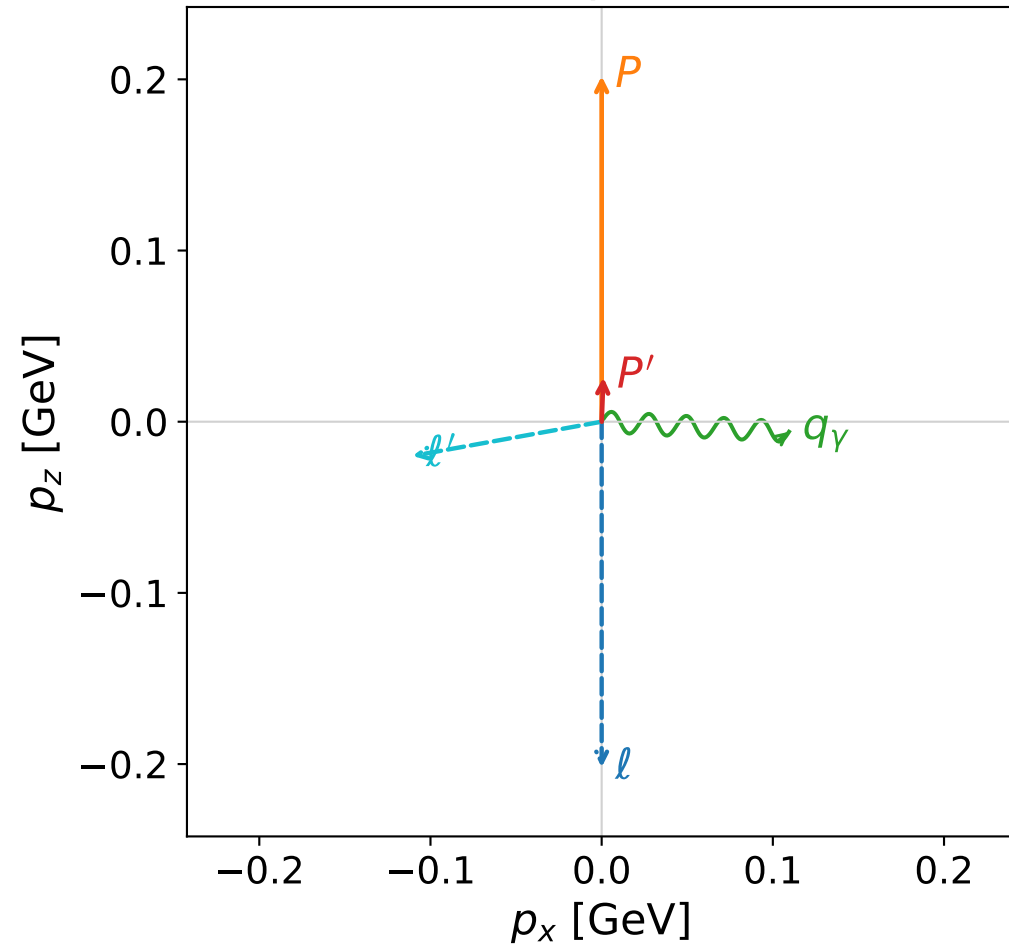


Leading normalized final-state amplitudes
 (N=3, retained=97.6%)

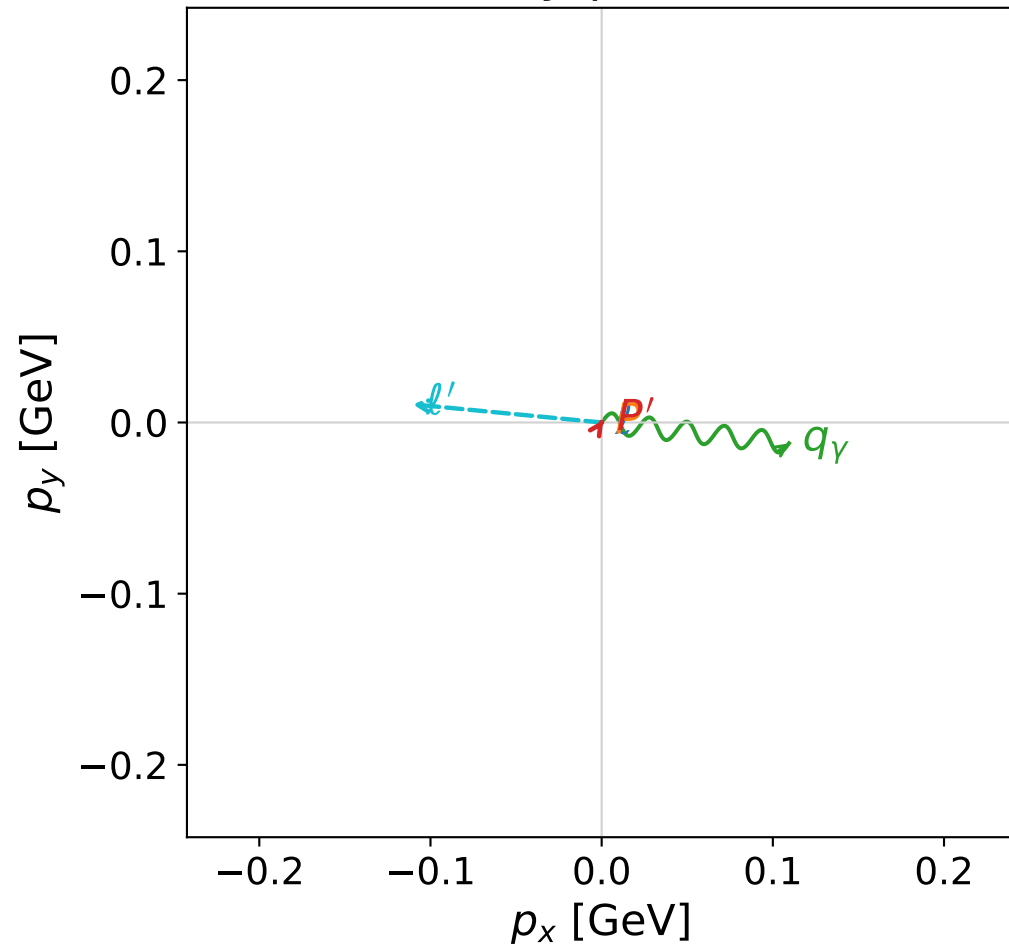


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



x--y plane



dw_mixing_angles_polarization_06_local_minimum_933
 $D_W=0.0151282$
 region: polarization_cluster_06_local_minimum_933
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0933
 lepton mixing angle: $\alpha_e=2.389794$ rad
 incoming lepton: $-0.730462|+\rangle + 0.682954|-\rangle$
 proton mixing angle: $\alpha_p=1.3723144$ rad
 incoming proton: $0.197181|+\rangle + 0.980367|-\rangle$

$s=1.34877$, $\sqrt{s}=1.16137$, $|\vec{P}|=0.201885$, $|\vec{P}'|=0.0259237$
 $\theta_{p'}=0.0702731$, $\theta_\gamma=1.62406$, $\phi_{p'}=0.988205$, $\phi_\gamma=6.17259$
 $E_\gamma=0.110269$, $\theta_{\ell'}=1.74905$, $\phi_{\ell'}=3.04561$
 $Q^2=0.0374493$, $x_B=0.153158$, $t=-0.0305422$, $W^2=1.08691$, $y=0.521433$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.2019$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2019$
 P $E= 0.9595$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.2019$
 ℓ' $E= 0.1127$ $p_x= -0.1104$ $p_y= 0.0106$ $p_z= -0.0200$
 P' $E= 0.9384$ $p_x= 0.0010$ $p_y= 0.0015$ $p_z= 0.0259$
 q_γ $E= 0.1103$ $p_x= 0.1094$ $p_y= -0.0122$ $p_z= -0.0059$

Leading normalized final-state amplitudes
 (N=3, retained=97.5%)

